

Research Methodology and Quantitative Techniques

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Donald Preston



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Preface

The system of collecting data for research projects is known as research methodology. The data may be collected for either theoretical or practical research for example management research may be strategically conceptualized along with operational planning methods and change management. Some important factors in research methodology include validity of research data, Ethics and the reliability of measures most of your work is finished by the time you finish the analysis of your data.

All research begins with a question. Intellectual curiosity is often the foundation for scholarly inquiry. Some questions are not testable. The classic philosophical example is to ask, “How many angels can dance on the head of a pin?” While the question might elicit profound and thoughtful revelations, it clearly cannot be tested with an empirical experiment. Prior to Descartes, this is precisely the kind of question that would engage the minds of learned men. Their answers came from within. The modern scientific method precludes asking questions that cannot be empirically tested. If the angels cannot be observed or detected, the question is considered inappropriate for scholarly research. A paradigm is maintained as much by the process of formulating questions as it is by the answers to those questions. By excluding certain types of questions, we limit the scope of our thinking. It is interesting to note, however, that modern physicists have begun to ask the same kinds of questions posed by the Eastern philosophers. “Does a tree falling in the forest make a sound if nobody is there to hear it?” This seemingly trivial question is at the heart of the observer/observed dichotomy. In fact, quantum mechanics predicts that this kind of question cannot be answered with complete certainty. It is the beginning of a new paradigm.

Quantitative research is used widely in social sciences such as psychology, sociology, anthropology, and political science. Research in mathematical sciences such as physics

is also ‘quantitative’ by definition, though this use of the term differs in context. In the social sciences, the term relates to empirical methods, originating in both philosophical positivism and the history of statistics, which contrast qualitative research methods. Qualitative methods produce information only on the particular cases studied, and any more general conclusions are only hypotheses. Quantitative methods can be used to verify, which of such hypotheses are true. A comprehensive analysis of 1274 articles published in the top two American sociology journals between 1935 and 2005 found that roughly two thirds of these objects used quantitative methods.

Qualitative methodology recognizes that the subjectivity of the researcher is intimately involved in scientific research. Subjectivity guides everything from the choice of topic that one studies, to formulating hypotheses, to selecting methodologies, and interpreting data. In qualitative methodology, the researcher is encouraged to reflect on the values and objectives he brings to his research and how these affect the research project. Other researchers are also encouraged to reflect on the values that any particular investigator utilizes.

Scientific research methodology was very successful at explaining natural phenomena. It provided a systematic way of knowing. Western philosophers embraced this new structure of inquiry. Eastern philosophy continued to stress the importance of the one seeking knowledge. By the beginning of the twentieth century, a complete schism had occurred. Western and Eastern philosophies were mutually exclusive and incompatible.

Then something remarkable happened. Einstein’s proposed that the observer was not separate from the phenomena being studied. Indeed, his theory of relativity actually stressed the role of the observer. Quantum mechanics carried this a step further and stated that the performance of observation could change the thing being observed. The researcher was not simply an observer, but in fact, was an integral part of the process. In physics, Western and Eastern philosophies have met.

This idea has not been incorporated into the standard social science research model, and today’s social science community see themselves as objective observers of the phenomena being studied. However, “it is an established principle of measurement that instruments react with the things they measure.” The concept of instrument reactivity states that an instrument itself can disturb the thing being measured.

The book is designed to be used both as a collection of readings and as an introductory research methods book in its own right.

– *Author*

Research Methodology and Quantitative Methods

RESEARCH METHODOLOGY

The system of collecting data for research projects is known as research methodology. The data may be collected for either theoretical or practical research for example management research may be strategically conceptualized along with operational planning methods and change management. Some important factors in research methodology include validity of research data, Ethics and the reliability of measures most of your work is finished by the time you finish the analysis of your data.

Formulating of research questions along with sampling whether probable or non-probable is followed by measurement that includes surveys and scaling.

This is followed by research design, which may be either experimental or quasi-experimental. The last two stages are data analysis and finally writing the research paper, which is organised carefully into graphs and tables so that only important relevant data is shown.

DEVELOPING THE HYPOTHESIS

The hypothesis is directly related to a theory but contains operationally defined variables and is in testable form. Hypotheses allow us to determine, through research,

if our theory is correct. In other words, does prior work experience result in better grades? When doing research, we are typically looking for some type of difference or change between two or more groups. In our study, we are testing the difference between having work experience and not having work experience on college grades. Every study has two hypotheses; one stated as a difference between groups and one stated as no difference between groups. When stated as a difference between groups, our hypothesis would be, “students with prior work experience earn higher grades than students without prior work experience.” This is called our research or scientific hypothesis. Because most statistics test for no difference, however, we must also have a null hypothesis. The null hypothesis is always written with the assumption that the groups do not differ. In this study, our null hypothesis would state that, “students with work experience will not receive different grades than students with no work experience.”

The null hypothesis is what we test through the use of statistics and is abbreviated H_0 . Since we are testing the null, we can assume then that if the null is not true then some alternative to the null must be true. The research hypothesis stated earlier becomes our alternative, abbreviated H_1 . In order to make research as specific as possible we typically look for one of two outcomes, either the null or the alternative hypothesis. To conclude that there is no difference between the two groups means we are accepting our null hypothesis. If we, however, show that the null is not true then we must reject it and therefore conclude that the alternative hypothesis must be true. While there may be a lot of gray area in the research itself, the results must always be stated in black and white.

STANDARDIZATION

Standardization refers to methods used in gathering and treating subjects for a specific study. In order to compare the results of one group to the results of a second group, we must assure that each group receives the same opportunities to succeed. Standardized tests, for instance, painstakingly assure that each student receives the same questions in the same order and is given the same amount of time, the same resources, and the same type of testing environment. Without standardization, we could never adequately compare groups.

For example, imagine that one group of students was given a particular test and allowed four hours to complete it in a quiet and well lit room. A second group was given the same test but only allowed 30 minutes to complete it while sitting in a busy school lunchroom full of laughing and talking children. If group 1 scored higher than group 2 could we truly say that they did better? The answer is obviously ‘no.’ To make sure we can compare results, we must make everything equal between the two or more groups. Only then could we say that group 1 performed better than group 2.

Standardization of the research methods is often a lengthy process. The same directions must be read to each student, the same questions must be given, and the

same amount of time must be assured. All of these factors must be decided before the first subject can be tested. While standardization refers mainly to the testing situation itself, these principles of ‘sameness’ involve the selection of subjects as well.

SELECTING SUBJECTS

If we want to know if Billy performed better than Sally, or if boys scored higher than girls in our class, or even if Asian children receive higher grades in our school than Caucasian children, the selection of subjects is rather simple. When we are testing the entire population of possible subjects, we are adequately assured that no subject bias has occurred. A population refers to the entire pool of possible subjects. In a classroom or other setting where the entire population is relatively small, testing all subjects may be simple. However, if we are attempting to understand or gain knowledge related to a large population, such as all third grade children, all depressed adults, or all retail employees, gathering and testing everyone would be relatively impossible. In this situation, we would need to gather a sample of the population, test this sample, and then make inferences aimed at the entire population of which they represent.

When determining which potential subjects from a large population to include in our study there are several approaches to choose from. Each of these sampling techniques have its own strengths and, of course, its own weaknesses. The idea behind adequate sampling, however, remains the same: to gather a sample of subjects that is representative of the greater population. The ideal research sample is therefore often referred to as a representative sample.

SIMPLE RANDOM SAMPLE

To assure that the sample of subjects taken from a known population truly represents the population, we could test every subject in the population and choose only those who fall around the mean of the entire population. This technique is usually pointless because doing so means we could just as easily have tested the entire population on our independent and dependent variables. Therefore in order to make sure all possible subjects have an equal opportunity to be chosen, simple random sampling is most often the selection method used.

To choose a random group of 10 students from a class of 30, for example, we could put everyone’s name in a hat and use the first ten names drawn as our sample. In this method, subjects are chosen just as ‘B6’ is chosen in a game of BINGO. This technique can work well with a small population but can be time consuming and archaic when the population size is large. To choose 30 students from a class of 250 students would be easier utilizing technology and what is referred to as a random number table.

A random number table is a computer generated list of numbers placed in random order. Each of the 250 students would be randomly assigned a number between one and 250. Then the groups would be formed once again using a random number generator.

SYSTEMATIC SAMPLE

When a population is very large, assigning a number to each potential subject could also be tiresome and time consuming. A systematic sample is a random sample compiled in a systematic manner. If you had a list of all licensed teachers, for example, and wanted to mail a survey to 200 of them, systematic sampling might be the sampling method of choice. For this example, a page and a teacher number on that page are determined at random. This would represent the first subject and the starting point for choosing the remaining subjects.

A random number would be generated, for example 150. Then every 150th teacher would become a subject until you have selected enough for your study. If you complete the list before selecting enough subjects, you would continue back at the beginning of the list. Once the subjects are selected, the technique of random assignment can be used to assign subjects to particular groups.

STRATIFIED RANDOM SAMPLE

The use of a stratified sample refers to the breaking down of the population into specific subsets before choosing which ones will take part in the study. For example, if you are studying all third grade students in your state, you may want to make sure that every county in your state is represented in your study. If you used a simple random sampling technique, you could conceivably end up with many subjects from one county and no subjects from other counties. A stratified sample allows you to choose your subject pool randomly from a predetermined set of subsets. In this example, we may want to choose 10 subjects at random from each county within the state.

Other subsets can also be used, such as age, race, or socioeconomic background. If you wanted to make sure that there were an equal number of males and females, you could use sex as your subset and then randomly choose the same number of subjects from each subset. This type of sampling is useful when the population has some known differences that could result in different outcomes. For instance, if you already know that 80% of the students are male, you may want to select 40 male students and 10 female students so that your sample represents the breakdown of sex within the population.

CLUSTER SAMPLE

Cluster sampling could be considered a more specific type of stratified sample. When this technique is used, potential subsets of subjects are first randomly eliminated and then the remaining subsets are used to randomly select the sample of subjects to be used in the study. For example, if you are measuring the effect of prior work experience on college grades in a particular state, you may first make a list of all colleges in the state. Then you would randomly select a number of colleges to either include or eliminate in the selection process. Once you have a subset of colleges, you could use the same

technique to randomly include or eliminate the specific classes. From the remaining classes, you would then randomly select a group of students with work experience and a group of students with no work experience to be placed in your two groups.

NON-PROBABILITY SAMPLE

Non-probability refers to a group of subjects chosen based on their availability rather than their degree of representativeness of the population. Surveys are often done in this manner. Imagine going to the local mall to gather information about the buying habits of mall shoppers. Your subject pool does not represent all mall shoppers but rather those mall shoppers who happen to walk by your location on that day. The same would hold true for a survey over the phone or via mail. Those who respond to your questions or return the mailed survey do not necessarily represent the population at large. Instead, they represent the population who was home and was willing to respond to your questions or those who took the time to complete and return the survey.

While at first glance this method seems unprofessional, it allows for the gathering of information in a short amount of time. It is not considered standardized research and would be scrutinized if submitted to a professional journal, but it does have its place. If you've ever visited a web site and seen a survey, you might have felt compelled to click on the results link. When watching a news programme you may have not changed channels because you are waiting for the results of a survey that will be reported at the end of the programme. We are highly interested in these 'informal' polls and using a Non-probability sample is a quick way to gather large amounts of information in a relatively short amount of time.

TESTING SUBJECTS

Once you determined your variables, applied the concept of standardization, and selected your subjects, you are almost ready to begin the testing process. The concept of testing refers to the application or analysis of your independent and dependent variables. If there is any manipulation of the subjects in your study, it occurs during this phase. Before testing any human subject, however, some type of consent form is necessary. Consent forms basically describe the study, how the results will be used, and any possible negative effects that may occur. They also give the subject the right to withdrawal from the study at any time without consequence. Your specific hypothesis does not need to be disclosed but each subject must be made aware of any general concerns and be able to ask questions before testing can begin..

If your hypothesis, for example, asked if there is a difference in effectiveness of different treatments for depression, you might assign your subjects to one of several different groups: cognitive therapy, dynamic therapy, humanistic therapy and possibly no therapy. Each subject would likely be tested prior to the study to determine a baseline for his or her level of depression and would then begin a predetermined and standardized

treatment plan. Because you are standardizing your study, each subject should get identical treatment short of the independent variable. In other words, the only thing you want to be different between the groups is the type of therapy received.

The no therapy group would be considered a control group and may participate in some type of non-therapy related activity while the other subjects receive therapy. This group is used to determine if time plays a role in the decrease of depressive symptoms. Without a control group you couldn't say that any particular therapy was more helpful than no therapy because subjects may have improved merely because of some outside factor unrelated to treatment. These factors are called extraneous variables, and control groups, along with randomization, help to keep the impact of these variables to a minimum.

ANALYZING RESULTS

The specific analysis performed on the subjects depends on the type of subjects, the type of questions being asked, and the purpose of the research. When we gathered and tested all possible subjects from a known population we would use descriptive statistics to analyse our results. Descriptive statistics require the testing of everyone in the population and are used to describe the qualities of the population in numerical format. For example, we could say that the mean score on an IQ test for all third graders at Jefferson High School is 102. We could also state that there is no difference between the IQs of boys and girls within our subjects if the data support these claims.

When we are using a sample of subjects smaller than the entire population, we must make some inferences using what we call inferential statistics. Like any inferences, we also assume a certain degree of error when making determinations about a population based on a sample of that population. Because of this, the results of inferential statistics are often stated within a predetermined level of confidence.

If we found that the mean of one group was 10 points higher than the mean of a second group in our work experience and college grades study, we could not assume that the population means are identical. We could, however, state that the means of the entire population are likely to differ by five to 15 points or that there is a 95% probability that the means of the entire population differs by ten points. In this sense, we are predicting the scores of the entire population based on the scores of our sample and stating them within a range or a predetermined level of confidence. This allows us to include the likely error that occurs whenever an inference is made.

DETERMINING SIGNIFICANCE

The term significance when related to research has a very specific role. Significance refers to the level of certainty in the results of a study. We can say that our subjects differed by an average of ten points with 100% certainty because we personally witnessed

this difference. To say that the population will differ is another story. To do this, we must determine how valid our results are based on a statistical degree of error. If we find, through the use of inferential statistics, that the grades of those with and without work experience are different we must state the estimated error involved in this inference. While the standard acceptable error is 5%, it can be as high as 20% or as low as 0.1%.

The amount of error to be accepted in any study must be determined prior to beginning the study. In other words, if we want to be 95% confident in our results, we set the significance level at .05. If we want to be 99% confident, our significance level is set at .01.

We can then state that there is a difference in the population means at the 95% significance level or at the 99% significance level if our statistics support this statement. If our statistics estimate that there is 10% error and we said we would accept only 5%, the results of our study would be stated as 'not significant.'

When determining significance, we are saying that a difference exists within our acceptable level of error and we must therefore reject the null hypothesis. When results are found to be not significant, the only option available is to accept the null hypothesis.

COMMUNICATING RESULTS

Results of a study are disseminated in many forms. The highest level of communicating results is often in a peer-reviewed professional journal. Peer-reviewed refers to a group of professionals in a particular field who read all submissions and publish only those that meet the highest degree of scrutiny and applicability. When errors are found in the sampling of subjects, the statistical analysis, or the inferences made, the study will often be rejected or returned to the author for revisions. Published substances in peer-reviewed would likely be the best source for research when you begin looking into your theory.

Results of research studies are also disseminated through textbooks, book chapters, conferences, presentations, and newsletters. For example, a study comparing the average salary in a particular county might be published in the local newspaper or in a brochure for the chamber of commerce. Our study of non-traditional students and work experience might be summarized in a board meeting of the college's department of student retention or published in a trade journal such as the "Journal of Higher Education."

Some studies are never released, especially if the results do not add to the already available research. Other studies are meant only to provide direction for larger studies. Our study of college students may be used only to determine if a larger study is likely to result in important findings. If we get significant results then a larger study, including a broader subject pool, may then be conducted. These types of studies are often called pilot studies because the goal is not to gather knowledge about the population, but rather to guide further research in a particular area.

REPLICATION

Replication is the key to the support of any worthwhile theory. Replication involves the process of repeating a study using the same methods, different subjects, and different experimenters. It can also involve applying the theory to new situations in an attempt to determine the generalizability to different age groups, locations, races, or cultures. For example, our study of non-traditional students may be completed using students from another college or from another state. It may be changed slightly to add additional variables such as age, sex, or race to determine if these variables play any role in our results.

Replication, therefore, is important for a number of reasons, including:

- Assurance that results are valid and reliable;
- Determination of generalizability or the role of extraneous variables;
- Application of results to real world situations; and
- Inspiration of new research combining previous findings from related studies.

PUTTING IT ALL TOGETHER

We asked the question, 'do college students with work experience earn better grades than those without work experience.' Knowing the steps involved in doing research and now having a basic understanding of the process, we could design our experiment and with fictional results could determine our ceases and how to report our findings to the world. To do this, lets start with our theory and progress through each of the ten steps.

DETERMINING A THEORY

Theories are developed through our interaction with our environment. For our particular theory, we observed that older college students tend to perform better on classroom tests than younger students.

As we attempt to explain why, we developed our theory that real world work experience creates a motivation in students that allows them to perform better than students without this motivation. Our theory, therefore, states that prior work experience will result in higher grades.

Defining Variables

Every experiment has an independent and a dependent variable. The independent variable is what we start with; it refers to the separation of our groups. In our case, we want to look at prior work experience so the presence or absence of this would constitute our experimental groups. We may place those students who have been in the workforce for more than one year in group 1 and those with less than one year in group 2.

Our dependent variable is our outcome measure so in our case we are looking for a difference in class grades. To operationally define the variable grades, we might use the

final course average as our outcome measure. If the independent and dependent variable(s) are difficult to determine, you can always complete the following statement to help narrow them down: The goal of this study is to determine what effect _____ (IV) has on _____ (DV). For us, the goal is to determine what effect one year or more of prior work experience has on course average.

DETERMINING HYPOTHESIS

When we plug our variables into our original theory we get our research hypothesis. Simply stated, Students with one or more years of prior work experience will receive higher final course averages than students with less than one year of prior work experience. Since statistical analysis often tests the null hypothesis or the idea that there is no difference between groups, our null hypothesis could be stated as: Final course averages of students with one or more years of prior work experience will not differ from final course averages of students with less than one year of prior work experience.

STANDARDIZATION

To make sure that each subject, no matter which group they belong to, receives the same treatment, we must standardize our research. While this may be difficult in the real world, our goal is to get as close as possible to the ideal.

Therefore, we may choose to gather subjects from a general psychology class since this is a class required of most students and will not be affected by college major. We may further decide to research only those students who have a specific instructor to keep the instruction between the two groups as similar as possible. Remember, our goal is to assure, at least as much as possible, that the only difference between the two groups is the independent variable.

SELECTING SUBJECTS

Because our population consists of all college students, it will be impossible to include everyone in the study. Therefore we need to apply some type of random selection. Since we want to use only those students who have the same instructor, we may ask all of this instructors students, prior to any teaching, how much work experience they have had.

Those who report a year or more become the potential subject pool for group 1 and those who have less than one year become the subject pool for group 2.

We could, at this point decide to include all of these subjects or to further reduce the subjects randomly. To reduce the subject pool we could assign each student in each group a random number and then choose, at random, a specific number of students to become subjects in our study. For the purpose of this example, as suggested, randomly choose 20 students in each group to participate in our study.

TESTING SUBJECTS

Since we are not applying any type of treatment to our subjects, this phase in the procession can be omitted. If we were determining if the teaching styles of different instructors played a role in grades, we would randomly assign each student to a teacher. In that case, teaching style would become an independent variable in our study.

ANALYZING RESULTS

Our original question asked if final averages would be different between our two groups. To determine this as suggested, look at the mean of each group. Therefore, as suggested, add up the averages of the 20 subjects in each group and divide each of these by 20.

If, after comparing the means of each group, we find that group 1 has a mean of 88 and group 2 has a mean of 82 then we can descriptively state that there is a six-point difference between the means of the two groups. Based on this statistic, we would then begin to show support for our alternative hypothesis and can progress to the next step.

DETERMINATION OF SIGNIFICANCE

Our goal was not to describe what their averages were, but rather to make inferences about what is likely happening in the entire population. We must therefore apply inferential statistics to our results to determine the significant or lack of significant findings. As suggested, set our confidence level at 95 per cent and then apply statistical analysis to our results to see if the difference of six points with a sample size of 40 is significant.

Imagine that we did find a significant difference. In this case we could say that with a 95% confidence level, students with one year of more work experience receive higher averages than those with less than one year of work experience. Since the null hypothesis, which stated that no difference exists between the two groups, was not correct, we must reject it. And by rejecting the null, we automatically accept our alternative hypothesis.

COMMUNICATING RESULTS

When communicating the results of our study we need to do several things. We need to make a case for why we did this research, which is often based on our literature search. We then need to report the process we took in gathering our sample and applying the treatment. We can then report our results and argue that there is a difference between the two groups and that this difference is significant enough to infer it will be present in the entire population.

Finally, we must evaluate our research in terms of its strengths, weaknesses, applicability, and needs for further study. In terms of strengths, we might include the rigors of gathering subjects and the fact that we used a random sample of students.

We may argue that the statistical methods used were ideal for the study or that we considered the recommendations of previously completed studies in this area. Weaknesses might include the small sample size, the limited pool from which our sample was gathered, or the reliance on self-reported work experience.

To discuss applicability and needs for further studies we could suggest that more studies be completed that use a broader base of subjects or different instructors. We could recommend that other variables be investigated such as student age, type and location of college, family educational history, sex, race, or socioeconomic background.

We might even suggest that while our findings were significant they are not yet applicable until these other variables are investigated.

REPLICATION

The final step in any research is replication. This can be done by us but is most often completed by other researchers based on their own review of the literature and the recommendations made by previous researchers. If others complete a similar study, or look at different variables and continue to find the same results, our results become stronger. When ten other studies agree with ours, the chances are greatly improved that our results were accurate.

If ten other studies disagree with our findings then the validity of our study will be, and most certainly should be, called into question.

By replicating studies and using previously gained knowledge to search for new answers, our profession continues to move forward. After all, we used the ideas of other researchers to design our research, and future researchers may incorporate our findings to make recommendations in their research. The cycle is never ending and allows for perpetual seeking of new knowledge.

SCOPE OF RESEARCH

NATIONAL INNOVATIVE CAPACITY: MODELING, MEASURING AND COMPARING NATIONAL CAPACITIES

National innovative capacity is the ability of a country to produce and commercialise a flow of innovative technology over the long term.

It depends on:

- The strength of a nation's common infrastructure;
- The cluster-specific innovation environment.
- The quality of linkages.

This research strand aims at building innovation indexes and measuring various dimensions of national innovation capacities.

For instance:

- *Strategic capacity:* It deals with the ability to mobilise and concentrate resources under some centralised decision making processes to achieve a critical scientific or technological objective.
- *Revolutionary capacities:* It deals with the ability to shift resources out of areas of lower and into areas of higher productivity and greater yield. This is a capacity to manage transitions. The difficulty is that such a capacity involves various dimensions which can be conflicting.

DESIGNING EFFICIENT INCENTIVE SYSTEMS FOR INVENTION AND INNOVATION: INTELLECTUAL PROPERTY RIGHTS, PRIZES, PUBLIC SUBSIDIES

One central problem in the economics of knowledge is the design of incentive systems that both reward inventors/knowledge producers and encourage dissemination of their output. Several scholars have described the two regimes that allocate resources for the creation of new knowledge: one is the system of granting intellectual property rights, as exemplified by modern patent and copyright systems, the other is the open science regime, as often found in the realm of pure scientific research and sometimes in the realm of commercial technological innovation, often in infant industries.

A large range of issues have to be addressed to elucidate the problem of designing efficient incentive systems:

- What is the best solution in case of particular kind of new technologies?
- What is the nature of the tension that arises when the two systems come up against each other?
- How designing proper incentive systems to encourage research and innovation in areas of high social return and low private profitability?
- In what condition a prize-based reward system provides a more efficient solution than granting intellectual property rights?
- Is there an economic case for granting intellectual property rights in the domain of research tools, instruments, basic knowledge?

RESEARCH IN EPFL LABS: NEW ECONOMICS OF SCIENCE

CEMI will be at the forefront of the College to develop and undertake research in the field of “economics of science” with EPFL as the main case. In this perspective, several topics are obvious:

- Assessing the impact of organizational practice on the productivity of university technology transfer offices
- Measuring the social value of basic research and the local spillovers. Accounting for the effects associated with mobility
- Scale, scope and spillovers: the determinants of research productivity in several fields

- Exploring the role of patents in knowledge transfer from EPFL
- Exploring the effect of the patenting of research tools and biomedical innovation: transfer opportunities and social costs
- Access policy for large scale research instrument, data bases.

All these topics should give rise to research design in close collaboration with the other EPFL Schools in order to benefit from the great opportunity to be located in an Institute of Technology.

These projects will be designed in close collaboration with Jan-Anders Manson, vice-president for Innovation and Knowledge Transfer.

NEW R&D METHODS AND THE PRODUCTION OF RELIABLE KNOWLEDGE IN SECTORS WHICH LAGGED BEHIND

Unequal access to pertinent knowledge bases may well constitute an important condition underlying perceptible differences in the success with which different areas of endeavour are pursued within the same society, and the pace at which productivity advances in different sectors of the economy during a given historical epoch. Today, it remains astonishing to observe the contrast between fields of economic activity where improvements in practice are closely reflecting rapid advances in human knowledge - such as is the case for information technologies, transportation, and certain areas of medical care —and other areas where the state of knowledge appears to be far more constraining.

The fact is that knowledge is not being developed to the same degree in every sector. A major policy concern is to understand the factors at the origin of such uneven development, and to implement a proper strategy in order to fill the gap between sectors with fast knowledge accumulation processes and those in which these processes remain weak.

To summarise, rapid and effective creation of know-how is most likely to occur when the following conditions converge:

- Practice in the field needs to be well specified, sustainable, replicable, imitable;
- There needs to be ability to learn from experience and experiment;
- The ability to experiment offline, with less expense than that would be involved in online experimentation, and to gain reliable information relevant to online use, greatly facilitates progress.
- A strong body of “scientific” knowledge greatly facilitates effective offline experimentation, and also quick and reliable evaluation of varying practice online.

Part of the problem in sectors which are lagging behind deals with the limited ability to conduct experiments. The main research issue here is to analyse the impact of new experimental methods and design, which have the potential to profoundly transform the way reliable knowledge is produced in these sectors. For instance, one of the most significant developments in modern medicine has been the randomised controlled trial,

the significance and use of which grew rapidly after its application to tuberculosis in the 1940s. Today the RCT is widely treated as the evidential ‘gold standard’ for demonstrating ‘what works’ and what is medical ‘best practice’. Education might be the next sector to be profoundly transformed through the application of RCTs. The growth of RCTs as an approach in educational research has been pushed forward by three important factors: computers, statistical techniques and demand for accountability in both practice and research. There is, therefore, a favourable context. The question is whether this new feature can change and transform the way knowledge is produced and distributed in a sector like education.

NEW MODELS OF INNOVATION: OPEN, DISTRIBUTED SYSTEMS AND THE ROLE OF USERS

This project involves the contribution of users in the innovation process not only in terms of sending market signals but also in terms of actively contributing to the modification of the product.

This project emphasizes, therefore, the functional source of innovation: while an innovation is considered a manufacturer’s innovation when the developer expects to benefit by selling it, an innovation is a user innovation when the user expects to benefit by using it.

This research aims at understanding the capabilities and limitations of user innovation processes, which involve quite often an open and distributed system. Its advocates claim that user innovation, involving freely revealing, is an efficient means of producing socially desirable innovation and maximizing “spillovers,” or knowledge transfer/leakage. The generation of innovation by users may be a complement or it may compete with innovations produced by manufacturers. In its role as a complement, user innovation may extend the diversity of products without endangering market positions of manufacturers and may help manufacturing firms to mitigate information asymmetry problems vis-à-vis future market needs. As a competitor, user innovation may offer products that better meet user needs.

The model involves two major deviations from the private investment model of innovation, which assumes that manufacturers innovate in products and processes to improve their competitive position and that returns to innovation result from excluding other manufacturers from adopting it. First, users of technologies, rather than manufacturers, are often the innovators. Second, user-innovators often freely reveal the proprietary knowledge they have developed at their private expense.

A host of empirical studies, mainly conducted by Eric von Hippel, his research group at MIT and his colleagues, show that user innovation is an important economic phenomenon. It constitutes the main source of knowledge in some sectors or an important contributor in others.

Deepening our understanding of the conditions leading to user innovation and of its economic impact is, therefore, a relevant issue:

- For a better assessment both of intangibles and intellectual capital at the firm level and of innovation capacities at the national level; and
- For a better understanding of some new organizational forms, such as user communities, which appear to be becoming more relevant in a knowledge society.

Thus our main research questions are the following:

- What are the different channels through which user innovations influence the economy and how should manufacturers adapt and respond to user innovations?
- What kinds of learning processes/ dynamic capabilities do user innovations enable across product/ technological generations?
- What kind of policy issues and challenges pertain to user innovation? Given the fact that user innovations contribute significantly to productivity growth and national competitiveness, what kinds of policy should be devised to promote them.

OTHER ISSUES TO BE DEVELOPED

- *The economics of knowledge policy:* While it is relatively easy to provide a long list of policy recommendations which are of some relevance in the context of the knowledge economy, it is far more difficult to develop the welfare economics of knowledge investment in order to build a framework for addressing policy issues.
- *Methodology for the optimal allocation of R&D funds to new technologies:* How does the R&D manager maximize the probability of developing a commercializable technology over a specific period.

LIMITATIONS OF RESEARCH

The limitations of the research are first addressed by discussing the purposes of the research, then discussing the limitations of the data management, data analysis, and the validity of the analysis.

PURPOSES OF RESEARCH

The “trustworthiness” of research depends on “What counts as knowledge?”. The general purposes of research can include knowledge production, understanding, and prediction.

This study has focused on the first two purposes: production of applied knowledge and process knowledge, both of which are suitable for developmental research. Applied knowledge is context-specific, useful for the solution of practical problems, while process knowledge is usually specified in terms of models. A qualitative approach is called for in the study of context and contextual influences. A second purpose of research is

promoting understanding of a process; in this case, the instructional approach and the underlying phenomena, including participation and dialogue. A secondary aspect of understanding is a better understanding of what it requires in using the development cycle as a form of inquiry.

DATA MANAGEMENT

Data management involves the procedures used for a systematic, coherent process of data collection, storage and retrieval for the purpose of high quality, accessible data, the documentation of analysis, and retention of data. The design decisions, implementation, and evaluation of the instructional design course were documented in eight data sources: working logs, electronic mail, syllabi, conferences, draft ID projects, completed ID projects, course evaluations, and self-evaluations. These data sources, which were based on observations, interviews, or documents, were event-driven, meaning that they served our instructional needs to watch, ask, and examine.

These observations, interviews, and documents were in place prior to the conceptual framework of the study. As a result, the data sources were not as complete, tightly defined, or structured across the six cases if they had been researcher-driven. Some data sources, such as syllabi, course evaluations, and self evaluations, evolved to suit the learning needs of the students.

However, because we had presented at research conferences, we had collected and stored data for each case, as well as conducted analysis with most of the data sources, although using different methodologies. These research efforts can be regarded as interim analyses in which we became familiar with procedures in recording observations and personal conferences, as well as retaining and analyzing documents. Over the six cases, we came to have a better understanding of the instructional setting, being sensitive to research opportunities and more systematic in our data collection and management efforts, but also retaining instruction and responsiveness to learners as our top priority.

DATA ANALYSIS

Within-Case Analysis

Data displays, structured summaries, and charts allowed a condensed view of the data sources and revealed that some further analysis was needed, such as coding of structured summaries to reveal themes as well as to identify exceptions and differences.

Across-Case Analysis

A means to report the changes in design decisions, implementation, and evaluation of the model across the six Cases. "Each case has a specific history-which we discard at our peril-but it is a history contained within the general principles that influence its

development". This summary attempted to preserve the uniqueness of each case, yet also make comparisons along the developmental cycle. In an effort to extend external validity, what participants' "did, said, or designed," were examined in multiple settings.

The processes of participation and dialogue within participation were examined in six different, developmental configurations and can be viewed as a replication of the focus of the study. A set of generalizations on how the model was implemented, as well as conditions necessary for its use. The danger to this generalization is that "multiple cases will be analysed at high levels of inference, aggregating out the local webs of causality and ending with a smoothed set of generalizations that may not apply to any single case". The goal was to better understand the overall processes at work across the cases; in this case, teacher and student thinking, participation and dialogue, and teacher responsiveness, represented within design decisions, class implementation, and model evaluation. We did not average, for example, course evaluation results, as one way to avoid misinterpretation and superficiality and to preserve case uniqueness.

Analytic Validity

In traditional instances of qualitative data collection and analysis, the research "shifts between cycles of inductive data collection and analysis to deductive cycles of testing and verification".

In this study, sources of data were already in place prior to conceptualizing a conceptual study framework. However, in this study the details of the conceptual framework and the subsequent data analysis of the six cases cycled back and forth to realise more appropriate matches of methodology and method to existing data sources and research objectives. The analytic cycle for this study could be better described as one which moved between conceptual framework, case analysis, and being clear as to the purpose of the study. Although being clear as to the purpose of the study is preferable before constructing a methodology, such purposes are not always clear based on the complexity of processes to be studied, amount of data, and personal involvement over time.

Potential shortcomings in this research that are sources for bias include the large amount of data which may have led to missing important information or overweighting some findings due to focusing on a particular and large set of data. Personal involvement with the course also increases the possibility that our recorded observations in working logs highlighted particular incidents while ignoring others. On the other hand, the working log recorded observations or design decisions that would have been lost to our collective memories over the five years of involvement with the course. Personal involvement as a co-instructor also implied a danger in being selective and overconfident with some data. Another shortcoming was not checking descriptions with each case of students and additional peer review outside of the co-instructor.

To address these shortcomings, we used multiple data sources for triangulation to achieve an agreement of one data source with another. Multiple sources of data, such as

working logs, e-mail, and syllabi, also provided different strengths and complemented each other. Syllabi compactly recorded design decisions, while working logs and e-mail documented our thinking that influenced these design decisions. The data sources were a mix of student-generated and instructor-generated data. A significant amount of time and effort was spent in carefully assembling the data reduction notebooks, which included data displays of structured summaries and charts.

The effort was made towards a goal of being able to track in the documents the reasoning behind descriptions and summary and the generalizations made about the model.

During the analysis of these data sources we looked for contrasts, comparisons, and exemplars and reported these during the data reduction so as not to filter out outliers and extreme instances. For example, in the ID project analysis we added a column to record any unique features of the project that might not be addressed by the completeness, consistency, and coherency criteria. Replication of the conceptual framework across multiple cases helped to provide evidence that what was described in one Case was based upon the details of the instructional approach and uniqueness of the setting and participants.

We attempted to remain "descriptive" and to report what decisions were made for the ID course, what occurred during the implementation of the reflexive approach, and systematically report the evaluation of the model in terms of student performance on ID projects, their perceptions of their learning and the course and data informing our responsiveness to learner needs. Another means of addressing verification of findings and ceases was an "auditing" by the dissertation chair who was also co-instructor in the cases under study. Through periodic reviews of methodology and analysis, numerous inconsistencies in design decisions, for example, were identified and prompted for clarification. Feedback also prompted me to clarify procedures used to analyse the different data sources. This feedback, although one of the responsibilities of a doctoral chairperson, is another aspect of the reflexive stance that we had used on previous research: the need to assume regular, ongoing, and self-conscious documentation of our teaching. The working logs also served as a "reflexivity journal" to record these efforts.

TYPES OF RESEARCH

Research can be classified by purpose or by method. If we categorise it by purpose, it would fall into two major categories: Basic Research and Applied Research, while in case of method, it would be deductive research and inductive research.

BASIC RESEARCH

Also called Pure or fundamental Research, it is undertaken for increase in knowledge. There is no direct benefit as it is a research for the sake of research.

It is conducted to satisfy any curiosity such as:

- What makes things happen,
- Why society changes and
- Why social relations are in a certain way.

In fact, it is the source of most new theories, principles and ideas. Basic research rarely helps anyone directly. It only stimulates new ways of thinking. The main motivation is to *expand man's knowledge*. There is absolutely no commercial value to the discoveries resulting from such research.

However, in the long run, it forms the basis of applied research or development commercial products. If basic work is done first, then applied spin-offs often eventually result from this research.

As Dr. George Smoot of LBNL says, "People cannot foresee the future well enough to predict what's going to develop from basic research. If we only did "applied research", we would still be making better spears."

To sum up, basic research is purely theoretical to increase our understanding of certain phenomena or behaviour but does not seek to solve any existing problem.

APPLIED RESEARCH

It is use of basic research or past theories, knowledge and methods for solving an existing problem. It deals with practical problems. It is opposed to pure research which is not problem-oriented but for the increase in knowledge which may or may not be used in future.

In the present world situation, more emphasis is being given to applied research to solve problems arising out of overpopulation and scarcity of natural resources. Applied research should not be treated the same as Research and Development which is involved in developing products demanded by the existing clients.

Applied Research, on the other hand, focuses on uncovering what needs are not being met and use that information in designing products or services that would create their own demand.

Thus, applied research brings in new customers and also provides better products and services to the existing customers. In old days, the mobile phone was expensive, bulky and had a short range. Applied Research foresaw that this product would have a limited market and stressed on cost-cutting, reduced weight and long-distance communication. Such measures caused a heavy demand.

METHODS OR APPROACHES

In research, ceases are based two methods known as the *deduction* and *induction*. Both are widely used in research projects. This helps the researchers to understand, explain, or predict business phenomena. Deduction follows an approach which is "top-down" or "from general to specific." On the other hand, induction is "bottom-up" in

nature or from specific to general. In deduction, we start from a theory and try to prove it right with the help of available information. In induction, we observe some happening, deduct a pattern and draw a conclusion.

Example of Deduction

- All men are mortal
- Socrates is a man
- Socrates is mortal

Example of Induction

- This ice is cold.
- All ice is cold.

Take another example: $3 + 5 = 8$ and eight is an even number. Therefore, an odd number added to another odd number will result in an even number.

Example of Both Deduction and Induction

- You push the light switch and find no light
- You ask the question, Why no light?
- You infer a reason to answer the question and explain the fact that the bulb is burned out.
- You use this hypothesis to conclude that the light will not go on when we push the switch. We know from experience that a burned-out bulb will not light.

DEDUCTIVE METHOD

Like Sherlock Holmes, the starting point is a given situation and moving to a specific conclusion. All relevant information is assembled, studies to solve the mystery. Aristotle, a Greek philosopher, described deduction as “drawing conclusions by applying rules or principles; logically moving from a general rule or principle to a specific solution”. It is the process of reaching a conclusion that is guaranteed to follow, if the evidence provided is true and the reasoning used to reach the conclusion is correct.

An investigator is distracted with the noise and thinks that “noise adversely affects mental problem solving.” It can be turned into a research question like “would it be more conducive to solve mental puzzle when there is no noise.”

In order to test it, two puzzles are given to the same group, one in noiseless conditions and other in a blaring noise. If the results of test conducted under noisy-environment are poor it can be deduced or concluded that controlling the aversive noise is helpful in solving mental puzzles. Someone may say “why research, it is apparent?” It is logical to say it but it had been previously proved and awareness has been created of the noise pollution. In fact, noise can be measured in terms of decibel

(dB) and by research, it has been established how much noise is tolerable under what conditions. Zero dB describes perfect silence while 140 dB would reflect a gunshot.

In deductive method, the premises provide a guarantee of the true cease.

Consider the following example:

- There are 32 books on the top-shelf of the book case,
- There are 12 books on the lower-shelf of the book case.
- There are no books anywhere on the bookcase

It can be concluded there are 44 books in the bookcase.

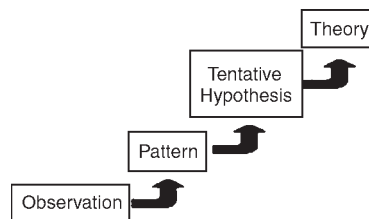


Fig. Inductive Approach.

INDUCTIVE METHOD

Also known as Inductive reasoning or logic, this method involves moving from a specific situation to a general cease. This is a way of “theory-building”, whereby “specific facts are used to create a theory that explains relationships between the facts and allows prediction of future knowledge”. This method is more open-ended and exploratory. This does not give 100% guarantee of truth but probability of being true.

Here is an example:

- Socrates was Greek.
- Most Greeks eat fish.
- Socrates ate fish.

While the premises were true, it is possible that cease was false. Maybe Socrates was allergic to fish, for example. It can be said that inductive approach is weaker than deductive approach as there is possibility of arriving at false cease. However, it is not entirely true.

TYPES OF RESEARCH METHODS

- Qualitative
- Quantitative
- Mixed
- Critical and action oriented.

Data Collection

Data collection depends on the research design (quantitative or qualitative design). Tutors India helps in a survey tool validation and also online and face to face data

collection process. We help you to conduct surveys [in person Interviews: Formal to informal; structured to unstructured; focus group discussion, observations, self-administered questionnaire, diaries, citizen report cards, Delphi techniques, expert judgement, online surveys, secondary sources such as journals, newspaper articles, annual reports, government sources such as census, budgets, policies, procedures, *etc.*

Quantitative Consulting

The preparation of a successful dissertation involves conducting effective research, analysing data and results presentation all which require a high level of statistical expertise. We at statswork provides solution from formulating methodology to the results presentation. You can approach statswork with any or all of the following steps:

- Framing your Research Methodology
- Study design
- Sample size calculation and justification
- Development of questionnaire
- Statistical techniques.

Appropriate selection of statistical tools (parametric and non-parametric) depends on the study design and sample size. The following are some of the statistical tools used by our expertise..

- Exploratory data analysis
- T-test, ANOVA, ANCOVA
- Chi-square
- Linear regression
- Dichotomous logistic regression
- Ordinal logistic regression
- Multinomial logistic regression
- Quantile regression
- Poisson regression
- Survival analysis
- LogLinear models
- Factor analysis
- Principal component analysis
- Multidimensional scaling
- Forecasting
- Decision Tree
- Neural Network
- Cluster analysis
- Multiple imputation
- An extensive array of graphical procedures.

QUANTITATIVE METHODS

- *Unit of Analysis*: The most elementary part of what is being studied or observed. Some examples include individuals, households, court cases, countries, states, firms, industries, *etc.*
- *Variables*: Concepts, characteristics, or properties that can vary, or change, from one unit of analysis to another. Please note that all variables must vary, if there is no variation among the different cases then it is not a variable. Some examples of variables include gender, social class, education, age, level of public enforcement, type of bankruptcy, *etc.*
 - *Dependent Variable*: Variables whose change the researcher wishes to explain
 - *Independent Variable*: Variables that help explain the change in the dependent variable
- *Hypothesis*: An empirical statement which seeks to test the relationship between at least two variables. For instance, As levels of public enforcement increases, levels of stock development also increases. This hypothesis has two variables:
 - Public enforcement{independent variable, and
 - Stock development{dependent variable.
- Levels of Measuring Variables
 - *Nominal*: A nominal variable has qualitative categories that cannot be ranked in a meaningful way in terms of degree or magnitude. Examples of nominal variables include race, type of bankruptcy, type of corporation, name. All of these variables have qualitative categories that cannot be ordered in terms of magnitude or degree. This is the least powerful type of variable.
 - *Ordinal*: An ordinal variable has qualitative categories that are ordered in terms of degree or magnitude. Examples of a nominal variable include class or degree obtained. The variable DEGREE OBTAINED may include the following categories: None, High School Diploma, College/University Degree, Masters, Advanced Degree (JD/PHD/MD). All of these categories are qualitative and are ordered in terms of the amount of education each individual has completed.
 - *Interval/Ratio*: An interval variable has quantitative values (or numbers). Some examples of interval variables include AGE (in years), number of shares outstanding, and Amount in DEBT (in dollars). For all of these variables the response is going to be a number or value. This is most powerful type of variable because you can do the most with it statistically. Note that if a variable has qualitative categories that ARE ordered and there are numerical values assigned to each category which are also ordered, we can treat this variable like an interval level variable. An example would be questionnaire that asks respondents about their feelings

towards President Obama's handling of the economy on a scale of 1 to 5 where (1=very bad job, 2=bad job, 3=neither bad nor good, 4=good job, and 5=very good job). The respondents are asked to choose a category that is ordered, but since it has ordered numbers attached to the categories, we can treat it as an interval level variable with some restrictions.

- *Dichotomous/Dummy*: A dichotomous variable is a variable with two (and only two) categories. These categories can be qualitative or quantitative values.

QUANTITATIVE RESEARCH

In the social sciences, quantitative research refers to the systematic empirical investigation of quantitative properties and phenomena and their relationships. The objective of quantitative research is to develop and employ mathematical models, theories and/or hypotheses pertaining to phenomena. The process of measurement is central to quantitative research because it provides the fundamental connection between empirical observation and mathematical expression of quantitative relationships.

Quantitative research is used widely in social sciences such as psychology, sociology, anthropology, and political science. Research in mathematical sciences such as physics is also 'quantitative' by definition, though this use of the term differs in context.

In the social sciences, the term relates to empirical methods, originating in both philosophical positivism and the history of statistics, which contrast qualitative research methods. Qualitative methods produce information only on the particular cases studied, and any more general conclusions are only hypotheses. Quantitative methods can be used to verify, which of such hypotheses are true. A comprehensive analysis of 1274 articles published in the top two American sociology journals between 1935 and 2005 found that roughly two thirds of these objects used quantitative methods:

Quantitative research is generally made using scientific methods, which can include:

- Collection of empirical data
- Evaluation of results
- Experimental control and manipulation of variables
- Modeling and analysis of data
- The development of instruments and methods for measurement
- The generation of models, theories and hypotheses

In the social sciences particularly, quantitative research is often contrasted with qualitative research which is the examination, analysis and interpretation of observations for the purpose of discovering underlying meanings and patterns of relationships, including classifications of types of phenomena and entities, in a manner that does not involve mathematical models.

Approaches to quantitative psychology were first modelled on quantitative approaches in the physical sciences by Gustav Fechner in his work on psychophysics, which built on the work of Ernst Heinrich Weber. Although a distinction is commonly drawn between qualitative and quantitative aspects of scientific investigation, it has been argued that

the two go hand in hand. For example, based on analysis of the history of science, Kuhn concludes that “large amounts of qualitative work have usually been prerequisite to fruitful quantification in the physical sciences”.

Qualitative research is often used to gain a general sense of phenomena and to form theories that can be tested using further quantitative research. For instance, in the social sciences qualitative research methods are often used to gain better understanding of such things as intentionality and meaning. Although quantitative investigation of the world has existed since people first began to record events or objects that had been counted, the modern idea of quantitative processes have their roots in Auguste Comte’s positivist framework. Statistics is the most widely used branch of mathematics in quantitative research outside of the physical sciences, and also finds applications within the physical sciences, such as in statistical mechanics. Statistical methods are used extensively within fields such as economics, social sciences and biology. Quantitative research using statistical methods starts with the collection of data, based on the hypothesis or theory. Usually a big sample of data is collected - this would require verification, validation and recording before the analysis can take place.

Software packages such as SPSS and R are typically used for this purpose. Causal relationships are studied by manipulating factors thought to influence the phenomena of interest while controlling other variables relevant to the experimental outcomes.

In the field of health, for example, researchers might measure and study the relationship between dietary intake and measurable physiological effects such as weight loss, controlling for other key variables such as exercise. Quantitatively based opinion surveys are widely used in the media, with statistics such as the proportion of respondents in favour of a position commonly reported. In opinion surveys, respondents are asked a set of structured questions and their responses are tabulated.

In the field of climate science, researchers compile and compare statistics such as temperature or atmospheric concentrations of carbon dioxide. Empirical relationships and associations are also frequently studied by using some form of General linear model, non-linear model, or by using factor analysis. A fundamental principle in quantitative research is that correlation does not imply causation. This principle follows from the fact that it is always possible a spurious relationship exists for variables between which covariance is found in some degree. Associations may be examined between any combination of continuous and categorical variables using methods of statistics. Views regarding the role of measurement in quantitative research are somewhat divergent. Measurement is often regarded as being only a means by which observations are expressed numerically in order to investigate causal relations or associations.

However, it has been argued that measurement often plays a more important role in quantitative research. For example, Kuhn argued that within quantitative research, the results that are shown can prove to be strange. This is because accepting a theory based on results of quantitative data could prove to be a natural phenomenon. He argued that such abnormalities are interesting when done during the process of obtaining data:

- When measurement departs from theory, it is likely to yield mere numbers, and their very neutrality makes them particularly sterile as a source of remedial suggestions. But numbers register the departure from theory with an authority and finesse that no qualitative technique can duplicate, and that departure is often enough to start a search.

In classical physics, the theory and definitions which underpin measurement are generally deterministic in nature. In contrast, probabilistic measurement models known as the Rasch model and Item response theory models are generally employed in the social sciences. Psychometrics is the field of study concerned with the theory and technique for measuring social and psychological attributes and phenomena. This field is central to much quantitative research that is undertaken within the social sciences. Quantitative research may involve the use of proxies as stand-ins for other quantities that cannot be directly measured.

Tree-ring width, for example, is considered a reliable proxy of ambient environmental conditions such as the warmth of growing seasons or amount of rainfall. Although scientists cannot directly measure the temperature of past years, tree-ring width and other climate proxies have been used to provide a semi-quantitative record of average temperature in the Northern Hemisphere back to 1000 A.D. When used in this way, the proxy record only reconstructs a certain amount of the variance of the original record. The proxy may be calibrated to determine how much variation is captured, including whether both short and long term variation is revealed.

In the case of tree-ring width, different species in different places may show more or less sensitivity to, say, rainfall or temperature: when reconstructing a temperature record there is considerable skill in selecting proxies that are well correlated with the desired variable.

Quantitative methods are research techniques that are used to gather quantitative data — information dealing with numbers and anything that is measurable. Statistics, tables and graphs, are often used to present the results of these methods. They are therefore to be distinguished from qualitative methods. In most physical and biological sciences, the use of either quantitative or qualitative methods is uncontroversial, and each is used when appropriate. In the social sciences, particularly in sociology, social anthropology and psychology, the use of one or other type of method has become a matter of controversy and even ideology, with particular schools of thought within each discipline favouring one type of method and pouring scorn on to the other.

Advocates of quantitative methods argue that only by using such methods can the social sciences become truly scientific; advocates of qualitative methods argue that quantitative methods tend to obscure the reality of the social phenomena under study because they underestimate or neglect the non-measurable factors, which may be the most important. The modern tendency is to use eclectic approaches. Quantitative methods might be used with a global qualitative frame. Qualitative methods might be used to understand the meaning of the numbers produced by quantitative methods. Using

quantitative methods, it is possible to give precise and testable expression to qualitative ideas. This combination of quantitative and qualitative data gathering is often referred to as mixed-methods research.

“PARTICIPATORY” TECHNIQUES IN QUALITATIVE RESEARCH

The development and use of PRA has changed considerably over the past 30 years. As Chambers (1994a:959) has noted, “in the early days, RRA seemed to be largely organized common sense” on the part of field researchers. In particular, RRA emphasized the use of secondary sources, semistructured interviews, and observation. At the time researchers understood that RRA was a strategy intended to make the best use of limited time for fieldwork that offered little scope for the rural poor to set the research agenda. However, this appreciation appears to have been at odds with the work of others who sought to devise and promote a field methodology capable of maximizing local participation in development such that proposed projects would better fit the needs of local people and vice versa. The techniques have been modified and adapted by researchers and development practitioners in a variety of novel ways, with the result that PRA techniques are widely seen as substitutable. Indeed, there now exists a variety of combinations of techniques that have their own name— participatory and learning methods (PALM); participatory learning and action (PLA); rapid assessment procedures (RAP)-and which are used for quite different purposes, including: 1) health research; 2) natural resource planning; 3) pest control; 4) what might be called experiential learning and communication; 5) development planning; and 6) project or programme evaluation.

The popularization of PRA, the constant modifying of techniques, the variability in PRA training, and the apparent variation in how they are used, highlight why the academic community should be concerned with PRA and arguments that techniques are similar or substitutable, and indeed about the value of sequencing techniques.

The use of interviewing methods has primarily focused on semistructured interviews (SSIs) as developed in anthropological fieldwork. However, there is a tension between the focus of PRA training that redefines the technique as a team activity-and hinges upon the development and use of an interview guide. Though manuals point out the importance of context and of sensitive listening for the effective use of SSIs, little attention is given to who should be selected to be interviewed or to the need and means to corroborate or cross-check interview data with other information. Furthermore, the short period spent in the field— typically less than one week per site-and the emphasis on facilitating local learning in effect deemphasizes the role of SSI, of observation, and of obtaining data from secondary sources to better understand complex social relationships.

Many PRA techniques are modelled on or borrowed directly from qualitative social research, though practitioners pay little attention to the published literature discussing the limitations of the techniques. In particular, the prominent role of group interviews (*i.e.*, structured interviewing or focus groups) is especially problematic because of the

need to carefully select participants and to manage the group interview. As textbooks on qualitative method attest, structured interviewing is intended to “expose every informant in a sample to the same stimuli.... The idea is to control the input that triggers each informant’s responses so that the output can be reliably compared”. The techniques that rely on this approach include questionnaires, the construction of free lists, sentence frame techniques, triad tests, pile sorts, rankings, paired comparisons, and ratings. With the exception of questionnaires, the techniques were developed to explore what people think, how they think it, and how they organize the material. In short, such techniques attempt to explore specific cultural domains in terms of things that can be listed.

The task of producing data that are valid and reliable requires in part that all participants in a group interview are competent or knowledgeable in the specific domain to be investigated. This is difficult to achieve.

As Frey and Fontana (1991:185) observe, “group responses can be effected by the size of the group, by the group members view of the purpose of the interview, and by the differences in the background of the members.” Failure to carefully select participants who are socially homogeneous (and who are strangers) is fraught with problems that include participants who feel under pressure to conform to peers or dominant individuals (the group effect), a stifling of discussion, interpersonal conflict, the production of irrelevant information, and posturing.

A related issue arises from the role of moderators in group interviews because their performance critically affects the data. At issue here is the extent to which group interviewing is more or less directed by the moderator and whether interviewing is more or less structured by the question format, the purpose of the discussion, and skill of the moderator. However, PRA practitioners are primarily concerned with issues of group management (with team building, creating consensus) and give limited attention to the selection of group members, the composition of groups, or the role of the moderator. When other PRA techniques are combined with the group interview, additional problems arise.

Furthermore, there is a tendency to use the information generated by a group interview as if the account speaks for itself. There are two interrelated issues involved: an accurate record of the group’s account and the need to analyse the account. Many PRA practitioners appear unaware that a detailed record of group discussions is an important source of data. The record of a focus group is the basis for an analysis of individual statements, group interaction, and the role of the moderator. In PRA, note taking has an undefined role in daily review sessions during fieldwork and in writing up the final report. In this approach, note taking is intended to record process issues-how interviews were conducted, identification of problems-and is not seen as providing data for the research report.

As Maynard (1998) makes clear regarding her use of focus groups to study widowhood, the accounts of the women required analysis because they are historical constructions. Maynard grounded her account in terms of “a gendered analysis of

marriage, family/household, domestic career and ideas about aging.” Thus, the accounts produced in a group interview require theoretical analysis. Only in this way is it possible to make sense of the perspective of individuals and groups, and only with this type of analysis is it possible to move beyond a focus on discourse to an understanding of the constraints facing individuals and their society.

Visualization techniques are used in group interviews and are intended to enable local people to “conduct their own analysis, and often to plan and take action” through a shift from verbally to visually oriented methods that are not dependent on literacy, which could exclude many people. Thus, Pictures can be a means of focusing on problems and memories that are too painful or too complicated to be expressed in words. They can help individuals come to an accommodation with their past experiences. Pictures can be shown by the individual, the community or the state to project an image or an idea which would be difficult to explain in any other way. They can bridge languages and cultures, they can educate the world and inform a friend.

PRA uses visual techniques in group interviews to explore aspects of the social world that are amenable to diagramming or mapping. Experience is said to be the best guide to the use of these techniques, and facilitators have little more to go on than general guidelines and a list of do’s and don’ts. Once again, and despite injunctions-to be nondirective, to facilitate, to listen closely, probe, and question-there appears to be limited awareness about the roles facilitators assume in group interviews or of the importance of carefully selecting participants. Instead, emphasis is placed on comprehending cognitive processes and identifying “simple, schematic devices which present information in a readily understandable visual form” such that, the act of constructing the diagrams is in itself an analytical procedure and helps the constructors to think through the dynamics they are trying...[to represent in visual form, JC). Second the diagrams become a means of creating communication and discussion between different people.

In this way, and supposedly without direction from outsiders, a group: 1) maps a locality; 2) constructs a seasonal calendar, locating itself and agricultural and social practices within it; 3) devises a profile of its use of time; 4) devises a longitudinal time line that indicates key dates, occurrences, and events; and 5) uses VENN diagrams to identify and analyse relationships between key areas and institutions.

PRA practitioners seem unaware of the kind of skills needed by local people to take part in visualization exercises. Drawing on Streets’ (1993) research on literacy, Robinson—Pant (1996) has argued that many PRA visualization techniques appear to impose Western cultural practices and require local people to learn new cultural skills. In particular, a distinction must be made between visual literacy and numeracy.

The required literacy may not exist locally. While some degree of visual literacy can be picked up in PRA exercises-reflected in the fun or playfulness experienced by participants in the exercises-there is every reason to believe that each individual (including facilitators and researchers) interprets a map or matrix differently. The only

way to address the issue would be to: 1) select individuals for inclusion in a group interview who are likely to share a similar worldview because of gender, age, religion, or caste, but this requires prior knowledge of the community; and 2) undertake research on local literacy and social cognition (Block 1998). PRA must adapt its techniques to accommodate the unevenly distributed skills and competencies that exist in a population.

Ranking and scoring techniques are problematic for the same reasons, namely their use cross-culturally and with groups. The techniques include exercises that solicit information, usually of a very particular type, that is listed, discussed, and ranked. The major types of ranking techniques are: well-being or wealth ranking; scoring or ranking of different options or opportunities; pair-wise ranking; and preference or problem ranking.

One of the defining features of PRA is that during the group interview facilitators should query members about the product of the visual ranking exercise (*i.e.*, the map, the matrix). “Interviewing the diagram” is the principal mechanism used to incorporate the full range of social perspectives. However, the concept is understood in two contradictory ways: that there should be no ambiguity or “blank spaces” and that all perspectives should be accurately represented and incorporated into a group “product.” The key concern with maps or matrices is expressed in the observation that “informants tell you something by drawing the diagram, and what they draw, and omit, tells you something about the informants”. Interrogating maps and diagrams involves asking what has changed, exploring blank or empty spaces, clarifying the symbols used, and discussing ambiguous or unclear issues. Maps and diagrams are not supposed to be an end in themselves; rather, it is the process of focused discussion that is important. Even so, the number of times a map or diagram has become the objective of PRA must be enormous since it is often seen as a summation of the exercise.

An additional way to clarify and explore this issue would be to repeat the same exercise with other groups. Repetition is important for two reasons. First, the decision to use PRA techniques is presumably because very little is known about a particular problem and its amenability to study by PRA. Second, the decision of who to recruit into groups is related directly to the nature of the problem being investigated. Because development will differentially effect a population, each issue needs to be assessed separately with different sections of the community. Also, it is not possible to generalize findings from a small number of group interviews that are likely to be unrepresentative of the wider community. However, there is little evidence in the literature that repetition is utilized to crosscheck results.

While the emphasis is to build into the map or matrix the full range of local knowledge, the limitations of the technique and the limited knowledge of participants are unlikely to emerge until the document is used and clarified in a second exercise or is put to practical use. Thus, while it is possible to interrogate a map or matrix as it is being constructed, ideally facilitators need to link it to a second technique, to repeat the exercise with a different group, or to make use of other qualitative techniques.

PRA in Practice

Careful attention to method can help define a basis for transparent accounts of the research process and a more informed discussion of the validity and credibility of RRA and PRA.

There is a need to distinguish between the process of participatory research, and claims regarding the nature of local people participating in it, and a range of distinct and logically separate outcomes of research.

The role or contribution of participation in social research is a matter of considerable debate. Participation has been defined to include passive participation, participation in information giving, participation in consultation, participation for material incentives, functional participation, interactive participation, and self-mobilization. Pretty, Thompson, and Kiara (1994) argue that the first four types are superficial and have no lasting impact on people's lives. In any event, claims that participation plays a central role in research require a transparent account of the research process that distinguishes between: 1) the act of bringing people together for research that may or may not involve locals learning from or discussing their situation; and 2) the outcome of the research. Research outcomes may or may not include: 1) increasing and codifying local knowledge in maps or documents; 2) agreeing upon local development priorities; 3) planning and undertaking joint action; and 4) production of a research report. PRA may or may not involve data extraction, and it may or may not facilitate community action or empowerment.' To the extent that PRA produces research data, such information may serve important functions at different levels-facilitate local action, inform policy—and in different ways, including establishing dialogue between communities and government. Each output needs to be documented and assessed separately in research reports before the issue of participation can be fruitfully pursued.

There is a need to provide better documentation on the use and type of group interview utilized, including details on the techniques) it was combined with, its objective, and the information it produced.

Group interviews may serve a variety of purposes other than generating data (*e.g.*, therapy, decision making, education), and they may follow organizational procedures that influence discussion. Most importantly, the role of the facilitator, the research instruments, and the methods used provide differing degrees of structure to the discussions. In PRA, insufficient information is provided about how individuals are recruited into groups and about what actually occurs in them. While general rules of thumb exist to avoid obvious biases, too little attention is given to such issues.

From the perspective of the researcher, a well-designed group interview attempts to elicit local respondents' "hierarchy of importance" in the form of emic understandings expressed in the local language. This task is facilitated by recording and transcribing the discussion for subsequent analysis. A further concern is that visualization, ranking, and scoring techniques have been added to the group-interview technique without a clear appreciation of their limitations. It seems clear that basic research on these

techniques-used separately or in combination-is required. There is a need to distinguish the data produced in a group interview from the knowledge and social competencies of local people.

PRA is premised on the assumption that its techniques effectively access local knowledge, but due to their proliferation it seems certain they are not consistently used. Additionally, there is the presumption that all members of a society are competent in the domain of inquiry that is the focus of research. In such circumstances, it seems reasonable to conclude that the techniques and the data they generate need to be more carefully assessed and that these issues should be addressed in research reports.

A related concern arises from the public nature of PRA exercises that allows a dominant group to construct a normative, unitary narrative that emphasizes consensus. Furthermore, the knowledge produced by PRA “is a construct of the planning context, behind which is concealed a complex micro-politics of knowledge production and use”. Local knowledge production has four aspects: 1) the shaping of knowledge by local relations of power; 2) the expression of outsider agendas as local knowledge; 3) local participation in the production of a consensus; and 4) the direct manipulation of peoples’ planning by development agents and projects. The complex nature of local knowledge production is unlikely to be adequately grasped by short field visits, assuming that researchers do pursue ambiguous findings and seek to explore minority views.

Finally, “backsliding” in the application of PRA creates problems. The pressure to produce results may lead researchers to skip key preparatory work, resulting in insufficient time spent building rapport with local people and poor or inconsistent training of facilitators, which tends to produce poor and unreliable data. This is particularly the case where divergent local interests are insufficiently explored, a situation that arises in the context of gender and ethnic inequality when insufficient attention is given to strategies of organizational and cultural change as opposed to merely including women or minorities as participants. There is a need to identify the strengths and limitations of an individual PRA technique prior to integrating or combining it with another PRA technique or with quantitative and qualitative techniques.

While the principles of PRA emphasise iterative learning, the extent to which this occurs is unclear since many reports fail to clarify their techniques or procedures. At the same time, the extent and manner in which a technique is used to cross-check as opposed to obtaining discrete types of information is also unclear. In reality, little or no crosschecking may occur. Equally problematic is the unexamined assumption that different techniques, and the information they produce, dovetail-as opposed to pointing in different, possibly contradictory directions-to form a coherent analysis or a harmonious and conflict-free society.

Are PRA techniques compatible with qualitative and quantitative techniques? How should the data produced by combining PRA with other techniques be assessed? These issues have yet to be examined, but their resolution would seem to involve: 1) an assessment of a PRA technique— individually and in combination with a second

technique—in obtaining specific types of data; and 2) awareness of the possibility that different methodological orientations may render certain techniques and the data they produce incompatible.

Of the techniques used by PRA, only two have been the subject of work that examines the strategy used to collect, analyse, and present data. Focus groups (group interviews) provide the framework in which many PRA techniques are utilized, yet little research exists on the value and problems of combining it with other techniques. Morgan (1996) identifies four ways focus groups have been used in combination with individual interviews or surveys. All the applications are recent, and nearly all raise significant and unanswered questions about sampling and the analysis of data derived from different “paradigms.”

Ward, Bertrand, and Brown (1991) compared focus groups to surveys on the extent to which data from these techniques converged or diverged. They found that:

Overall, for 28 percent of variables the results were similar; for 42 percent the results were similar but focus groups provided additional detail; for 17 percent the results were similar, but the survey provided more detail. And in only 12 percent of the variables were the results dissimilar.

The key finding was that to the extent that a focus group was appropriate to the problem being investigated, and was properly utilized, it functioned effectively in combination with surveys, individual interviews, or as a stand-alone technique. Note, however, that PRA utilizes focus groups in combination with visual and verbal techniques that have not been examined to ascertain their limitations, and that the use of group interviews in PRA diverges sharply from their use by trained focus group facilitators.

The only PRA technique that has been the focus of basic research regarding its strategy of collecting data is wealth ranking, which was compared to data derived from household surveys to determine whether it provided a valid measure of socioeconomic stratification. Wealth ranking can focus either on locally defined household assets, or it can solicit local terms and understandings about household wellbeing or wealth. The solicited terms are then used as the basis for an exercise that ranks all households in a community or setting where the individuals undertaking the ranking know one another. The technique is used in the group interview format; therefore, the caveats identified earlier apply.

Scoones (1995) undertook research in southern Zimbabwe to compare the value of a wealth-ranking exercise that began by eliciting local wealth criteria that were used in group interviews with local households. This is a carefully designed piece of research in which various definitions of household were teased out so that the economic and social relationships between households could be explored: historical and ecological changes were addressed, gender issues were considered, and a differentiated set of indicators about assets, consumption, and perceptions were applied. The study’s attention to context, culture, and method is excellent and it successfully demonstrates that when properly conducted, wealth ranking “is a cost-effective research tool for examining issues of wealth and poverty in a rural context”.

A second study has examined wealth ranking as a means of analysing social stratification. Adams et al. (1997) attempted the more limited task of comparing findings from a wealth-ranking exercise to data obtained simultaneously from a survey of a small number of households. Unlike Scoones, this research relied upon methodological shortcuts. In particular, local interviewers were trained to use one wealth— ranking technique, and they made use of criteria derived from a previous study of a different area than the one to be ranked. In addition, the ranking exercise was performed by a small number of key informants in each administrative subdivision who were asked as a group to assign a household to one of three predefined wealth groups. Fifty-five villages were ranked; then a random sample of 30 households was selected for in-depth interviews to obtain a household census and collect information against which the results of the ranking exercise were compared.

The authors examined whether wealth ranking provided an adequate empirical measure of stratification; however, there are important caveats in their use of the technique:

1. The study fails to examine the nature of the social unit (households) being ranked;
2. The ranking criteria were derived from a study that may not adequately reflect local social variability;
3. It relied upon an unrepresentative number of key informants to do the ranking; and finally,
4. Ranking was undertaken as a group exercise. Had it been undertaken by individuals, an independent check on the bias of the exercise would have been possible.

It may be, as Adams et al. (1997:1170) argue, that measured differences across households “are consistent with our hypotheses and support the construct validity of the wealth— ranking technique.” It may also be that their use of the technique does not measure what it purports to measure. Given the absence of comparable studies, the validity of the method as demonstrated by these authors must remain in doubt.

Finally, the extent to which different methodological orientations may be incompatible is suggested by Jodha’s (1988) study of rural poverty in Rajasthan over a 20-year period. Jodha was unable to reconcile quantitative data on household income which showed that for 38 percent of households in two villages per capita income had substantially declined, with qualitative indicators suggested by villagers that indicated dramatic improvements in well-being, including reduced reliance on patrons, reduced dependence on low-pay options, improved mobility and household liquidity, improvements in consumption, and improved acquisition of consumer durables. Jodha (1988:2421) explains this paradox as one that arises from factors underlying rural poverty that “are too detailed and at times too complex to be captured by standard and simplistic methods”.

Jodha’s study illustrates a major disagreement between those who advocate a consumption-based approach to measuring income and expenditure (IDS 1996) and

those who argue that local people's perceptions are fundamental to identifying what poverty is and how it is experienced. However, epistemological differences need not prevent research from exploring both positions by, for instance, focusing on issues of agreement and overlap or by integrating PRA with quantitative and qualitative techniques. The integration of different techniques and measures of poverty in a carefully designed study should increase the scope for a more nuanced analysis and appreciation of policy issues than is possible through studies that are confined to one methodology or set of techniques. Such a study would also have greater validity.

SUBJECTIVITY AND OBJECTIVITY IN QUALITATIVE METHODOLOGY

Qualitative methodology recognizes that the subjectivity of the researcher is intimately involved in scientific research. Subjectivity guides everything from the choice of topic that one studies, to formulating hypotheses, to selecting methodologies, and interpreting data. In qualitative methodology, the researcher is encouraged to reflect on the values and objectives he brings to his research and how these affect the research project. Other researchers are also encouraged to reflect on the values that any particular investigator utilizes.

A key issue that arises with the recognition of subjectivity is how it affects objectivity. Two positions have been articulated. Many qualitative researchers counterpoise subjectivity and objectivity. Objectivity is said to negate subjectivity since it renders the observer a passive recipient of external information, devoid of agency. And the researcher's subjectivity is said to negate the possibility of objectively knowing a social psychological world.

The investigator's values are said to define the world that is studied. One never really sees or talks about the world, *per se*. One only sees and talks about what one's values dictate. A world may exist beyond values, but it can never be known as it is, only as values shape our knowledge of it.

GERGEN (2001) outlines this subjectivist position under the moniker of postmodernism, which he contrasts with modernism. However modernists such as BERKELEY and HUME proposed exactly the same doctrine centuries ago. GERGEN states: "For modernists, the world simply is out there, available for observation. Within the texts of postmodernism, however, there are no grounds for such a presumption. There is no means of declaring that the world is either out there or reflected objectively by an 'in here'".

"To tell the truth, on this account, is not to furnish an accurate picture of what actually happened but to participate in a set of social conventions...To be objective is to play by the rules within a given tradition of social practices...To do science is not to hold a mirror to nature but to participate actively in the interpretive conventions and practices of a particular culture. The major question that must be asked of scientific accounts, then, is not whether they are true to nature but what these accounts... offer to the culture more generally".

“A postmodern empiricism would replace the ‘truth game’ with a search for culturally useful theories and findings with significant cultural meaning”. “Arguments about what is really real are futile”.

Perception, cognition, and communication are mired in a web of values that prevents getting beyond these. Consequently, the criteria of truth and objectivity denote congruence with cultural values, symbols, and terms; they do not denote apprehending a world beyond the individual (and his culture). The limitations of this perspective are obvious. If Catholics accept Vatican dogma that Christ was born from a virgin mother and that his body levitated to heaven after his death, they are objective and even scientific according to GERGEN’S definition, since they are playing by the rules of the game. Germans who accepted Nazi propaganda that Jews were genetically inferior to Aryans would also have been objective and scientific! Science, religion, and ideology are indistinguishable once all values, cognition, and social relations are construed as distorting mechanisms rather than as potentially augmenting one’s understanding of social and physical matters. In subjectivism, all viewpoints are simply another way of approaching a thing. But none of them delivers any information about the thing itself. My view that Santa Claus is a myth and your view that he is a real person have equal truth value—namely, none at all. Whether one favors one or the other is simply a matter of how interesting they appear as cultural expressions.

Similarly, I may think women enjoy being raped while you think they hate it. Neither indicates what women truly feel. They are simply our views of the topic. They are to be judged according to what meanings they contribute to the culture. If my opinion imparts a cultural meaning that women are happy and well adjusted while yours contributes a cultural meaning that women are frightened and angry and maladjusted, we select among the two views on that level.

Subjectivism is often regarded as the *sine qua non* of qualitative methodology. However, this is untrue. Qualitative methodology has an objectivist strand as well. Objectivism states that the researcher’s subjectivity can enable her to accurately comprehend the world as it exists in itself. Of course, subjectivity can bias the researcher and preclude objectively understanding a subject’s psychological reality. However, this is not inevitable. In fact, one of the advantages of recognizing subjectivity is to reflect on whether it facilitates or impedes objective comprehension. Distorting values can then be replaced by values that enhance objectivity. Objectivism integrates subjectivity and objectivity because it argues that objective knowledge requires active, sophisticated subjective processes—such as perception, analytical reasoning, synthetic reasoning, logical deduction, and the distinction of essences from appearances. Conversely, subjective processes can enhance objective comprehension of the world. Objectivism was enunciated by DILTHEY (1833-1911) in his articulation of the cultural sciences (*Geisteswissenschaften*).

The key procedure in the cultural sciences was a qualitative hermeneutic interpretation of life expressions (*Lebensäußerungen*). Hermeneutic interpretation requires that the

researcher employ an active, sophisticated subjectivity to objectively comprehend subjective experience (Erlebnis) in life expressions. Verstehen is to achieve Allgemeingültigkeit, or valid interpretations, of Erlebnis. DILTHEY'S objectivist approach to hermeneutics continued its earlier use during the Reformation when theologians employed it to identify the true original meaning of biblical texts. The objectivist sense is also contained in the etymological origin of hermeneutics. The term derives from Hermes, the Greek messenger god who helped humans understand what the gods were trying to say. Hermeneutics in social science is similarly to help an observer clearly understand (disambiguate) what the subjective experience of another is, *i.e.*, what their expressions mean.

Objectivity presupposes an independent reality that can be grasped. If there is no independent reality, or if reality cannot be apprehended, or if reality is merely the concoction of the observer, then the notion of objectivity is moot. The psychology of people is independent of the observer just as physical objects are. Just as the moon is there and has certain characteristics independently of the astronomer, so my wife has certain emotions about her mother independently of me. It is incumbent on me to understand my wife's emotions as they exist for her. If I do not understand my wife's emotional state as it is—if I try to construct it as meaningful in my terms rather than as meaningful for her—I will pay dearly. Of course, some of her emotions are generated in part by my acts. Nevertheless, her emotions are HER Erlebnis and I must objectively understand them as they are FOR HER, as her psychological reality.

The observer may be correct or incorrect in assessing peoples' psychology. Just as the observer may accurately perceive or misperceive an optical illusion, so he may perceive or misperceive peoples' psychology. There must be empirical criteria to establish the degree of accuracy/objectivity with which one perceives peoples' psychology, just as there are empirical criteria to determine the degree of illusion in the perception of objects.

Of course people must agree about what constitutes evidence. And they use socially constructed instruments to obtain evidence. However, their agreement and measurement concerns what is real.

When physicists agree that their instruments have measured certain properties of atoms, they are not simply talking about their measurement operations, *per se*, or how the properties are mere reflections of those operations. They are not trapped in a web of subjective processes that distance and distort phenomena.

Whereas postmodernists are fascinated by the fact that their data reflect their theories and methods, realists emphasize how their theories, methods, and data reflect real things. The objective information about real things which social and subjective processes (including measurement) generate is reflected in real results such as producing nuclear weapons, which really kill people, and medical procedures, which really cure disease (cf. MERTON, 1972 for an incisive refutation of the subjectivist position that one's values insulate oneself from the world). Theories, methods, and conclusions which are

incompatible with real results are winnowed out so that better ones can be cultivated. In contrast, postmodernism smugly assumes that social constructions are self-confirming, reality beyond social constructions does not exist or (what amounts to the same thing) is unknowable and inconsequential, and any construction is tenable as long as it commands interest.

Objectivism is the highest form of respect for the subjects we are studying. It respects their psychological reality as something meaningful and important which must be accurately comprehended. Subjectivism either denies a psychological reality to subjects, or else makes it unknowable. The psychology of other people is clouded by the subjectivity of the observer and is not recognized for what it (truly) is. To objectively comprehend peoples' psychology, the researcher must organize his subjectivity appropriately. Hypothetical concepts must be well-defined so that they can be identified unambiguously. An appropriate methodology must be adopted in order to solicit complete, meaningful evidence that can be used to test the validity of hypothetical concepts. And the evidence must be analysed through sensitive, systematic procedures which can detect its features and compare them to the characteristics of hypothetical concepts. In this way, the researcher can be warranted in believing that her concepts illuminate the true nature of peoples' psychology. Nebulous hypothetical concepts, insufficient or inappropriate behavioral evidence, and arbitrary analyses vitiate objectivity and allow the researcher to impose her theoretical constructs on the data.

QUALITATIVE RESEARCH: OBJECTIVISTIC AND SUBJECTIVISTIC

The meaning units of the statement on moral reasoning can be represented by the following central themes:

- No moral obligation to drive/help—(meaning unit: "I don't think he has any obligation")
- Don't help distant social relations—(meaning unit: "don't know him well")
- Self-gratification—(meaning unit: "sleep late"; "put yourself first")
- Self-responsibility—(meaning unit: "everyone responsible for own self"; "not responsible for others")
- Help if it's convenient for self — (meaning unit: "if he and I were going to the same place")
- People use each other—(meaning unit: "people think as long as you have a car they have a ride")
- Private property can be used as one desires without obligation—(meaning unit: "it's my car")
- Self-protection—(meaning unit: "don't know people well"; "out of protection for myself")

The central themes are the significant psychological elements expressed in the narrative. Although central themes are constructions of the researcher that go beyond the subject's literal words, they are consistent with these and represent their significance.

In this sense, the central themes objectively summarize the psychological meanings that the subject expresses in the narrative (cf. RATNER 2002). This procedure avoids impressionism that is common in qualitative methodology, whereby the researcher simply declares meanings without grounding these in empirical evidence in the form of the subject's statements.

It will be illustrative to examine an instance of qualitative research in which the researcher's values were allowed to generate a conclusion that was not empirically grounded. The contrast with the foregoing example will highlight what objectivity consists in and how it can be achieved.

The impressionistic research was conducted by ROWE, WERTSCH, and KOSYAEVA (2002).

It was guided by the theoretical orientation that individuals construct personal meanings about things rather than reflect social meanings. Social meanings are said to be impersonal, reified, monolithic, and static. Individuals are said to be active and to never merely receive social meanings. Instead, individuals always transform social meanings into personal significations. This testifies to their activity, creativity, and agency. The authors present a brief conversation to document this point of view. Let us examine it to see whether their conclusions are empirically validated. The authors observed patrons in the Winter Palace museum, St. Petersburg, looking at a 19th century painting that depicts the Winter Palace and its locale. One conversation between two patrons went as follows:

"K: See here? It's the Winter Palace, and in 1985 I lived in St. Petersburg for a summer with a friend in her apartment down this street here.

S: You lived right there?

K: Yes, well, not right in that building but down the street here a little way and I would walk down to the square everyday."

From this minimal interchange, the authors conclude that the two patrons have transformed social meanings into personal ones. They state:

"Instead of bringing autobiographical narratives into contact with official culture as part of an attempt to enrich the latter, it seems to us that this [narrative] involves an escape from the public memory sphere...These visitors are refusing to engage in the museum's public memory space... It is meaning making on one's own terms".

The question is, do the patrons say, or even imply, this? To refuse and escape from something is to actively reject it. One has to specifically express a dislike and a deliberate avoidance. Patron K. simply made a casual remark that she lived on a street that appeared in the painting. This in no way implies that she is escaping from the public memory sphere, refusing to engage in public memory, or making an idiosyncratic meaning.

Patron S. was even less expressive. She simply asked a single, simple clarifying question of K. There is no evidence that the strong terms the authors use (escape, refusal, making own meanings) correspond to K's or S's words or intentions. Much more substantial evidence is necessary to justify an interpretation that an escape or refusal is

being signified. Even making a personal remark about the painting is not necessarily making one's own meaning. The modern era is highly individualistic and people often look for personal issues in social, political, religious, and economic events.

It is common for Americans to focus upon the personalities and sex lives of politicians or the cooking recipes of their wives. Such obfuscation of important political issues is encouraged by the posturing of politicians and the superficial, sensationalistic reporting of the news media. Consequently, there is nothing novel, creative, or idiosyncratic about people raising personal issues in relation to public phenomena such as paintings. The contrast between the two examples I have presented illustrates the flaws of subjectivism and a step towards making interpretation objective. It illustrates how the active subjectivity of the researcher can elicit and work on complex data (statements) to elucidate their psychological significance. Such objective interpretation will be corroborated by behavioral results in the way that individuals eventually act. Subjectivism will be also exposed by behavioural results which contradict its arbitrary conclusions. If qualitative methodology emphasizes this distinction and strives to direct researchers' subjective processes to objectively study the psychology of subjects, it will make a great contribution to social science.

QUALITATIVE, QUANTITATIVE AND MIXED-METHOD RESEARCH

There are two overarching ways of gathering data, or fields of research. These are qualitative research methods and quantitative research methods. Qualitative research is concerned with developing knowledge where little or none exists and uses words, observations, and descriptions to develop this knowledge. Quantitative research is concerned with expanding knowledge that already exists and using numerical data to report the findings from the research. But, perhaps you want to use both qualitative and quantitative methods, or a mixed-method design, in your research. A mixed-method design allows researchers to design a study using both qualitative and quantitative methods by using numerical and textual data.

Qualitative Research

Social work is a profession that owes a large debt of gratitude to many other disciplines. Anthropology, psychology, sociology, and medicine have all contributed to the development of our profession. One of the areas in which this becomes exceedingly clear is the field of qualitative research. Qualitative research has deep roots in the fields of anthropology and sociology, where the development of rigorous and exact methods for fieldwork has long been fostered.

The use of qualitative research methods is debated among social work practitioners, faculty, researchers, and other professionals. It is generally agreed that qualitative research is employed when little or nothing is known about a subject, or when the researcher wants to gain an in-depth understanding of a person's experience. Some

may argue that qualitative methods are better suited for studies on complicated topics such as a person's comfort level with death, how it feels to be unemployed, or how a child views the drinking habits of an alcoholic parent.

Qualitative research primarily relies on information generated from observations of the researcher and discussions and interviews with study participants. However, researchers engaged in qualitative research might also gather some descriptive information such as the demographics of participants and their settings in order to place their experiences within a context. In their simplest form, qualitative research methods are used to help us understand the characteristics of a phenomenon. Often this type of research uncovers these characteristics by focusing on the ideas of the people involved.

As an example, let us imagine for a moment that you are a case manager at a homeless shelter and the year is 1982. You have noticed that a large number of your consumers who report being intravenous drug users are also suffering from a strange new illness that seems to impair their immune system. You may be aware that AIDS was a relatively unknown disease in 1982 and that scientists were just beginning to understand the causes of the transmission of this disease. As a case manager, you may want to design a qualitative study that will help you explore the experiences of those who are suffering from this disease by interviewing people living with AIDS (recording their own words). You may also want to collect some demographic information such as sex, age, race, and length of illness to describe their experiences within the context of the research population.

Quantitative Research

Advocates of quantitative research argue that it is only through the use of methods that report numerical representation that the social sciences can become truly valid. Quantitative research seeks to explain relationships between two or more factors. The aim of quantitative research is to determine how one thing (a variable) affects another in a population. A variable is any attribute or characteristic that changes or assumes different values. Variables can represent subject characteristics (*e.g.*, age, race, sex) or the things you are really interested in (*e.g.*, agency performance; rate of relapse in addiction treatment; physiological, psychological, or sociological causes of child maltreatment). Variables can also represent the effect of any intervention that subjects receive, such as a cultural sensitivity training.

Mixed-Method Research

Mixed-method research uses both qualitative and quantitative research designs. Using more than one research method while collecting and analysing data in a study is called *concurrent mixed-method research*. When data collected through the use of one type of research design provides a basis for the collection of data using the other type, this is

called *sequential mixed-method research*. When data are analysed one way and then analysed again in another way, it is called *conversion mixed-method research*.

There are several reasons to use a mixed-method design. Among these are that it can test the consistency of findings obtained through different forms of data collection. This is referred to as *triangulation*; this means that the findings from the methods used are consistent and support each other. Or a researcher might use a mixed-method design because it allows him or her to use qualitative methods to add richness and detail to the results obtained from the use of quantitative methods. Researchers may also choose a mixed-method design so they can use results from one method to shape subsequent methods or steps in the research process. This is frequently seen when a qualitative study is used to shape a quantitative study. In addition, mixed-method research can be used as a means to develop new research questions or to use one method to challenge results obtained through another method.

Developing your Research Questions

You may be asking yourself at this point, “Where do research questions originate?” Research questions may arise from your personal experience. Thus, a person who was adopted may feel compelled to study the factors that make adoptions work well for children.

Research questions may develop out of research articles or theories you are studying. A theory is a statement or set of statements designed to explain a phenomenon based upon observations and experiments and often agreed upon by most experts in a particular field. For example, you may want to test the credibility of the claims put forth by a developmental theory on aging that you learned about in one of your human behaviour classes. Research questions may arise out of your own practice experience. Regardless of the source, most questions are born out of the researcher’s personal interest in a subject.

To illustrate this process, we may begin with an observation (“This person smiles at me and goes out of her way to help me”), then we have an idea (“This person would make a good friend”), and then we develop a question (“Does this person like me?”). We can examine this question by drawing from our past experiences, by consulting others, or by asking the person directly.

When you are developing research questions, there are some issues to keep in mind. The first thing to consider is whether the question is empirical. This means the researcher must decide whether it can be quantified. For example, a question such as “What is the best religion?” is both value laden and subjective (“the best”). As a researcher, you need to be careful to remember that we can study values in order to understand what others think, but we cannot conduct research on values in order to evaluate them. Therefore, we can approach value-laden issues through qualitative methods that are meant to deal with the subjective questions that we have—this eliminates any objectivity from the research. “How many people cheat on their partner?” or “Has having an

abortion prevented further unwanted pregnancies?” are both examples of questions that attempt to quantify issues of moral worth and can be measured through quantitative methods.

What is a Hypothesis?

A hypothesis is a research statement about relationships between variables that is testable and that can be accepted or rejected based on the evidence. Therefore, you can only develop hypotheses that are quantifiable. To design a study to test your hypothesis, you use quantitative research methods. Hypotheses are divided into two categories: research hypotheses and null hypotheses. The research hypothesis asserts that there is a relationship between the variables, and the null hypothesis claims that the relationship between the variables can be rejected. In other words, the null hypothesis is what the researcher is attempting to reject. For example, we may have a null hypothesis that no difference exists between a treatment group and a nontreatment group after intervention. If this is rejected, then the research hypothesis that the treatment group will be different from the nontreatment group after intervention (*e.g.*, less sick or more educated) is supported. Hypotheses are typically abbreviated as H_0 (null hypothesis), H_a (research hypothesis), and H_1 , H_2 , H_3 (a number is used when there is more than one research hypothesis).

Imagine that you are working at an emergency shelter with a consumer named Joe. Joe is in need of permanent housing (he has been living on the streets for the past two years). While you are collecting assessment history with Joe, he discloses that he has a long history of drug abuse. One initial hypothesis may be “A history of substance abuse is related to not having stable housing.” In further discussions with Joe, you explore this hypothesis with him, and he confirms that his substance abuse has interfered with his ability to seek and keep a job—a strong factor in his being homeless. You then decide to design a research study to determine if this relationship between substance abuse and homelessness exists beyond your client. You can also test a second hypothesis that looks at the relationship between substance abuse and unemployment.

Quantitative and Qualitative Research

Similarly, designs are often equated with qualitative and quantitative research methods. Social surveys and experiments are frequently viewed as prime examples of quantitative research and are evaluated against the strengths and weaknesses of statistical, quantitative research methods and analysis. Case studies, on the other hand, are often seen as prime examples of qualitative research ± which adopts an interpretive approach to data, studies ‘things’ within their context and considers the subjective meanings that people bring to their situation. It is erroneous to equate a particular research design with either quantitative or qualitative methods. Yin, a respected authority on case study design, has stressed the irrelevance of the quantitative/qualitative distinction for case studies. He points out that:

- A point of confusion... has been the unfortunate linking between the case study method and certain types of data collection $\pm$ for example those focusing on qualitative methods, ethnography, or participant observation. People have thought that the case study method required them to embrace these data collection methods... On the contrary, the method does not imply any particular form of data collection $\pm$ which can be qualitative or quantitative.

Similarly, Marsh argues that quantitative surveys can provide information and explanations that are 'adequate at the level of meaning'. While recognizing that survey research has not always been good at tapping the subjective dimension of behaviour, she argues that:

- Making sense of social action... is... hard and surveys have not traditionally been very good at it. The earliest survey researchers started a tradition... of bringing the meaning from outside, either by making use of the researcher's stock of plausible explanations... or by bringing it from subsidiary in-depth interviews sprinkling quotes... liberally on the raw correlations derived from the survey. Survey research became much more exciting... when it began including meaningful dimensions in the study design. [This has been done in] two ways, firstly [by] asking the actor either for her reasons directly, or to supply information about the central values in her life around which we may assume she is orienting her life. [This] involves collecting a sufficiently complete picture of the context in which an actor finds herself that a team of outsiders may read off the meaningful dimensions.

Adopting a Sceptical Approach to Explanations

The need for research design stems from a sceptical approach to research and a view that scientific knowledge must always be provisional. The purpose of research design is to reduce the ambiguity of much research evidence. We can always find some evidence consistent with almost any theory. However, we should be sceptical of the evidence, and rather than seeking evidence that is consistent with our theory we should seek evidence that provides a compelling test of the theory. There are two related strategies for doing this: eliminating rival explanations of the evidence and deliberately seeking evidence that could disprove the theory.

THE QUALITY OF QUALITATIVE RESEARCH

Probably many qualitative researchers have been criticised by critics who have claimed that their research is too subjective, or the number of cases is too small, or that mere talking is never a scientific method, and so on. Qualitative inquiry, moreover, has a short history in Japanese social work research and therefore Japanese social work researchers may need to strongly defend the validity of their qualitative research. However, they should do so in the confidence that qualitative research is no less rigorous

than more traditional forms of inquiry. Although “validity” and “reliability” of qualitative research is discussed by many researchers, the most often quoted concept of the problem of establishing validity is probably the notion of trustworthiness that was developed mainly by Lincoln and Guba (1985). The notion of trustworthiness has four elements: credibility, transferability, dependability, and confirmability. These are analogous to “internal validity”, “external validity”, “reliability”, and “objectivity” in conventional criteria. Because their philosophical base is constructivism, these criteria are not applied to all the qualitative research methods. We believe, however, that these criteria are useful for novice researchers to understand the validity and reliability issues of qualitative research and so we will describe them and some of the techniques related to them.

Credibility, Prolonged Engagement and Peer Debriefing

Credibility is analogous to “internal validity” in conventional criteria. It relates to how the reconstruction of the researchers fits the realities and views the participants express in the process of the inquiry. To establish credibility, researchers use a variety of techniques, but because we have limited space, we will discuss only the following.

One is “prolonged engagement.” That is “the investment of sufficient time to achieve certain purposes; learning the ‘culture [of the participants],’ testing for misinformation introduced by distortions either of the self or of the respondents, and building trust [with the participants]”. Another is “peer debriefing.” This is the process of “allowing a peer who is a professional outside the context and who has some general understanding of the study to analyse materials, test working hypotheses and emerging designs, and listen to the researcher’s ideas and concerns”.

Transferability and Thick Descriptions

Transferability refers to the possibility that what was found in one context by a piece of qualitative research is applicable to another context. As Lincoln and Guba (1985) point out: If there is to be transferability, the burden of proof lies less with the original investigator than with the person seeking to make an application elsewhere. The original inquirer cannot know the sites to which transferability might be sought, but the applicators can and do.... The responsibility of the original investigator ends in providing sufficient descriptive data to make such similarity judgements possible.

Providing sufficient descriptive data is often called “thick description”. For example, if you conduct qualitative research on a day centre for people with mental disabilities, you should describe the context so thickly that users of your research report can judge whether the findings are applicable to their own settings.

Dependability, Confirmability and Auditing

Dependability is the qualitative researcher’s equivalent of the conventional term “reliability”, which is equal to replicability. In quantitative research, reliability means

that the same tests should produce the same results. For qualitative researchers, this kind of replicability is impossible to realise because the research design is so flexible and the research findings are produced by constantly changing interactions between researchers and participants. Therefore, as Guba and Lincoln (1989) states, Far from being threats to dependability, such changes and shifts are hallmarks of a maturing-and successful-inquiry. But such changes and shifts need to be both tracked and trackable. This idea leads to a technique of “auditing,” mentioned below.

On the other hand, confirmability is parallel to “objectivity” in conventional criteria, and is “concerned with establishing the fact that the data and interpretations of an inquiry were not merely figments of the inquirer’s imagination.” Researchers need to link “assertions, findings, and interpretations, and so on to the data themselves in readily discernible ways”.

Although dependability and confirmability correspond to different notions in conventional criteria, both are realised by similar techniques in qualitative research. Of these techniques, “auditing” is emphasised strongly by many authors. Auditing in qualitative research is analogous to a fiscal audit. Schwandt (1997) states that auditing is “a procedure whereby a third-party examiner systematically reviews the audit trail maintained by the inquirer”. In the case of qualitative interview research, the audit trail includes recorded materials such as cassette tapes, interview transcripts, interview guides, lists of interviewees, lists of categories and hypotheses the researcher used while analysing the data, notes about research procedures, and so on.

Lincoln and Guba (1985) claim that “even for a complex project, a week to ten days will be sufficient” to complete auditing. However, that may sound too expensive for our readers, most of whom may be novices, and they might be relieved to see what other researchers have written:

The researcher’s work in preparing an “audit trail” and the auditor’s analysis, with its very detailed procedures, are at least as expensive.... We should probably expect that detailed documentation and auditing will continue to be restricted to high-stakes studies, or to those in which the researcher has a special interest in documentation or auditing as such.

Huberman and Miles (1994) also warn that this sort of re-analysis through auditing “raises questions about invasion of privacy and about potential harm to informants”. In spite of these risks and limitations, simpler types of auditing could be useful tools for improving the quality of qualitative research.

Research Process and Design

AN OVERVIEW

We understand the world by asking questions and searching for answers. Our construction of reality depends on the nature of our inquiry.

Until the sixteenth century, human inquiry was primarily based on introspection. The way to know things was to turn inward and use logic to seek the truth. This paradigm had endured for a millennium and was a well-established conceptual framework for understanding the world. The seeker of knowledge was an integral part of the inquiry process.

A profound change occurred during the sixteenth and seventeenth centuries. Copernicus, Kepler, Galileo, Descartes, Bacon, Newton, and Locke presented new ways of examining nature. Our method of understanding the world came to rely on measurement and quantification. Mathematics replaced introspection as the key to supreme truths. The Scientific Revolution was born.

Objectivity became a critical component of the new scientific method. The investigator was an observer, rather than a participant in the inquiry process. A mechanistic view of the universe evolved. We believed that we could understand the whole by performing an examination of the individual parts. Experimentation and deduction became the tools of the scholar. For two hundred years, the new paradigm slowly evolved to become part of the reality framework of society. The Age of Enlightenment had arrived.

Scientific research methodology was very successful at explaining natural phenomena. It provided a systematic way of knowing. Western philosophers embraced this new structure of inquiry. Eastern philosophy continued to stress the importance of the one seeking knowledge. By the beginning of the twentieth century, a complete schism had occurred. Western and Eastern philosophies were mutually exclusive and incompatible.

Then something remarkable happened. Einstein's proposed that the observer was not separate from the phenomena being studied. Indeed, his theory of relativity actually stressed the role of the observer.

Quantum mechanics carried this a step further and stated that the performance of observation could change the thing being observed. The researcher was not simply an observer, but in fact, was an integral part of the process. In physics, Western and Eastern philosophies have met.

This idea has not been incorporated into the standard social science research model, and today's social science community see themselves as objective observers of the phenomena being studied. However, "it is an established principle of measurement that instruments react with the things they measure." The concept of instrument reactivity states that an instrument itself can disturb the thing being measured.

DEFINITION AND PROBLEM RECOGNITION

All research begins with a question. Intellectual curiosity is often the foundation for scholarly inquiry. Some questions are not testable. The classic philosophical example is to ask, "How many angels can dance on the head of a pin?" While the question might elicit profound and thoughtful revelations, it clearly cannot be tested with an empirical experiment. Prior to Descartes, this is precisely the kind of question that would engage the minds of learned men. Their answers came from within. The modern scientific method precludes asking questions that cannot be empirically tested. If the angels cannot be observed or detected, the question is considered inappropriate for scholarly research. A paradigm is maintained as much by the process of formulating questions as it is by the answers to those questions. By excluding certain types of questions, we limit the scope of our thinking. It is interesting to note, however, that modern physicists have begun to ask the same kinds of questions posed by the Eastern philosophers. "Does a tree falling in the forest make a sound if nobody is there to hear it?" This seemingly trivial question is at the heart of the observer/observed dichotomy. In fact, quantum mechanics predicts that this kind of question cannot be answered with complete certainty. It is the beginning of a new paradigm.

Defining the goals and objectives of a research project is one of the most important steps in the research process. Clearly stated goals keep a research project focused. The process of goal definition usually begins by writing down the broad and general goals of the study. As the process continues, the goals become more clearly defined and the research issues are narrowed.

Exploratory research goes hand-in-hand with the goal clarification process. The literature review is especially important because it obviates the need to reinvent the wheel for every new research question. More importantly, it gives researchers the opportunity to build on each others work.

The research question itself can be stated as a hypothesis. A hypothesis is simply the investigator's belief about a problem. Typically, a researcher formulates an opinion during the literature review process. The process of reviewing other scholar's work often clarifies the theoretical issues associated with the research question. It also can help to elucidate the significance of the issues to the research community.

The hypothesis is converted into a null hypothesis in order to make it testable. "The only way to test a hypothesis is to eliminate alternatives of the hypothesis." Statistical techniques will enable us to reject a null hypothesis, but they do not provide us with a way to accept a hypothesis. Therefore, all hypothesis testing is indirect.

EXPERIMENTAL QUASI-EXPERIMENTAL DESIGN

The design of any experiment is of utmost importance because it has the power to be the most rigid type of research. The design, however, is always dependent on feasibility. The best approach is to control for as many confounding variables as possible in order to eliminate or reduce errors in the assumptions that will be made. It is also extremely desirable that any threats to internal or external validity be neutralized. In the perfect world, all research would do this and the results of research would be accurate and powerful. In the real world, however, this is rarely the case. We are often dealing with human subjects, which in itself confounds any study. We are also dealing with the restraints of time and situation, often resulting in less than perfect conditions in which to gather information.

There are three basic experimental designs, each containing subsets with specific strengths and weaknesses.

These three basic designs include:

1. Pre-experimental design;
2. Quasi-experimental design; and
3. True experimental design.

PRE-EXPERIMENTAL DESIGN

Pre-experimental designs are so named because they follow basic experimental steps but fail to include a control group. In other words, a single group is often studied but no comparison between an equivalent non-treatment group is made. Examples include the following:

The One-Shot Case Study

In this arrangement, subjects are presented with some type of treatment, such as a semester of college work experience, and then the outcome measure is applied, such as

college grades. Like all experimental designs, the goal is to determine if the treatment had any effect on the outcome. Without a comparison group, it is impossible to determine if the outcome scores are any higher than they would have been without the treatment. And, without any pre-test scores, it is impossible to determine if any change within the group itself has taken place.

One Group Pretest Posttest Study

A benefit of this design over the previously discussed design is the inclusion of a pretest to determine baseline scores. To use this design in our study of college performance, we could compare college grades prior to gaining the work experience to the grades after completing a semester of work experience. We can now at least state whether a change in the outcome or dependent variable has taken place. What we cannot say is if this change would have occurred even without the application of the treatment or independent variable. It is possible that mere maturation caused the change in grades and not the work experience itself.

The Static Group Comparison Study

This design attempts to make up for the lack of a control group but falls short in relation to showing if a change has occurred. In the static group comparison study, two groups are chosen, one of which receives the treatment and the other does not. A posttest score is then determined to measure the difference, after treatment, between the two groups. As you can see, this study does not include any pre-testing and therefore any difference between the two groups prior to the study are unknown.

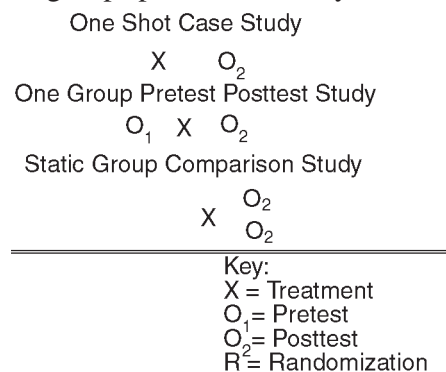


Fig. Diagrams of Pre-Experimental Designs.

QUASI-EXPERIMENTAL DESIGN

Quasi designs fair better than pre-experimental studies in that they employ a means to compare groups. They fall short, however on one very important aspect of the experiment: randomization.

Pretest Posttest Nonequivalent Group

With this design, both a control group and an experimental group is compared, however, the groups are chosen and assigned out of convenience rather than through randomization. This might be the method of choice for our study on work experience as it would be difficult to choose students in a college setting at random and place them in specific groups and classes. We might ask students to participate in a one-semester work experience programme. We would then measure all of the students' grades prior to the start of the programme and then again after the programme. Those students who participated would be our treatment group; those who did not would be our control group.

Time Series Designs

Time series designs refer to the pretesting and posttesting of one group of subjects at different intervals. The purpose might be to determine long term effect of treatment and therefore the number of pre- and posttests can vary from one each to many. Sometimes there is an interruption between tests in order to assess the strength of treatment over an extended time period. When such a design is employed, the posttest is referred to as follow-up.

Nonequivalent Before-After Design

This design is used when we want to compare two groups that are likely to be different even before the study begins. In other words, if we want to see how a new treatment affects people with different psychological disorders, the disorders themselves would create two or more nonequivalent groups.

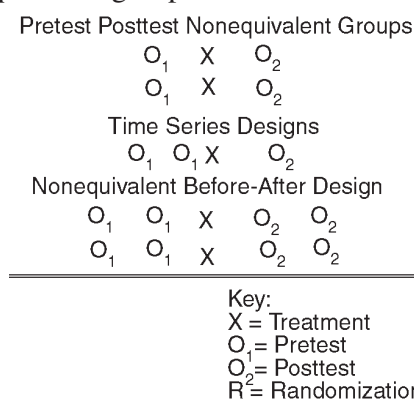


Fig. Diagrams of Quasi Experimental Designs.

Once again, the number of pretests and posttests can vary from one each to many. The obvious concern with all of the quasi-experimental designs results from the method of choosing subjects to participate in the experiment. While we could compare grades and determine if there was a difference between the two groups before and after the

study, we could not state that this difference is related to the work experience itself or some other confounding variable. It is certainly possible that those who volunteered for the study were inherently different in terms of motivation from those who did not participate. Whenever subjects are chosen for groups based on convenience rather than randomization, the reason for inclusion in the study itself confounds our results.

TRUE EXPERIMENTAL DESIGN

True experimental design makes up for the shortcomings of the two designs previously discussed. They employ both a control group and a means to measure the change that occurs in both groups. In this sense, we attempt to control for all confounding variables, or at least consider their impact, while attempting to determine if the treatment is what truly caused the change. The true experiment is often thought of as the only research method that can adequately measure the cause and effect relationship.

Posttest Equivalent Groups Study

Randomization and the comparison of both a control and an experimental group are utilized in this type of study. Each group, chosen and assigned at random is presented with either the treatment or some type of control. Posttests are then given to each subject to determine if a difference between the two groups exists. While this is approaching the best method, it falls short in its lack of a pretest measure. It is difficult to determine if the difference apparent at the end of the study is an actual change from the possible difference at the beginning of the study. In other words, randomization does well to mix subjects but it does not completely assure us that this mix is truly creating an equivalency between the two groups.

Pretest Posttest Equivalent Groups Study

Of those discussed, this method is the most effective in terms of demonstrating cause and effect but it is also the most difficult to perform. The pretest posttest equivalent groups design provides for both a control group and a measure of change but also adds a pretest to assess any differences between the groups prior to the study taking place.

To apply this design to our work experience study, we would select students from the college at random and then place the chosen students into one of two groups using random assignment.

We would then measure the previous semester's grades for each group to get a mean grade point average. The treatment, or work experience would be applied to one group and a control would be applied to the other.

It is important that the two groups be treated in a similar manner to control for variables such as socialization, so we may allow our control group to participate in some activity such as a softball league while the other group is participating in the work experience programme.

At the end of the semester, the experiment would end and the next semester's grades would be gathered and compared. If we found that the change in grades for the experimental group was significantly different than the change in the grades of our control group, we could reasonably argue that one semester of work experience compared to one semester of non-work related activity results in a significant difference in grades.

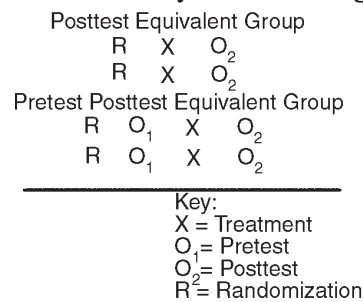


Fig. Diagrams of True Experimental Designs.

NOW WE CAN SAY

The experiment, especially the true experimental design is often the measure of choice when attempting to determine a cause and effect relationship. Utilizing randomization and the pre-testing and post-testing of both an experimental group and a control group allows us to control for more confounding variables than any other research method. These confounding variables, when not addressed, can often result in inaccurate findings.

Controlling for confounding variables is important in research and especially important in the experimental designs. This process helps us assure valid results both internally and externally. The threats to internal validity, those that apply to the experimental situation itself, and external validity, those relating to the generalizability of our results to the real world are also issues of great concern to researchers. As the saying goes: garbage in, garbage out. If we start with a flawed design as suggested, end up with flawed results.

As the degree of control for each of the designs discussed increases, the difficulty in performing the research also increases. Feasibility is always an issue and even when the most stringent control is used, the mere fact that the subjects have agreed to participate in the experiment may have a negative effect on the study's generalizability. Are volunteer subjects truly representative of the population at large? As you can see, there are varying degrees of experimental research, but there is no perfect experiment that controls for all possible variables and assures us of 100% generalizability.

CORRELATIONAL DESIGNS

Alternative Explanatory Variables:

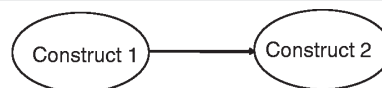
- A developmental psychologist has noticed that children with bigger feet tend to have greater vocabularies than children with smaller feet?

- This is an example of a simple correlational design. Can you think of a powerful alternative explanation?

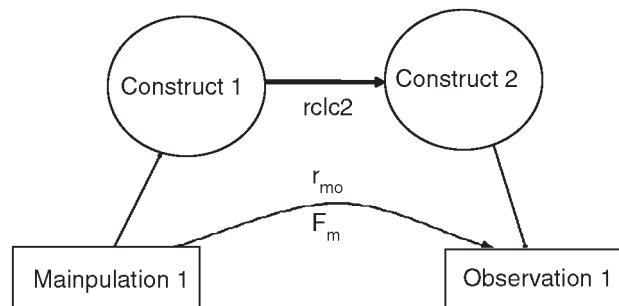
TYPES OF DATA AND CORRELATIONAL DESIGNS

- Varieties of data
 - Direct
 - i. Self report of personality/attitudes/values
 - ii. Peer/supervisor/subordinate ratings of performance
 - iii. Ability scales
 - Indirect
 - i. Implicit measures
 - ii. Unobtrusive measures
- Historical, archival

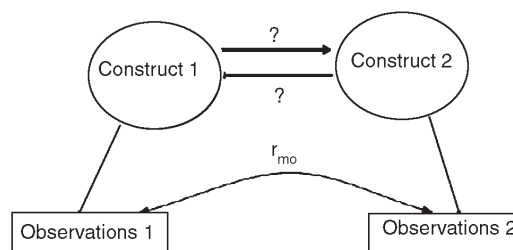
THEORY AND THEORY TESTING I: THEORY



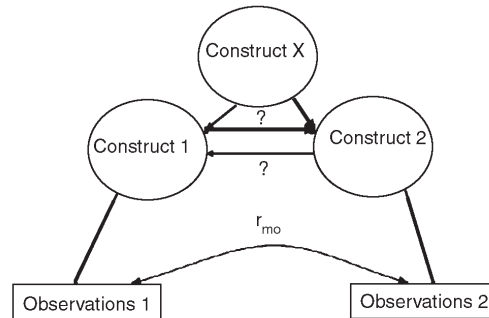
THEORY AND THEORY TESTING II: EXPERIMENTAL MANIPULATION



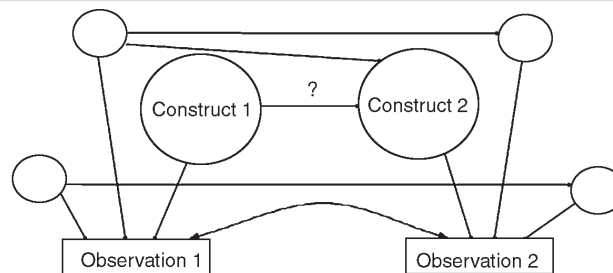
THEORY AND THEORY TESTING III: CORRELATIONAL INFERENCE



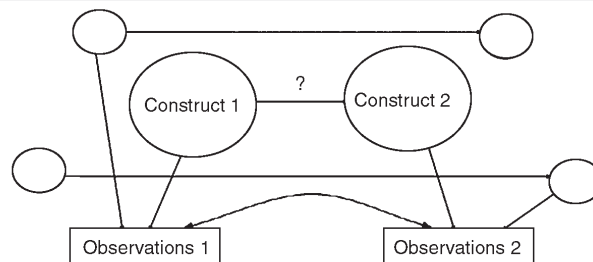
THEORY AND THEORY TESTING IV: CORRELATIONAL INFERENCE



THEORY AND THEORY TESTING V: ALTERNATIVE EXPLANATIONS



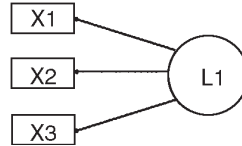
THEORY AND THEORY TESTING VI: ELIMINATE ALTERNATIVE EXPLANATIONS



STEPS IN CORRELATIONAL INFERENCE

- Estimate the reliability of the variables
 - Magnitude of correlation is influenced by the reliability of the correlation
 - Varieties of reliability
- Estimate the construct validity of the measures
 - Convergent, Discriminant, Incremental validity
- Consider alternative explanatory variables

CLASSIC RELIABILITY THEORY: HOW WELL DO WE MEASURE WHAT EVER WE ARE MEASURING



Classic Reliability Theory

How well do we measure what ever we are measuring and what is the relationships between latent variables.

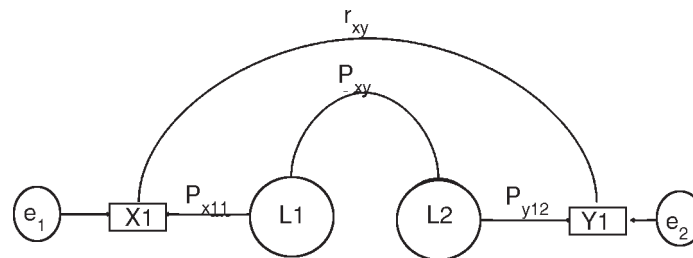
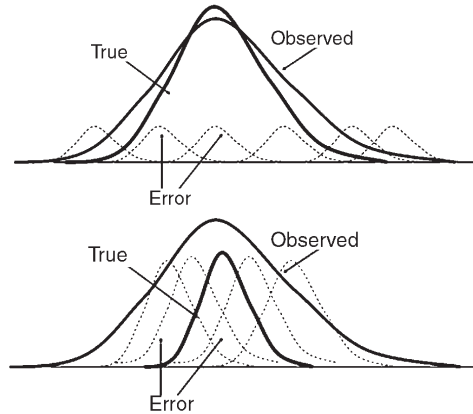
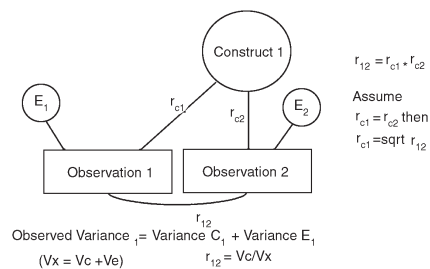


Fig. Observed= True + Error

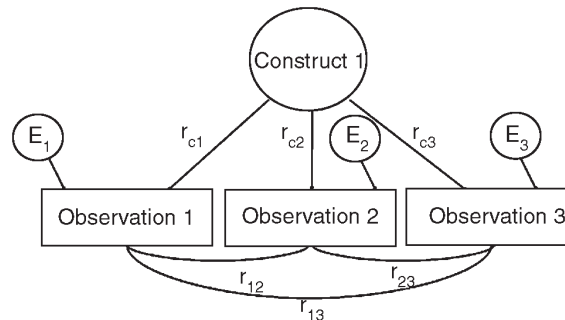


RELIABILITY OF MEASUREMENT

(How well does an observation reflect the construct)



(How well does an observation reflect the construct)



DOMAIN SAMPLING THEORY

Consider a domain (D) of k items relevant to a construct.. Let D_i represent the number of items in D which the subject can pass given all D items. Call this the domain score for subject I. What is the correlation of scores on an item j with the domain scores?

THE NATURE OF CORRELATION

Correlational research studies almost always use the correlation coefficient to indicate the degree of relationship between two variables. The correlation coefficient is a number ranging from 1 through 0 to -1 . It is tempting to think of a correlation coefficient as indicating the proportion of sameness between the two variables.

But this is not true: A correlation of .90 does not mean that the two variables are 90% the same. In fact, the proportion of common variance for a correlation of .90 is .81 or 81%.

The proportion of common variance is the square of the correlation coefficient. If we have a correlation of .50 between two variables the proportion of common variance is only .25 or 25% ($.50 \times .50 = .2500$). We could say that the two variables demonstrate 25% sameness but 75% ($100 - 25$) differentness.

ACTION RESEARCH DESIGN

Action Research (AR) represents a growing field of educational research whose chief identifying characteristic is the recognition of the pragmatic requirements of educational practitioners for organized reflective inquiry into classroom instruction. AR is an process designed to empower all participants in the educational process (students, instructors and other parties) with the means to improve the practices conducted within the educational experience. All participants were knowing, active members of the research process.

Action research has been described as an informal, qualitative, formative, subjective, interpretive, reflective and experiential model of inquiry in which all individuals involved

in the study are knowing and contributing participants. Action research has the primary intent of providing a framework for qualitative investigations by teachers and researchers in complex working classroom situations. *Some of the most widely accepted definitions of Action Research include following:*

- *Action Research:* Aims to contribute both to the practical concerns of people in an immediate problematic situation and to the goals of social science by joint collaboration within a mutually acceptable ethical framework.
- Action Research is a form of self-reflective enquiry undertaken by participants in social (including educational) situations in order to improve the rationality and justice of
 - Their own social or educational practices,
 - Their understanding of these practices, and
 - The situations in which the practices are carried out. It is most rationally empowering when undertaken by participants collaboratively... \sometimes in cooperation with outsiders.
- *Action Research:* Is the systematic study of attempts to improve educational practise by groups of participants by means of their own practical actions and by means of their own reflection upon the effects of those actions.

The action research framework is most appropriate for participants who recognize the existence of shortcomings in their educational activities and who would like to adopt some initial stance in regard to the problem, formulate a plan, carry out an intervention, evaluate the outcomes and develop further strategies in an iterative fashion. In short, action research is characterized by those constraints and strengths given a research methodology intended to be a workable technique for working classroom teachers.

ACTION RESEARCH DESIGN

The essentials of action research design are considered by Elliott as per the following characteristic cycle:

- Initially an exploratory stance is adopted, where an understanding of a problem is developed and plans are made for some form of interventionary strategy.
- Then the intervention is carried out.
- During and around the time of the intervention, pertinent observations are collected in various forms.
- The new interventional strategies are carried out, and the cyclic process repeats, continuing until a sufficient understanding of the problem is achieved.

The protocol is iterative or cyclical in nature and is intended to foster deeper understanding of a given situation, starting with conceptualizing and particularizing the problem and moving through several interventions and evaluations. A representation of an AR protocol by Kemmis is provided in Figure.

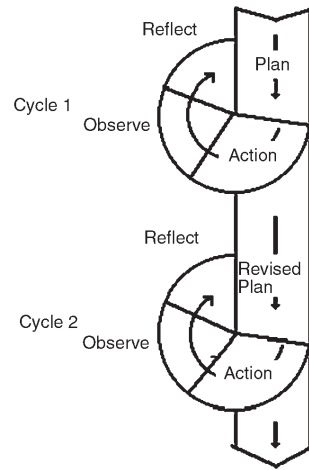


Fig. Action Research Protocol after Kemmis.

Figure clearly displays the iterative nature of AR along with the major steps of planning, action, observation and reflection before revising the plan. This may be thought of as similar in nature to the numerical computing technique known as successive approximation - the idea is to close in upon a final goal or outcome by repeated iterations.

Later protocols reflect changes in the goal as determined via experience during the reflections of earlier iterations of AR. For instance, Figure reflects the evolution of the general idea or main topic of interest throughout the process.

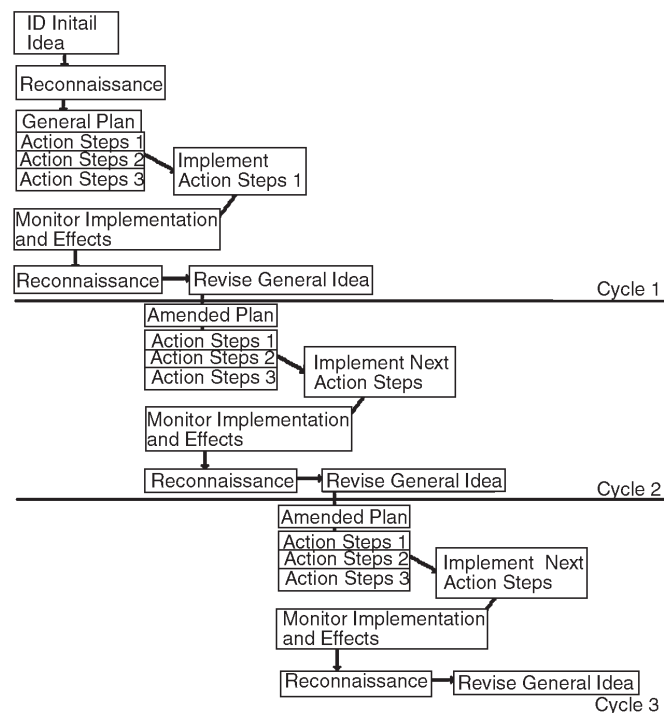


Fig. Action Research after Elliott.

Elliott's model emphasizes constant evolution and redefinition of the original goal through a series of reconnaissances recurring every cycle. The reconnaissance necessarily includes some degree of analysis. This design permits much greater flexibility, and seeks to "...recapture some of the 'messiness' which the Kemmis version tends to gloss [over] ". Ebbutt further emphasizes the evolution of the overall plan through a spiral analogy.

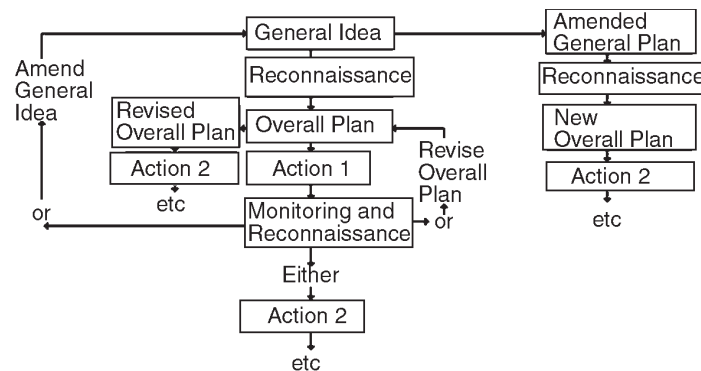


Fig. Action Research Protocol after Ebbutt.

THE ROLE OF COMMUNICATION

Another distinguishing characteristic of Action Research is the degree of empowerment given to all participants. Involvement is of a knowing nature, with no hidden controls or preemption of direction by the researcher. All participants negotiate meaning from the data and contribute to the selection of interventionary strategies, including the university researchers, the teachers and the students.

THE ROLE OF REFLECTION

Another distinguishing characteristic of action research is the degree of empowerment given to all participants. Involvement is of a knowing nature, with no hidden controls or preemption of direction by the researcher. All participants including the university researchers, the teachers and the students negotiate meaning from the data and contribute to the selection of interventionary strategies.

Elliott considers the need for communication between all participants to be of paramount importance:

Since action research looks at a problem from the point of view of those involved it can only be validated in unconstrained dialogue with them....Since action research involves unconstrained dialogue between "researcher" (whether he be an outsider or teacher/researcher) and the participants, there must be free information flow between them.

Perhaps the key component involved in action research is the notion of praxis. Action research is intended to be the reflective counterpart of practical diagnosis. Schon

describes the use of reflection to generate models from a body of previous knowledge. These models are used to re-frame a problem; then experiments are performed to bring about outcomes which are subjected to further analysis. This model (called reflection-in-action) frames means and ends interdependently and recognizes that there is little or no separation of research from practice, little or no separation of knowing and doing. Schon's model of reflection-in-action compliments the iterative and investigative natures of action research.

CASE STUDY RESEARCH DESIGN

The case study research design have evolved over the past few years as a useful tool for investigating trends and specific situations in many scientific disciplines, especially social science, psychology, anthropology and ecology. This method of study is especially useful for trying to test theoretical models by using them in real world situations. For example, if an anthropologist were to live amongst a remote tribe, whilst their observations might produce no quantitative data, they are still useful to science.

WHAT IS A CASE STUDY?

Basically, a case study is an in depth study of a particular situation rather than a sweeping statistical survey. It is a method used to narrow down a very broad field of research into one easily researchable topic. Whilst it will not answer a question completely, it will give some indications and allow further elaboration and hypothesis creation on a subject. The case study research design is also useful for testing whether scientific theories and models actually work in the real world. You may come out with a great computer model for describing how the ecosystem of a rock pool works but it is only by trying it out on a real life pool that you can see if it is a realistic simulation.

For psychologists, anthropologists and social scientists they have been regarded as a valid method of research for many years. Scientists are sometimes guilty of becoming bogged down in the general picture and it is sometimes important to understand specific cases and ensure a more holistic approach to research.

THE ARGUMENT FOR AND AGAINST THE CASE STUDY RESEARCH DESIGN

Some argue that because a case study is such a narrow field that its results cannot be extrapolated to fit an entire question and that they show only one narrow example. On the other hand, it is argued that a case study provides more realistic responses than a purely statistical survey. The truth probably lies between the two and it is probably best to try and synergize the two approaches. It is valid to conduct case studies but they should be tied in with more general statistical processes. For example, a statistical

survey might show how much time people spend talking on mobile phones, but it is case studies of a narrow group that will determine why this is so. The other main thing to remember during case studies is their flexibility.

Whilst a pure scientist is trying to prove or disprove a hypothesis, a case study might introduce new and unexpected results during its course, and lead to research taking new directions. The argument between case study and statistical method also appears to be one of scale. Whilst many 'physical' scientists avoid case studies, for psychology, anthropology and ecology they are an essential tool. It is important to ensure that you realise that a case study cannot be generalized to fit a whole population or ecosystem. Finally, one peripheral point is that, when informing others of your results, case studies make more interesting topics than purely statistical surveys, something that has been realised by teachers and magazine editors for many years. The general public has little interest in pages of statistical calculations but some well placed case studies can have a strong impact.

HOW TO DESIGN AND CONDUCT A CASE STUDY

The advantage of the case study research design is that you can focus on specific and interesting cases. This may be an attempt to test a theory with a typical case or it can be a specific topic that is of interest. Research should be thorough and note taking should be meticulous and systematic. The first foundation of the case study is the subject and relevance. In a case study, you are deliberately trying to isolate a small study group, one individual case or one particular population.

For example, statistical analysis may have shown that birthrates in African countries are increasing. A case study on one or two specific countries becomes a powerful and focused tool for determining the social and economic pressures driving this.

In the design of a case study, it is important to plan and design how you are going to address the study and make sure that all collected data is relevant. Unlike a scientific report, there is no strict set of rules so the most important part is making sure that the study is focused and concise; otherwise you will end up having to wade through a lot of irrelevant information. It is best if you make yourself a short list of 4 or 5 bullet points that you are going to try and address during the study. If you make sure that all research refers back to these then you will not be far wrong. With a case study, even more than a questionnaire or survey, it is important to be passive in your research. You are much more of an observer than an experimenter and you must remember that, even in a multi-subject case, each case must be treated individually and then cross case ceases can be drawn.

HOW TO ANALYSE THE RESULTS

Analyzing results for a case study tends to be more opinion based than statistical methods. The usual idea is to try and collate your data into a manageable form and construct a narrative around it.

Use examples in your narrative whilst keeping things concise and interesting. It is useful to show some numerical data but remember that you are only trying to judge trends and not analyse every last piece of data. Constantly refer back to your bullet points so that you do not lose focus.

It is always a good idea to assume that a person reading your research may not possess a lot of knowledge of the subject so try to write accordingly.

In addition, unlike a scientific study which deals with facts, a case study is based on opinion and is very much designed to provoke reasoned debate. There really is no right or wrong answer in a case study.

CREATING A RESEARCH DESIGN

DESIGN AND THE PURPOSE OF THE RESEARCH

Your research *design* is your plan for providing a sound, and, if possible, conclusive answer to your research question. Designs vary greatly, depending on the research question being addressed and the methods of data collection the investigator chooses to use. Creating a design starts with the purpose of the research.

Most research is done for one of three purposes. The most common purpose is to *describe* some set of variables or relationships among them as accurately as possible. Other purposes of research are to *explore* a topic to learn more about it or to *explain* why certain social patterns or relationships occur as they do.

Some studies combine more than one purpose. For example, while the Sudan Fertility Study was designed primarily as a descriptive investigation, we showed how data from this study were analysed to explain why future fertility rates may decline in northern Sudan. Nevertheless, as suggested, describe each approach separately to show how each purpose affects design decisions. Exploratory research generally requires less rigorous design decisions; so as suggested, start with it.

Exploratory Research

Exploratory research, as the name suggests, is a way of gaining some initial information about a problem or topic. You may be interested in a certain problem, but don't have enough information to write a clear research question. Perhaps you are not sure what the critical variables are or what methods of data collection may work best. Before beginning an investigation, you might decide to explore the problem by doing some informal interviews or by living for a short time with the group you want to study. These techniques are often used in exploratory research.

Example. Julia and Ridha explored experiences of Kuwaiti women during the Iraqi invasion and following occupation of Kuwait in 1990. Twenty women who played various roles, including active opposition to the Iraqi invaders, told of their experiences before, during, and after the invasion. The women reported that they actively sought

their own self-development and liberation, but once the conflict ended, their choices and rights tended to become restricted again by traditional male prerogatives.

Exploratory studies, however, seldom provide satisfactory answers to research questions. One reason is that most exploratory studies are based on samples too small to permit generalizing the results to a larger population.

This is certainly true of the study of Kuwaiti women. But exploratory studies can provide valuable, even critical, information for designing larger scale descriptive or explanatory studies.

Descriptive Research

Descriptive research is more specific and focused than exploratory research. The researcher starts with a well defined problem or research question and a clearly defined plan for collecting and analyzing data. Descriptive research is intended to produce clear, well-founded answers to some question or specific, factual information. Surveys are frequently used in descriptive research.

Example. Fattah conducted interviews, using a questionnaire, with 200 randomly selected farm families in Iraq. The questionnaire contained over 200 items about husband-wife decision-making related to the operation of the farm, financial matters, social life, entertainment, training of children, and childcare. She also obtained information on whether the families had a “modern” or “traditional” view of things. These data provided a detailed description of the degree to which wives participated in decision-making with their husbands. Fattah concluded that factors associated with modernism will continue to increase and will enhance the position of women in rural Iraq, which, today, remains male dominated.

Explanatory Research

Explanatory research goes beyond exploratory or descriptive research by trying to find the reasons why certain relationships occur. It seeks to provide explanations for what has been observed. Explanations are based on interpretation of findings in terms of broader concepts and accepted theory.

Example. Using extensive interviews and observation, Dei developed an explanation for how Ghanian villagers survived prolonged drought and related hardships. He found they got through hard times by shifting from farm production for a market economy to shared production with other villagers for household consumption.

A strong sense of community solidarity developed and allowed the villages to survive with what they could produce together. The explanation for their survival was the successful return to earlier farming methods.

Regardless of the purpose of your research, you will also need to decide on a technique for data collection. There are two contrasting yet complimentary techniques - quantitative and qualitative means of collecting data.

QUANTITATIVE AND QUALITATIVE DATA

Many fields of science, particularly the physical sciences, report observations in terms of numbers. Observations reported in numbers are referred to as quantitative data. Social scientists obtain and analyse quantitative data, as you have seen with various studies we have cited. Social scientists also obtain and analyse data in qualitative form. These are observations recorded in words - descriptions of what persons said or did, how they interacted with one another, or what a researcher observed by watching their behaviour. Each technique is useful; each also has certain strengths and weaknesses; each also presents a different set of design decisions.

Some researchers draw a sharp distinction between quantitative and qualitative research. A different view is presented in Types of Data. This discussion shows how quantitative data is based in part on qualitative judgments and that qualitative data can also be described and analysed numerically.

Quantitative Research

A quantitative approach requires a well-developed research design.

Quantitative research is usually based on:

- Careful and precise specification of the question to be answered;
- Identification, definition, and measurement of the key variables;
- Selection and specification of one or more methods of collecting data;
- Development of a sampling plan; and
- Numerical analysis of the data, including use of appropriate statistical tests.

Numbers used in quantitative studies may be as simple as counting the number of "Yes" versus "No" responses to some question. More complex forms of measurement, however, are generally used. The descriptive research by Fattah, for example, used a quantitative design. The many decision-making variables were defined and measures were developed for each. These measures, in the form of questions, were then combined in a questionnaire that was used in interviewing a random sample of farm families.

Quantitative data has important strengths. Using numerical measures provides more precise descriptions of variables. In addition, quantitative research permits use of larger samples. Also, quantitative data can be combined and analysed using various statistical techniques. You are certainly familiar with averages and percentages as ways of summarizing data.

Most of this site focuses on planning and conducting quantitative research and analyzing data based on quantitative variables.

Qualitative Research

Qualitative research, in contrast to quantitative research, seeks to dig deeper into the reasons for behaviour we observe. Generally, researchers doing qualitative research start with a broad idea of what they are looking for. In qualitative studies, researchers often

combine informal interviews, in the form of long conversations with people, with systematic observation of their daily activities. Data are derived continuously from what the persons in the group being studied do and say, how they say they feel about things, and the reasons they give the researcher for what they do. The researcher accumulates a large amount of data, generally in the form of long sets of notes and written descriptions of what was observed. As data are recorded, the researcher tries to interpret what the data mean and to develop an explanation for what has been observed. In this sense, analysis of data proceeds with data collection. Qualitative research is generally a much more flexible process, allowing researcher to take advantage of new lines of inquiry as they develop.

The research studies cited earlier by Julia and Ridha and by Dei were based on qualitative research methods. Research by ElSayed and Ahmed on the “Socio-Cultural Aspect of Kala-azar among the Masalit and Hansa Tribes” further emphasizes use of qualitative techniques.

The authors used small group discussions with villagers, personal interviews with patients, and direct observation of health and sanitation practices in the villages. They found that lack of knowledge about the disease is the main reason for its spread.

Combining Quantitative and Qualitative Techniques

Researchers frequently use both quantitative and qualitative techniques in the same investigation. In this way, researchers benefit from the strengths of each approach and minimize their respective limitations. Altareb, for example, investigated attitudes towards Muslims held by undergraduate students at an American university. He began by conducting focus groups, a qualitative technique, to get an initial understanding of students’ attitudes. After this exploratory phase, he developed a quantitative measure of attitudes for the main part of the investigation.

Davidson also combined qualitative and quantitative methods in his investigation of ways families in the Nuba Mountains of the Sudan were adapting to rapidly changing socio-economic conditions in their area. He began by interviewing village elders and then conducted a survey to identify the characteristics of each of the 12 villages in his study area. Following the survey, he carried out in-depth interviews with selected households. This combination of methods provided a rich mix of quantitative data for describing adaptive techniques used by villagers and detailed and qualitative data for understanding how they were adapting to changes.

UNIT OF ANALYSIS

In designing research you also have to decide on what basis you will analyse the data you will collect. This requires a decision about the part *of analysis* for the study. Units of analysis may consist of individuals, groups, organizations, geographical places, or other entities. Generally the way you state your research question indicates which part will be used as the basis of analyzing the data.

It is easy to confuse part of analysis with the entities you collect data from. Frequently, they are the same. In the Sudan Fertility Survey, for example, the data were collected from women and the data were analysed in terms of the same women.

In this case, the decision was clear. In another study, a researcher might interview factory managers to get information to compare the efficiency of different sized factories. In this case, the part of analysis is factories, not managers because the purpose of the research is to compare factories and managers happened to be the source of data. If you have any doubt, think about the question you are trying to answer. How do the data have to be organized to answer the question?

Individuals as the Part of Analysis

When the purpose of research is to say something about individuals, they become the part of analysis. Individuals are the most frequent part of analysis. They may be any age, either gender, or become the units of analysis because they have a common characteristic. Primary, intermediate and secondary students were the units of analysis in a trend study by Badri and Burchinal.

Many studies are based on collecting data from university students, who then become the part of analysis. Cook, for example, investigated the attitudes of Egyptian university students towards Islam and higher education; Had-Elzein and Ahmed obtained data on perceptions and vision of peace from Sudanese university students.

Married women were the part of analysis in the Sudan Fertility Survey. Mageed, Sulima and Kawther collected data about attitudes towards female genital mutilation from adult men and women.

Agricultural technical workers were the part of analysis in a study of worker satisfaction in Kenya. Muneer selected farmers, who became the part of analysis, in his study of the role of cooperatives in western Sudan. Khalafalla selected policy-makers as her sample who also became her part of analysis.

Groups as the Part of Analysis

Research is often conducted by collecting and analyzing data for groups of people. When the purpose is to describe two or more persons as a unit, these groups become the part of analysis. Households, consisting of various numbers of persons, often are used as the part of analysis. Examples include the research by Julia and Ridha on Iraqi farm families and research by Davidson on changes in household life in the Nuba Mountains. Households were the part of analysis in an extensive study of year-long spending patterns among farm families in Sierra Leon. The sample used by Grotberg and Badri consisted of intact families with a child not older than five and one-half years. The mother-child pairs in these families became the part of analysis in this study. Other units of analysis at the group level include friendship groups, clubs, social groups, and groups of street children.

Organizations as the Part of Analysis

Research often is directed at learning about organizations, such as businesses, firms, ministries of government, universities, political parties, religious bodies, or military units. Although data about each organization may be obtained by interviewing members of an organization, the data would be used to describe and compare features of each organization. For example, Gimbel and associates studied job satisfaction in community-based organizations. They collected data from employees but analysed their data in terms of characteristics of the organizations. Mohamed analysed operations in a sample of 150 government divisions in the United Arab Emirates.

In a study of safety programmes, Vredenburg collected data about practices in 62 hospitals. These are only a few of the many management studies based on organizations as the part of analysis.

Geographical Places as the Part of Analysis

Places, such as where people live, can also become the focus of study and therefore be used as the part of analysis. Fishing villages in Uganda were the part of analysis in a socio-economic investigation by Namara. Cities, provinces, and rural versus urban areas are used as the part of analysis in most reports issued by international organizations. Countries frequently are used as the part of analysis. Bulluck compared the extent to which developing countries meet the social needs of citizens. Countries were also used in comparing infant mortality rates throughout sub-Saharan Africa. Wang examined the extent to which women's reproductive rights were protected in 101 countries.

Other Units of Analysis

Although most social research is based on the kinds of units of analysis just mentioned, other things may be selected for analysis. These include the contents of newspapers and magazines; rural development projects; and folk tales.

Summarizes frequently used units of analysis.

1. *Individuals*: By far the most frequently used
2. *Groups*: Husband-wife pairs, social groups
3. *Organizations*—businesses, offices, factories, government ministries
4. *Classrooms, schools, university student bodies or groups*
5. *Geographical areas*: Villages, communities, cities, regions of a country, entire countries
6. *Mass media materials*: Newspapers, magazines, television shows

Deciding on the Part of Analysis

Most likely the part of analysis you will use in a study will be clear from the way you defined your problem. If you have any doubts about the part you should use, you will

need to resolve this confusion before going on. Otherwise, you may not analyse your data properly and your entire study could be jeopardized. Ask yourself what part you will base your analysis on. This is, as the term implies, your part of analysis. Also, in the process of analyzing your data, you may shift from one part to another.

In the study by Gimbel and associates mentioned under organizations as the part of analysis, the authors used organizations as the part of analysis. However, since they collected data from employees, they could have undertaken other analyses as well. For example, they might have become interested in the level of morale among employees and the length of time each worked for one of the organizations. In this case, they would have shifted to employees as the part of analysis. Our point is simple: just be sure of your part of analysis and that you have data appropriate to that unit.

Collecting Data at the Lowest Possible Part

Also, remember an important rule: Regardless of what you are studying, always obtain data in terms of the lowest part of analysis possible. There is a simple reason for this. You can usually combine data collected at a lower level into a higher level for analysis, but it won't work the other way around. For example, if you intended to analyse data about the production of things by households, collect data about the production activities for each member of the household. Then you can combine the data from each household member for analysis at the level of the entire household. This way you can still do other analyses at the level of members of the household, such as what wives and daughters produced in contrast to husbands and sons, as well as for the entire household. But if you asked only for production of the household as a whole, you would not be able to describe production by separate members of the household.

Design Alternatives

Designing research is a creative process. Designs can be put together in a number of ways. By describing some of

the more frequently used designs, we hope you will get an idea of the various kinds of research questions you can ask and designs you can use in answering them.

To start, designs can be developed for:

- Describing variables as they currently exist;
- For describing changes that have occurred; and
- For describing changes as they occur.

Cross-sectional Designs

Surveys are designed to describe some set of variables as they exist at the time of data collection. Data are collected from a sample over a relatively short period of time. The design used for a survey is also called a *cross-sectional* design because the sample is carefully selected to represent the cross section of some population. Either quantitative

or qualitative techniques can be used to collect survey data, although, in general, quantitative techniques are used. The frequently cited Sudan Fertility Survey was based on a survey or cross-sectional design.

The survey design can be expanded to include more than one sample, as suggested, show shortly as part of the discussion of seeking differences between groups with respect to some variable.

SAMPLING

It is incumbent on the researcher to clearly define the target population. There are no strict rules to follow, and the researcher must rely on logic and judgement. The population is defined in keeping with the objectives of the study. Sometimes, the entire population will be sufficiently small, and the researcher can include the entire population in the study. This type of research is called a census study because data is gathered on every member of the population.

Usually, the population is too large for the researcher to attempt to survey all of its members. A small, but carefully chosen sample can be used to represent the population. The sample reflects the characteristics of the population from which it is drawn.

Sampling methods are classified as either probability or Non-probability. In probability samples, each member of the population has a known probability of being selected. Probability methods include random sampling, systematic sampling, and stratified sampling. In Non-probability sampling, members are selected from the population in some nonrandom manner.

These include convenience sampling, judgement sampling, quota sampling, and snowball sampling. The other common form of Non-probability sampling occurs by accident when the researcher inadvertently introduces nonrandomness into the sample selection process. The advantage of probability sampling is that sampling error can be calculated. Sampling error is the degree to which a sample might differ from the population. When inferring to the population, results are reported plus or minus the sampling error.

In Non-probability sampling, the degree to which the sample differs from the population remains unknown:

- *Random sampling:* Is the purest form of probability sampling. Each member of the population has an equal chance of being selected. When there are very large populations, it is often difficult or impossible to identify every member of the population, so the pool of available subjects becomes biased. Random sampling is frequently used to select a specified number of records from a computer file.
- *Systematic sampling:* Is often used instead of random sampling. It is also called an Nth name selection technique. After the required sample size has been calculated, every Nth record is selected from a list of population members.

As long as the list does not contain any hidden order, this sampling method is as good as the random sampling method. Its only advantage over the random sampling technique is simplicity.

- *Stratified sampling*: Is commonly used probability method that is superior to random sampling because it reduces sampling error. A stratum is a subset of the population that share at least one common characteristic. The researcher first identifies the relevant strata and their actual representation in the population. Random sampling is then used to select subjects for each stratum until the number of subjects in that stratum is proportional to its frequency in the population.
- *Convenience sampling*: Is used in exploratory research where the researcher is interested in getting an inexpensive approximation of the truth. As the name implies, the sample is selected because they are convenient. This Non-probability method is often used during preliminary research efforts to get a gross estimate of the results, without incurring the cost or time required to select a random sample.
- *Judgement sampling*: Is a common Non-probability method. The researcher selects the sample based on judgement. This is usually an extension of convenience sampling. For example, a researcher may decide to draw the entire sample from one “representative” city, even though the population includes all cities. When using this method, the researcher must be confident that the chosen sample is truly representative of the entire population.
- *Quota sampling*: Is the Non-probability equivalent of stratified sampling. Like stratified sampling, the researcher first identifies the strata and their proportions as they are represented in the population. Then convenience or judgement sampling is used to select the required number of subjects from each stratum. This differs from stratified sampling, where the strata are filled by random sampling.
- *Snowball sampling*: Is a special Non-probability method used when the desired sample characteristic is rare. It may be extremely difficult or cost prohibitive to locate respondents in these situations. Snowball sampling relies on referrals from initial subjects to generate additional subjects. While this technique can dramatically lower search costs, it comes at the expense of introducing bias because the technique itself reduces the likelihood that the sample will represent a good cross section from the population.

DATA COLLECTION

Data collection is a term used to describe a process of preparing and collecting data - for example as part of a process improvement or similar project. The purpose of data collection is to obtain information to keep on record, to make decisions about important issues, to pass information on to others. Primarily, data is collected to provide information regarding a specific topic.

Data collection usually takes place early on in an improvement project, and is often formalised through a data collection plan which often contains the following activity.

- *Pre collection activity:* Agree goals, target data, definitions, methods
- *Collection:* Data collection
- *Present Findings:* Usually involves some form of sorting analysis and/or presentation.

Prior to any data collection, pre-collection activity is one of the most crucial steps in the process. It is often discovered too late that the value of their interview information is discounted as a consequence of poor sampling of both questions and informants and poor elicitation techniques. After pre-collection activity is fully completed, data collection in the field, whether by interviewing or other methods, can be carried out in a structured, systematic and scientific way.

A formal data collection process is necessary as it ensures that data gathered is both defined and accurate and that subsequent decisions based on arguments embodied in the findings are valid. The process provides both a baseline from which to measure from and in certain cases a target on what to improve.

Types of data collection:

- 1-By mail questionnaires
- 2-By personal interview.

Other main types of collection include census, sample survey, and administrative by-product and each with their respective advantages and disadvantages. A census refers to data collection about everyone or everything in a group or population and has advantages, such as accuracy and detail and disadvantages, such as cost and time. A sample survey is a data collection method that includes only part of the total population and has advantages, such as cost and time and disadvantages, such as accuracy and detail. Administrative by-product data is collected as a byproduct of an organization's day-to-day operations and has advantages, such as accuracy, time simplicity and disadvantages, such as no flexibility and lack of control.

DATA ANALYSIS

Data analysis is a practice in which raw data is ordered and organized so that useful information can be extracted from it. The process of organizing and thinking about data is key to understanding what the data does and does not contain. There are a variety of ways in which people can approach data analysis, and it is notoriously easy to manipulate data during the analysis phase to push certain ceases or agendas. For this reason, it is important to pay attention when data analysis is presented, and to think critic

Raw data can take a variety of forms, including measurements, survey responses, and observations. In its raw form, this information can be incredibly useful, but also overwhelming. Over the course of the data analysis process, the raw data is ordered in a way which will be useful. For example, survey results may be tallied, so that people can see at a glance how many people answered the survey, and how people responded

to specific questions. In the course of organizing the data, trends often emerge, and these trends can be highlighted in the writeup of the data to ensure that readers take note. In a casual survey of ice cream preferences, for example, more women than men might express a fondness for chocolate, and this could be a point of interest for the researcher. Modeling the data with the use of mathematics and other tools can sometimes exaggerate such points of interest in the data, making them easier for the researcher to see.

Charts, graphs, and textual writeups of data are all forms of data analysis. These methods are designed to refine and distill the data so that readers can glean interesting information without needing to sort through all of the data on their own. Summarizing data is often critical to supporting arguments made with that data, as is presenting the data in a clear and understandable way. When people encounter summarized data and ceases, they should view them critically. Asking where the data is from is important, as is asking about the sampling method used to collect the data, and the size of the sample. If the source of the data appears to have a conflict of interest with the type of data being gathered, this can call the results into question. Likewise, data gathered from a small sample or a sample which is not truly random may be of questionable utility. Reputable researchers will always provide information about the data gathering techniques used, the source of funding, and the point of the data collection in the beginning of the analysis so that readers can think about this information while they review the analysis.

VALIDITY AND RELIABILITY

Validity refers to the accuracy or truthfulness of a measurement. Are we measuring what we think we are? “Validity itself is a simple concept, but the determination of the validity of a measure is elusive”.

Face validity is based solely on the judgement of the researcher. Each question is scrutinized and modified until the researcher is satisfied that it is an accurate measure of the desired construct. The determination of face validity is based on the subjective opinion of the researcher.

Content validity is similar to face validity in that it relies on the judgement of the researcher. However, where face validity only evaluates the individual items on an instrument, content validity goes further in that it attempts to determine if an instrument provides adequate coverage of a topic. Expert opinions, literature searches, and pretest open-ended questions help to establish content validity.

Criterion-related validity can be either predictive or concurrent. When a dependent/independent relationship has been established between two or more variables, criterion-related validity can be assessed. A mathematical model is developed to be able to predict the dependent variable from the independent variable(s). Predictive validity refers to the ability of an independent variable to predict a future value of the dependent variable. Concurrent validity is concerned with the relationship between two or more variables at the same point in time.

Construct validity refers to the theoretical foundations underlying a particular scale or measurement. It looks at the underlying theories or constructs that explain a phenomena. This is also quite subjective and depends heavily on the understanding, opinions, and biases of the researcher. Reliability is synonymous with repeatability. A measurement that yields consistent results over time is said to be reliable. When a measurement is prone to random error, it lacks reliability.

The reliability of an instrument places an upper limit on its validity. A measurement that lacks reliability will necessarily be invalid. There are three basic methods to test reliability: test-retest, equivalent form, and internal consistency.

A test-retest measure of reliability can be obtained by administering the same instrument to the same group of people at two different points in time. The degree to which both administrations are in agreement is a measure of the reliability of the instrument. This technique for assessing reliability suffers two possible drawbacks. First, a person may have changed between the first and second measurement. Second, the initial administration of an instrument might in itself induce a person to answer differently on the second administration.

The second method of determining reliability is called the equivalent-form technique. The researcher creates two different instruments designed to measure identical constructs. The degree of correlation between the instruments is a measure of equivalent-form reliability. The difficulty in using this method is that it may be very difficult to create a totally equivalent instrument.

The most popular methods of estimating reliability use measures of internal consistency. When an instrument includes a series of questions designed to examine the same construct, the questions can be arbitrarily split into two groups. The correlation between the two subsets of questions is called the split-half reliability. The problem is that this measure of reliability changes depending on how the questions are split. A better statistic, known as Chronbach's alpha, is based on the mean interitem correlation for all possible variable pairs. It provides a conservative estimate of reliability, and generally represents "the lower bound to the reliability of an unweighted scale of items". For dichotomous nominal data, the KR-20 is used instead of Chronbach's alpha.

VARIABILITY AND ERROR

Most research is an attempt to understand and explain variability. When a measurement lacks variability, no statistical tests can be performed. Variability refers to the dispersion of scores. Ideally, when a researcher finds differences between respondents, they are due to true difference on the variable being measured. However, the combination of systematic and random errors can dilute the accuracy of a measurement. Systematic error is introduced through a constant bias in a measurement. It can usually be traced to a fault in the sampling procedure or in the design of a questionnaire. Random error does not occur in any consistent pattern, and it is not controllable by the researcher.

Scientific research involves the formulation and testing of one or more hypotheses. A hypothesis cannot be proved directly, so a null hypothesis is established to give the researcher an indirect method of testing a theory. Sampling is necessary when the population is too large, or when the researcher is unable to investigate all members of the target group. Random and systematic sampling are the best methods because they guarantee that each member of the population will have a known non-zero chance of being selected.

The mathematical reliability of a measurement, or group of measurements, can be calculated, however, validity can only be implied by the data, and it is not directly verifiable. Social science research is generally an attempt to explain or understand the variability in a group of people

HYPOTHESIS TESTING

Questions like the ones fall into a domain called *hypothesis testing*. Hypothesis testing is a way of systematically quantifying how certain you are of the result of a statistical experiment. In the coin example the “experiment” was flipping the coin 100 times. There are two questions you can ask. One, assuming the coin was fair, how likely is it that you’d observe the results we did? Two, what is the likelihood that the coin is fair given the results you observed? Of course, an experiment can be much more complex than coin flipping. Any situation where you’re taking a random sample of a population and measuring something about it is an experiment, and for our purposes this includes A/B testing.

Let’s focus on the coin flip example understand the basics.

THE NULL HYPOTHESIS

The most common type of hypothesis testing involves a *null hypothesis*. The null hypothesis, denoted H_0 , is a statement about the world which can plausibly account for the data you observe.

Don’t read anything into the fact that it’s called the “null” hypothesis — it’s just the hypothesis we’re trying to test. For example, “the coin is fair” is an example of a null hypothesis, as is “the coin is biased.” The important part is that the null hypothesis be able to be expressed in simple, mathematical terms. We’ll see how to express these statements mathematically in just a bit.

The main goal of hypothesis testing is to tell us whether we have enough evidence to reject the null hypothesis. In our case we want to know whether the coin is biased or not, so our null hypothesis should be “the coin is fair.” If we get enough evidence that contradicts this hypothesis, say, by flipping it 100 times and having it come up heads only once, then we can safely reject it.

All of this is perfectly quantifiable, of course. What constitutes “enough” and “safely” are all a matter of statistics.

THE STATISTICS, INTUITIVELY

So, we have a coin. Our null hypothesis is that this coin is fair. We flip it 100 times and it comes up heads 51 times. Do we know whether the coin is biased or not?

Our gut might say the coin is fair, or at least probably fair, but we can't say for sure. The expected number of heads is 50 and 51 is quite close. But what if we flipped the coin 100,000 times and it came up heads 51,000 times? We see 51% heads both times, but in the second instance the coin is more likely to be biased. Lack of evidence to the contrary is not evidence that the null hypothesis is true. Rather, it means that we don't have sufficient evidence to conclude that the null hypothesis is false. The coin might actually have a 51% bias towards heads, after all. If instead we saw 1 head for 100 flips that would be another story. Intuitively we know that the chance of seeing this if the null hypothesis were true is so small that we would be comfortable rejecting the null hypothesis and declaring the coin to be biased.

THE COIN FLIP

Formally the flip of a coin can be represented by a Bernoulli trial. A Bernoulli trial is a random variable X such that,

$$\Pr(X = 1) = 1 - \Pr(X = 0) = 1 - q = p$$

That is, X takes on the value 1 with probability p , and 0 with probability $1 - p$. Now, let's say we have 100 coin flips. Let X_i represent the i^{th} coin flip. Then the random variable,

$$Y = \sum_{i=1}^{100} X_i$$

represents the run of 100 coin flips.

THE STATISTICS, MATHEMATICALLY

Say you have a set of observations O and a null hypothesis H_0 . In the coin example we were trying to calculate,

$$P(O \setminus H_0)$$

i.e., the probability that we observed what we did given the null hypothesis. If that probability is sufficiently small we're confident concluding the null hypothesis is false. We can use whatever level of confidence we want before rejecting the null hypothesis, but most people choose 90%, 95%, or 99%. For example if we choose a 95% confidence level we reject the null hypothesis if,

$$P(O \setminus H_0) \leq 1 - 0.95 = 0.05$$

The Central Limit Theorem is the main piece of math here. Briefly, the Central Limit Theorem says that the sum of any number of re-averaged identically distributed random variables approximates a normal distribution.

Remember our random variables from before? If we let,

$$p = \frac{Y}{N}$$

then p is the proportion of heads in our sample of 100 coin flips. In our case, it is equal to 0.51, or 51%. But by the central limit theorem we also know that p approximates a normal distribution. This means we can estimate the standard deviation of p as,

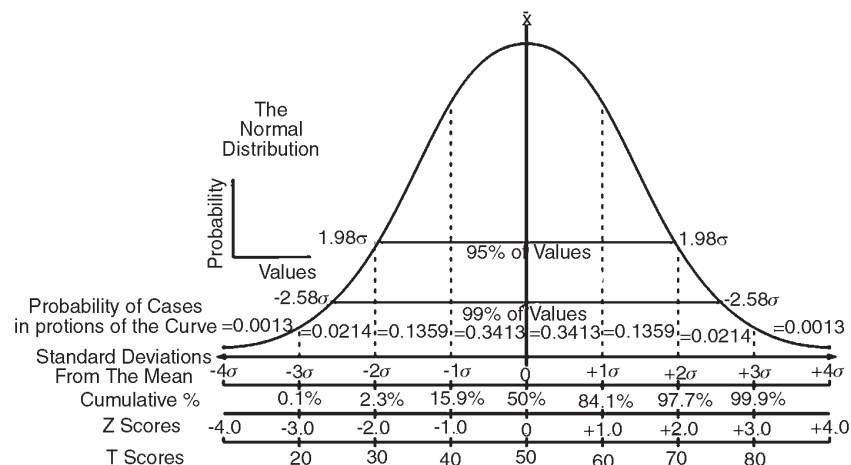
$$\sigma = \sqrt{\frac{p(1-p)}{N}}$$

WRAPPING IT UP

The coin is fair. Mathematically we're saying

$$H_0 : p_0 = 0.50$$

Here's the normal curve:



A 95% level of confidence means we reject the null hypothesis if p falls outside 95% of the area of the normal curve. Looking at that chart we see that this corresponds to approximately 1.98 standard deviations.

The so-called “z-score” tells us how many standard deviations away from the mean our sample is, and it's calculated as,

$$z = \frac{p - 0.50}{\sqrt{\frac{0.50(1 - 0.50)}{N}}}$$

The numerator is “ $p - 0.50$ ” because our null hypothesis is that $p = 0.50$. This measures how far the sample mean, p , diverges from the expect mean of a fair coin, 0.50.

THE DATA

Let's say we flipped three coins 100 times each and got the following data.

Table. Data for 100 Flips of a Coin.

Coin	Flips	Pct. Heads	Z-score
Coin 1	100	51%	0.20
Coin 2	100	60%	2.04
Coin 3	100	75%	5.77

Using a 95% confidence level we'd conclude that Coin 2 and Coin 3 are biased using the techniques we've developed so far. Coin 2 is 2.04 standard deviations from the mean and Coin 3 is 5.77 standard deviations.

When your test statistic meets the 95% confidence threshold we call it *statistically significant*.

This means there's only a 5% chance of observing what you did assuming the null hypothesis was true. Phrased another way, there's only a 5% chance that your observation is due to random variation.

RECAP

Hypothesis testing is a way of systematically quantifying how certain you are of the result of a statistical experiment. You start by forming a null hypothesis, *e.g.*, "this coin is fair," and then calculate the likelihood that your observations are due to pure chance rather than a real difference in the population.

The confidence interval is the level at which you reject the null hypothesis. If there is a 95% chance that there's a real difference in your observations, given the null hypothesis, then you are confident in rejecting it. This also means there is a 5% chance you're wrong and the difference is due to random fluctuations.

The null hypothesis can be any mathematical statement and the test you use depends on both the underlying data and your null hypothesis. In our coin flipping example the underlying data approximated a normal distribution and we wanted to test whether the observed proportion of heads was different enough to be significant. In this case we were measuring the *sample mean*.

We can measure anything, though: the sample variance, correlation, *etc.* Different tests need to be used to determine whether these are statistically significant.

Evaluation Research Method

One specific form of social research — evaluation research— is of particular interest here. The Introduction to Evaluation Research presents an overview of what evaluation is and how it differs from social research generally. We also introduce several evaluation models to give you some perspective on the evaluation endeavour. Evaluation should not be considered in a vacuum. Here, we consider evaluation as embedded within a larger Planning-Evaluation Cycle.

Evaluation can be a threatening activity. Many groups and organizations struggle with how to build a good evaluation capability into their everyday activities and procedures.

This is essentially an organizational culture issue. Here we consider some of the issues a group or organization needs to address in order to develop an evaluation culture that works in their context.

INTRODUCTION TO EVALUATION

Evaluation is a methodological area that is closely related to, but distinguishable from more traditional social research. Evaluation utilizes many of the same methodologies used in traditional social research, but because evaluation takes place within a political and organizational context, it requires group skills, management ability, political dexterity, sensitivity to multiple stakeholders and other skills that social research in general does not rely on as much. Here we introduce the idea of evaluation and some of the major terms and issues in the field.

DEFINITIONS OF EVALUATION

Probably the most frequently given definition is:

Evaluation is the systematic assessment of the worth or merit of some object

This definition is hardly perfect. There are many types of evaluations that do not necessarily result in an assessment of worth or merit — descriptive studies, implementation analyses, and formative evaluations, to name a few. Better perhaps is a definition that emphasizes the information-processing and feedback functions of evaluation. For instance, one might say:

Evaluation is the systematic acquisition and assessment of information to provide useful feedback about some object Both definitions agree that evaluation is a systematic endeavour and both use the deliberately ambiguous term ‘object’ which could refer to a programme, policy, technology, person, need, activity, and so on. The latter definition emphasizes *acquiring and assessing information* rather than *assessing worth or merit* because all evaluation work involves collecting and sifting through data, making judgements about the validity of the information and of inferences we derive from it, whether or not an assessment of worth or merit results.

THE GOALS OF EVALUATION

The generic goal of most evaluations is to provide “useful feedback” to a variety of audiences including sponsors, donors, client-groups, administrators, staff, and other relevant constituencies. Most often, feedback is perceived as “useful” if it aids in decision-making. But the relationship between an evaluation and its impact is not a simple one — studies that seem critical sometimes fail to influence short-term decisions, and studies that initially seem to have no influence can have a delayed impact when more congenial conditions arise. Despite this, there is broad consensus that the major goal of evaluation should be to influence decision-making or policy formulation through the provision of empirically-driven feedback.

EVALUATION STRATEGIES

‘Evaluation strategies’ means broad, overarching perspectives on evaluation. They encompass the most general groups or “camps” of evaluators; although, at its best, evaluation work borrows eclectically from the perspectives of all these camps.

Scientific-experimental models are probably the most historically dominant evaluation strategies. Taking their values and methods from the sciences — especially the social sciences — they prioritize on the desirability of impartiality, accuracy, objectivity and the validity of the information generated. Included under scientific-experimental models would be: the tradition of experimental and quasi-experimental designs; objectives-based research that comes from education; econometrically-oriented perspectives including cost-effectiveness and cost-benefit analysis; and the recent articulation of theory-driven evaluation.

The second class of strategies are *management-oriented systems models*. Two of the most common of these are PERT, the Programme Evaluation and Review Technique, and CPM, the Critical Path Method. Both have been widely used in business and government in this country. It would also be legitimate to include the Logical Framework or “Logframe” model developed at U.S. Agency for International Development and general systems theory and operations research approaches in this category. Two management-oriented systems models were originated by evaluators: the UTOS model where U stands for Units, T for Treatments, O for Observing Observations and S for Settings; and the CIPP model where the C stands for Context, the I for Input, the first P for Process and the second P for Product. These management-oriented systems models emphasize comprehensiveness in evaluation, placing evaluation within a larger framework of organizational activities.

The third class of strategies are the *qualitative/anthropological models*. They emphasize the importance of observation, the need to retain the phenomenological quality of the evaluation context, and the value of subjective human interpretation in the evaluation process. Included in this category are the approaches known in evaluation as naturalistic or ‘Fourth Generation’ evaluation; the various qualitative schools; critical theory and art criticism approaches; and, the ‘grounded theory’ approach of Glaser and Strauss among others.

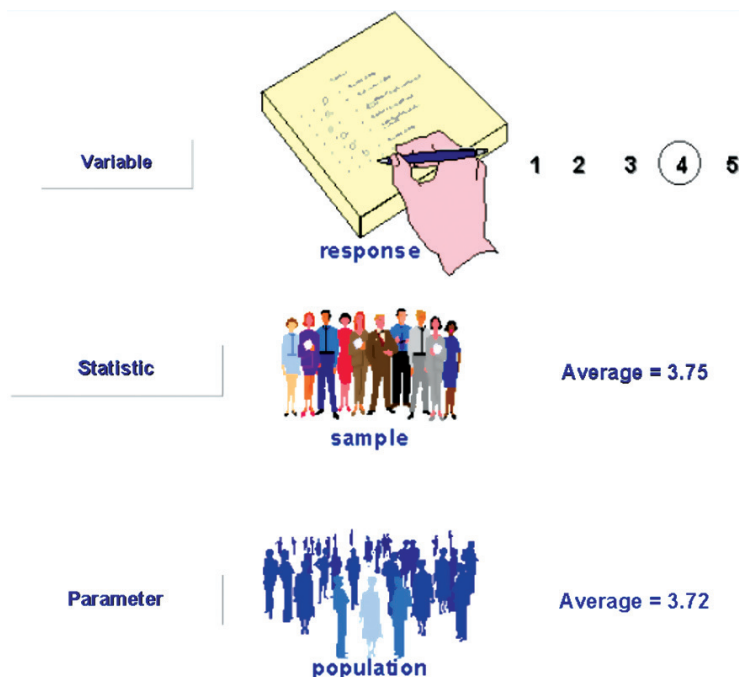
Finally, a fourth class of strategies is termed *participant-oriented models*. As the term suggests, they emphasize the central importance of the evaluation participants, especially clients and users of the programme or technology. Client-centred and stakeholder approaches are examples of participant-oriented models, as are consumer-oriented evaluation systems. With all of these strategies to choose from, how to decide? Debates that rage within the evaluation profession — and they do rage — are generally battles between these different strategists, with each claiming the superiority of their position. In reality, most good evaluators are familiar with all four categories and borrow from each as the need arises.

There is no inherent incompatibility between these broad strategies — each of them brings something valuable to the evaluation table. In fact, in recent years attention has increasingly turned to how one might integrate results from evaluations that use different strategies, carried out from different perspectives, and using different methods. Clearly, there are no simple answers here. The problems are complex and the methodologies needed will and should be varied.

STATISTICAL TERMS IN SAMPLING

Let’s begin by defining some very simple terms that are relevant here. First, let’s look at the results of our sampling efforts. When we sample, the units that we sample — usually people — supply us with one or more responses. In this sense, a response is a specific measurement value that a sampling unit supplies. In the figure, the person is

responding to a survey instrument and gives a response of '4'. When we look across the responses that we get for our entire sample, we use a statistic. There are a wide variety of statistics we can use— mean, median, mode, and so on. In this example, we see that the mean or average for the sample is 3.75. But the reason we sample is so that we might get an estimate for the population we sampled from. If we could, we would much prefer to measure the entire population. If you measure the entire population and calculate a value like a mean or average, we don't refer to this as a statistic, we call it a parameter of the population.



THE SAMPLING DISTRIBUTION

So how do we get from our sample statistic to an estimate of the population parameter? A crucial midway concept you need to understand is the sampling distribution. In order to understand it, you have to be able and willing to do a thought experiment. Imagine that instead of just taking a single sample like we do in a typical study, you took three independent samples of the same population. And furthermore, imagine that for each of your three samples, you collected a single response and computed a single statistic, say, the mean of the response. Even though all three samples came from the same population, you wouldn't expect to get the exact same statistic from each. They would differ slightly just due to the random "luck of the draw" or to the natural fluctuations or vagaries of drawing a sample. But you would expect that all three samples would yield a similar statistical estimate because they were drawn from the same population. Now, for the leap of imagination! Imagine that you did an *infinite* number of samples from the same

population and computed the average for each one. If you plotted them on a histogram or bar graph you should find that most of them converge on the same central value and that you get fewer and fewer samples that have averages farther away up or down from that central value. In other words, the bar graph would be well described by the *bell curve* shape that is an indication of a “normal” distribution in statistics. The distribution of an infinite number of samples of the same size as the sample in your study is known as the sampling distribution.



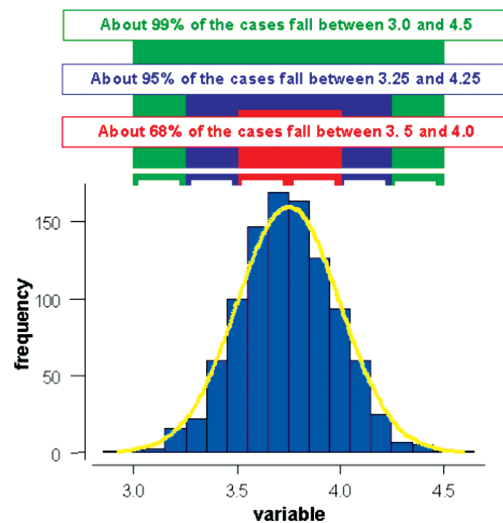
We don't ever actually construct a sampling distribution. Why not? You're not paying attention! Because to construct it we would have to take an *infinite* number of samples and at least the last time I checked, on this planet infinite is not a number we know how to reach. So why do we even talk about a sampling distribution? Now that's a good question! Because we need to realize that our sample is just one of a potentially infinite number of samples that we could have taken. When we keep the sampling distribution in mind, we realize that while the statistic we got from our sample is probably near the center of the sampling distribution (because most of the samples would be there) we could have gotten one of the extreme samples just by the luck of the draw. If we take the average of the sampling distribution — the average of the averages of an infinite number of samples — we would be much closer to the true population average — the parameter of interest. So the average of the sampling distribution is essentially equivalent to the parameter.

But what is the standard deviation of the sampling distribution (OK, never had statistics? There are any number of places on the web where you can learn about them or even just brush up if you've gotten rusty. This isn't one of them. I'm going to assume that you at least know what a standard deviation is, or that you're capable of finding out relatively quickly). The standard deviation of the sampling distribution tells us something about how different samples would be distributed.

In statistics it is referred to as the standard error (so we can keep it separate in our minds from standard deviations. Getting confused? Go get a cup of coffee and come back in ten minutes...OK, let's try once more... A standard deviation is the spread of the scores around the average in a single sample. The standard error is the spread of the averages around the average of averages in a sampling distribution. Got it?)

SAMPLING ERROR

In sampling contexts, the standard error is called sampling error. Sampling error gives us some idea of the precision of our statistical estimate. A low sampling error means that we had relatively less variability or range in the sampling distribution. But here we go again — we never actually see the sampling distribution! So how do we calculate sampling error? We base our calculation *on the standard deviation of our sample*. The greater the sample standard deviation, the greater the standard error (and the sampling error). The standard error is also related to the sample size. The greater your sample size, the smaller the standard error.

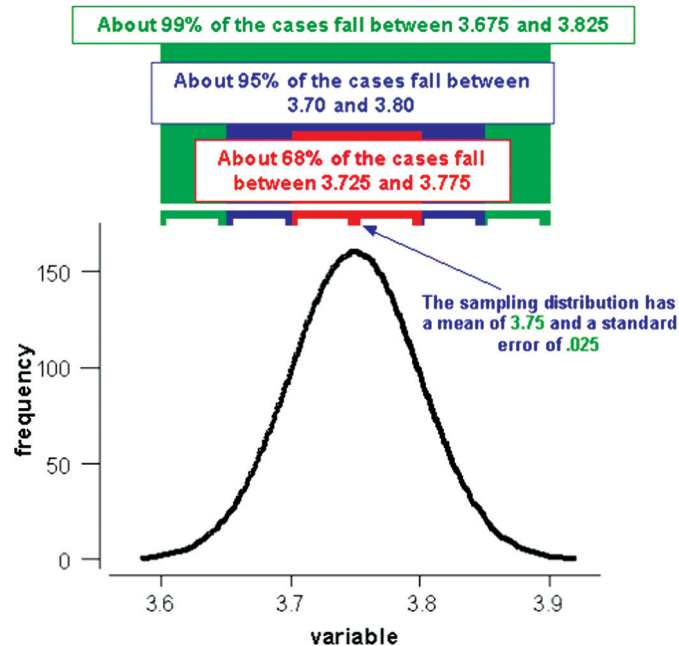


Why? Because the greater the sample size, the closer your sample is to the actual population itself. If you take a sample that consists of the entire population you actually have no sampling error because you don't have a sample, you have the entire population. In that case, the mean you estimate is the parameter.

THE 68, 95, 99 PERCENT RULE

You've probably heard this one before, but it's so important that it's always worth repeating... There is a general rule that applies whenever we have a normal or bell-shaped distribution. Start with the average — the center of the distribution. If you go up and down (*i.e.*, left and right) one standard unit, you will include approximately 68% of the cases in the distribution (*i.e.*, 68% of the area under the curve). If you go up and

down two standard units, you will include approximately 95% of the cases. And if you go plus-and-minus three standard units, you will include about 99% of the cases. Notice that I didn't specify in the previous few sentences whether I was talking about standard deviation units or standard error units. That's because the same rule holds for both types of distributions (*i.e.*, the raw data and sampling distributions).



For instance, in the figure, the mean of the distribution is 3.75 and the standard unit is .25 (If this was a distribution of raw data, we would be talking in standard deviation units. If it's a sampling distribution, we'd be talking in standard error units). If we go up and down one standard unit from the mean, we would be going up and down .25 from the mean of 3.75. Within this range — 3.5 to 4.0 — we would expect to see approximately 68% of the cases. This section is marked in red on the figure. I leave to you to figure out the other ranges. But what does this all mean you ask? If we are dealing with raw data and we know the mean and standard deviation of a sample, we can *predict* the intervals within which 68, 95 and 99% of our cases would be expected to fall. We call these intervals the— guess what — 68, 95 and 99% confidence intervals.

Now, here's where everything should come together in one great aha! experience if you've been following along. If we had a *sampling distribution*, we would be able to predict the 68, 95 and 99% confidence intervals for where the population parameter should be! And isn't that why we sampled in the first place? So that we could predict where the population is on that variable? There's only one hitch. We don't actually have the sampling distribution (now this is the third time I've said this in this essay)! But we do have the distribution for the sample itself. And we can from that distribution estimate the standard error (the sampling error) because it is based on the standard deviation and we have that. And, of course, we don't actually know the population

parameter value — we're trying to find that out — but we can use our best estimate for that — the sample statistic. Now, if we have the mean of the sampling distribution (or set it to the mean from our sample) and we have an estimate of the standard error (we calculate that from our sample) then we have the two key ingredients that we need for our sampling distribution in order to estimate confidence intervals for the population parameter.

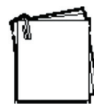
Perhaps an example will help. Let's assume we did a study and drew a single sample from the population. Furthermore, let's assume that the average for the sample was 3.75 and the standard deviation was .25. This is the raw data distribution depicted above. now, what would the sampling distribution be in this case? Well, we don't actually construct it (because we would need to take an infinite number of samples) but we *can* estimate it. For starters, we assume that the mean of the sampling distribution is the mean of the sample, which is 3.75. Then, we calculate the standard error. To do this, we use the standard deviation for our sample and the sample size (in this case $N=100$) and we come up with a standard error of .025 (just trust me on this). Now we have everything we need to estimate a confidence interval for the population parameter. We would estimate that the probability is 68% that the true parameter value falls between 3.725 and 3.775 (*i.e.*, 3.75 plus and minus .025); that the 95% confidence interval is 3.700 to 3.800; and that we can say with 99% confidence that the population value is between 3.675 and 3.825. The real value (in this fictitious example) was 3.72 and so we have correctly estimated that value with our sample.

PROBABILITY SAMPLING

A probability sampling method is any method of sampling that utilizes some form of *random selection*. In order to have a random selection method, you must set up some process or procedure that assures that the different units in your population have equal probabilities of being chosen. Humans have long practiced various forms of random selection, such as picking a name out of a hat, or choosing the short straw. These days, we tend to use computers as the mechanism for generating random numbers as the basis for random selection.

SOME DEFINITIONS

List of clients



Random subsample



Before I can explain the various probability methods we have to define some basic terms. These are:

N = the number of cases in the sampling frame

n = the number of cases in the sample

${}_NC_n$ = the number of combinations (subsets) of n from N

$f = n/N$ = the sampling fraction

That's it. With those terms defined we can begin to define the different probability sampling methods.

SIMPLE RANDOM SAMPLING

The simplest form of random sampling is called simple random sampling. Pretty tricky, huh? Here's the quick description of simple random sampling:

Objective: To select n units out of N such that each ${}_NC_n$ has an equal chance of being selected.

Procedure: Use a table of random numbers, a computer random number generator, or a mechanical device to select the sample.

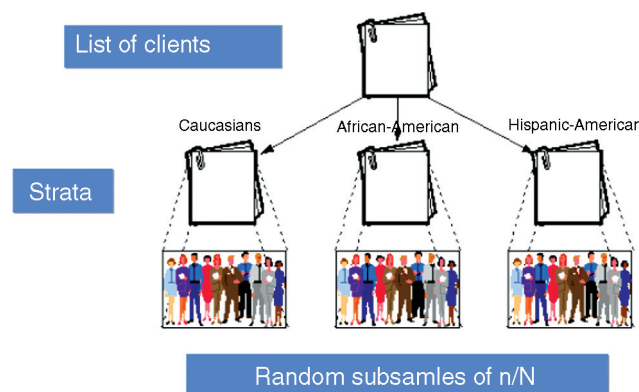
A somewhat stilted, if accurate, definition. Let's see if we can make it a little more real. How do we select a simple random sample? Let's assume that we are doing some research with a small service agency that wishes to assess client's views of quality of service over the past year. First, we have to get the sampling frame organized. To accomplish this, we'll go through agency records to identify every client over the past 12 months. If we're lucky, the agency has good accurate computerized records and can quickly produce such a list. Then, we have to actually draw the sample. Decide on the number of clients you would like to have in the final sample. For the sake of the example, let's say you want to select 100 clients to survey and that there were 1000 clients over the past 12 months. Then, the sampling fraction is $f = n/N = 100/1000 = .10$ or 10%. Now, to actually draw the sample, you have several options. You could print off the list of 1000 clients, tear them into separate strips, put the strips in a hat, mix them up real good, close your eyes and pull out the first 100. But this mechanical procedure would be tedious and the quality of the sample would depend on how thoroughly you mixed them up and how randomly you reached in. Perhaps a better procedure would be to use the kind of ball machine that is popular with many of the state lotteries. You would need three sets of balls numbered 0 to 9, one set for each of the digits from 000 to 999 (if we select 000 we'll call that 1000). Number the list of names from 1 to 1000 and then use the ball machine to select the three digits that selects each person. The obvious disadvantage here is that you need to get the ball machines. (Where do they make those things, anyway? Is there a ball machine industry?).

Neither of these mechanical procedures is very feasible and, with the development of inexpensive computers there is a much easier way. Here's a simple procedure that's especially useful if you have the names of the clients already on the computer. Many computer programmes can generate a series of random numbers. Let's assume you can copy and paste the list of client names into a column in an EXCEL spreadsheet. Then, in the column right next to it paste the function `=RAND()` which is EXCEL's way of

putting a random number between 0 and 1 in the cells. Then, sort both columns — the list of names and the random number — by the random numbers. This rearranges the list in random order from the lowest to the highest random number. Then, all you have to do is take the first hundred names in this sorted list pretty simple. You could probably accomplish the whole thing in under a minute.

Simple random sampling is simple to accomplish and is easy to explain to others. Because simple random sampling is a fair way to select a sample, it is reasonable to generalize the results from the sample back to the population. Simple random sampling is not the most statistically efficient method of sampling and you may, just because of the luck of the draw, not get good representation of subgroups in a population. To deal with these issues, we have to turn to other sampling methods.

STRATIFIED RANDOM SAMPLING



Stratified Random Sampling, also sometimes called *proportional* or *quota* random sampling, involves dividing your population into homogeneous subgroups and then taking a simple random sample in each subgroup.

In more formal terms:

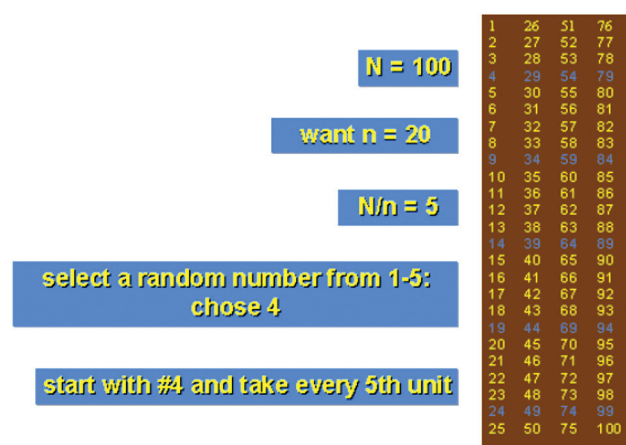
Objective: Divide the population into non-overlapping groups (*i.e.*, *strata*) $N_1, N_2, N_3, \dots, N_i$, such that $N_1 + N_2 + N_3 + \dots + N_i = N$. Then do a simple random sample of $f = n/N$ in each strata.

There are several major reasons why you might prefer stratified sampling over simple random sampling. First, it assures that you will be able to represent not only the overall population, but also key subgroups of the population, especially small minority groups. If you want to be able to talk about subgroups, this may be the only way to effectively assure you'll be able to. If the subgroup is extremely small, you can use different sampling fractions (f) within the different strata to randomly over-sample the small group (although you'll then have to weight the within-group estimates using the sampling fraction whenever you want overall population estimates). When we use the same sampling fraction within strata we are conducting *proportionate* stratified random sampling. When we use different sampling fractions in the strata, we call this *disproportionate* stratified

random sampling. Second, stratified random sampling will generally have more statistical precision than simple random sampling. This will only be true if the strata or groups are homogeneous. If they are, we expect that the variability within-groups is lower than the variability for the population as a whole. Stratified sampling capitalizes on that fact.

For example, let's say that the population of clients for our agency can be divided into three groups: Caucasian, African-American and Hispanic-American. Furthermore, let's assume that both the African-Americans and Hispanic-Americans are relatively small minorities of the clientele (10% and 5% respectively). If we just did a simple random sample of $n=100$ with a sampling fraction of 10%, we would expect by chance alone that we would only get 10 and 5 persons from each of our two smaller groups. And, by chance, we could get fewer than that! If we stratify, we can do better. First, let's determine how many people we want to have in each group. Let's say we still want to take a sample of 100 from the population of 1000 clients over the past year. But we think that in order to say anything about subgroups we will need at least 25 cases in each group. So, let's sample 50 Caucasians, 25 African-Americans, and 25 Hispanic-Americans. We know that 10% of the population, or 100 clients, are African-American. If we randomly sample 25 of these, we have a within-stratum sampling fraction of $25/100 = 25\%$. Similarly, we know that 5% or 50 clients are Hispanic-American. So our within-stratum sampling fraction will be $25/50 = 50\%$. Finally, by subtraction we know that there are 850 Caucasian clients. Our within-stratum sampling fraction for them is $50/850 = \text{about } 5.88\%$. Because the groups are more homogeneous within-group than across the population as a whole, we can expect greater statistical precision (less variance). And, because we stratified, we know we will have enough cases from each group to make meaningful subgroup inferences.

SYSTEMATIC RANDOM SAMPLING



Here are the steps you need to follow in order to achieve a systematic random sample: number the units in the population from 1 to N

decide on the n (sample size) that you want or need

$k = N/n$ = the interval size

randomly select an integer between 1 to k

then take every k th unit.

All of this will be much clearer with an example. Let's assume that we have a population that only has $N=100$ people in it and that you want to take a sample of $n=20$. To use systematic sampling, the population must be listed in a random order. The sampling fraction would be $f = 20/100 = 20\%$ in this case, the interval size, k , is equal to $N/n = 100/20 = 5$. Now, select a random integer from 1 to 5. In our example, imagine that you chose 4. Now, to select the sample, start with the 4th unit in the list and take every k -th unit (every 5th, because $k=5$). You would be sampling units 4, 9, 14, 19, and so on to 100 and you would wind up with 20 units in your sample.

For this to work, it is essential that the units in the population are randomly ordered, at least with respect to the characteristics you are measuring. Why would you ever want to use systematic random sampling? For one thing, it is fairly easy to do. You only have to select a single random number to start things off. It may also be more precise than simple random sampling. Finally, in some situations there is simply no easier way to do random sampling. For instance, I once had to do a study that involved sampling from all the books in a library. Once selected, I would have to go to the shelf, locate the book, and record when it last circulated. I knew that I had a fairly good sampling frame in the form of the shelf list (which is a card catalogue where the entries are arranged in the order they occur on the shelf). To do a simple random sample, I could have estimated the total number of books and generated random numbers to draw the sample; but how would I find book #74,329 easily if that is the number I selected? I couldn't very well count the cards until I came to 74,329! Stratifying wouldn't solve that problem either. For instance, I could have stratified by card catalogue drawer and drawn a simple random sample within each drawer. But I'd still be stuck counting cards. Instead, I did a systematic random sample. I estimated the number of books in the entire collection. Let's imagine it was 100,000. I decided that I wanted to take a sample of 1000 for a sampling fraction of $1000/100,000 = 1\%$. To get the sampling interval k , I divided $N/n = 100,000/1000 = 100$.

Then I selected a random integer between 1 and 100. Let's say I got 57. Next I did a little side study to determine how thick a thousand cards are in the card catalogue (taking into account the varying ages of the cards). Let's say that on average I found that two cards that were separated by 100 cards were about .75 inches apart in the catalogue drawer. That information gave me everything I needed to draw the sample. I counted to the 57th by hand and recorded the book information. Then, I took a compass. (Remember those from your high-school math class? They're the funny little metal instruments with a sharp pin on one end and a pencil on the other that you used to draw circles in geometry class.) Then I set the compass at .75", stuck the pin end in at the 57th card and pointed with the pencil end to the next card (approximately 100 books

away). In this way, I approximated selecting the 157th, 257th, 357th, and so on. I was able to accomplish the entire selection procedure in very little time using this systematic random sampling approach. I'd probably still be there counting cards if I'd tried another random sampling method. (Okay, so I have no life. I got compensated nicely, I don't mind saying, for coming up with this scheme.)

CLUSTER (AREA) RANDOM SAMPLING

The problem with random sampling methods when we have to sample a population that's disbursed across a wide geographic region is that you will have to cover a lot of ground geographically in order to get to each of the units you sampled. Imagine taking a simple random sample of all the residents of New York State in order to conduct personal interviews. By the luck of the draw you will wind up with respondents who come from all over the state. Your interviewers are going to have a lot of travelling to do. It is for precisely this problem that cluster or area random sampling was invented.

In cluster sampling, we follow these steps:

- Divide population into clusters (usually along geographic boundaries)
- Randomly sample clusters
- Measure all units within sampled clusters.

For instance, in the figure we see a map of the counties in New York State. Let's say that we have to do a survey of town governments that will require us going to the towns personally. If we do a simple random sample state-wide we'll have to cover the entire state geographically. Instead, we decide to do a cluster sampling of five counties. Once these are selected, we go to *every* town government in the five areas. Clearly this strategy will help us to economize on our mileage. Cluster or area sampling, then, is useful in situations like this, and is done primarily for efficiency of administration. Note also, that we probably don't have to worry about using this approach if we are conducting a mail or telephone survey because it doesn't matter as much (or cost more or raise inefficiency) where we call or send letters to.

MULTI-STAGE SAMPLING

The four methods we've covered so far — simple, stratified, systematic and cluster — are the simplest random sampling strategies. In most real applied social research, we would use sampling methods that are considerably more complex than these simple variations. The most important principle here is that we can combine the simple methods described earlier in a variety of useful ways that help us address our sampling needs in the most efficient and effective manner possible. When we combine sampling methods, we call this multi-stage sampling.

For example, consider the idea of sampling New York State residents for face-to-face interviews. Clearly we would want to do some type of cluster sampling as the first stage of the process. We might sample townships or census tracts throughout the state.

But in cluster sampling we would then go on to measure everyone in the clusters we select. Even if we are sampling census tracts we may not be able to measure *everyone* who is in the census tract. So, we might set up a stratified sampling process within the clusters.

In this case, we would have a two-stage sampling process with stratified samples within cluster samples. Or, consider the problem of sampling students in grade schools. We might begin with a national sample of school districts stratified by economics and educational level.

Within selected districts, we might do a simple random sample of schools. Within schools, we might do a simple random sample of classes or grades. And, within classes, we might even do a simple random sample of students. In this case, we have three or four stages in the sampling process and we use both stratified and simple random sampling. By combining different sampling methods we are able to achieve a rich variety of probabilistic sampling methods that can be used in a wide range of social research contexts.

TYPES OF EVALUATION

There are many different types of evaluations depending on the object being evaluated and the purpose of the evaluation. Perhaps the most important basic distinction in evaluation types is that between *formative* and *summative* evaluation. Formative evaluations strengthen or improve the object being evaluated — they help form it by examining the delivery of the programme or technology, the quality of its implementation, and the assessment of the organizational context, personnel, procedures, inputs, and so on.

Summative evaluations, in contrast, examine the effects or outcomes of some object — they summarize it by describing what happens subsequent to delivery of the programme or technology; assessing whether the object can be said to have caused the outcome; determining the overall impact of the causal factor beyond only the immediate target outcomes; and, estimating the relative costs associated with the object.

Formative evaluation includes several evaluation types:

- *Needs assessment* determines who needs the programme, how great the need is, and what might work to meet the need
- *Evaluability assessment* determines whether an evaluation is feasible and how stakeholders can help shape its usefulness
- *Structured conceptualization* helps stakeholders define the programme or technology, the target population, and the possible outcomes
- *Implementation evaluation* monitors the fidelity of the programme or technology delivery
- *Process evaluation* investigates the process of delivering the programme or technology, including alternative delivery procedures

Summative evaluation can also be subdivided:

- *Outcome evaluations* investigate whether the programme or technology caused demonstrable effects on specifically defined target outcomes
- *Impact evaluation* is broader and assesses the overall or net effects — intended or unintended — of the programme or technology as a whole
- *Cost-effectiveness and cost-benefit analysis* address questions of efficiency by standardizing outcomes in terms of their dollar costs and values
- *Secondary analysis* reexamines existing data to address new questions or use methods not previously employed
- *Meta-analysis* integrates the outcome estimates from multiple studies to arrive at an overall or summary judgement on an evaluation question

EVALUATION QUESTIONS AND METHODS

Evaluators ask many different kinds of questions and use a variety of methods to address them. These are considered within the framework of formative and summative evaluation as presented above.

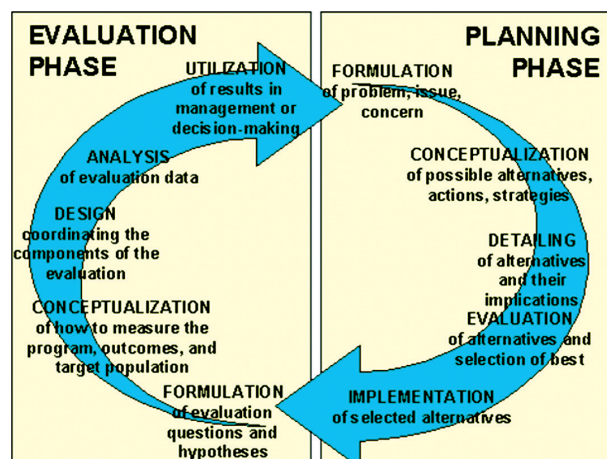
In formative research the major questions and methodologies are:

What is the definition and scope of the problem or issue, or what's the question?

Formulating and conceptualizing methods might be used including brainstorming, focus groups, nominal group techniques, Delphi methods, brainwriting, stakeholder analysis, synectics, lateral thinking, input-output analysis, and concept mapping.

Where is the problem and how big or serious is it?

The most common method used here is “needs assessment” which can include: analysis of existing data sources, and the use of sample surveys, interviews of constituent populations, qualitative research, expert testimony, and focus groups.



How should the programme or technology be delivered to address the problem?

Some of the methods already listed apply here, as do detailing methodologies like simulation techniques, or multivariate methods like multiattribute utility theory or

exploratory causal modelling; decision-making methods; and project planning and implementation methods like flow charting, PERT/CPM, and project scheduling.

How well is the programme or technology delivered?

Qualitative and quantitative monitoring techniques, the use of management information systems, and implementation assessment would be appropriate methodologies here.

The questions and methods addressed under summative evaluation include:

What type of evaluation is feasible?

Evaluability assessment can be used here, as well as standard approaches for selecting an appropriate evaluation design.

What was the effectiveness of the programme or technology?

One would choose from observational and correlational methods for demonstrating whether desired effects occurred, and quasi-experimental and experimental designs for determining whether observed effects can reasonably be attributed to the intervention and not to other sources.

What is the net impact of the programme?

Econometric methods for assessing cost effectiveness and cost/benefits would apply here, along with qualitative methods that enable us to summarize the full range of intended and unintended impacts. Clearly, this introduction is not meant to be exhaustive. Each of these methods, and the many not mentioned, are supported by an extensive methodological research literature. This is a formidable set of tools. But the need to improve, update and adapt these methods to changing circumstances means that methodological research and development needs to have a major place in evaluation work.

THE PLANNING-EVALUATION CYCLE

Often, evaluation is construed as part of a larger managerial or administrative process. Sometimes this is referred to as the *planning-evaluation cycle*. The distinctions between planning and evaluation are not always clear; this cycle is described in many different ways with various phases claimed by both planners and evaluators. Usually, the first stage of such a cycle—the planning phase—is designed to elaborate a set of potential actions, programmes, or technologies, and select the best for implementation. Depending on the organization and the problem being addressed, a planning process could involve any or all of these stages: the *formulation* of the problem, issue, or concern; the broad *conceptualization* of the major alternatives that might be considered; the *detailing* of these alternatives and their potential implications; the *evaluation* of the alternatives and the selection of the best one; and the *implementation* of the selected alternative.

Although these stages are traditionally considered planning, there is a lot of evaluation work involved. Evaluators are trained in needs assessment, they use methodologies—like the concept mapping one presented later—that help in conceptualization and detailing, and they have the skills to help assess alternatives and make a choice of the best one.

The evaluation phase also involves a sequence of stages that typically includes: the *formulation* of the major objectives, goals, and hypotheses of the programme or technology; the *conceptualization* and operationalization of the major components of the evaluation — the programme, participants, setting, and measures; the *design* of the evaluation, *detailing* how these components will be coordinated; the *analysis* of the information, both qualitative and quantitative; and the *utilization* of the evaluation results.

AN EVALUATION CULTURE

I took the idea of an evaluation culture from a wonderful paper written by Donald Campbell in 1969 entitled ‘Methods for an Experimenting Society.’ Following in the footsteps of that paper, this one is considerably more naive and utopian. And, I have changed the name of this idealized society to reflect terminology that is perhaps more amenable to the climate of the 1990s.

For the term *experimenting*, I have substituted the softer and broader term *evaluating*. And for the term *society*, I have substituted the more internationally-flavored term *culture*. With these shifts in emphasis duly noted, I want you to know that I see the evaluation culture as one that a member of the experimenting society would feel comfortable visiting, and perhaps even thinking of taking as a permanent residence.

What would an evaluation culture look like? What should its values be? You should know at the outset that I fully hope that some version of this fantasy will become an integral part of twenty-first century thought. There is no particular order of importance to the way these ideas are presented — I’ll leave that ordering to subsequent efforts.

First, our evaluation culture will embrace an *action-oriented* perspective that actively seeks solutions to problems, trying out tentative ones, weighing the results and consequences of actions, all within an endless cycle of supposition-action-evidence-revision that characterizes good science and good management. In this activist evaluation culture, we will encourage innovative approaches at all levels. But well-intentioned activism by itself is not enough, and may at times be risky, dangerous, and lead to detrimental consequences. In an evaluation culture, we won’t act for action’s sake — we’ll always attempt to assess the effects of our actions.

This evaluation culture will be an accessible, *teaching-oriented* one that emphasizes the unity of formal evaluation and everyday thought. Most of our evaluations will be simple, informal, efficient, practical, low-cost and easily carried out and understood by non-technicians. Evaluations won’t just be delegated to one person or department — we will encourage everyone in our organizations to become involved in evaluating what they and their organizations do. Where technical expertise is needed we will encourage the experts to also educate us about the technical side of what they do, demanding that they try to find ways to explain their techniques and methods adequately for nontechnicians. We will devote considerable resources to teaching others about evaluation principles.

Our evaluation culture will be *diverse, inclusive, participatory, responsive and fundamentally non-hierarchical*. World problems cannot be solved by simple “silver bullet” solutions. There is growing recognition in many arenas that our most fundamental problems are systemic, interconnected, and inextricably linked to social and economic issues and factors. Solutions will involve husbanding the resources, talents and insights of a wide range of people.

The formulation of problems and potential solutions needs to involve a broad range of constituencies. More than just “research” skills will be needed. Especially important will be skills in negotiation and consensus-building processes. Evaluators are familiar with arguments for greater diversity and inclusiveness — we’ve been talking about stakeholder, participative, multiple-constituency research for nearly two decades. No one that I know is seriously debating anymore whether we should move to more inclusive participatory approaches. The real question seems to be how such work might best be accomplished, and despite all the rhetoric about the importance of participatory methods, we have a long way to go in learning how to do them effectively.

Our evaluation culture will be a *humble, self-critical* one. We will openly acknowledge our limitations and recognize that what we learn from a single evaluation study, however well designed, will almost always be equivocal and tentative. In this regard, I believe we too often undervalue cowardice in research. I find it wholly appropriate that evaluators resist being drawn into making decisions for others, although certainly the results of our work should help inform the decision makers. A cowardly approach saves the evaluator from being drawn into the political context, helping assure the impartiality needed for objective assessment, and it protects the evaluator from taking responsibility for making decisions that should be left to those who have been duly-authorized — and who have to live with the consequences.

Most programme decisions, especially decisions about whether to continue a programme or close it down, must include more input than an evaluation alone can ever provide. While evaluators can help to elucidate what has happened in the past or might happen under certain circumstances, it is the responsibility of the organization and society as a whole to determine what ought to happen. The debate about the appropriate role of an evaluator in the decision-making process is an extremely intense one right now in evaluation circles, and my position advocating a cowardly reluctance of the evaluator to undertake a decision-making role may very well be in the minority. We will need to debate this issue vigorously, especially for politically-complex, international-evaluation contexts.

Our evaluation culture will need to be an *interdisciplinary* one, doing more than just grafting one discipline onto another through constructing multi-discipline research teams. We’ll need such teams, of course, but I mean to imply something deeper, more personally internalized — we need to move towards being nondisciplinary, consciously putting aside the blinders of our respective specialties in an attempt to foster a more whole view of the phenomena we study. As we consider the programmes we are evaluating,

we each should be able to speculate about a broad range of implementation factors or potential consequences. We should be able to anticipate some of the organizational and systems-related features of these programmes, the economic factors that might enhance or reduce implementation, their social and psychological dimensions, and especially whether the ultimate utilizers can understand or know how to utilize and be willing to utilize the results of our evaluation work. We should also be able to anticipate a broad spectrum of potential consequences — system-related, production-related, economic, nutritional, social, environmental.

This evaluation culture will also be an honest, *truth-seeking* one that stresses accountability and scientific credibility. In many quarters in contemporary society, it appears that many people have given up on the ideas of truth and validity. Our evaluation culture needs to hold to the goal of getting at the truth while at the same time honestly acknowledging the revisability of all scientific knowledge. We need to be critical of those who have given up on the goal of “getting it right” about reality, especially those among the humanities and social sciences who argue that truth is entirely relative to the knower, objectivity an impossibility, and reality nothing more than a construction or illusion that cannot be examined publicly. For them, the goal of seeking the truth is inappropriate and unacceptable, and science a tool of oppression rather than a road to greater enlightenment. Philosophers have, of course, debated such issues for thousands of years and will undoubtedly do so for thousands more. We in the evaluation culture need to check in on their thinking from time to time, but until they settle these debates, we need to hold steadfastly to the goal of getting at the truth — the goal of getting it right about reality.

Our evaluation culture will be prospective and *forward-looking*, anticipating where evaluation feedback will be needed rather than just reacting to situations as they arise. We will construct simple, low-cost evaluation and monitoring information systems when we first initiate a new programme or technology — we cannot wait until a programme is complete or a technology is in the field before we turn our attention to its evaluation.

Finally, the evaluation culture I envision is one that will emphasize fair, open, *ethical and democratic* processes. We should move away from private ownership of and exclusive access to data.

The data from all of our evaluations needs to be accessible to all interested groups allowing more extensive independent secondary analyses and opportunities for replication or refutation of original results. We should encourage open commentary and debate regarding the results of specific evaluations.

Especially when there are multiple parties who have a stake in such results, it is important for our reporting procedures to include formal opportunities for competitive review and response. Our evaluation culture must continually strive for greater understanding of the ethical dilemmas posed by our research. Our desire for valid, scientific inference will at times put us in conflict with ethical principles.

The situation is likely to be especially complex in international-evaluation contexts where we will often be dealing with multiple cultures and countries that are at different stages of economic development and have different value systems and morals. We need to be ready to deal with potential ethical and political issues posed by our methodologies in an open, direct, and democratic manner.

Do you agree with the values I'm describing here? What other characteristics might this evaluation culture have? You tell me. There are many more values and characteristics that ought to be considered. For now, the ones mentioned above, and others in the literature, provide us with a starting point at which we can all join the discussion. I hope you will add to the list, and I encourage each of you to criticize these tentative statements I've offered about the extraordinary potential of the evaluation culture that we are all in the process of creating today.

SAMPLING

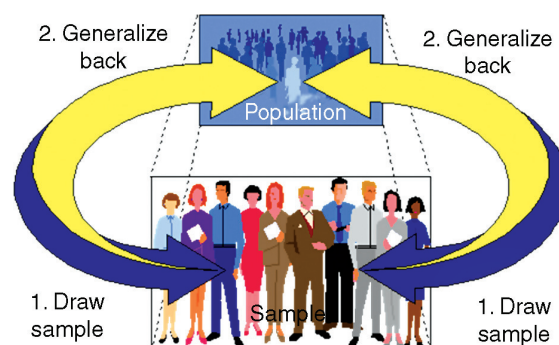
Sampling is the process of selecting units (*e.g.*, people, organizations) from a population of interest so that by studying the sample we may fairly generalize our results back to the population from which they were chosen.

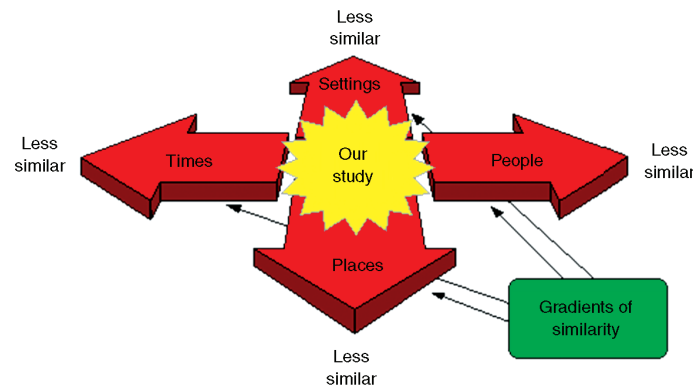
Let's begin by covering some of the key terms in sampling like "population" and "sampling frame." Then, because some types of sampling rely upon quantitative models, we'll talk about some of the statistical terms used in sampling. Finally, we'll discuss the major distinction between probability and Non-probability sampling methods and work through the major types in each.

EXTERNAL VALIDITY

External validity is related to generalizing. That's the major thing you need to keep in mind. Recall that validity refers to the approximate truth of propositions, inferences, or conclusions.

So, *external* validity refers to the approximate truth of conclusions the involve generalizations. Put in more pedestrian terms, external validity is the degree to which the conclusions in your study would hold for other persons in other places and at other times.





In science there are two major approaches to how we provide evidence for a generalization. I'll call the first approach the Sampling Model. In the sampling model, you start by identifying the population you would like to generalize to. Then, you draw a fair sample from that population and conduct your research with the sample. Finally, because the sample is representative of the population, you can automatically generalize your results back to the population. There are several problems with this approach. First, perhaps you don't know at the time of your study who you might ultimately like to generalize to. Second, you may not be easily able to draw a fair or representative sample. Third, it's impossible to sample across all times that you might like to generalize to (like next year).

I'll call the second approach to generalizing the Proximal Similarity Model. 'Proximal' means 'nearby' and 'similarity' means... well, it means 'similarity'. The term *proximal similarity* was suggested by Donald T. Campbell as an appropriate relabeling of the term *external validity* (although he was the first to admit that it probably wouldn't catch on!). Under this model, we begin by thinking about different generalizability contexts and developing a theory about which contexts are more like our study and which are less so. For instance, we might imagine several settings that have people who are more similar to the people in our study or people who are less similar. This also holds for times and places. When we place different contexts in terms of their relative similarities, we can call this implicit theoretical a *gradient of similarity*. Once we have developed this proximal similarity framework, we are able to generalize. How? We conclude that we can generalize the results of our study to other persons, places or times that are more like (that is, more proximally similar) to our study. Notice that here, we can never generalize with certainty — it is always a question of more or less similar.

Threats to External Validity

A threat to external validity is an explanation of how you might be wrong in making a generalization. For instance, you conclude that the results of your study (which was done in a specific place, with certain types of people, and at a specific time) can be generalized to another context (for instance, another place, with slightly different people,

at a slightly later time). There are three major threats to external validity because there are three ways you could be wrong — people, places or times. Your critics could come along, for example, and argue that the results of your study are due to the unusual type of people who were in the study. Or, they could argue that it might only work because of the unusual place you did the study in (perhaps you did your educational study in a college town with lots of high-achieving educationally-oriented kids). Or, they might suggest that you did your study in a peculiar time. For instance, if you did your smoking cessation study the week after the Surgeon General issues the well-publicized results of the latest smoking and cancer studies, you might get different results than if you had done it the week before.

Improving External Validity

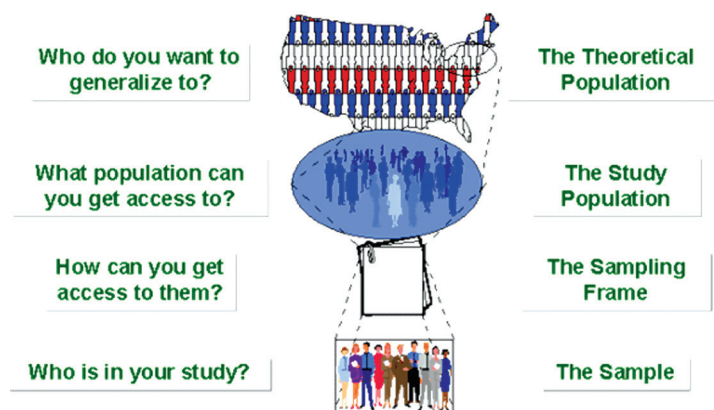
How can we improve external validity? One way, based on the sampling model, suggests that you do a good job of drawing a sample from a population. For instance, you should use random selection, if possible, rather than a nonrandom procedure. And, once selected, you should try to assure that the respondents participate in your study and that you keep your dropout rates low. A second approach would be to use the theory of proximal similarity more effectively. How? Perhaps you could do a better job of describing the ways your contexts and others differ, providing lots of data about the degree of similarity between various groups of people, places, and even times. You might even be able to map out the degree of proximal similarity among various contexts with a methodology like concept mapping. Perhaps the best approach to criticisms of generalizations is simply to show them that they're wrong — do your study in a variety of places, with different people and at different times. That is, your external validity (ability to generalize) will be stronger the more you replicate your study.

SAMPLING TERMINOLOGY

As with anything else in life you have to learn the language of an area if you're going to ever hope to use it. Here, I want to introduce several different terms for the major groups that are involved in a sampling process and the role that each group plays in the logic of sampling.

The major question that motivates sampling in the first place is: "Who do you want to generalize to?" Or should it be: "To whom do you want to generalize?" In most social research we are interested in more than just the people who directly participate in our study. We would like to be able to talk in general terms and not be confined only to the people who are in our study. Now, there are times when we aren't very concerned about generalizing. Maybe we're just evaluating a programme in a local agency and we don't care whether the programme would work with other people in other places and at other times. In that case, sampling and generalizing might not be of interest. In other cases, we would really like to be able to generalize almost universally. When

psychologists do research, they are often interested in developing theories that would hold for all humans. But in most applied social research, we are interested in generalizing to specific groups. The group you wish to generalize to is often called the population in your study. This is the group you would like to sample from because this is the group you are interested in generalizing to. Let's imagine that you wish to generalize to urban homeless males between the ages of 30 and 50 in the United States. If that is the population of interest, you are likely to have a very hard time developing a reasonable sampling plan. You are probably not going to find an accurate listing of this population, and even if you did, you would almost certainly not be able to mount a national sample across hundreds of urban areas. So we probably should make a distinction between the population you would like to generalize to, and the population that will be accessible to you. We'll call the former the theoretical population and the latter the accessible population. In this example, the accessible population might be homeless males between the ages of 30 and 50 in six selected urban areas across the U.S.



Once you've identified the theoretical and accessible populations, you have to do one more thing before you can actually draw a sample — you have to get a list of the members of the accessible population. (Or, you have to spell out in detail how you will contact them to assure representativeness). The listing of the accessible population from which you'll draw your sample is called the sampling frame. If you were doing a phone survey and selecting names from the telephone book, the book would be your sampling frame.

That wouldn't be a great way to sample because significant subportions of the population either don't have a phone or have moved in or out of the area since the last book was printed. Notice that in this case, you might identify the area code and all three-digit prefixes within that area code and draw a sample simply by randomly dialing numbers (cleverly known as *random-digit-dialing*). In this case, the sampling frame is not a list *per se*, but is rather a procedure that you follow as the actual basis for sampling. Finally, you actually draw your sample (using one of the many sampling procedures). The sample is the group of people who you select to be in your study. Notice that I didn't say that the sample was the group of people who are actually *in* your study. You

may not be able to contact or recruit all of the people you actually sample, or some could drop out over the course of the study. The group that actually completes your study is a subsample of the sample — it doesn't include nonrespondents or dropouts. The problem of nonresponse and its effects on a study will be addressed when discussing "mortality" threats to internal validity.

People often confuse what is meant by random selection with the idea of random assignment. You should make sure that you understand the distinction between random selection and random assignment.

At this point, you should appreciate that sampling is a difficult multi-step process and that there are lots of places you can go wrong. In fact, as we move from each step to the next in identifying a sample, there is the possibility of introducing systematic error or bias.

For instance, even if you are able to identify perfectly the population of interest, you may not have access to all of them. And even if you do, you may not have a complete and accurate enumeration or sampling frame from which to select. And, even if you do, you may not draw the sample correctly or accurately. And, even if you do, they may not all come and they may not all stay. Depressed yet? This is a very difficult business indeed. At times like this I'm reminded of what Donald Campbell used to say (I'll paraphrase here): "Cousins to the amoeba, it's amazing that we know anything at all!"

NON-PROBABILITY SAMPLING

The difference between Non-probability and probability sampling is that Non-probability sampling does not involve *random* selection and probability sampling does. Does that mean that Non-probability samples aren't representative of the population? Not necessarily. But it does mean that Non-probability samples cannot depend upon the rationale of probability theory.

At least with a probabilistic sample, we know the odds or probability that we have represented the population well. We are able to estimate confidence intervals for the statistic. With Non-probability samples, we may or may not represent the population well, and it will often be hard for us to know how well we've done so. In general, researchers prefer probabilistic or random sampling methods over nonprobabilistic ones, and consider them to be more accurate and rigorous. However, in applied social research there may be circumstances where it is not feasible, practical or theoretically sensible to do random sampling. Here, we consider a wide range of nonprobabilistic alternatives.

We can divide Non-probability sampling methods into two broad types: *accidental* or *purposive*. Most sampling methods are purposive in nature because we usually approach the sampling problem with a specific plan in mind. The most important distinctions among these types of sampling methods are the ones between the different types of purposive sampling approaches.

Accidental, Haphazard or Convenience Sampling

One of the most common methods of sampling goes under the various titles listed here. I would include in this category the traditional “man on the street” (of course, now it’s probably the “person on the street”) interviews conducted frequently by television news programmes to get a quick (although nonrepresentative) reading of public opinion. I would also argue that the typical use of college students in much psychological research is primarily a matter of convenience. (You don’t really believe that psychologists use college students because they believe they’re representative of the population at large, do you?). In clinical practice, we might use clients who are available to us as our sample. In many research contexts, we sample simply by asking for volunteers. Clearly, the problem with all of these types of samples is that we have no evidence that they are representative of the populations we’re interested in generalizing to — and in many cases we would clearly suspect that they are not.

Purposive Sampling

In purposive sampling, we sample with a *purpose* in mind. We usually would have one or more specific predefined groups we are seeking. For instance, have you ever run into people in a mall or on the street who are carrying a clipboard and who are stopping various people and asking if they could interview them? Most likely they are conducting a purposive sample (and most likely they are engaged in market research). They might be looking for Caucasian females between 30-40 years old. They size up the people passing by and anyone who looks to be in that category they stop to ask if they will participate. One of the first things they’re likely to do is verify that the respondent does in fact meet the criteria for being in the sample. Purposive sampling can be very useful for situations where you need to reach a targeted sample quickly and where sampling for proportionality is not the primary concern. With a purposive sample, you are likely to get the opinions of your target population, but you are also likely to overweight subgroups in your population that are more readily accessible.

All of the methods that follow can be considered subcategories of purposive sampling methods. We might sample for specific groups or types of people as in modal instance, expert, or quota sampling. We might sample for diversity as in heterogeneity sampling. Or, we might capitalize on informal social networks to identify specific respondents who are hard to locate otherwise, as in snowball sampling. In all of these methods we know what we want — we are sampling with a purpose.

Modal Instance Sampling

In statistics, the *mode* is the most frequently occurring value in a distribution. In sampling, when we do a modal instance sample, we are sampling the most frequent case, or the “typical” case. In a lot of informal public opinion polls, for instance, they interview a “typical” voter. There are a number of problems with this sampling approach.

First, how do we know what the “typical” or “modal” case is? We could say that the modal voter is a person who is of average age, educational level, and income in the population. But, it’s not clear that using the averages of these is the fairest (consider the skewed distribution of income, for instance). And, how do you know that those three variables — age, education, income — are the only or even the most relevant for classifying the typical voter? What if religion or ethnicity is an important discriminator? Clearly, modal instance sampling is only sensible for informal sampling contexts.

Expert Sampling

Expert sampling involves the assembling of a sample of persons with known or demonstrable experience and expertise in some area. Often, we convene such a sample under the auspices of a “panel of experts.” There are actually two reasons you might do expert sampling. First, because it would be the best way to elicit the views of persons who have specific expertise.

In this case, expert sampling is essentially just a specific subcase of purposive sampling. But the other reason you might use expert sampling is to provide evidence for the validity of another sampling approach you’ve chosen. For instance, let’s say you do modal instance sampling and are concerned that the criteria you used for defining the modal instance are subject to criticism. You might convene an expert panel consisting of persons with acknowledged experience and insight into that field or topic and ask them to examine your modal definitions and comment on their appropriateness and validity. The advantage of doing this is that you aren’t out on your own trying to defend your decisions — you have some acknowledged experts to back you. The disadvantage is that even the experts can be, and often are, wrong.

Quota Sampling

In quota sampling, you select people nonrandomly according to some fixed quota. There are two types of quota sampling: *proportional* and *non-proportional*. In proportional quota sampling you want to represent the major characteristics of the population by sampling a proportional amount of each. For instance, if you know the population has 40% women and 60% men, and that you want a total sample size of 100, you will continue sampling until you get those percentages and then you will stop. So, if you’ve already got the 40 women for your sample, but not the sixty men, you will continue to sample men but even if legitimate women respondents come along, you will not sample them because you have already “met your quota.” The problem here (as in much purposive sampling) is that you have to decide the specific characteristics on which you will base the quota. Will it be by gender, age, education race, religion, *etc.*?

Nonproportional quota sampling is a bit less restrictive. In this method, you specify the minimum number of sampled units you want in each category here, you’re not concerned with having numbers that match the proportions in the population. Instead,

you simply want to have enough to assure that you will be able to talk about even small groups in the population. This method is the nonprobabilistic analogue of stratified random sampling in that it is typically used to assure that smaller groups are adequately represented in your sample.

Heterogeneity Sampling

We sample for heterogeneity when we want to include all opinions or views, and we aren't concerned about representing these views proportionately. Another term for this is sampling for *diversity*. In many brainstorming or nominal group processes (including concept mapping), we would use some form of heterogeneity sampling because our primary interest is in getting broad spectrum of ideas, not identifying the "average" or "modal instance" ones. In effect, what we would like to be sampling is not people, but ideas. We imagine that there is a universe of all possible ideas relevant to some topic and that we want to sample this population, not the population of people who have the ideas. Clearly, in order to get all of the ideas, and especially the "outlier" or unusual ones, we have to include a broad and diverse range of participants. Heterogeneity sampling is, in this sense, almost the opposite of modal instance sampling.

Snowball Sampling

In snowball sampling, you begin by identifying someone who meets the criteria for inclusion in your study. You then ask them to recommend others who they may know who also meet the criteria. Although this method would hardly lead to representative samples, there are times when it may be the best method available. Snowball sampling is especially useful when you are trying to reach populations that are inaccessible or hard to find.

For instance, if you are studying the homeless, you are not likely to be able to find good lists of homeless people within a specific geographical area. However, if you go to that area and identify one or two, you may find that they know very well who the other homeless people in their vicinity are and how you can find them.

DIFFERENT RESEARCH METHODS

TRUE EXPERIMENTAL DESIGN

True experimental design is regarded as the most accurate form of experimental research, in that it tries to prove or disprove a hypothesis mathematically, with statistical analysis.

For some of the physical sciences, such as physics, chemistry and geology, they are standard and commonly used. For social sciences, psychology and biology, they can be a little more difficult to set up.

For an experiment to be classed as a true experimental design, it must fit all of the following criteria.

The sample groups must be assigned randomly.

There must be a viable control group.

Only one variable can be manipulated and tested. It is possible to test more than one, but such experiments and their statistical analysis tend to be cumbersome and difficult.

The tested subjects must be randomly assigned to either control or experimental groups.

Advantages

The results of a true experimental design can be statistically analyzed and so there can be little argument about the results.

It is also much easier for other researchers to replicate the experiment and validate the results.

For physical sciences working with mainly numerical data, it is much easier to manipulate one variable, so true experimental design usually gives a yes or no answer.

Disadvantages

Whilst perfect in principle, there are a number of problems with this type of design. Firstly, they can be almost too perfect, with the conditions being under complete control and not being representative of real world conditions.

For psychologists and behavioral biologists, for example, there can never be any guarantee that a human or living organism will exhibit 'normal' behaviour under experimental conditions.

True experiments can be too accurate and it is very difficult to obtain a complete rejection or acceptance of a hypothesis because the standards of proof required are so difficult to reach.

True experiments are also difficult and expensive to set up. They can also be very impractical.

While for some fields, like physics, there are not as many variables so the design is easy, for social sciences and biological sciences, where variations are not so clearly defined it is much more difficult to exclude other factors that may be affecting the manipulated variable.

True experimental design is an integral part of science, usually acting as a final test of a hypothesis. Whilst they can be cumbersome and expensive to set up, literature reviews, qualitative research and descriptive research can serve as a good precursor to generate a testable hypothesis, saving time and money.

Whilst they can be a little artificial and restrictive, they are the only type of research that is accepted by all disciplines as statistically provable.

QUASI-EXPERIMENTAL DESIGN

Quasi-experimental design is a form of experimental research used extensively in the social sciences and psychology

Whilst regarded as unscientific and unreliable, by physical and biological scientists, the method is, nevertheless, a very useful method for measuring social variables. The inherent weaknesses in the methodology do not undermine the validity of the data, as long as they are recognized and allowed for during the whole experimental process. Quasi experiments resemble quantitative and qualitative experiments, but lack random allocation of groups or proper controls, so firm statistical analysis can be very difficult.

Design

Quasi-experimental design involves selecting groups, upon which a variable is tested, without any random pre-selection processes.

For example, to perform an educational experiment, a class might be arbitrarily divided by alphabetical selection or by seating arrangement. The division is often convenient and, especially in an educational situation, causes as little disruption as possible.

After this selection, the experiment proceeds in a very similar way to any other experiment, with a variable being compared between different groups, or over a period of time.

Advantages

Especially in social sciences, where pre-selection and randomization of groups is often difficult, they can be very useful in generating results for general trends.

E.g. if we study the effect of maternal alcohol use when the mother is pregnant, we know that alcohol does harm embryos. A strict experimental design would include that mothers were randomly assigned to drink alcohol. This would be highly illegal because of the possible harm the study might do to the embryos.

So what researchers does is to ask people how much alcohol they used in their pregnancy and then assign them to groups.

Quasi-experimental design is often integrated with individual case studies; the figures and results generated often reinforce the findings in a case study, and allow some sort of statistical analysis to take place.

In addition, without extensive pre-screening and randomization needing to be undertaken, they do reduce the time and resources needed for experimentation.

Disadvantages

Without proper randomization, statistical tests can be meaningless. For example, these experimental designs do not take into account any pre-existing factors (as for the

mothers: what made them drink or not drink alcohol), or recognize that influences outside the experiment may have affected the results. A quasi experiment constructed to analyse the effects of different educational programmes on two groups of children, for example, might generate results that show that one programme is more effective than the other.

These results will not stand up to rigorous statistical scrutiny because the researcher also need to control other factors that may have affected the results. This is really hard to do properly.

One group of children may have been slightly more intelligent or motivated. Without some form of pre-testing or random selection, it is hard to judge the influence of such factors.

Disadvantages aside, as long as the shortcomings of the quasi-experimental design are recognized, these studies can be a very powerful tool, especially in situations where ‘true’ experiments are not possible.

They are very good way to obtain a general overview and then follow up with a case study or quantitative experiment, to focus on the underlying reasons for the results generated.

DOUBLE BLIND EXPERIMENT

A double blind experiment is an experimental method used to ensure impartiality, and avoid errors arising from bias.

It is very easy for a researcher, even subconsciously, to influence experimental observations, especially in behavioral science, so this method provides an extra check. For example, imagine that a company is asking consumers for opinions about its products, using a survey.

There is a distinct danger that the interviewer may subconsciously emphasize the company’s products when asking the questions. This is the major reason why market research companies generally prefer to use computers, and double blind experiments, for gathering important data.

The Blind Experiment

The blind experiment is the minimum standard for any test involving subjects and opinions, and failure to adhere to this principle may result in experimental flaws.

The idea is that the groups studied, including the control, should not be aware of in which group they are placed. In medicine, when researchers are testing a new medicine, they ensure that the placebo looks, and tastes, the same as the actual medicine.

There is strong evidence of a placebo effect with medicine, where, if people believe that they are receiving a medicine, they show some signs of improvement in health. A blind experiment reduces the risk of bias from this effect, giving an honest baseline for the research, and allowing a realistic statistical comparison.

The Double Blind Experiment

The double blind experiment takes this precaution against bias one-step further, by ensuring that the researcher does not know in which group a patient falls.

Whilst the vast majority of researchers are professionals, there is always a chance that the researcher might subconsciously tip off a patient about the pill they were receiving. They may even favour giving the pill to patients that they thought had the best chance of recovery, skewing the results.

Whilst nobody likes to think of scientists as dishonest, there is often pressure, from billion dollar drug companies and the fight for research grants, to generate positive results.

This always gives a chance that a scientist might manipulate results, and try to show the research in a better light. Proving that the researcher carried out a double blind experiment reduces the chance of criticism.

Other Applications

Whilst better known in medicine, double blind experiments are often used in other fields. Surveys, questionnaires and market research all use this technique to retain credibility.

If you wish to compare two different brands of washing powder, the samples should be in the same packaging. A consumer might have an inbuilt brand identity awareness, and preference, which will lead to favouritism and bias.

An example of the weakness of single blind techniques is in police line-ups, where a witness picks out a suspect from a group. Many legal experts are advocating that these line-ups should be unsupervised, and unprompted.

If the police are fixed on bringing a particular subject to justice, they may consciously, or subconsciously, tip off the witness. Humans are very good at understanding body language and unconscious cues, so the chance of observer's bias should be minimized.

DESCRIPTIVE RESEARCH DESIGN

Descriptive research design is a scientific method which involves observing and describing the behaviour of a subject without influencing it in any way. Many scientific disciplines, especially social science and psychology, use this method to obtain a general overview of the subject.

Some subjects cannot be observed in any other way; for example, a social case study of an individual subject is a descriptive research design and allows observation without affecting normal behaviour. It is also useful where it is not possible to test and measure the large number of samples needed for more quantitative types of experimentation.

These types of experiments are often used by anthropologists, psychologists and social scientists to observe natural behaviors without affecting them in any way. It is also used by market researchers to judge the habits of customers, or by companies

wishing to judge the morale of staff. The results from a descriptive research can in no way be used as a definitive answer or to disprove a hypothesis but, if the limitations are understood, they can still be a useful tool in many areas of scientific research.

Advantages

The subject is being observed in a completely natural and unchanged natural environment. A good example of this would be an anthropologist who wanted to study a tribe without affecting their normal behaviour in any way. True experiments, whilst giving analyzable data, often adversely influence the normal behaviour of the subject.

Descriptive research is often used as a precursor to more quantitatively research designs, the general overview giving some valuable pointers as to what variables are worth testing quantitatively. Quantitative experiments are often expensive and time-consuming so it is often good sense to get an idea of what hypotheses are worth testing.

Disadvantages

Because there are no variables manipulated, there is no way to statistically analyse the results. Many scientists regard this type of study as very unreliable and ‘unscientific’. In addition, the results of observational studies are not repeatable, and so there can be no replication of the experiment and reviewing of the results.

Descriptive research design is a valid method for researching specific subjects and as a precursor to more quantitative studies. Whilst there are some valid concerns about the statistical validity, as long as the limitations are understood by the researcher, this type of study is an invaluable scientific tool. Whilst the results are always open to question and to different interpretations, there is no doubt that they are preferable to performing no research at all.

LITERATURE REVIEW

Many students are instructed, as part of their research programme, to perform a literature review, without always understanding what a literature review is.

Most are aware that it is a process of gathering information from other sources and documenting it, but few have any idea of how to evaluate the information, or how to present it.

A literature review can be a precursor in the introduction of a research paper, or it can be an entire paper in itself, often the first stage of large research projects, allowing the supervisor to ascertain that the student is on the correct path.

A literature review is a critical and in depth evaluation of previous research. It is a summary and synopsis of a particular area of research, allowing anybody reading the paper to establish why you are pursuing this particular research programme. A good literature review expands upon the reasons behind selecting a particular research question.

What is a Literature Review Not

It is not a chronological catalog of all of the sources, but an evaluation, integrating the previous research together, and also explaining how it integrates into the proposed research programme. All sides of an argument must be clearly explained, to avoid bias, and areas of agreement and disagreement should be highlighted.

It is not a collection of quotes and paraphrasing from other sources. A good literature review should also have some evaluation of the quality and findings of the research. A good literature review should avoid the temptation of impressing the importance of a particular research programme. The fact that a researcher is undertaking the research programme speaks for its importance, and an educated reader may well be insulted that they are not allowed to judge the importance for themselves. They want to be re-assured that it is a serious paper, not a pseudo-scientific sales advertisement.

Whilst some literature reviews can be presented in a chronological order, it is best avoided. For example, a review of Victorian Age Physics, could present J.J. Thomson's famous experiments in a chronological order. Otherwise, this is usually perceived as being a little lazy, and it is better to organize the review around ideas and individual points.

As a general rule, certainly for a longer review, each paragraph should address one point, and present and evaluate all of the evidence, from all of the differing points of view.

Conducting a Literature Review

Evaluating the credibility of sources is one of the most difficult aspects, especially with the ease of finding information on the internet. The only real way to evaluate is through experience, but there are a few tricks for evaluating information quickly, yet accurately. There is such a thing as 'too much information,' and Google does not distinguish or judge the quality of results, only how search engine friendly a paper is. This is why it is still good practice to begin research in an academic library. Any journals found there can be regarded as safe and credible.

The next stage is to use the internet, and this is where the difficulties start. It is very difficult to judge the credibility of an online paper. The main thing is to structure the internet research as if it were on paper. Bookmark papers, which may be relevant, in one folder and make another subfolder for a 'shortlist.'

The easiest way is to scan the work, using the abstract and introduction as guides. This helps to eliminate the non-relevant work and also some of the lower quality research.

If it sets off alarm bells, there may be something wrong, and the paper is probably of a low quality. Be very careful not to fall into the trap of rejecting research just because it conflicts with your hypothesis. Failure to do this will completely invalidate the literature review and potentially undermine the research project. Any research that may be relevant should be moved to the shortlist folder.

The next stage is to critically evaluate the paper and decide if the research is sufficient quality. Think about it this way: The temptation is to try to include as many sources as possible, because it is easy to fall into the trap of thinking that a long bibliography equates to a good paper. A smaller number of quality sources is far preferable than a long list of irrelevance.

Check into the credentials of any source upon which you rely heavily for the literature review. The reputation of the University or organization is a factor, as is the experience of the researcher.

If their name keeps cropping up, and they have written many peoples, the source is usually OK. Look for agreements. Good research should have been replicated by other independent researchers, with similar results, showing that the information is usually fairly safe to use.

If the process is proving to be difficult, and in some fields, like medicine and environmental research, there is a lot of poor science, do not be afraid to ask a supervisor for a few tips. They should know some good and reputable sources to look at. It may be a little extra work for them, but there will be even more work if they have to tear apart a review because it is built upon shaky evidence.

Conducting a good literature review is a matter of experience, and even the best scientists have fallen into the trap of using poor evidence. This is not a problem, and is part of the scientific process; if a research programme is well constructed, it will not affect the results.

SURVEY RESEARCH DESIGN

The survey research design is often used because of the low cost and easy accessible information. Conducting accurate and meaningful surveys is one of the most important facets of market research in the consumer driven 21st century.

Businesses, governments and media spend billions of dollars on finding out what people think and feel.

Accurate research can generate vast amounts of revenue; bad or inaccurate research can cost millions, or even bring down governments. The survey research design is a very valuable tool for assessing opinions and trends. Even on a small scale, such as local government or small businesses, judging opinion with carefully designed surveys can dramatically change strategies.

Television chat-shows and newspapers are usually full of facts and figures gleaned from surveys but often no information is given as to where this information comes from or what kind of people were asked.

A cursory examination of these figures usually shows that the results of these surveys are often manipulated or carefully sifted to try and reflect distort the results to match the whims of the owners.

Businesses are often guilty of carefully selecting certain results to try and portray themselves as the answer to all needs. When you decide to enter this minefield and

design a survey, how do you avoid falling into the trap of inaccuracy and bias? How do you ensure that your survey research design reflects the views of a genuine cross-section of the population?

The simple answer is that you cannot; even with unlimited budget, time and resources, there is no way of achieving 100% accuracy. Opinions, on all levels, are very fluid and can change on a daily or even hourly basis.

Despite this, surveys are still a powerful tool and can be an extremely powerful research tool. As long as you design your survey well and are prepared to be self-critical, you can still obtain an accurate representation of opinion.

Establishing the Aims of Your Research

This is the single most important step of your survey research design and can make or break your research; every single element of your survey must refer back to this design or it will be fatally flawed.

If your research is too broad, you will have to ask too many questions; too narrow and you will not be researching the topic thoroughly enough.

Researching and Determining Your Sample Group

This is the next crucial step in determining your survey and depends upon many factors.

The first is accuracy; you want to try and interview as broad a base of people as possible. Quantity is not always the answer; if you were researching a detergent, for example, you would want to target your questions at those who actually use such products. For a political or ethical survey, about which anybody can have a valid opinion, you want to try and represent a well balanced cross section of society.

It is always worth checking beforehand what quantity and breadth of response you need to provide significant results or your hard work may be in vain.

Before you start the planning, it is important that you consult somebody about the statistical side of your survey research design. This way, you know what number and type of responses you need to make it a valid survey and prevent inaccurate results.

Methodology

How do you make sure that your questionnaire reaches the target group? There are many methods of reaching people but all have advantages and disadvantages. For a college or university study it is unlikely that you will have the facilities to use internet, e-mail or phone surveying so we will concentrate on only the likely methods you will use.

FACE TO FACE

This is probably the most traditional method of the survey research design. It can be very accurate. It allows you to be selective about to whom you ask questions and you

can explain anything that they do not understand. In addition, you can make a judgement about who you think is wasting your time or giving stupid answers.

There are a few things to be careful of with this approach; firstly, people can be reluctant to give up their time without some form of incentive. Another factor to bear in mind is that is difficult to ask personal questions face to face without embarrassing people. It is also very time consuming and difficult to obtain a representative sample.

Finally, if you are going to be asking questions door-to-door, it is essential to ensure that you have some official identification to prove who you are.

Mail

This does not necessarily mean using the postal service; putting in the legwork and delivering questionnaires around a campus or workplace is another method.

This is a good way of targeting a certain section of people and is excellent if you need to ask personal or potentially embarrassing questions. The problems with this method are that you cannot be sure of how many responses you will receive until a long time period has passed. You must also be wary of collecting personal data; most countries have laws about how much information you can keep about people so it is always wise to check with somebody more knowledgeable.

Structuring and Designing the Questionnaire

The design of your questionnaire depends very much upon the type of survey and the target audience.

If you are asking questions face to face it is easy to explain if people are unsure of a question. On the other hand, if your questionnaire is going to include many personal questions then mailing methods are preferable (but may violate local legislation).

You must keep your questionnaire as short as possible; people will either refuse to fill in a long questionnaire or get bored halfway through.

Cover Note

It is also polite, especially with mailed questionnaires, to send a short cover note explaining what you are doing and how the subject should return the surveys to you.

You should introduce yourself; explain why you are doing the research, what will happen with the results and who to contact if the subject has any queries.

Types of Question

Multiple choice questions allow many different answers, including don't know, to be assessed. The main strength of this type of question is that the form is easy to fill in and the answers can be checked easily and quantitatively; this is useful for large sample groups.

Rating, on some scale, is a tried and tested form of question structure. This way is very useful when you are seeking to be a little more open-ended than is possible with multiple choice questions. It is a little harder to analyse your responses. It is important to make sure that the scale allows extreme views.

Questions asking for opinions must be open-ended and allow the subject to give their own response; you should avoid entrapment and appear to be as neutral as possible during the procedure. The major problem is that you have to devise a numerical way of analyzing and statistically evaluating the responses which can lead to a biased view, if care is not taken. These types of question should really be reserved for experienced researchers.

The order in which you ask the questions can be important. Try to start off with the most relevant questions first. Also friendly and non-threatening questions put the interviewee at ease. Questions should be simple and straightforward using everyday language rather than perfect grammar.

Try and group questions about similar topics together; this makes it a lot quicker for people to answer questions more quickly and easily.

Some researchers advocate mixing up and randomizing questions for accuracy but this approach tends to be more appropriate for advanced market research. For this type of survey the researcher is trying to disguise the nature of the research and filter out preconceptions.

It is also a good idea to try out a test survey; ask a small group to give genuine and honest feedback so that you can make adjustments. Common mistakes when doing the survey research design.

Analyzing Your Results

This is where the fun starts and it will depend upon the type of questions used.

For multiple choice questions it is a matter of counting up the answers to each question and using statistics to 'crunch the numbers' and test relevance.

Rating type question require a little more work but they follow broadly the same principle.

For opinion questions you have to devise some way of judging the responses numerically.

The next step is to devise which statistical test you are going to use and start to enter some numbers to judge the significance of your data.

This is where you have to analyse the results. Be self critical whether your results showed what you expected or not. Any survey has flaws in its method so it is always a good idea to show that you are aware of these.

For example, a university represents only a narrow cross section of society; as long as you are aware of this then your results are valid. If your survey gave unexpected results explain the possible reasons for why this happened and suggestions for refining the techniques and structure of your survey next time.

Identical Twins Study

The identical twins study has been used for a long time, to study the effects of environment and genetics on human development.

Some studies have tried to determine how genetics and environmental factors contribute to intelligence, aggression or substance addictions.

Most of the twin's studies compare identical twins, having 100% genetic similarity, with non-identical twins, with about 50% genetic similarity. The researcher compares the occurrence of an individual trait between identical and fraternal twins. If the identical twins show more similarity for this trait than the non-identical twins, then the excess is assumed to be down to genetic factors.

This type of analysis would then allow the researchers to estimate the heritability of specific traits and quantify the effect of genetic factors on the individual trait. Psychologists have long known that a twin study is not a true experimental design, but it has led to some interesting insights into the influence of genes on human behaviour.

For this method, a number of assumptions have to be made; that the identical twins share identical DNA profiles, and that the environmental factors are the same for all participants.

Criticisms

There have been few criticisms of identical twin studies over the years. By their nature, and because of small sample sizes, it is very difficult to quantitatively analyse the results and so all experimentation tends to be observational; the sample groups cannot be random so statistical analysis is impossible.

The experimental methods assume that there is little difference in the environmental factors between fraternal and identical twins, but there is a criticism that the tendency of adults to treat identical twins in exactly the same way makes this assumption invalid. Parents tend to dress identical twins the same way and encourage them to pursue the same interests.

The distinction between environmental factors and genetic influences may not be as black and white as the identical twins study assumes. There is probably an interaction between genes and environment and so the whole picture may be a lot more complex.

In addition, the experiment tends to assume that one gene affects one behavioral trait. Modern genetic research is showing that many different genes can influence behaviour.

The above criticisms all have some validity, but the main point is that twin studies have never claimed to be anything other than observational, identifying and trying to explain trends rather than prove a hypothesis.

Whilst there are some concerns about the validity of the identical twins study, such experiments are certainly better than performing no research at all.

Twins studies are now trying to analyse the environmental factors more. Instead of assuming that the environmental factors are the same, they are now contrasting shared family environment with the individual events suffered by the individual twin.

In addition, identical twins study is constantly evolving into more complex forms, now taking into account whole families and other siblings in addition to the twins.

Research into the human genome is now resurrecting the studies of twins; hereditary trends observed in an identical twins study can now be studied quantitatively in the laboratory. It is now standard practice, when conducting twin's research to analyse DNA from all participants and this is bypassing many of the concerns about the twin study.

META ANALYSIS

Meta analysis is a statistical technique developed by social scientists, who are very limited in the type of experiments they can perform.

Social scientists have great difficulty in designing and implementing true experiments, so meta-analysis gives them a quantitative tool to analyse statistically data drawn from a number of studies, performed over a period of time.

Medicine and psychology increasingly use this method, as a way of avoiding time-consuming and intricate studies, largely repeating the work of previous research.

What is Meta-analysis

Social studies often use very small sample sizes, so any statistics used generally give results containing large margins of error.

This can be a major problem when interpreting and drawing conclusions, because it can mask any underlying trends or correlations. Such conclusions are only tenuous, at best, and leave the research open for criticism.

Meta analysis is the process of drawing from a larger body of research, and using powerful statistical analyzes on the conglomerated data. This gives a much larger sample population and is more likely to generate meaningful and usable data.

The Advantages of Meta Analysis

Meta analysis is an excellent way of reducing the complexity and breadth of research, allowing funds to be diverted elsewhere. For rare medical conditions, it allows researchers to collect data from further afield than would be possible for one research group.

As the method becomes more common, database programmes have made the process much easier, with professionals working in parallel able to enter their results and access the data. This allows constant quality assessments and also reducing the chances of unnecessary repeat research, as papers can often take many months to be published, and the computer records ensure that any researcher is aware of the latest directions and results.

The field of meta study is also a lot more rigorous than the traditional literature review, which often relies heavily upon the individual interpretation of the researcher.

When used with the databases, a meta study allows a much wider net to be cast than by the traditional literature review, and is excellent for highlighting correlations and links between studies that may not be readily apparent as well as ensuring that the compiler does not subconsciously infer correlations that do not exist.

The Disadvantages of Meta Analysis

There are a number of disadvantages to meta analysis, of which a researcher must be aware before relying upon the data and generated statistics.

The main problem is that there is the potential for publication bias and skewed data.

Research generating results not refuting a hypothesis may tend to remain unpublished, or risks not being entered into the database. If the meta study is restricted to the research with positive results, then the validity is compromised.

The researcher compiling the data must make sure that all research is quantitative, rather than qualitative, and that the data is comparable across the various research programmes, allowing a genuine statistical analysis.

It is important to pre-select the studies, ensuring that all of the research used is of a sufficient quality to be used.

One erroneous or poorly conducted study can place the results of the entire meta analysis at risk. On the other hand, setting almost unattainable criteria and criteria for inclusion can leave the meta study with too small a sample size to be statistically relevant.

Striking a balance can be a little tricky, but the whole field is in a state of constant development, incorporating protocols similar to the scientific method used for normal quantitative research. Finding the data is rapidly becoming the real key, with skilled meta analysts developing a skill-set of library based skills, finding information buried in government reports and conference data, developing the knack of assessing the quality of sources quickly and effectively.

Conclusions and the Future

Meta analysis is here to stay, as an invaluable tool for research, and is rapidly gaining momentum as a stand-alone discipline, with practitioners straddling the divide between statisticians and librarians.

The conveniences, as long as the disadvantages are taken into account, are too apparent to ignore, and a meta study can reduce the need for long, expensive and potentially intrusive repeated research studies.

SYSTEMATIC REVIEWS

Heavily used by the health care sector, systematic reviews are a powerful way of isolating and critically evaluating previous research.

Modern medical research generates so much literature, and fills so many journals, that a traditional literature review could take months, and still be out of date by the time that the research is designed and performed.

In addition, researchers are often guilty of selecting the research best fitting their pre-conceived notions, a weakness of the traditional 'narrative' literature review process.

To help medical professionals, specialist compilers assess and condense the research, entering it into easily accessible research databases. They are an integral part of the research process, and every student of medicine routinely receives a long and extensive training in the best methods for critically evaluating literature.

Addressing the Deficiencies in Narration

The problems with narrative literature came to light a couple of decades ago, when critics realized that reviewers looking at the same body of evidence often generated completely different findings. They drew conclusions based upon their specialty, rather than the compelling evidence contained within the body of research.

It is unclear whether this was a case of conscious or subconscious manipulation (bias), but this particular finding was worrying, especially in a research area where life and death could be at stake. To address this issue, medical authorities developed a new protocol of systematic reviewing, based upon a structure as strict as the scientific method governing empirical research programmes.

The Protocols Underpinning Systematic Reviews

Define a research question, in a similar way to formulating a research question for a standard research design

Locate and select relevant previous research studies, with no attempt at evaluation at this stage. Ideally, research in languages other than English should be used, and the researcher should try to find papers and reports unpublished in journals, such as conference speeches or company reports.

Critically evaluate the studies. The reviewer should assess each study upon criteria based upon quality, strength of the findings and validity. For safety, this process should include at least two independent reviewers, although a greater number is advisable. Combine the results. This is the process of combining all of the findings, sometimes qualitatively, but usually quantitatively, using meta-analysis.

Publish the results. As with any research, the results have to be written and published, usually with a system of independent review. Discussion of the conclusions, as with any research, allows the validity of the findings to be verified.

The Reasoning Behind Systematic Reviews

The principle behind the systematic reviews process is that the researcher critically evaluates previous studies, in a much more comprehensive and systematic way than a

standard literature review. In many cases, statistical meta-analysis tools are used to give the review a quantitative foundation, allowing correlations to be documented and conclusions to be drawn. Whilst the techniques are mainly used by medicine and psychology, there is a growing trend towards using systematic reviews in other disciplines. Many branches of science are becoming increasingly fragmented and anarchic, so this layer of analysis aggregates all of the disparate elements.

Systematic reviews, and meta-analysis, are regarded as a cornerstone of health care research, essential where it is impractical or unethical to keep repeating old research.

In addition to the potential risks of repeated research upon patients and volunteers, there are now laws in many countries prohibiting excessive research using animals. Systematic reviews are a great way of reducing the amount of suffering caused by vivisection.

Disadvantages of Systematic Reviews

As with most systems, despite the protocols, systematic reviews do have some inherent weaknesses. The main problem is the rapid advancement of medical research and technology, often meaning that many reviews are out of date before they are even published, forcing researchers to update their findings constantly. The development of specialist organizations for finding and evaluating data minimizes the effects of this particular shortcoming.

As with any subjective review, there is the problem of selection bias, where contradictory research is jettisoned, although most medical researchers are adept at following the proper procedures.

Funding and research grants cause researchers to try to find results that suit their paymasters, a growing problem in many areas of science, not just medicine. The specialist reviewers sidestep this problem, to a certain extent, by producing independent research, uncorrupted by governmental or private health care funding, curbing the worst excesses.

Often, a blind system is used, and reviewers are unaware of where the papers they are reviewing came from, or who they are written by. This lessens allegations of favouritism and judging research by the reputation of the researcher rather than on merit. Ultimately, the onus is on the reader to draw their own assessments, using their own experience to judge the quality of the systematic review. Whilst not a perfect system, systematic reviews are far superior to the traditional narrative approach, which often allows a lot of good research to fall through the cracks.

OBSERVATIONAL STUDY

In the fields of social science, psychology and medicine, amongst others, observational study is an essential tool.

This type of research draws a conclusion by comparing subjects against a control group, in cases where the researcher has no control over the experiment.

A research study comparing the risk of developing lung cancer, between smokers and non-smokers, would be a good example of an observational study.

The main reason for performing any observational research is due to ethical concerns.

With the smoking example, a scientist cannot give cigarettes to non-smokers for 20 years and compare them with a control group. This also brings up the other good reason for such studies, in that few researchers can study the long-term effects of certain variables, especially when it runs into decades.

For this study of long-term and subtle effects, they have to use pre-existing conditions and medical records. The researcher may want to study an extremely small sample group, so it is easier to start with known cases and works backwards. The thalidomide cases, for example, are an example of an observational study where researchers had to work backwards, and establish that the drug was the cause of disabilities.

The main problem with observational studies is that the experimenter has no control over the composition of the control groups, and cannot randomize the allocation of subjects. This can create bias, and can also mask cause and effect relationships or, alternatively, suggest correlations where there are none (error in research).

For example, in the smoking example, if the researcher found that there is a correlation between smoking and increased rates of lung cancer, without knowing the full and complete background of the subjects, there is no way of determining whether other factors were involved, such as diet, occupation or genetics. Randomization is assumed to even out external causal effects, but this is impossible in an observational study. There is no independent variable, so it is dangerous to assume cause and effect relationships, a process often misunderstood by the mass media lauding the next wonder food, or sensationalizing a political debate with unfounded results and pseudo-science.

Despite the limitations, an observational study allows a useful insight into a phenomenon, and sidesteps the ethical and practical difficulties of setting up a large and cumbersome medical research project.

Naturalistic Observation

In many scientific disciplines, naturalistic observation is a useful tool for expanding knowledge about a specific phenomenon or species.

In fields such as anthropology, behavioral biology and ecology, watching a person or organism in a natural environment is essential.

Most naturalistic observation is unobtrusive, such as a researcher setting up a camera to film the behaviour of a badger underground. Most nature documentaries are examples of naturalistic observational study, where days, weeks or even years of film are analyzed and edited, to give an overview of the life cycle of the organism.

There is often little attempt at analysis, quantitative or qualitative, but the observational study does uncover unknown phenomena and behaviors.

Obtrusive naturalistic observational study is often used in anthropology, where a researcher lives with a remote tribe for a period of time and records their behaviour.

By living there, she is influencing their social interactions and habits, but can still make some excellent observations. Often, anthropologists will adopt the lifestyle of a particular group of people, in an attempt to understand why they have certain customs and beliefs.

In technical terms, it would be difficult to follow people without discovery, and it would also be unethical to observe without consent, so obtrusive naturalistic observation is the only method that can be used with human subjects.

Many of the producers of the recent glut of reality shows try to claim that their shows are psychological experiments, based around observational study. This is stretching the idea too far, as there are very few people who would not change their behaviour when they are aware that a camera is watching.

In these cases, it is difficult to make any realistic and valid observations about their lifestyle.

Most criticisms of naturalistic observation are based around this principle, and an anthropologist or social scientist has to ensure that they intervene as little as possible.

Field Experiments

For geologists, social scientists and environmental biologists, amongst others, field experiments are an integral part of the discipline.

As the name suggests, a field study is an experiment performed outside the laboratory, in the 'real' world. Unlike case studies and observational studies, a field experiment still follows all of the steps of the scientific process, addressing research problems and generating hypotheses.

The obvious advantage of a field study is that it is practical and also allows experimentation, without artificially introducing confounding variables.

A population biologist examining an ecosystem could not move the entire environment into the laboratory, so field experiments are the only realistic research method in many fields of science.

In addition, they circumvent the accusation leveled at laboratory experiments of lacking external or ecological validity, or adversely affecting the behaviour of the subject.

Social scientists and psychologists often used field experiments to perform blind studies, where the subject was not even aware that they were under scrutiny.

A good example of this is the Piliavin and Piliavin experiment, where the propensity of strangers to help blood covered 'victims' was measured. This is now frowned upon, under the policy of informed consent, and is only used in rare and highly regulated circumstances.

Field experiments can suffer from a lack of a discrete control group and often have many variables to try to eliminate.

For example, if the effects of a medicine are studied, and the subject is instructed not to drink alcohol, there is no guarantee that the subject followed the instructions, so field studies often sacrifice internal validity for external validity.

For fields like biology, geology and environmental science, this is not a problem, and the field experiment can be treated as a sound experimental practice, following the steps of the scientific method.

A major concern shared by all disciplines is the cost of field studies, as they tend to be very expensive.

For example, even a modestly sized research ship costs many thousands of dollars every day, so a long oceanographical research programme can run into the millions of dollars.

Pilot studies are often used to test the feasibility of any long term or extensive research programme before committing vast amounts of funds and resources. The changeable nature of the external environment and the often-prohibitive investment of time and money mean that field experiments are rarely replicable, so any generalization is always tenuous.

COHORT STUDY

A cohort study is a research programme investigating a particular group with a certain trait, and observes over a period of time.

Some examples of cohorts may be people who have taken a certain medication, or have a medical condition. Outside medicine, it may be a population of animals that has lived near a certain pollutant or a sociological study of poverty.

A cohort study can delve even further and divide a cohort into sub-groups, for example, a cohort of smokers could be sub-divided, with one group suffering from obesity. In this respect, a cohort study is often interchangeable with the term naturalistic observation.

There are two main sub-types of cohort study, the retrospective and the prospective cohort study. The major difference between the two is that the retrospective looks at phenomena that have already happened, whilst the prospective type starts from the present.

Retrospective Cohort Study

The retrospective case study is historical in nature. Whilst still beginning with the division into cohorts, the researcher looks at historical data to judge the effects of the variable.

For example, it might compare the incidence of bowel cancer over time in vegetarians and meat eaters, by comparing the medical histories. It is a lot easier than the prospective, but there is no control, and confounding variables can be a problem, as the researcher cannot easily assess the lifestyle of the subject.

A retrospective study is a very cheap and effective way of studying health risks or the effects of exposure to pollutants and toxins. It gives results quickly, at the cost of validity, because it is impossible to eliminate all of the potentially confounding variables from historical records and interviews alone.

Prospective Cohort Study

In a prospective cohort study, the effects of a certain variable are plotted over time, and the study becomes an ongoing process. To maintain validity, all of the subjects must be initially free of the condition tested for.

For example, an investigation, over time, into the effects of smoking upon lung cancer must ensure that all of the subjects are free of the disease. It is also possible to subgroup and try to control variables, such as weight, occupation type or social status.

They are preferable to a retrospective study, but are expensive and usually require a long period of time to generate useful results, so are very expensive and difficult.

Some studies have been running for decades, but are generating excellent data about underlying trends in a population. The prospective cohort study is a great way to study long-term trends, allowing the researcher to measure any potential confounding variables, but the potential cost of error is high, so pilot studies are often used to ensure that the study runs smoothly.

Ambidirectional Cohort Study

The ambidirectional cohort study is the ultimate method, combining retrospective and prospective aspects. The researcher studies and analyzes the previous history of the cohorts and then continues the research in a prospective manner.

This gives the most accurate results, but is an extremely arduous undertaking, costing time and a great deal of money.

The ambidirectional study shares one major drawback with the prospective study, in that it is impossible to guarantee that any data can be followed up, as participants may decline to participate or die prematurely. These studies need to look at very large samples to ensure that any attributional losses can be absorbed by the statistics.

Longitudinal Study

A longitudinal study is observational research performed over a period of years or even decades, and allows social scientists and economists to study long-term effects in a human population. A cohort study is a subset of the longitudinal study because it observes the effect on a specific group of people over time. Quite often, a longitudinal study is an extended case study, observing individuals over long periods, and is a purely qualitative undertaking. The lack of quantitative data means that any observations are speculative, as with many case studies, but they allow a unique and valuable perspective on some aspects of human culture and sociology.

The 'up' Series

The groundbreaking television documentary 'UP' is probably the most famous example of a long-term longitudinal study, a case study of a group of British people from birth.

The original producer, Michael Apted, proposed the hypothesis that children born into a certain social class would remain entrenched in that class throughout their life.

In 1967, he selected children from the rich, poor and middle classes, and proceeded to interview and film them every seven years. The highly acclaimed series is still running, with the next set of interviews to be performed in 2011/2012, and it has provided a unique insight into the development of British culture since the 1960's. Even this series highlights one of the major flaws of a longitudinal study, the problem that there can be no retesting or restart. Apted, with hindsight, wished that he had used more female subjects, showing the importance of the initial planning stage of a longitudinal study. Once a course of action is decided, the clock cannot be turned back, and the results must stand as tested.

The Cross Sectional Study

The cross sectional study looks at a different aspect than the standard longitudinal study.

The longitudinal study uses time as the main variable, and tries to make an in depth study of how a small sample changes and fluctuates over time. A cross sectional study, on the other hand, takes a snapshot of a population at a certain time, allowing conclusions about phenomena across a wide population to be drawn.

An example of a cross-sectional study would be a medical study looking at the incidence of breast cancer in a population. The researcher can look at a wide range of ages, ethnicities and social backgrounds. If a significant number of women from a certain social background are found to have the disease, then the researcher can investigate further. This is a relatively easy way to perform a preliminary experiment, allowing the researcher to focus on certain population groups and understand the wider picture.

Of course, researchers often use both methods, using a cross section to take the snapshot and isolate potential areas of interest, and then conducting a longitudinal study to find the reason behind the trend.

This is called panel data, or time series cross-sectional data, but is generally a complicated and expensive type of research, notoriously difficult to analyse.

Such programmes are rare, but can give excellent data, allowing a long-term picture of phenomena to be ascertained.

FACTORIAL DESIGN

A factorial design is often used by scientists wishing to understand the effect of two or more independent variables upon a single dependent variable.

Traditional research methods generally study the effect of one variable at a time, because it is statistically easier to manipulate. However, in many cases, two factors may be interdependent, and it is impractical or false to attempt to analyse them in the traditional way.

Social researchers often use factorial designs to assess the effects of educational methods, whilst taking into account the influence of socio-economic factors and background.

Agricultural science, with a need for field-testing, often uses factorial designs to test the effect of variables on crops. In such large-scale studies, it is difficult and impractical to isolate and test each variable individually.

Factorial experiments allow subtle manipulations of a larger number of interdependent variables. Whilst the method has limitations, it is a useful method for streamlining research and letting powerful statistical methods highlight any correlations.

The Basics

Imagine an aquaculture research group attempting to test the effects of food additives upon the growth rate of trout.

A traditional experiment would involve randomly selecting different tanks of fish and feeding them varying levels of the additive contained within the feed, for example none or 10%. However, as any fish farmer knows, the density of stocking is also crucial to fish growth; if there are not enough fish in a tank, then the wasted capacity costs money. If the density is too high, then the fish grow at a slower rate.

Rather than the traditional experiment, the researchers could use a factorial design and co-ordinate the additive trial with different stocking densities, perhaps choosing four groups. The factorial experiment then needs 4×2 , or eight treatments.

The traditional rules of the scientific method are still in force, so statistics require that every experiment be conducted in triplicate.

This means 24 separate treatment tanks. Of course, the researchers could also test, for example, 4 levels of concentration for the additive, and this would give 4×4 or 16 tanks, meaning 48 tanks in total. Each factor is an independent variable, whilst the level is the subdivision of a factor. Assuming that we are designing an experiment with two factors, a 2×2 would mean two levels for each, whereas a 2×4 would mean two subdivisions for one factor and four for the other. It is possible to test more than two factors, but this becomes unwieldy very quickly. In the fish farm example, imagine adding another factor, temperature, with four levels into the mix. It would then be $4 \times 4 \times 4$, or 64 runs. In triplicate, this would be 192 tanks, a huge undertaking.

There are a few other methods, such as fractional factorial designs, to reduce this, but they are not always statistically valid. This lies firmly in the realm of advanced statistics and is a long, complicated and arduous undertaking.

The Pros and Cons of Factorial Design

Factorial designs are extremely useful to psychologists and field scientists as a preliminary study, allowing them to judge whether there is a link between variables, whilst reducing the possibility of experimental error and confounding variables.

The factorial design, as well as simplifying the process and making research cheaper, allows many levels of analysis. As well as highlighting the relationships between variables, it also allows the effects of manipulating a single variable to be isolated and analyzed singly.

The main disadvantage is the difficulty of experimenting with more than two factors, or many levels. A factorial design has to be planned meticulously, as an error in one of the levels, or in the general operationalization, will jeopardize a great amount of work. Other than these slight detractions, a factorial design is a mainstay of many scientific disciplines, delivering great results in the field.

Case Control Study

A case control study is a method extensively used by the medical profession, as an easy and quick way of comparing treatments, or investigating the causes of disease. Longitudinal studies are the preferred method, but are often expensive, time consuming and difficult. Whilst this method does suffer from some weaknesses, it is quick, and delivers results quickly.

The case control study uses groups of patients stricken with a disease, and compares them with a control group of patients not suffering symptoms. Medical records and interviews are used to try to build up a historical picture of the patient's life, allowing cross-reference between patients and statistical analysis. Any trends can then be highlighted and action can be taken.

Statistical analysis allows the researcher to draw a conclusion about whether a certain situation or exposure led to the medical condition. For example, a scientist could compare a group of coal miners suffering from lung cancer with those clear of the disease, and try to establish the underlying cause. If the majority of the cases arose in collieries owned by one company, it might indicate that the company's safety equipment and procedures were at fault.

Possibly the most famous case control study using this method was a study into whether bicycle helmets reduce the chance of cyclists receiving bad head injuries in an accident.

Obviously, the researcher could not use standard experimentation and compare a control group of non-helmet wearers with helmet wearers, measuring the chances of head injury, as this would be unethical. A case study control was utilized, and the researchers looked at medical records, comparing the number of head injury sufferers wearing helmets against those without. This generated a statistical result, showing that wearing a cycle helmet made it 88% less likely that head injury would be suffered in an accident.

The main weakness of the case control study is that it is very poor at determining cause and effect relationships.

In the cycle helmet example, it could be argued that a cyclist who bothered wearing a helmet may well have been a safer cyclist anyway, and less likely to have accidents.

Evidence showed that children wearing helmets were more likely to be from a more affluent class, more used to cycling through parks than city streets. The study also showed that helmets were of little use to adults.

Whilst most agree that cycle helmets are probably a good thing for children, there is not enough evidence to suggest that they should be mandatory for adults outside extreme cycling. These problems serve as a warning that the results of any case control study should not be relied upon, instead acting as a guide, possibly allowing deeper and more rigorous methods to be utilized.

Pilot Study

A pilot study is a standard scientific tool for ‘soft’ research, allowing scientists to conduct a preliminary analysis before committing to a full-blown study or experiment.

A small chemistry experiment in a college laboratory, for example, costs very little, and mistakes or validity problems easily rectified. At the other end of the scale, a medical experiment taking samples from thousands of people from across the world is expensive, often running into the millions of dollars.

Finding out that there was a problem with the equipment or with the statistics used is unacceptable, and there will be dire consequences. A field research project in the Amazon Basin costs a lot of time and money, so finding out that the electronics used do not function in the humid and warm conditions is too late.

To test the feasibility, equipment and methods, researchers will often use a pilot study, a small-scale rehearsal of the larger research design. Generally, the pilot study technique specifically refers to a smaller scale version of the experiment, although equipment tests are an increasingly important part of this sub-group of experiments.

For example, the medical researchers may conduct a smaller survey upon a hundred people, to check that the protocols are fine.

The Amazon Researchers may perform an experiment, in similar conditions, sending a small team either to the Amazon to test the procedures, or by using something like the tropical bio-dome at the Eden Project.

Pilot studies are also excellent for training inexperienced researchers, allowing them to make mistakes without fear of losing their job or failing the assignment.

Logistical and financial estimates can be extrapolated from the pilot study, and the research question, and the project can be streamlined to reduce wastage of resources and time.

Pilots can be an important part of attracting grants for research as the results can be placed before the funding body.

Generally, most funding bodies see research as an investment, so are not going to dole out money unless they are certain that there is a chance of a financial return.

Unfortunately, and paper reporting the preliminary pilot study, especially if problems were reported, is often stigmatized and sidelined. This is unfair, and punishes researchers for being methodical, so these attitudes are under a period of re-evaluation.

Discouraging researchers from reporting methodological errors, as found in pilot studies, means that later researchers may make the same mistakes.

The other major problem is deciding whether the results from the pilot study can be included in the final results and analysis, a procedure that varies wildly between disciplines.

Pilots are rapidly becoming an essential precursor to many research projects, especially when universities are constantly striving to reduce costs. Whilst there are weaknesses, they are extremely useful for driving procedures in an age increasingly dominated by technology, much of it untested under field conditions.

Research Methods of Statistics

To understand the use of statistics, one needs to know a little bit about experimental design or how a researcher conducts investigations. A little knowledge about methodology will provide us with a place to hang our statistics. In other words, statistics are not numbers that just appear out of nowhere. Rather, the numbers (data) are generated out of research. Statistics are merely a tool to help us answer research questions. As such, an understanding of methodology will facilitate our understanding of basic statistics.

KEY CONCEPT OF VALIDITY

A key concept relevant to a discussion of research methodology is that of validity. When an individual asks, “Is this study valid?”, they are questioning the validity of at least one aspect of the study. There are four types of validity that can be discussed in relation to research and statistics. Thus, when discussing the validity of a study, one must be specific as to which type of validity is under discussion. Therefore, the answer to the question asked above might be that the study is valid in relation to one type of validity but invalid in relation to another type of validity. Each of the four types of validity will be briefly defined and described below. Be aware that this represents a cursory discussion of the concept of validity. Each type of validity has many threats which can pose a problem in a research study. Examples, but not an exhaustive discussion, of threats to each validity will be provided. For a comprehensive discussion of the four types of validity, the threats associated with each type of validity, and additional validity

issues see Cook and Campbell (1979). **Statistical Conclusion Validity:** Unfortunately, without a background in basic statistics, this type of validity is difficult to understand. According to Cook and Campbell (1979), “statistical conclusion validity refers to inferences about whether it is reasonable to presume covariation given a specified alpha level and the obtained variances.” Essentially, the question that is being asked is- “Are the variables under study related?” or “Is variable A correlated (does it covary) with Variable B?”. If a study has good statistical conclusion validity, we should be relatively certain that the answer to these questions is “yes”. Examples of issues or problems that would threaten statistical conclusion validity would be random heterogeneity of the research subjects (the subjects represent a diverse group- this increases statistical error) and small sample size (more difficult to find meaningful relationships with a small number of subjects).

Internal Validity: Once it has been determined that the two variables (A & B) are related, the next issue to be determined is one of causality. Does A cause B? If a study is lacking internal validity, one can not make cause and effect statements based on the research; the study would be descriptive but not causal. There are many potential threats to internal validity. For example, if a study has a pretest, an experimental treatment, and a follow-up posttest, history is a threat to internal validity. If a difference is found between the pretest and posttest, it might be due to the experimental treatment but it might also be due to any other event that subjects experienced between the two times of testing.

Construct Validity: One is examining the issue of construct validity when one is asking the questions “Am I really measuring the construct that I want to study?” or “Is my study confounded (Am I confusing constructs)?”. For example, if I want to know a particular drug (Variable A) will be effective for treating depression (Variable B), I will need at least one measure of depression. If that measure does not truly reflect depression levels but rather anxiety levels (Confounding Variable X), then my study will be lacking construct validity. Thus, good construct validity means the we will be relatively sure that Construct A is related to Construct B and that this is possibly a causal relationship. Examples of other threats to construct validity include subjects apprehension about being evaluated, hypothesis guessing on the part of subjects, and bias introduced in a study by expectancies on the part of the experimenter.

External Validity: External validity addresses the issue of being able to generalize the results of your study to other times, places, and persons. For example, if you conduct a study looking at heart disease in men, can these results be generalized to women?

Therefore, one needs to ask the following questions to determine if a threat to the external validity exists: “Would I find these same results with a different sample?”, “Would I get these same results if I conducted my study in a different setting?”, and “Would I get these same results if I had conducted this study in the past or if I redo this study in the future?” If I can not answer “yes” to each of these questions, then the external validity of my study is threatened.

DESCRIPTIVE STATISTICS

Descriptive statistics are often used to describe variables. Descriptive statistics are performed by analyzing one variable at a time (univariate analysis). All researchers perform these descriptive statistics before beginning any type of data analysis.

FREQUENCY TABLES

- Frequency tables are a detailed description of the categories/values for one variable. A frequency table most often includes all of the following: Absolute frequency (or just frequency): This tells you how many times a particular category in your variable occurs. This is a tally, count, or frequency of occurrence of each individual category/value.
- *Relative frequency (or per cent)*: This tells you the percentage of each category/value relative to the total number of cases.
- *Cumulative frequency*: This is simply a cumulation of the relative frequency for each category/value.
- *Cross tabulations*: This is also referred to as a grouped frequency table for two variables. A crosstab simply presents the absolute frequency broken down by categories of two or more variables. It is also possible to find percentages in these types of tables. For instance, using the example below, we can find the percentage of young people that listen to music.

MEASURES OF CENTRAL TENDENCIES

Measures of central tendencies provide the most occurring or middle value/category for each variable.

METHODS OF CENTRAL TENDENCY OF AVERAGES

Condensation of data is necessary for a proper statistical analysis. A large number of big numbers are not only confusing to mind but also difficult to analyse. After a thorough scrutiny of collected data, classification which is a process of arranging data into different homogenous classes according to resemblances and similarities is carried out first. Then of course tabulation of data is resorted to.

The classification and tabulation of the collected data besides removing the complexity render condensation and comparison. An average is defined as a value which should represent the whole mass of data. It is a typical or central value summarizing the whole data. It is also called a measure of central tendency for the reason that the individual values in the data show some tendency to centre about this average. It will be located in between the minimum and the maximum of the values in the data.

There are five types of average which are;

1. Arithmetic Mean
2. Median
3. Mode
4. Geometric Mean and
5. Harmonic Mean.

ARITHMETIC MEAN

The Arithmetic mean or simply the mean is the best known easily understood and most frequently used average in any statistical analysis. It is defined as the sum of all the values in the data.

MEDIAN

Median is another widely known and frequently used average. It is defined as the most central or the middle most value of the data given in the form of an array. By an array, we mean an arrangement of the data either in ascending order or descending order of magnitude. In the case of ungrouped data one has to form an array first and then locate the middle most value which is the median.

RELATIVE MERITS AND DEMERITS OF MEAN, MEDIAN AND MODE

Mean: The mean is the most commonly and frequently used average. It is a simple average, understandable even to a layman. It is based on all the values in a given data. It is easy to calculate and is basic to the calculation of further statistical measures of dispersion, correlation *etc.* Of all the averages, it is the most stable one. However it has some demerits.

It gives undue weightages to extreme value. In other words it is greatly influenced by extreme values. Moreover; it cannot be calculated for data with open-ended classes at the extreme. It cannot be fixed graphically unlike the median or the mode. It is the most useful average of analysis when the analysis is made with full reference to the nature of individual values of the data. In spite of a few shortcomings; it is the most satisfactory average.

Median: The median is another well-known and widely used average. It is well-defined formula and is easily understood. It is advantageously used as a representative value of such factors or qualities which cannot be measured. Unlike the mean, median can be located graphically. It is also possible to find the median for data with open ended classes at the extreme. It is amenable for further algebraic processes. However, it is an average, not based on all the values of the given data. It is not as stable as the mean. It has only a limited use in practice.

Mode: It is a useful measure of central tendency, as a representative of the majority of values in the data. It is a practical average, easily understood by even laymen. Its

calculations are not difficult. It can be ascertained even for data with open-ended classes at the extreme. It can be located by graphical means using a frequency curve. The mode is not based on all the values in the data. It becomes less useful when the data distribution is not uni-modal. Of all the averages, it is the most unstable average.

MEASURES OF VARIABILITY

Measures of variability is defined as the dispersion (or deviation) away from the mean for each variable. Measures of variability only exist for interval level variables. There are three measures of variability {range, standard deviation, and variance.

A discussion of each can be found below followed by a summary table:

- **Range:** The range is found by taking the highest value of a variable minus the lowest value of that variable.
- **Standard deviation:** The standard deviation exists for all interval variables. It is the average distance of each value away from the sample mean. The larger the standard deviation, the farther away the values are from the mean; the smaller the standard deviation the closer, the values are to the mean. Suppose you passed out a questionnaire asking randomly selected individuals to rate President Obama's job performance on a scale from 1 to 10. You find that on average these individuals give the President a rating of 5.8, and this variable has a standard deviation of 1.2. This means that on average, each rating of the President is approximately 1.2 points away from 5.8 (the sample mean).
- **Variance:** The variance is always going to be the standard deviation squared. The variance cannot be interpreted as meaning anything other than the standard deviation squared.

RESEARCH APPROACH AND METHOD

SOCIAL MAPPING

Social mapping is a research method which involves asking people to plot out where they see the boundaries of their 'space'. This will be used to develop and refine our understanding of community and place. This involves walking with and talking to people as they move through defined spaces, and seeing how their understandings and shaping of community is informed via their interactions and movements. Such social mapping in the first instance will be geared towards the central life-world themes of the project (for example, where people work, meet, eat, and play).

These are then mapped against the social themes of the project (for example, what a given life-world theme such as work has meant in the past and now), and then interpreted in terms of a series of layers of social analysis that form the theoretical level of our methodology (for example, broader implications about social formations at work in

how people define a given thematic). In summary, the initial stage is to build up profiles of the different communities and places that will be involved in the project.

This will be done by drawing on a range of sources through a variety of strategies. This is then mapped against increasingly abstract modes of analysis. Our intention is to move from the empirical to the abstract and back again in a constant journey of return, testing each level against the others.

GATHERING THE EVIDENCE

Because of the diversity of the communities that will take part in the project, a flexible toolbox of methods is required for gathering research material. Our toolbox ranges across the following techniques:

- Writing background social histories for each of the research locales
- Writing thematic essays that provide general background to local communities or background to pressing social issues
- Conducting interviews and strategic conversations
- Collecting stories and life-histories
- Eliciting photographic narratives or artistic representations of the locale
- Conducting a survey questionnaire
- Gathering quantitative data from official and unofficial sources
- Collecting policy documents relevant and contextualizing official discourses.

WRITING SOCIAL HISTORIES

To complement these social maps, social histories will be written to develop a sense of the larger composition of the community. This includes finding out when and how the community came to be, as well as understanding the impact of different formative events on the community. One point of reference here will be, for example, conflict and violence, including the way in which both frontier wars and/or involvement in overseas conflicts have affected and shaped the community.

WRITING THEMATIC ESSAYS

Thematic essays involve the writer exploring some focussed aspect of social history or contemporary social life. It is a way of providing context for understanding contemporary social issues. For instance, in a city such as Sarajevo a thematic essay could take into account all kinds of celebrations and rituals from medieval times to the present associated with the social theme of ‘identity-difference’.

In relation to a town such as Broadmeadows a thematic essay might take contemporary questions of ‘sport and leisure’ as its focus. In short, in focusing upon a locale (or locales) a thematic essay should directly address at least one of our social themes or life-world themes listed below. This might range, for example, from the social theme of ‘belonging-mobility’—perhaps discussed in relation to pressing social issue of

refugees—to the life-world theme of ‘arts and symbolism’, perhaps discussed in relation to births and marriages. A thematic essay might stretch beyond any one locale to explore issues related to a life-world theme such as ‘place and environment’ or open up a discussion into dialectical issues contained in the social theme of ‘authority-participation’. Thematic essays can present the outcomes of thematic research and/or they can include elements of creative or lyrical writing on a theme.

CONDUCTING INTERVIEWS AND STRATEGIC CONVERSATIONS

We will use two particular kinds of semi-structured interviews to explore specific topics and themes with relevant community members; sometimes to capture deeper and more nuanced information about topics that are included in our survey questionnaires and sometimes to get a deep understanding of particular developments or projects within the community concerned.

In the first kind of interview—strategic conversations—considerable thought needs to go into the choice of people to be interviewed in relation to the nature of the topic. The taped interviews thus need to be preceded by background research and preliminary discussions with the interviewees. In such cases the interviewer plays a proactive and strategic role in the discussion of the researched topic. The term ‘strategic conversation’ indicates that an active dialogue has taken place in which the interviewer and interviewee have pushed each other, based on some prior understanding of each other’s views on the subject. A strategic conversation in this sense goes beyond the usual research interview where an interviewer faces an unknown respondent and asks them to answer a series of set questions on the designated topic.

Strategic conversations are supplemented by a second kind interview— response interviews. These are shorter discussions conducted with other people involved in the projects or developments that are relevant to the strategic interviews. These response interviews allow us to cross-reference the experiences of a range of people beyond those involved in the key strategic conversations.

COLLECTING LIFE-HISTORIES AND STORIES

Collecting Stories and Life Histories All kinds of stories already circulate in local communities and some untold stories deserve to go into broader public circulation. These can range from local histories and myths to oral histories to recent experiences and events. We are interested in eliciting local stories that are well-crafted and communicated as concisely as possible without losing their narrative richness. Such stories can be collected by community members, by ‘outside’ researchers, or by a combination of both. They can be collected in the form of written accounts or as ‘digital stories’ that combine images and audio. In many cases they will touch on more than one of the life-world or social themes listed below. Life histories are organized around individual life narratives that provide background and context to contemporary

community life. They involve background research, lengthy interviews and collaboration with the subject to ensure that the story is told accurately and with a degree of depth and reflection relating to the life-world or social themes. They provide an opportunity to explore the ‘lived experience’ of changes over time and to capture dynamically-contextualized stories of local community life.

ELICITING PHOTOGRAPHIC NARRATIVES

Another way of drawing community participants further into the project is to use photography as tool. The approach that will be used is known as reflexive photography. In this approach, community participant observers are given a disposable camera and invited to take photographs of people, places and things in their communities around the life-world themes outlined below. Reflexive photography assumes that community members possess a great deal of ‘inside knowledge’ about the communities to which they belong.

Community participant observers will also be invited to supplement their photos with meaningful photographs from their own collections as well as other personal artefacts that they believe expresses something about their community. They will also be given a mini-photo album and will be asked to arrange their photos and to think about the connections between them. The purpose of this is to encourage the community researchers to begin to construct reflexively meaningful narratives about the places and events depicted in the photos. Reflexive photography supplements one of the other research tools—interviews.

COLLECTING COMMUNITY LIFE-PROFILES

Community life-profiles are developed as a very concise version of a life history and/or a snapshot of local community life as it is experienced by individuals. They begin as interviews of ten to fifteen minutes that follow a schedule of questions relating to the subject’s direct experiences of the complexities and dynamics of local community life (relevant to the life-world or social themes listed below). The interviewer turns this into a concise narrative that is returned to the subject for amendment and approval. Because this process is not very time-consuming it is possible to collect a large number of such community life-profiles over time and they can be used as background data for a wide range of research interests.

CONDUCTING A SURVEY QUESTIONNAIRE

We have developed a questionnaire that is used as a quantitative indicator drawing upon some of the life-world themes and social themes of the project. The questionnaire allows for comparative analysis across the different places of research. Here we provide a printable copy of the questionnaire that you can either open or download in English, Spanish (Español) and Bosnian.

The questionnaire uses a combination of 10-point Likert scales, multiple-choice questions and open-ended questions, and is designed to gather data from a large number of respondents. It can be used both as a mail-out self-administered questionnaire and as a structured interview schedule. The questionnaire will be analysed by applying a commonly-used statistical analysis programme.

GATHERING QUANTITATIVE DATA

To provide context for other methods of data collection, a number of quantitative strategies are being used. We are developing systematic social profiles of each of the communities across the globe using community statistics or small-area data. These include demographic data at local government level, including population, age, culture, education, employment and income. This will be supplemented with data relating to mortality, morbidity rates and population health.

COLLECTING RELEVANT POLICY DOCUMENTS AND CONTEXTUALIZING OFFICIAL DISCOURSES

This refers to the ways in which communities are constituted via official documents and reports, including those put out by civic and professional organizations and representative bodies. These might include tourist brochures and pamphlets, information regarding cultural activities and events in the communities, business planning documents, health reports and information and the like. Official discourse might also include official mappings of community against which our social mappings can be compared.

LIFE-WORLD THEMES

We have chosen a range of ‘life-world’ themes, social facts and activities in everyday life. They are treated as broad categories around which to base all our qualitative and quantitative research. These are the primary categories that will be used directly in working with community participants. They are as follows:

- Arts and symbolism
- Celebration and ritual
- Food and drink
- Sport and leisure
- Learning and education
- Health-care and medicine
- Work and money
- Governance and law
- Place and environment
- Technology and resources
- Conflict and violence.

SOCIAL THEMES

We have, in turn, chosen a range of ‘social’ themes, over-arching issues that inform daily life over time and space. These will also inform the direct work with community participants, but at a secondary level of reflection on the research outcomes based on the life-world themes. They are as follows:

- Identity-difference
- Past-present
- Wellbeing-wretchedness
- Local knowledges-expert systems
- Authority-participation
- Belonging-mobility
- Mediation-disconnectedness
- Freedom-obligation
- Equality-wealth distribution
- Security-risk.

SOCIAL ANALYSIS

At the tertiary level, we move to a much more theoretical level of academic reflection, based on a particular theorization of ‘modes of practice’ (James 1996)—that is established sociological, anthropological and political categories of analysis or ways of framing the ‘things in the world’.

At the level of conjunctural analysis, categories of production, exchange, organization, communication and inquiry will be used in order to organize and give shape to the information developed in the community profiles. We are interested here in finding out how communities are structured and constituted through these different ‘modes of practice’. By such an exploration this project will examine how communities embedded within specific locales are linked into large flows of people, resources and information that are prominent characteristics of the contemporary globalization.

Further, we will determine how the sustainability of such communities is affected—either adversely or positively—by such flows. The analysis will be drawn into forms of social integration that are at work in the community. The notion of integration here refers to the different ways in which individuals are bound into community, ranging from the most concrete, face-to-face forms of social integration to more highly abstracted modes, via information technologies, for example.

While different modes of integration may co-exist with one another—for example, the face-to-face co-exists with e-mail communication—they do not necessarily do so in equal measure. One mode of integration can tend to dominate other forms of integration, framing those other modes in the process. Our hypothesis is that different forms of integration affect the sustainability of communities through time, with communities that are structured primarily through face-to-face social integration facing particular challenges to sustainability in the contemporary conditions of globalization.

METHODOLOGY OF THE RESEARCH PROJECT

In 1998, our research team undertook a qualitative study on 24 pedophiles in treatment at a specialized clinic for sex offenders. This study was to be my doctoral thesis. The initial goals of this study were to understand (1) What motivates pedophile sex offenders to enter and remain in a specialized treatment programme, and (2) How they experience the treatment on a daily basis. Non-directive semi-structured interviews (Kandel, 1972; Legras, 1971) were used to gather the subjects' thoughts about their experience of the treatment as well as what they considered to have motivated them to enter and remain in the treatment programme. The interviews were analysed using diverse qualitative methods, including text-based comparative analysis (Maykut and Morehouse, 1994), Multiple-case depth analysis (Schneider, 1999), and psychoanalytically-based qualitative analysis.

In order to examine specific questions of interest, Plan Analysis (Caspar, 1995, 1997) was also used on a selected number of subjects. Plan Analysis was done by independent judges based on qualitative theme analysis. It involves gathering clinically relevant information about an individual's behaviour and experience through careful observation and synthesizing it into a meaningful whole. As Caspar writes (1997), "the fundamental question that guides Plan Analysis is as follows: which purpose, conscious or unconscious, underlies an individual's behaviors and experiences?" The different components of a Plan Analysis are then interrelated and hierarchically organized. They include a patient's goals or desires but also the means by which these goals are attained. Some obvious advantages of Plan Analysis are that it takes into consideration many different theories, it is idiosyncratic and robust and it is pragmatic and relies on concrete methodology.

Finally, after using the qualitative methods mentioned above, some specific pre-established hypotheses derived from the literature on sex abusers were tested using quantitative methods on selected cases. The first method used was Perry's Wish and Fear List (1996). The Wish and Fear List is based on the notion that behaviour is motivated. These motives can be both biologically and psychologically determined in varying degrees and consist, in their simplest form, of wishes and fears. According to Perry (1996), "a dynamic *Wish* differs from a simple desire by playing a causal role in a variety of behaviors, fantasies and experiences". A *Fear* can be defined as "a negative belief, expectation, or aversive experience which the subject wishes to avoid". The method developed by Perry encompasses notions of Erikson's developmental stages and of self-psychology and reintroduces an object-relations perspective via an interest for motives. Such a perspective, although avowedly psychodynamic, has the advantage of being free of strict psychoanalytic jargon and of not violating any basic propositions from learning theory.

Finally, the last instrument used was the Core Conflictual Relationship Theme (CCRT) (Luborsky and Crits-Christoph, 1990). Amongst others things, the CCRT identifies three

components: a subject's wishes (W), the responses from others (RO), and the responses of self (RS). There is much evidence that these components describe core repetitive relationship patterns and share many characteristics with Freud's transference theory. The CCRT was used by one independent judge in order to examine our subjects' relationship patterns.

A research topic like sex abuse—even more so when it concerns children and pedophilic behaviour—presents many major difficulties. The first one a researcher is confronted to is the question of definition of sex abuse and, more specifically, of sexual abuse of children. Despite the spectacular increase of research in this field, no consensus has yet been reached. Between the writings of Krafft-Ebing (1912) and the more recent definition of pedophilia in the DSM-IV (APA, 1994) can be found a multitude of definitions and concepts often revealing strong legal and moral implications. Some of these definitions, still in use today, include an individual masturbating while thinking or fantasizing about children, a child being victim of an exhibitionist or a more clearly defined sexual contact between a child and an adult. In that sense, some definitions are still the result of strong prejudice or personal conflicts. It is then not surprising to see that sexology has often been infiltrated by demonology and that many examples mentioned in the literature clearly are limited to the most provocative cases.

This has often led to certain exaggerations such as considering all sexual activity of or between children as abusive (Finkelhor, 1979). The problem of studying sexual abuse of children and pedophilic behaviour is even greater when one considers the reactions a research project may bring up. Since many studies are often used in court in order to obtain either a guilty or a non-guilty verdict, some researchers have been thought of in the public opinion as being pro-pedophile. The least that can be said is that the study of sexual abuse is an emotionally charged one. As my study was on the motivation for treatment of pedophilic sexual abusers, I could not help but confront these questions as well as what brought me to give attention to this topic. But my main concern (and what I wish to emphasize here) was what could be understood from interviewing pedophilic subjects. With all of this in mind and as I was in a psychoanalytic cure at the time, I proceeded to bring up these questions with my analyst, who later on acted as "third party".

SUBJECTIVITY IN UNDERSTANDING THE SUBJECTS

My first subject was a man in his sixties. Prior his arrest, he had held an important and very well considered professional position. He was serving a long-term federal sentence after being condemned for sexually abusing young boys over a period of many years. He showed up at the interview very well dressed and looking confident. He answered all questions without hesitation, giving much detail in a way a professor would lecture his students. After 70 minutes of interviewing, the meeting was interrupted by a guard asking that the subject return in his cell for inspection. I then proceeded to note down my subjective impressions in a notebook I reserved for this use. These close to two pages of notes included the following:

“(Subject) seems very educated and definitely very articulate. His strong accent gives him a sort of aristocratic twist. It felt as if he was actually lecturing me or teaching me something that I should know. (I) felt like a kid who’s begging for information and desperately waiting for his next word. Seemed like I was learning something !!! (...) The contact was good, maybe even a little too good? Something fascinating about him?, almost father or grandfather like (...) What is it? (...) At the same time, it felt intrusive, as if I should better watch out and not get screwed. Don’t know what to make of it (...) Very confusing, ambivalent.”

The complete notes were later on brought in an analytic session and a total of close to four sessions were actually used to try to understand what felt during this interview. The most striking aspect was felt like twenty or twenty-five years younger, almost like a little boy. It was felt charmed and fascinated by this person as if could learn much from him. Through analysis, anyhow managed to establish clear relations between this and my own history, more precisely with my father. In that sense, it seemed as if was longing for something that was missing and that this subject could actually give me, that is: a feeling of being an “adult, mature man”.

As such, during the interview, there were *fantasies* of establishing contact with him once his imprisonment would be over, hoping that what was missed during the interview could be obtained later through some kind of contact. While having these feelings, it was also felt as if it should better make sure one don’t get “screwed”. This last term brought up much anxiety, as it could easily be related to sexual abuse. And somehow, it did feel like rape. It then became clear that it was felt ambivalent towards the subject, both fearing and wanting him to give me the feeling of being an “adult man”. Something the subject lived somehow resonated in me, while at the same time not being completely mine. This led to the idea that, although the subject and was shared certain common fantasies, differences could be found. It so appeared that through different processes, namely projective identification and identification to the aggressor, the subject had led me on a *playground* where his rules prevailed and where he could actually bring me to feel like he might have felt while maintaining himself in a superior-fatherly- position. Spontaneously, this leads to many hypotheses regarding the subject’s functioning:

1. He might have missed having a strong-if any- father figure capable of leading him to adult life;
2. He may have struggled with an authoritarian maternal figure; and
3. He may feel like a young boy and not entirely like an adult man and these feelings are intolerable for him.

It was later realized, many psychoanalytic theories have been developed around these ideas and identification to the aggressor is often a core component of them. According to Laplanche and Pontalis (1967), identification to the aggressor involves the subject identifying to his aggressor by making himself the perpetrator or making his certain characteristics of the aggressor. As such, the pedophilic act involves mastery by leaving the victim position to become the aggressor. More recent theories (Balier,

1993, 1996) also suggest that the pedophile is in many ways “his mother’s narcissistic doll”. Indirectly, this suggestion raises the question as to what place-if any- the father had in the child’s life. It also suggests that the pedophile struggles with symbiotic issues with the mother and lacks positive identification with the father. Through analysis with a “third party”, these hypotheses seemed applicable. But the question still remained as to how true or significant they were.

Verifying these hypotheses As quantitative measures had been used in order to examine other aspects of the research question, quantitative results were readily available for the subject presented above. Hence, it was interesting to triangulate hypotheses resulting from the examination of my subjectivity with other methods. Using Caspar’s Plan Analysis (1997), it appeared that some of the most central themes brought up by this subject were “a need to have a sense of mastery”, “a fear of being dominated”, and “a wish to put the therapists to test by confronting them in order to make sure they are strong, authoritarian and reliable “father” figures”. The Wish and Fear List was used by independent judges and suggested that the subject’s most prevalent motives were: a wish to be autonomous, a wish to control and dominate others, a wish to have and be a role model or a mentor, a fear of being dominated, and a fear of physical harm. Overall, most motives mentioned by the subject were related to what Erikson (1985) called autonomy versus shame and doubt issues. Finally, the CCRT indicated that the subject was most often motivated to have control over others, to be independent, and to be his own person. Others most often reacted to these wishes by being controlling (in the case of women) or being distant- feeling controlled (in the case of men). The subject then was left feeling helpless, incompetent and disappointed.

What is interesting about these methods is that they all seem to underline and empirically support different aspects of what had been *subjectively* felt during the interview. Hence, examining my subjectivity provided-in itself- a good indication of the subject’s functioning. But as it will be described next, examining one’s subjectivity can also be important when the time comes to analyse the data at hand as other factors also have an influence on the research process.

SUBJECTIVITY IN THE RESEARCH MENTOR

As it was intensive training in quantitative research and still worked in that field, chosen to do a Ph.D. where one could learn new and different research methods. This led me to a psychoanalysis professor who had been doing qualitative work for many years. Very early in the research project, a deep abyss could already be seen between the way this professor saw therapeutic work and the way patients were treated at the penitentiary. All pedophilic patients there had the opportunity to start a treatment programme in a specialized clinic associated to the penitentiary. This treatment, as most programmes for sexual abusers, was cognitive-behavioural oriented and included the use of aversive conditioning using ammonia or, occasionally, mild self-inflicted electroshocks. Although such treatments have been proven most effective with sexual

abusers and while psychoanalytically orientated therapies had largely failed in this matter, my professor was somewhat shocked and offended when it was first described the treatment to him.

In the months that followed, it was ended up being a moderator for both parties, that is the cognitive-behavioural and the old fashioned psychoanalytic models. Clearly enough, as my professor was the one who not only rated my work but had the final word on the thesis project, it was made clear efforts when analyzing the data in order to demonstrate his impressions were right: “people just cannot be satisfied with such a rigid treatment that gives them so little room!”. For example, it would be would over emphasize the fact that the subjects had little chance to talk (which is a central aspect of psychoanalytic work) despite their desire-and capacity- to do so. In that sense, when one first analysed the interviews, took special care to show that the subjects had a strong wish to talk, while looking for clues that they were not satisfied with the present treatment. For these reasons, it would be over emphasise a subject’s strengths and capacities in order to demonstrate that he could be amenable to psychoanalytic therapy but was deprived of it.

As work with my research mentor eventually became difficult and even counterproductive, one became more and more frustrated and then switched positions. In analyzing data, it would be given more attention to clues indicating that the subject was *not* amenable to psychoanalytic therapy. At the same time, it would be over emphasized the “cruelty” (as he put it) of the cognitive-behavioural treatment. It became clear that my efforts, although not entirely conscious at the time, aimed at contradicting and offending him.

This confrontation was motivated by technical as well as emotional motives. For instance, although it quickly became clear, through analysis, that it was deliberately confronted with research mentor in order to contradict him as a result of being frustrated for having been placed in a moderator position, there also appeared to be other reasons for doing so.

One of these reasons seems of particular importance and was the result of certain “similarities” between the relation had with my supervisor and the one my first subject had with his father (as seen in theme analysis): neither one would “put out”, that is give what they were supposed to give (*e.g.*, support and input). Because of this, it was identified easily with the subject and used his (most shocking) words to confront my supervisor, as the subject had done himself with his father. The only way to overcome these difficulties was to analyse them with the help of the “third party”.

The question then is for the researcher to determine just *how much-if at all*- he or she wishes to validate his findings with other methods. Although a study relying entirely and exclusively on one’s subjectivity *may* not respect the criteria established by science, it nonetheless can be a central part of the research project. Its advantage is clearly that, unlike in quantitative methods, it can *at least* make us cautious when analysing data. For example, through the use of a “third party”, it quickly became clear that the subject

interviewed had a major impact on me. Because of this, it later was clear that one had a tendency, when analysing the data, to over emphasize positive aspects of the subject's personality or to search for possible explanations for his behaviours (excuses) while overlooking comments which may have negatively influenced how one saw him. This general tendency could be explained by two possible reasons. The first one was motivation or desire to preserve the image or representation had of this subject. The second one was the result of the relationship that was established with my thesis supervisor and the (temporary) conflicts in which it resulted. As looking back at that time, I can only be convinced that without making use of a "third party", many of the conclusions which were suggested would have been clearly biased and not representative of the subjects' dynamic functioning. In other words, it was a belief that the use of a "third party" helped in being more objective about my own subjectivity and the impact it had on the data analysis.

Once again, such a stand reveals an effort to reconcile modern and postmodern epistemology. By seeking to explore my subjectivity and to accept that it may be richly informative about the subject's functioning, it suggests that the essence of meaningful understanding is in a subjective representation of the subject or, as psychoanalysts would put it, in a shared and approximate metaphor of his functioning.

At the same time, the use of a "third party" in order to better distinguish subjectivity from delusion implies that subjectivity can stand in the way of a better understanding of the subject.

Hence, it should then be penetrated as much as possible in order to better differentiate what belongs to the researcher from what belongs to the subject. As such, it reveals a definite postmodern epistemology where demonstration, as can also be seen in any effort to triangulate findings, is necessary. As my efforts were aimed at making use of both epistemological stand points, they can certainly be seen as an attempt, although certainly incomplete, to reconcile the best of both perspectives.

MEASURES OF DISPERSION

When a mass of quantitative data is collected for a statistical purpose, it is tabulated to a form with in view that its characteristics as a whole may be readily determined. A single significant and representative expression or a measure called average is then computed to summarize or explain as a whole the entire data.

This form of a comparing two or more groups or series but an average along is not a satisfactory criterion for such a purpose. Dispersion means the extent of values around some average. There are four measures of dispersion, namely

- Range
- Quartile Deviation or semi interquartile range
- Mean Deviation
- Standard Deviation.

RANGE

Range is the simplest measure of dispersion. It is defined as the positive difference between the largest and the smallest values in the given data. It is easily understood and computed but depends exclusively on the two extreme values while it is desirable to have a measure dependent on all the values. The range is a very useful measure in statistical quality control of products in industries wherein the interest lies in getting a quick rather than an accurate measure of variability.

QUARTILE DEVIATION OR SEMI-INTERQUARTILE RANGE

This measure is based on two measures called the lower or first quartile and the upper or third quartile. The lower quartile denoted as Q_1 is defined as the value which leaves $\frac{1}{4}$ of the values below it when the data forms an array and the third quartile denoted as Q_3 is the value which leaves $\frac{3}{4}$ s of values below it when the data forms an array. Once Q_1 and Q_3 are known, the quartile deviation Q-D is given by; $Q-D = \frac{Q_3 - Q_1}{2}$ and $Q_3 - Q_1$ is called the interquartile range

MEAN DEVIATION

The mean deviation of a data is defined as the means of the absolute deviation of values from some average especially the arithmetic mean or median. It is a better measure of dispersion than range and Q D as it takes into account, all the values in the given data for ungrouped data.

M. D about median M can also be defined by considering median instead of mean.

STANDARD DEVIATION

The standard deviation abbreviated as S. D and symbolically represents as sigma is the most important and wide used measure of dispersion. It is defined as the root mean square deviation of values from their mean *i.e.* it is the square root of the means of square deviation of values from their mean. The square of the s. d. is called the variance.

COEFFICIENT OF DISPERSION

For any data, it is always desirable that the measure of dispersion is less. A small value for the measure means that the values in the data are more or less consistent, centering on their average. Suppose it is required to compare the dispersion between two or more data.

For this purpose any measure of dispersion cannot be used as such for two reasons. The two data under study may pertain to different variables such as heights and weights of the individual. Number of marriages and ages of persons who have committed suicides etc and even if the variables in the two data are the same, their averages may be different.

Hence one is in need of a relative measure of dispersion which will be free from the two differences cited above. Such a measure is the coefficient of dispersion which is defined as the ratio between measure dispersion and an average. This coefficient is a constant, rendering ready comparison of dispersion between data.

There are four coefficients which are commonly referred, namely:

1. Quartile coefficient of dispersion = Quartile Deviation/Mean
2. Coefficient of variation (C. V) = Standard Deviation/Mean x 100
3. Mean Deviation/Mean
4. Mean Deviation/Median of which the first two are the most frequently used coefficients.

ASSOCIATION OF ATTRIBUTES

In social sciences, we come across certain phenomena which are incapable of quantitative measurement. Blindness, deafness, religion; juvenile delinquency, marital status etc are some phenomena which are not measurable. Such characteristics are called attributes. In these cases, one can make only counting of individuals who possess or do not possess these attributes. In other words what can do is to state so many individuals are blind or so many non-blind. While dealing with one attribute the classification of data is done on the basis of presence or absence of the attribute. It is also absolutely essential that a clear-cut definition of the attribute under study is made because only such a definition paves way for the counting of the individuals possessing or not possessing the attribute. Two attributes are said to be associated only if they appear together in a great number of cases than is to be expected if they are independent. On the other hand, if the number of observed cases is less than the expected, under assumption of independence, attributes are associated. In order to ascertain whether the attributes are associated or not the following methods can be used.

1. Comparison of observed and expected frequencies.
2. Proportion method
3. Yule's coefficient of Association
4. Coefficient of colligation
5. Coefficient of contingency.

REGRESSION

The dictionary meaning of the term 'regression' is the act of returning or going back. The term was first used by Sir Francis Galton in 1877 when he analysed the relationship between the height of fathers and sons. His study of the heights of fathers and sons revealed a very interesting relationship-tall fathers tend to have tall sons and short fathers, short sons, but the average height of the son of a group of tall fathers is less than that of the fathers and the average height of the sons of short fathers is greater than that of the fathers. The line describing this tendency to regress or step back was called by

Galton a Regression Line. The term continues to be in use to represent the trend present but no longer has it necessarily carried the original implication of stepping back that Galton intended.

The statistical device that enables the estimation or prediction of the unknown values of one variable based on the known values of another variable is termed regression. Regression is the measure of average relationship between variables.

LINEAR REGRESSION

In case there exists relationship between two variables x and y the plotted points in the scatter diagram lie concentrated around a curve and the relationship is said to be expressed by means of curvilinear regression. In the particular case, when the curve is a straight line, is called the line of regression and the regression is said to be linear. Once it is known that the relationship is expressible by means of straight line, a technique known as the Principles of Least Squares is adopted to fit the line on the basis of the given bivariate data.

A line of regression gives the most probable values of one variable for the given value of the other variable. For two series x and y there are two regression lines, one on assuming x as dependent and y dependent variable and the two on assuming x as dependent and independent variable.

INFERENCEAL STATISTICS

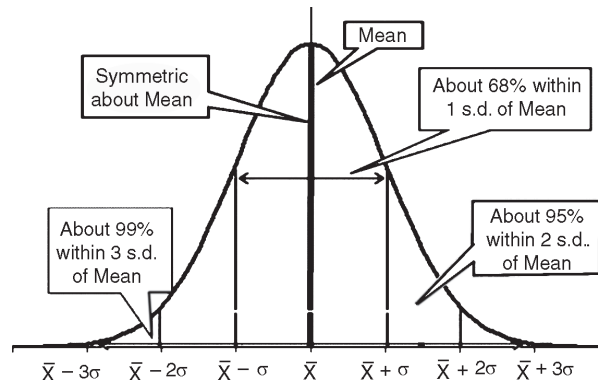
MORE DEFINITIONS AND TERMS

An interval variable is said to be normally distributed if it has all of the following characteristics:

- A bell shape curve.
- It is perfectly symmetrical.
- All measures of central tendencies (mode, median, and mean) lie in the middle of the curve. These measures of central tendencies divide the curve in half (where 50% of the values lie to the left of the mean, and 50% lie to the right).
- Approximately 95% of the values are found two standard deviations away from the mean (in both directions).

Variables that are determined by nature are normally distributed (graphically they have a normal curve) such as age, weight, height, *etc.* It is important to understand what a normal curve looks like and its characteristics because almost all methods described below assume normality. If this assumption is violated (*i.e.* a variable is not normally distributed) it can have an effect on the statistical results (resulting in significance when in reality it is not significant, or not resulting in statistical significance when it is

significant). If variables are not normally distributed, it is easy to make transformations, such as logging or taking the square root, in order to achieve normality.



CONFIDENCE INTERVALS

Confidence Intervals are used to estimate a range of the population based on some sample of any interval level variable. Confidence intervals are two numbers which represent the higher and lower limits of a statistic, coefficient, or parameter. Confidence intervals assume the interval level variable has a normal distribution, and uses the sample in order to find a range of the entire population. When dealing with confidence intervals, a confidence or an α level must be specified.

STATISTICAL SIGNIFICANCE

Statistical significance represents the results of some statistical test that is being performed. The statistical test varies depending on the levels of measurement of the variables, and the objective of the research or hypothesis. There are numerous different tests but they all have some similarities and include all of the following:

- *One Null Hypothesis:* The null hypothesis usually states there is no relationship between the variables being tested. The null hypothesis is already determined and based on the method being used. Most null hypotheses state that one statistic or number is equal to another statistic or number. This is usually displayed as:

$$H_0: a = b$$

- *One Alternative Hypothesis:* The alternative hypothesis usually states that the two or more variables are somehow related. Like the null hypothesis, the alternative hypothesis is also already determined based on the method being used. The alternative hypothesis is the opposite of the null hypothesis and usually states that one statistic or number is not equal to another statistic or number. Alternative hypotheses can be displayed using one of the following forms:

- i. $H_a: a \neq b$
- ii. $H_a: a > b$
- iii. $H_a: a < b$
- *Outcome of Statistical Test:* All tests will always have one of two outcomes:
 - Reject H_0 and Accept H_a , which means that the null is wrong and the alternative hypothesis is right. When this occurs, there is statistical significance, and a confidence or an α level is always specified.
 - Accept H_0 and Reject H_a , which means that the null is right and the alternative hypothesis is wrong. When this occurs, there is no statistical significance, and therefore no need to specify a confidence or an α level.
- *Confidence Level:* With every test that is being performed, the researcher must specify a confidence level if and only if the H_0 is rejected and the H_a is accepted. The confidence level literally translates into the percentage or probability of correctly rejecting the null hypothesis (or the level in which the outcome is correct). In social science and statistics, the most common confidence levels are AT LEAST 90%. This means that with each test, researchers want to be right at least 90% of the time. Usually researchers report confidence levels at 90%, 95%, 99%, or 99.9%, the higher the confidence level the better the results. The confidence level can also be displayed as a probability, where 90% confidence level translates into being right with a probability of 0.90.
- a level or *p—value*: In addition to specifying a confidence level when rejecting the H_0 , an α level must also be specified. Similar to the confidence level, the α level can be displayed as a percentage or a probability. The α level represents the probability or percentage of rejecting the null when it should not have been rejected. In other words, the α level is the probability or percentage of making a mistake, and the lower the α level the better the results. A statistical significance (or α level) of 1% is better than a statistical significance of 5%.
 - α (probability) = 1 – probability confidence level
 - α (percentage) = 100 – percentage confidence level
- What does a statistical test do: When performing any type of inferential statistics and any type of statistical testing, a value is generated based on the data, and this value is being compared to some corresponding critical value in order to determine statistical significance.

BASIC STATISTICS AND DESCRIPTIVE STATISTICS

“True” Mean and Confidence Interval. Probably the most often used descriptive statistic is the mean. The mean is a particularly informative measure of the “central tendency” of the variable if it is reported along with its confidence intervals. As mentioned earlier, usually we are interested in statistics (such as the mean) from our sample only

to the extent to which they can infer information about the population. The *confidence intervals* for the mean give us a range of values around the mean where we expect the “true” (population) mean is located. For example, if the mean in your sample is 23, and the lower and upper limits of the $p=.05$ confidence interval are 19 and 27 respectively, then you can conclude that there is a 95% probability that the population mean is greater than 19 and lower than 27. If you set the p -level to a smaller value, then the interval would become wider thereby increasing the “certainty” of the estimate, and vice versa; as we all know from the weather forecast, the more “vague” the prediction (*i.e.*, wider the confidence interval), the more likely it will materialize. Note that the width of the confidence interval depends on the sample size and on the variation of data values. The larger the sample size, the more reliable its mean. The larger the variation, the less reliable the mean. The calculation of confidence intervals is based on the assumption that the variable is normally distributed in the population. The estimate may not be valid if this assumption is not met, unless the sample size is large, say $n=100$ or more.

USERS OF STATISTICS

Official statistics are intended for a wide range of users including governments (central and local), research institutions, professional statisticians, journalists and the media, businesses, educational institutions and the general public. There are three types of users: those with a general interest, business interest or research interest. Each of these user groups has different needs for statistical information.

USERS WITH A GENERAL INTEREST

Users with a general interest include the media, schools and the general public. They use official statistics in order to be informed on a particular topic, to observe trends within the society of a local area, country, region of the world.

USERS WITH A BUSINESS INTEREST

Users with a business interest include decision makers and users with a particular interest. For them, official statistics are an important reference, providing information on the phenomena or circumstances their own work is focusing on. For instance, those users will take some official statistics into consideration before launching a product, or deciding on a specific policy or on a marketing strategy. As with the general interest users, this group does not usually have a good understanding of statistical methodologies.

USERS WITH A RESEARCH INTEREST

Users with a research interest are universities, consultants and government agencies. They generally understand something about statistical methodology and want to dig deeper into the facts and the statistical observations; they have an analytical purpose in

inventing or explaining interrelations of causes and effects of different phenomena. In this field, official statistics are also used to assess a government's policies.

One common point for all these users is their need to be able to trust the official information. They need to be confident that the results published are authoritative and unbiased. Producers of official statistics must maintain a reputation of professionalism and independence.

The statistical system must be free from interference that could influence decisions on the choice of sources, methods used for data collection, the selection of results to be released as official, and the timing and form of dissemination. Statistical business processes should be transparent and follow international standards of good practice. Statistical programmes are decided on an annual or multi-annual basis by governments in many countries. They also provide a way to judge the performance of the statistical system.

PRODUCERS AT THE NATIONAL LEVEL

Official statistics are collected and produced by national statistical institutes (NSIs) or national statistical offices (NSOs), depending on national administrations. They are responsible for producing and disseminating official statistical information, providing the highest quality data. The criteria for quality are: relevance and completeness, timeliness, accuracy, accessibility and clarity, cost efficiency, transparency, comparability and coherence, which are called quality principles. Their role is also to foster statistical literacy among important user groups and the general public.

The core tasks of NSOs, for both centralized and decentralized systems, are investing user needs and filtering these for relevance. Then they transform the relevant user needs into measurable concepts to facilitate data collection and dissemination. This is part of the production process.

Statistical agencies also determine the resources necessary for the various activities and ensure relevance by keeping permanent networks with various representatives of different types of users. Moreover statistical producers have to anticipate user needs when producing new types of statistics in order to speed up the process. It can be called the "antenna function". Another way to assess the quality of their services is to undertake periodically user satisfaction surveys.

Nevertheless, NSOs are usually not the only official and national institutes to produce official statistics. Central banks and some ministries or other central authorities may have statistical functions as well. Together, all producers of official statistics form the statistical system of a country. In some countries, especially those that have a federal structure, producers of official statistics exist also at regional or even municipal levels.

The NSO is in charge of the coordination between statistical producers and of ensuring the coherence and compliance of the statistical system to the principles. In countries with less than 25 million inhabitants, the NSO should also have the exclusive responsibility for all household surveys and all business surveys for official statistics. This should improve efficiency and ensure confidentiality.

The NSO has to pay particular attention to ensuring that the information materials, the terminology and the metadata of the statistical results disseminated are coherent and understandable for non-users especially in the case of diverging results compiled from different sources. The NSO also has to decide which ones are the official statistics. Moreover, it ensures the dissemination platforms for all official statistical registers from which contact information for sample surveys is extracted. Finally, the NSO should support and advise other producers of official statistics and organize meetings with all of them.

The NSO has a major responsibility as its President/Director General represents the entire system of official statistics, both at the national and at international levels. Statistical producers perform advocacy work on official statistics and can provide advice and services, such as training activities in order to broaden the know-how accumulated in the agency.

PRODUCTION PROCESS

The usual production process of official statistics includes 6 steps:

1. The *programming phase* starts the process with investigations into the information needs of users (topic of the information needed, its period, accuracy and timeliness). A lot of information is gathered, and then filtered in order to focus on one specific activity. Official statistics can generate results that fulfil a great number of user needs and not only target a unique user group. Filtering allows selecting relevant information. These needs are translated into the best way of collecting data from respondents.
2. The *design phase* is when tests and statistical surveys are designed or redesigned and tested. The first surveys are called *pilots*. Tools and resources are also prepared to conduct the surveys and to implement them fully. This phase includes the definition of the results to be published as official.
3. *Data collection* through statistical surveys can be done through different processes: by mail, face-to-face interviews, telephone interviews, internet, sample survey, sampling frame.
4. The *processing phase* includes data entry, control, coding and editing. This phase is highly IT-dependent; computer-assisted telephone interviewing (CATI) and computer-assisted personal interviewing (CAPI) techniques are really useful in terms of speed. The results and the quality parameters have to be analysed carefully before passing to the next phase. This monitoring can be systematically done by tallying the same phenomenon with other sources at the aggregate level.
5. *Dissemination* is more than the mere release of the results and statistical products such as press releases, electronic dissemination on the Internet or hard-copy publication, to customers. It may include publications related to the topic with more details or analytical content, or may target specific user

groups. It includes the generation of additional results for specific user requests such as statistical services. This means that consequently, microdata or the final sets of data have to be stored and well documented for a considerable period.

6. An *evaluation* of the whole process is necessary to identify and make improvements in efficiency and quality of the process launched by the statistical producer; this helps for the next programming exercise.

Sometimes statistical producers can make ad hoc surveys but this is not recommended in the case of official statistics. The methodological information should be used in most of the surveys to obtain relevant results. Another data life-cycle object model for statistical information systems was developed by the Statistical Office of the European Communities, Eurostat in 2003. A proposal for a new generic statistical business process model was submitted in 2007 by Statistics New Zealand and the United Nations Economic Commission for Europe Secretariat to a joint UNECE/Eurostat/OECD work session on statistical metadata and is already used by several official statistical agencies even if it is still being developed. The aim of the generic model is to provide a better basis for the production process, to make it more complete and detailed than the previous model.

DATA REVISION

Even after they have been published, official statistics are often revised. Policy-makers may need preliminary statistics quickly for decision-making purposes, but eventually it is important to publish the best available information, so official statistics are often published in several ‘vintages’. The first available version of the data is called real-time data, and subsequent corrected data are called revised data. In order to understand the accuracy of economic data and the possible impact of data errors on macroeconomic decision-making, the Federal Reserve Bank of Philadelphia has published a dataset that records both initial real-time data estimates, and subsequent data revisions, for a large number of macroeconomic series. A similar dataset for Europe has been developed by the Euro-Area Business Cycle Network.

STATISTICAL SIGNIFICANCE AND PRACTICE

In statistics, a result is called statistically significant if it is unlikely to have occurred by chance. The phrase *test of significance* was coined by Ronald Fisher. As used in statistics, *significant* does not mean *important* or *meaningful*, as it does in everyday speech. Research analysts who focus solely on significant results may miss important response patterns which individually may fall under the threshold set for tests of significance. Many researchers urge that tests of significance should always be accompanied by effect-size statistics, which approximate the size and thus the practical importance of the difference. The amount of evidence required to accept that an event

is unlikely to have arisen by chance is known as the significance level or critical p-value: in traditional Fisherian statistical hypothesis testing, the p-value is the probability of observing data at least as extreme as that observed, *given that the null hypothesis is true*. If the obtained p-value is small then it can be said either the null hypothesis is false or an unusual event has occurred. It is worth stressing that p-values do not have any repeat sampling interpretation.

An alternative statistical hypothesis testing framework is the Neyman–Pearson frequentist school which requires both a null and an alternative hypothesis to be defined and investigates the repeat sampling properties of the procedure, *i.e.* the probability that a decision to reject the null hypothesis will be made when it is in fact true and should not have been rejected (this is called a “false positive” or Type I error) and the probability that a decision will be made to accept the null hypothesis when it is in fact false (Type II error).

It is worth stressing that Fisherian p-values are philosophically different from Neyman–Pearson Type I errors. This confusion is unfortunately propagated by many statistics textbooks.

USE IN PRACTICE

The significance level is usually denoted by the Greek symbol α (lowercase alpha). Popular levels of significance are 10% (0.1), 5% (0.05), 1% (0.01), 0.5% (0.005), and 0.1% (0.001). If a *test of significance* gives a p-value lower than the α -level, the null hypothesis is thus rejected. Such results are informally referred to as ‘statistically significant’. For example, if someone argues that “there’s only one chance in a thousand this could have happened by coincidence,” a 0.001 level of statistical significance is being implied. The lower the significance level, the stronger the evidence required. Choosing level of significance is an arbitrary task, but for many applications, a level of 5% is chosen, for no better reason than that it is conventional. In some situations it is convenient to express the statistical significance as $1 - \alpha$. In general, when interpreting a stated significance, one must be careful to note what, precisely, is being tested statistically.

Different α -levels trade off countervailing effects. Smaller levels of α increase confidence in the determination of significance, but run an increased risk of failing to reject a false null hypothesis (a Type II error, or “false negative determination”), and so have less statistical power. The selection of an α -level thus inevitably involves a compromise between significance and power, and consequently between the Type I error and the Type II error. More powerful experiments – usually experiments with more subjects or replications – can obviate this choice to an arbitrary degree.

However, values might more easily be found using tabulated values which are often found in text books: see standard normal table. The use of σ implicitly assumes a normal distribution of measurement values. For example, if a theory predicts a parameter to have a value of, say, 109 ± 3 , and one measures the parameter to be 100, then one

might report the measurement as a “ 3σ deviation” from the theoretical prediction. In terms of α , this statement is equivalent to saying that “assuming the theory is true, the likelihood of obtaining the experimental result by coincidence is 0.27%” (since $1 - \text{erf}(3/\sqrt{2}) = 0.0027$).

Fixed significance levels such as those mentioned above may be regarded as useful in exploratory data analyses. However, modern statistical advice is that, where the outcome of a test is essentially the final outcome of an experiment or other study, the p-value should be quoted explicitly. And, importantly, it should be quoted whether the p-value is judged to be significant. This is to allow maximum information to be transferred from a summary of the study into meta-analyses.

PITFALLS AND CRITICISM

The scientific literature contains extensive discussion of the use of the concept of statistical significance and in particular of its potential misuse and criticism of its use: see potential misuse and criticism for details of these opinions.

SIGNAL-NOISE RATIO CONCEPTUALISATION OF SIGNIFICANCE

Statistical significance can be considered to be the confidence one has in a given result. In a comparison study, it is dependent on the relative difference between the groups compared, the amount of measurement and the noise associated with the measurement. In other words, the confidence one has in a given result being non-random (*i.e.* it is not a consequence of chance) depends on the signal-to-noise ratio (SNR) and the sample size.

MULTIVARIATE STATISTICS

Multivariate statistics is a form of statistics encompassing the simultaneous observation and analysis of more than one statistical variable. The application of multivariate statistics is multivariate analysis. Methods of bivariate statistics, for example simple linear regression and correlation, are special cases of multivariate statistics in which two variables are involved.

Multivariate statistics concerns understanding the different aims and background of each of the different forms of multivariate analysis, and how they relate to each other. The practical implementation of multivariate statistics to a particular problem may involve several types of univariate and multivariate analysis in order to understand the relationships between variables and their relevance to the actual problem being studied.

In addition, multivariate statistics is concerned with multivariate probability distributions, in terms of both:

- How these can be used to represent the distributions of observed data;
- How they can be used as part of statistical inference, particularly where several different quantities are of interest to the same analysis.

TYPES OF ANALYSIS

There are many different models, each with its own type of analysis:

1. Multivariate analysis of variance (MANOVA) extends the analysis of variance to cover cases where there is more than one dependent variable to be analysed simultaneously.
2. Multivariate regression analysis attempts to determine a formula that can describe how elements in a vector of variables respond simultaneously to changes in others. For linear relations, regression analyses here are based on forms of the general linear model.
3. Principal components analysis (PCA) creates a new set of orthogonal variables that contain the same information as the original set. It rotates the axes of variation to give a new set of orthogonal axes, ordered so that they summarize decreasing proportions of the variation.
4. Factor analysis is similar to PCA but allows the user to extract a specified number of synthetic variables, fewer than the original set, leaving the remaining unexplained variation as error. The extracted variables are known as latent variables or factors; each one may be supposed to account for covariation in a group of observed variables.
5. Canonical correlation analysis finds linear relationships among two sets of variables; it is the generalised (*i.e.* canonical) version of bivariate correlation.
6. Redundancy analysis is similar to canonical correlation analysis but allows the user to derive a specified number of synthetic variables from one set of (independent) variables that explain as much variance as possible in another (independent) set. It is a multivariate analogue of regression.
7. Correspondence analysis (CA), or reciprocal averaging, finds (like PCA) a set of synthetic variables that summarise the original set. The underlying model assumes chi-squared dissimilarities among records (cases). There is also canonical (or “constrained”) correspondence analysis (CCA) for summarising the joint variation in two sets of variables (like canonical correlation analysis).
8. Multidimensional scaling comprises various algorithms to determine a set of synthetic variables that best represent the pairwise distances between records. The original method is principal coordinates analysis (based on PCA).
9. Discriminant analysis, or canonical variate analysis, attempts to establish whether a set of variables can be used to distinguish between two or more groups of cases.
10. Linear discriminant analysis (LDA) computes a linear predictor from two sets of normally distributed data to allow for classification of new observations.
11. Clustering systems assign objects into groups (called clusters) so that objects (cases) from the same cluster are more similar to each other than objects from different clusters.

12. Recursive partitioning creates a decision tree that attempts to correctly classify members of the population based on a dichotomous dependent variable.
13. Artificial neural networks extend regression and clustering methods to non-linear multivariate models.

IMPORTANT PROBABILITY DISTRIBUTIONS

There is a set of probability distributions used in multivariate analyses that play a similar role to the corresponding set of distributions that are used in univariate analysis when the normal distribution is appropriate to a dataset. These multivariate distributions are:

- Multivariate normal distribution
- Wishart distribution
- Multivariate Student-t distribution.

The Inverse-Wishart distribution is important in Bayesian inference, for example in Bayesian multivariate linear regression.

Additionally, Hotelling's T-squared distribution is a univariate distribution, generalising Student's t-distribution, that is used in multivariate hypothesis testing.

SOFTWARE AND TOOLS

There are an enormous number of software packages and other tools for multivariate analysis, including:

- Calc
- MiniTab
- R
- SAS (software)
- sciPy for Python
- SPSS
- Stata
- STATISTICA
- TMVA- Toolkit for Multivariate Data Analysis in ROOT
- The Unscrambler.

THE SOFTWARE TOOLS OF RESEARCH

The software tools of research are typically more abundant than hardware tools in the social sciences. Software is usually thought of to mean computer programmes that tell the hardware what to do, but any tool not related to a physical device can be considered software. Included in this category is statistical software, consent forms, published tests, questionnaires, observation forms, and, to a lesser degree, the interview.

Statistical Software. Simple statistical problems, such as determining the mean or the median of a small data set, can easily be done with a calculator. Most formulas that will be used in a research report, however, are a lot more complex. While a calculator will work, a statistical programme can reduce the computation time by hours, days, or even weeks. Imagine trying to determine the mean, standard deviation, t-score, and z-score conversions of twelve data sets each containing 300 subjects. Even the best statistician will spend many hours on this project that could be done by a computer in a matter of minutes once the data is entered.

The most widely used statistical software used for social science research is the Statistical Package for the Social Sciences (SPSS) and is relatively easy to use if you have basic computer knowledge. SPSS can perform hundreds of statistical computations and even graph your data. Another programme, SAS, also performs these functions and is gaining popularity with many researchers. Both, however, can be expensive to purchase so it would be wise to use your school's software or look into a student version.

Consent Forms. The consent form is a necessity for anyone doing research with human subjects. The purpose of the consent form is to provide information to the potential subject regarding the experiment or study and to answer questions regarding their participation. Consent forms should always include both potential benefits and harm that may result from participation as well as the option to quit the study at any time without repercussions.

The title of the study, purpose, and researchers' names and affiliation are often included towards the top of the form. Contact information, such as phone numbers, should also be included. The bulk of the consent form involves the specific procedures that will be used, how the information gathered during the study will be used, and the expected experience that the subject will endure if he or she agrees to participate.

Procedures and Requirements. Before a potential subject can consent to any study, especially those involving invasive techniques, he or she must be made aware of the process that will take place. The procedures and requirements section explains to the subject what he or she will experience during the study. If a survey is involved, this section might inform the subject that they will be asked to respond to questions related to their past work experience or their college grades, for example.

Potential Benefits. There are often potential benefits to any study for the individual subject and should always be potential benefits to the population at large. This section allows the researcher to make statement such as, "the results of this research could lead to better teaching methods that may ultimately improve the way children learn." Benefits to the potential subject should also be stated, such as free medication and treatment for a medical study, or consultation with a psychologist. Even something as simple as self-growth could be included in this section.

Potential Harms. Perhaps, more important than benefits, are the potential harms that could result from participating in the study. The subject must be informed of any risks involved, including both physical and emotional. Even if these potential harms

are rare or farfetched, the subject has the right to know before consenting to participate. Often you will see statements that discuss the possibility of negative thoughts or feelings associated with the subject's responses or actions. For example, a study addressing depression after the death of a loved one could easily bring about sad thoughts and depressed feelings as the subject answers questions related to his or her experience. If this is a possibility, it should be addressed in the consent form.

Statement of Voluntary Consent. The statement of voluntary consent typically contains two main statements. First, the subject will be agreeing that he or she has (a) read this form, (b) understand the form, and (c) has had all questions answered. Second, the subject should be informed that he or she can drop out of the study at any point (if this is feasible without causing harm) without repercussions. This means that at any point from signing the form to the completion of the study, they could refuse to continue being a subject and will not be harassed, humiliated, or otherwise coerced.

Published Standardized Tests. Often researchers want to gather information related to a general area such as personality or intelligence. For these instances, the use of a standardized test may be the best choice. With already published tests you can be sure of both validity and reliability and can save a lot of time that might otherwise be spent on test construction. Standardized tests can be classified into five main categories: achievement, aptitude, interest, personality, and intelligence.

Achievement Tests. Achievement tests are designed specifically to measure an individual's previously learned knowledge or ability. They are available for many topic areas related to psychology, education, business, and other fields. Achievement tests require that prior learning take place and that this learning be demonstrated in order to pass.

Aptitude Tests. Aptitude tests attempt to predict an individual's performance in some activity at some point in the future. They do not require any specific prior learning although basic knowledge related to reading and writing is usually required and some preparation, such as studying up on math formulas or sentence structure, can be helpful. A well-known example of this type is the Scholastic Achievement Test (SAT), designed to predict future college performance.

Interest Inventories. Interest inventories also require only general knowledge but no preparation is needed. These tests look at an individual's subjective interests in order to make predictions about some future behaviour or activity. Perhaps the most used interest inventory is the Strong Interest Inventory, which compares interests related to specific careers in order to help guide an individual's career path. Endorsed interests are compared with the interests of successful individuals in various fields and predictions are made regarding the test-taker's fit with the various career fields.

Personality Tests. Typically designed to assess and diagnose personality and mental health related disorders, personality tests are used extensively by psychologists in clinical, educational, and business related settings. By far the most widely used test of this type is the Minnesota Multiphasic Personality Inventory, Second Edition (MMPI-2), which compares an individual's responses on a series of true-false items to those suffering

from various mental disorders such as depression, schizophrenia, and anxiety. The theory behind the test argues that if you endorse items similar to the items endorsed by those with depression, for example, then the chances that you are also depressed increases.

Intelligence Tests. Intelligence tests could be classified as aptitude tests since they are sometimes used to predict future performance. They could also be classified as personality tests since they can be used to diagnose disorders such as learning disabilities and mental retardation. However, because of their limited scope, we will place them in their own category. The purpose of an intelligence test is to attain a summary score or intelligence quotient (IQ) of an individual's intellectual ability. Scores are compared to each other and can be broken down into different subcategories depending on the intelligence test used. The most commonly used tests of this type are the Wechsler Scales, including the Wechsler Adult Intelligence Scale (WAIS), the Wechsler Intelligence Scale for Children (WISC), and the Wechsler Preschool and Primary Scale of Intelligence (WPPSI).

Self-Response Questionnaires. Self-response questionnaires are a great way to gather large amounts of information in a relatively short amount of time. A questionnaire, similar to a survey you might see on a web page, allows subjects to respond to questions, rate responses, or offer opinions. Their responses can then be used to place them in specific categories or groups or can be compared to other subjects for data analysis. A concern with self-report, however, is the accuracy of the responses. Unlike direct observation, there is no way of knowing if the subject has told the truth or whether or not the question was understood as intended. There are several different methods for gathering information on a questionnaire or survey, including a Likert scale, the Thurstone technique, and the semantic differential.

Likert Scale. The Likert scale is a popular method used in surveys because it allows the researcher to quantify opinion based items. Questions are typically grouped together and rated or responded to based on a five-point scale. This scale typically ranges in order from one extreme to the other, such as (1) very interested; (2) somewhat interested; (3) unsure; (4) not very interested; and (5) not interested at all. Items that might be rated with this scale representing the subject's level of interest could include a list of careers or academic majors, for example.

Observation Forms. Observation of subjects has been a longstanding means of gathering information. While this method works well for informal research or research involving only a single subject, it is not an easy task to generalize to the greater population. Scientific methods require that any observation be as standardized and objective as possible if generalization is to occur. Observation forms are often used to allow researchers to detail their observations on an agreed upon scale and observer ratings are often correlated to determine if all are measuring the same behaviours.

Human behaviour is complex and measuring it merely by watching behaviours presents unique challenges. Imagine measuring violent behaviour in a group of 3rd graders during recess.

What one observer sees as violent, pushing another child for example, a second researcher may view as aggressive but not violent. A third observer may miss the behaviour all together due to other behaviours or some distracter on the playground. To increase validity and reliability, the following steps are recommended:

1. Operationally define behaviours to be observed as much as possible
2. Practice observations before the study to correlate results in order to make sure all observers are measuring the same behaviours
3. Use at least two observers per subject when possible in order to minimize missed behaviours or misinterpretations
4. Retrain observers frequently and correlate these training observations.

Q-Sort. The Q-Sort is a technique adopted originally into humanistic psychology as a means for a client to self-evaluate his current status and then decide on treatment goals. Since the instrument is completed and interpreted by the client without the opinion of the therapist, it allows the client to be in complete control of what issues are to be worked on. This follows the humanistic view that the client, not the therapist, is the one with the answers to the client's problems.

The Q-Sort consists of a number of cards, often as many as 40 or 50, each consisting of a single trait, belief, or behaviour. The goal is to sort these cards into one of five columns ranging from statements such as, 'very much like me' to 'not at all like me.'

There are typically a specific number of cards allowed for each column, forcing the client to balance the cards evenly. The qualities in each column are then recorded and the results are used to assist the client in determining issues he or she wishes to work on in treatment. The Q-Sort can also be completed during and after treatment to assess changes and progress towards the self-determined goals.

The Interview. The interview is a well-known means of gathering information, especially in organizational settings such as hiring employees or gathering background information. Just like observations, interviews can be very subjective. It is therefore important to determine questions prior to the study and to develop a protocol that will be followed by all interviewers. This protocol will ask the same questions in the same order to every subject and responses should be recorded exactly. It is also helpful if possible responses to questions are kept as closed ended as possible. For example, "how many siblings do you have?" rather than "tell me about your siblings."

What makes the interview different than a self-administered questionnaire is the ability to judge behaviours, and to inspire or clarify responses. Interviews can be time consuming, especially if a large number of subjects are to be interviewed. And, if open-ended questions are used, the quantitative procedures of statistics become much more difficult.

Scientific Method

DEFINITION

The method of scientific investigation is nothing but the expression of the necessary mode of working of the human mind. It is simply the mode at which all phenomena are reasoned about, rendered precise and exact. There is no more difference, but there is just the same kind of difference, between the mental operations of a man of science and those of an ordinary person, as there is between the operations and methods of a baker or of a butcher weighing out his goods in common scales, and the operations of a chemist in performing a difficult and complex analysis by means of his balance and finely graduated weights.

It is not that the action of the scales in the one case, and the balance in the other, differ in the principles of their construction or manner of working; but the beam of one is set on an infinitely finer axis than the other, and of course turns by the addition of a much smaller weight.

You will understand this better, perhaps, if I give you some familiar example. You have all heard it repeated, I dare say, that men of science work by means of induction and deduction, and that by the help of these operations, they, in a sort of sense, wring from Nature certain other things, which are called natural laws, and causes, and that out of these, by some cunning skill of their own, they build up hypotheses and theories. And it is imagined by many, that the operations of the common mind can be by no means compared with these processes, and that they have to be acquired by a sort of

special apprenticeship to the craft. To hear all these large words, you would think that the mind of a man of science must be constituted differently from that of his fellow men; but if you will not be frightened by terms, you will discover that you are quite wrong, and that all these terrible apparatus are being used by yourselves every day and every hour of your lives.

There is a well-known incident in one of Moliere's plays, where the author makes the hero express unbounded delight on being told that he had been talking prose during the whole of his life.

In the same way, I trust, that you will take comfort, and be delighted with yourselves, on the discovery that you have been acting on the principles of inductive and deductive philosophy during the same period. Probably there is not one here who has not in the course of the day had occasion to set in motion a complex train of reasoning, of the very same kind, though differing of course in degree, as that which a scientific man goes through in tracing the causes of natural phenomena.

A very trivial circumstance will serve to exemplify this. Suppose you go into a fruiterer's shop, wanting an apple,—you take up one, and, on biting it, you find it is sour; you look at it, and see that it is hard and green. You take up another one, and that too is hard, green, and sour. The shopman offers you a third; but, before biting it, you examine it, and find that it is hard and green, and you immediately say that you will not have it, as it must be sour, like those that you have already tried.

Nothing can be more simple than that, you think; but if you will take the trouble to analyse and trace out into its logical elements what has been done by the mind, you will be greatly surprised. In the first place you have performed the operation of induction.

You found that, in two experiences, hardness and greenness in apples went together with sourness. It was so in the first case, and it was confirmed by the second. True, it is a very small basis, but still it is enough to make an induction from; you generalise the facts, and you expect to find sourness in apples where you get hardness and greenness.

You found upon that a general law that all hard and green apples are sour; and that, so far as it goes, is a perfect induction. Well, having got your natural law in this way, when you are offered another apple which you find is hard and green, you say, "All hard and green apples are sour; this apple is hard and green, therefore this apple is sour."

That train of reasoning is what logicians call a syllogism, and has all its various parts and terms,—its major premiss, its minor premiss and its conclusion. And, by the help of further reasoning, which, if drawn out, would have to be exhibited in two or three other syllogisms, you arrive at your final determination, "I will not have that apple."

So that, you see, you have, in the first place, established a law by induction, and upon that you have founded a deduction, and reasoned out the special particular case. Well now, suppose, having got your conclusion of the law, that at some time afterwards, you are discussing the qualities of apples with a friend: you will say to him, "It is a very curious thing,—but I find that all hard and green apples are sour!" Your friend says to you, "But how do you know that?" You at once reply, "Oh, because I have tried them

over and over again, and have always found them to be so.” Well, if we were talking science instead of common sense, we should call that an experimental verification. And, if still opposed, you go further, and say, “I have heard from the people in Somersetshire and Devonshire, where a large number of apples are grown, that they have observed the same thing.

It is also found to be the case in Normandy, and in North America. In short, I find it to be the universal experience of mankind wherever attention has been directed to the subject.” Whereupon, your friend, unless he is a very unreasonable man, agrees with you, and is convinced that you are quite right in the conclusion you have drawn. He believes, although perhaps he does not know he believes it, that the more extensive verifications are,—that the more frequently experiments have been made, and results of the same kind arrived at,—that the more varied the conditions under which the same results are attained, the more certain is the ultimate conclusion, and he disputes the question no further. He sees that the experiment has been tried under all sorts of conditions, as to time, place, and people, with the same result; and he says with you, therefore, that the law you have laid down must be a good one, and he must believe it.

In science we do the same thing;—the philosopher exercises precisely the same faculties, though in a much more delicate manner. In scientific inquiry it becomes a matter of duty to expose a supposed law to every possible kind of verification, and to take care, moreover, that this is done intentionally, and not left to a mere accident, as in the case of the apples. And in science, as in common life, our confidence in a law is in exact proportion to the absence of variation in the result of our experimental verifications.

For instance, if you let go your grasp of an article you may have in your hand, it will immediately fall to the ground. That is a very common verification of one of the best established laws of nature—that of gravitation. The method by which men of science establish the existence of that law is exactly the same as that by which we have established the trivial proposition about the sourness of hard and green apples.

But we believe it in such an extensive, thorough, and unhesitating manner because the universal experience of mankind verifies it, and we can verify it ourselves at any time; and that is the strongest possible foundation on which any natural law can rest.

So much, then, by way of proof that the method of establishing laws in science is exactly the same as that pursued in common life. Let us now turn to another matter (though really it is but another phase of the same question), and that is, the method by which, from the relations of certain phenomena, we prove that some stand in the position of causes towards the others.

I want to put the case clearly before you, and I will therefore show you what I mean by another familiar example. I will suppose that one of you, on coming down in the morning to the parlor of your house, finds that a tea-pot and some spoons which had been left in the room on the previous evening are gone,—the window is open, and you observe the mark of a dirty hand on the window-frame, and perhaps, in addition to that, you notice the impress of a hobnailed shoe on the gravel outside.

All these phenomena have struck your attention instantly, and before two seconds have passed you say, “Oh, somebody has broken open the window, entered the room, and run off with the spoons and the tea-pot!” That speech is out of your mouth in a moment. And you will probably add, “I know there has; I am quite sure of it!” You mean to say exactly what you know; but in reality you are giving expression to what is, in all essential particulars, an hypothesis. You do not KNOW it at all; it is nothing but an hypothesis rapidly framed in your own mind. And it is an hypothesis founded on a long train of inductions and deductions.

What are those inductions and deductions, and how have you got at this hypothesis? You have observed in the first place, that the window is open; but by a train of reasoning involving many inductions and deductions, you have probably arrived long before at the general law—and a very good one it is—that windows do not open of themselves; and you therefore conclude that something has opened the window. A second general law that you have arrived at in the same way is, that tea-pots and spoons do not go out of a window spontaneously, and you are satisfied that, as they are not now where you left them, they have been removed.

In the third place, you look at the marks on the windowsill, and the shoe-marks outside, and you say that in all previous experience the former kind of mark has never been produced by anything else but the hand of a human being; and the same experience shows that no other animal but man at present wears shoes with hob-nails in them such as would produce the marks in the gravel. I do not know, even if we could discover any of those “missing links” that are talked about, that they would help us to any other conclusion! At any rate the law which states our present experience is strong enough for my present purpose. You next reach the conclusion that, as these kind of marks have not been left by any other animal than man, or are liable to be formed in any other way than a man’s hand and shoe, the marks in question have been formed by a man in that way.

You have, further, a general law, founded on observation and experience, and that, too, is, I am sorry to say, a very universal and unimpeachable one,—that some men are thieves; and you assume at once from all these premisses—and that is what constitutes your hypothesis—that the man who made the marks outside and on the window-sill, opened the window, got into the room, and stole your tea-pot and spoons. You have now arrived at a *vera causa*;—you have assumed a cause which, it is plain, is competent to produce all the phenomena you have observed. You can explain all these phenomena only by the hypothesis of a thief. But that is a hypothetical conclusion, of the justice of which you have no absolute proof at all; it is only rendered highly probable by a series of inductive and deductive reasonings.

I suppose your first action, assuming that you are a man of ordinary common sense, and that you have established this hypothesis to your own satisfaction, will very likely be to go off for the police, and set them on the track of the burglar, with the view to the recovery of your property. But just as you are starting with this object, some person

comes in, and on learning what you are about, says, "My good friend, you are going on a great deal too fast. How do you know that the man who really made the marks took the spoons? It might have been a monkey that took them, and the man may have merely looked in afterwards." You would probably reply, "Well, that is all very well, but you see it is contrary to all experience of the way tea-pots and spoons are abstracted; so that, at any rate, your hypothesis is less probable than mine."

While you are talking the thing over in this way, another friend arrives, one of the good kind of people that I was talking of a little while ago. And he might say, "Oh, my dear sir, you are certainly going on a great deal too fast. You are most presumptuous. You admit that all these occurrences took place when you were fast asleep, at a time when you could not possibly have known anything about what was taking place. How do you know that the laws of Nature are not suspended during the night? It may be that there has been some kind of supernatural interference in this case." In point of fact, he declares that your hypothesis is one of which you cannot at all demonstrate the truth, and that you are by no means sure that the laws of Nature are the same when you are asleep as when you are awake.

Well, now, you cannot at the moment answer that kind of reasoning. You feel that your worthy friend has you somewhat at a disadvantage. You will feel perfectly convinced in your own mind, however, that you are quite right, and you say to him, "My good friend, I can only be guided by the natural probabilities of the case, and if you will be kind enough to stand aside and permit me to pass, I will go and fetch the police." Well, we will suppose that your journey is successful, and that by good luck you meet with a policeman; that eventually the burglar is found with your property on his person, and the marks correspond to his hand and to his boots. Probably any jury would consider those facts a very good experimental verification of your hypothesis, touching the cause of the abnormal phenomena observed in your parlor, and would act accordingly.

Now, in this supposititious case, I have taken phenomena of a very common kind, in order that you might see what are the different steps in an ordinary process of reasoning, if you will only take the trouble to analyse it carefully. All the operations I have described, you will see, are involved in the mind of any man of sense in leading him to a conclusion as to the course he should take in order to make good a robbery and punish the offender. I say that you are led, in that case, to your conclusion by exactly the same train of reasoning as that which a man of science pursues when he is endeavouring to discover the origin and laws of the most occult phenomena. The process is, and always must be, the same; and precisely the same mode of reasoning was employed by Newton and Laplace in their endeavours to discover and define the causes of the movements of the heavenly bodies, as you, with your own common sense, would employ to detect a burglar.

The only difference is, that the nature of the inquiry being more abstruse, every step has to be most carefully watched, so that there may not be a single crack or flaw in your hypothesis. A flaw or crack in many of the hypotheses of daily life may be of little or no moment as affecting the general correctness of the conclusions at which we may arrive;

but, in a scientific inquiry, a fallacy, great or small, is always of importance, and is sure to be in the long run constantly productive of mischievous if not fatal results.

Do not allow yourselves to be misled by the common notion that an hypothesis is untrustworthy simply because it is an hypothesis. It is often urged, in respect to some scientific conclusion, that, after all, it is only an hypothesis. But what more have we to guide us in nine-tenths of the most important affairs of daily life than hypotheses, and often very ill-based ones? So that in science, where the evidence of an hypothesis is subjected to the most rigid examination, we may rightly pursue the same course. You may have hypotheses, and hypotheses. A man may say, if he likes, that the moon is made of green cheese: that is an hypothesis.

But another man, who has devoted a great deal of time and attention to the subject, and availed himself of the most powerful telescopes and the results of the observations of others, declares that in his opinion it is probably composed of materials very similar to those of which our own earth is made up: and that is also only an hypothesis. But I need not tell you that there is an enormous difference in the value of the two hypotheses.

That one which is based on sound scientific knowledge is sure to have a corresponding value; and that which is a mere hasty random guess is likely to have but little value. Every great step in our progress in discovering causes has been made in exactly the same way as that which I have detailed to you.

A person observing the occurrence of certain facts and phenomena asks, naturally enough, what process, what kind of operation known to occur in Nature applied to the particular case, will unravel and explain the mystery? Hence you have the scientific hypothesis; and its value will be proportionate to the care and completeness with which its basis had been tested and verified.

It is in these matters as in the commonest affairs of practical life: the guess of the fool will be folly, while the guess of the wise man will contain wisdom. In all cases, you see that the value of the result depends on the patience and faithfulness with which the investigator applies to his hypothesis every possible kind of verification.

SCIENTIFIC AND NON-SCIENTIFIC METHOD

PURPOSE OF SCIENTIFIC METHOD

The scientific method is the means by which researchers are able to make conclusive statements about their studies with a minimum of bias. The interpretation of data, for example the result of a new drug study, can be laden with bias. The researcher often has a personal stake in the results of his work. As any skilled debater knows, just about any opinion can be justified and presented as fact. In order to minimize the influence of personal stakes and biased opinions, a standard method of testing a hypothesis is expected to be used by all members of the scientific community.

Scientific Method Working

The first step to using the scientific method is to have some basis for conducting your research. This is based on observed phenomena that is either directly or indirectly related to the specific subject matter of your proposed research. For example, you may have observed that Drug A is effective in treating an illness (Disease A) caused by Virus A. A new illness (Disease B) has arisen that mimics some symptoms of Disease A, but with variation (such as the patient with Disease B has swollen lymph nodes and a low grade fever instead of no swelling and a high grade fever). Outbreaks of Disease B occur near outbreaks of Disease A. These are the observations you make in your first step of using the Scientific Method.

The next step is to form a hypothesis to explain some aspect of your observations. You speculate that the virus that causes Disease B is either Virus A or it is related to Virus A. Your hypothesis is that the cause of Disease A and Disease B is the same virus.

Now that you have a hypothesis, you are ready to test it. You must now use your hypothesis to predict other phenomena that have not yet been observed. You know that Drug A will wipe out Disease A. If Disease B is caused by the same virus, you reason that the same drug should be effective.

The final step of the scientific method is to rigorously test your prediction. Remember, you cannot "prove" your hypothesis. You can only fail to disprove it. While this is an example of how the scientific method is used in everyday research and hypothesis testing, it is also the basis of creating theories and laws.

The scientific method requires a hypothesis to be eliminated if experiments repeatedly contradict predictions. No matter how great a hypothesis sounds, it is only as good as its ability to consistently predict experimental results. It should also be noted that a theory or hypothesis is not meaningful if it is not quantitative and testable. If a theory does not allow for predictions and experimental research to confirm these predictions, then it is not a scientific theory.

What is a Hypothesis

It is important to distinguish between a hypothesis, and a theory or law. Although in everyday language, people sometimes use these terms interchangeably, they have very distinct connotations in the scientific community. A hypothesis is a 'small' cause and effect statement about a specific set of circumstances. It represents a belief that a researcher possesses before conducting a satisfactory number of experiments that could potentially disprove that belief. For example, you open your refrigerator at home and are greeted with a horrible sour smell. You decide that the milk must have gone bad.

This is your hypothesis. It is based on the phenomena you are observing right now (sour smell) as well as knowledge from past experience (bad milk has a sour smell). You test your hypothesis by opening the container of milk and smelling it. You find that the milk doesn't smell sour after all, so you must come up with another hypothesis.

A theory or law in the world of science is a hypothesis, or many hypotheses, which have undergone rigorous tests and have never been disproved. There is no set number of tests or a set length of time in which a hypothesis can become a theory or a law. A hypothesis becomes a theory or law when it is the general consensus of the scientific community that it should be so. Theories and laws are not as easily discarded as hypotheses.

Misapplications of the Scientific Method

A common error encountered by people who claim to use the scientific method is a lack of testing. A hypothesis brought about by common observations or common sense does not have scientific validity. As stated above, even though a good debater may be quite convincing as he conveys the merits of his theory, logical arguments are not an acceptable replacement for experimental testing.

Although the purpose of the scientific method is to eliminate researcher bias, an investigation of the raw data from an experiment is always a good idea. Researchers sometimes toss out data that does not support their hypothesis. This isn't necessarily done with the intent of deception, it is sometimes done because the researcher so passionately believes in his hypothesis that he assumes unsupportive data must have been obtained in error. Other times, outside forces (such as the corporation sponsoring and conducting the research) may put extreme pressure on the researcher to get specific results.

The best way for the scientific community, and the general public, to deal with these errors is to promote multiple, independent experiments. We are all familiar with "breaking news" (that seems to break nearly every day!) about a new miracle drug or herbal remedy. In most cases, this "breaking news" was released by a single source—usually a source with financial stakes in the new miracle.

Look for multiple sources to confirm a hypothesis before you hand your money over for a new product. If possible, also try to discover where the funding came from in these experiments. You may have three different lab reports, all confirming that Drug A is the most effective cure, but if all three laboratories are funded by the same drug company—you may want to raise an eyebrow.

SCIENTIFIC METHOD – A HISTORY

It is often overlooked that the scientific method was actually developed in Christian Europe by men who believed that matter behaved in a rational manner because God had created an orderly universe. If the universe exists in a strictly materialistic vacuum, the product of random particle collisions, then there is no foundation upon which one can expect to observe order in nature. Many of the founders of the principle scientific disciplines (such as Newton, Galileo and Kepler), were Bible-believing Christians. Johannes Kepler — one of the founders of the discipline of astronomy — said that science was "thinking God's thoughts after him."

IMPORTANCE

The scientific method can be divided into two primary categories:

- Empirical science and
- Historical science.

Empirical science entails a systematic approach to epistemology that uses observable, testable, repeatable, and falsifiable experimentation to understand how nature commonly behaves. It finds its implementation in such disciplines as immunology, rocket science, molecular biology, *etc.* Historical science involves the interpretation of evidence and the deduction of past occurrences, which is normally based upon an underlying supportive paradigm.

Recognizing that every person has presuppositions that shape the way in which empirical evidence is interpreted is important. Theists and non-theists possess the same evidence; but that evidence is interpreted within a framework which corresponds to the individual's respective worldview. In its original form, 'science' could simply be defined as 'knowledge'.

Today, however, science — in the view of an outspoken part of the scientific enterprise — is the systematic method of gaining knowledge about the universe with reference to purely naturalistic or materialistic causation. Science in this sense automatically rules out the notion of God because supernatural claims — it is asserted — cannot be tested and repeated. If an idea is not testable, repeatable, observable and falsifiable, it is not considered scientific. The problem with the above definition of science is that, even though naturalistic science claims to be neutral and unbiased, it starts with a set of basic metaphysical axioms — that only matter and energy exist and all explanations and causes must be directly related to the laws that matter and energy follow. Many scientists have claimed that allowing supernatural explanations into our understanding of the universe would cause us to stop searching for answers and simply invoke a "god-of-the-gaps". This is, of course, false.

The design inference is not based upon a lack of knowledge (as some would contend), but rather it is based upon our uniform and repeated experience. While we know of no naturalistic cause by which complex and specified biological information can arise from inorganic matter, we do know that in all other circumstances information originates from intelligence. Moreover, many scientists now see evidence of intelligent design in the 'irreducible complexity' of molecular machines and circuits in the cell, the pattern of appearance of the major groups of organisms in the fossil record, the fine-tuning of the laws and constants of physics to support complex life, the fine tuning of our terrestrial environment, the information processing system of the cell, and even the phenomenon known as 'homology' (evidence previously thought to provide unequivocal support for neo-Darwinism).

While critics may disagree with the conclusions of the design arguments, they cannot reasonably deny that they are based upon commonly accepted observations of the natural world. The term 'science' commonly connotes an activity in which theories are developed

to explain observations of the natural world, the empirical, observational basis of the theory of intelligent design provides a good reason for regarding intelligent design as a scientific theory.

CONCEPTS OF RESEARCH IN ECONOMICS

In the first section, we start with the definition of various terms relating to research. Terms to be discussed are ‘research’, ‘research methods’ and ‘research methodology’ and, finally, a brief discussion of various types of research. In the second section, we will discuss what is economics and what economists do. This discussion is presented at the outset to illustrate and highlight various skills needed to carry out economic analysis. What is attempted here is to highlight the importance and interdependence of economic theory and measurement in the study of economics.

In the third section, we introduce basic concepts of research. As is well known, the method of research or analysis economists use in carrying out their task is the scientific method, which is used in all of science. Therefore, it is important to discuss science in general and its method, namely scientific method. We should note, however, that there is no such a thing as *the* scientific method, because there are many variations. Scientific method essentially refers to the general or generalized process called the “scientific approach” to obtaining new and reliable knowledge. What we attempt to do in this section is to discuss the key terms and concepts of the scientific method, before we delve into research procedures in the next chapter. These concepts include theory and model, variables, assumptions, parameters, the hypothesis, and the testing of hypothesis among others.

KEY TERMS IN RESEARCH

What is Research

The term “research” is often loosely defined and thus used in a similar way. This unfortunate development results from a misconception about what is research. To properly understand what is research, it is good to start with common misconceptions about research. First, fact transferal is not research. Consider a typical high school research project. The teacher assigns a “research project” on some topic. The students went to the library, checked out several books, and might have copied several pertinent pages from the book. The typical student organized collected information and wrote up the “research report”. What these students did is information gathering and organization; it is nothing more or nothing less. No doubt the student went through some motions associated with research. But finding fact and fact transferal alone is not research. Transfer of information from one source, namely books and pertinent pages, to another source, namely the so-called research report, is nothing more than fact transferal, but

not research. To my distress I find many college students repeat this same mistake by submitting a ‘research report’ which is nothing but fact transferal from one source to their report. A second misconception about research is that research is related to laboratory research (for example, in chemistry or biology in the natural sciences). When people hear term the “research”, they often conjure up this image. But research is not limited to certain fields of study; it is characterized by the methods used.

Research Methods and Methodology

Research methods provide the specific details of how one accomplishes a research task (procedures and methods); It provides specific and detailed procedures of how to initiate, carry out, and complete a research task by mainly focusing on how to do it.

Research methodology deals with general approaches or guidelines to conducting research. It provides the principles for organizing, planning, designing, and conducting research, but it cannot tell you in detail how to conduct a specific, individual research.

In carrying out an applied and quantitative economic research, there are several necessary backgrounds. The researcher should first have solid training in economic theory, quantitative methods (statistics and econometrics), data analysis techniques, and adequate training in micro-computer technology, as well as some training in research methods. Unfortunately, undergraduate students beginning their research most often do not have these backgrounds.

Therefore, they are going to be overwhelmed and intimidated by the lack of necessary skills. Many often give up in frustration, even if they are willing to persevere and to learn these skills.

What is sorely need is a practical guide to initiate, conduct, and complete an applied and quantitative economic research. One proven and effective way of learning these various skills which economists use is *learning by doing by example*.

TYPES OF RESEARCH

Basic vs. Applied Research: The distinction between basic and applied research is largely by the focus of its application. This distinction comes from basic science vs. applied science. Example: physics and engineering.

Basic research focuses on determining or establishing the basic or fundamental relationships within a discipline without paying attention to any practical applications to the real world. In contrast, applied research is usually conducted to solve a particular and concrete problem.

Descriptive vs. Analytical Research

The distinction between descriptive and analytical research is based on the question it asks. Descriptive research attempts to determine, describe, or identify *what* is, while analytical research attempts to establish *why* it is that way or how it came to be.

The descriptive research uses description, classification, measurement, and comparison to describe what phenomena are. The analytical research usually concerns itself with cause-effect relationships.

Examples: Examining the fluctuations of U. S. international trade balance during 1974-1995 is an example of descriptive research; while explaining *why* and *how* U.S. trade balance move in a particular way over time is an example of analytical research.

Another example: Starting from late 1986, the value of U.S. dollar value has steadily increased against the Japanese yen and German Mark. Examining the magnitude of this trend in the value of U.S. dollar is another example of descriptive research; while explaining *how* and *why* this surge in the value of the U.S. dollar is occurring. If one attempts to explain *how* and *why* this surge in the value of U.S. dollar is going to affect the U.S. economy, as well as the economies of Japan and Germany, this is another example of analytical research.

Disciplinary, Subject-Matter, and Problem-Solving Research

By disciplinary research, we mean research “designed to improve a discipline” as Johnson(1986) defines it. It dwells on theories, relationships, and analytical procedures and techniques within the discipline. Examples: Economic research or social research.

By subject-matter research, we mean research “on a subject of interest” within a discipline. Example: Research in resource economics or in international economics. By problem-solving research, we mean research “designed to solve a specific problem for a specific decision maker”. It is often multidisciplinary.

Example: A multidisciplinary study of on the demand for new mass transit involving economics, sociology, and civil engineering. Or a multidisciplinary study of new medical surgery involving medical doctors, engineers, and an economist.

ECONOMICS

What is Economics

Case and Fair define economics as “the study of how individuals and societies choose to use the scarce resources that nature and previous generations have provided” Since it deals with the behaviour of human beings and their interactions, it is a social science.

Stiglitz also defines it as a social science, but he adds that “it studies the social problem of choice from a scientific viewpoint, which means that it is built on a systematic exploration of the problem of choice”.

This systematic exploration involves both the formulation of *theories* and the examination of *data* to test the validity of theories. This latter definition of economics by Stiglitz is helpful in identifying two major tasks of economists: They are *observing* facts and trends in data and *explaining* and interpreting certain or whole aspect of the economy.

From the above definitions of economics, the importance of both theory and measurement is obviously the essential ingredients of meaningful and serious economic research. Theory without measurement is unfruitful and empty; measurement without theory is equally meaningless.

A theory consists of a set of assumptions (or hypotheses), and conclusions derived from those assumptions. Theories are logical exercises: if the assumptions hold, then the results follow.

Examples: When theory is formally presented, it is called model. Examples of model car, ship, or airplane to understand how economists use models. Bamoul and Bliner's definition of theory: "A theory is a deliberate simplification of relationships whose purpose is to explain how those relationships work. It is an explanation of the mechanism behind observed phenomena. The relationship is usually couched in terms of the relationship between or among variables. What is variable? A variable is anything which varies over time and space; then the measurement of variable in terms of numbers or facts expressed in quantitative terms are data.

The data are then quantitative information on various aspects of economics and business. We need to learn how to describe and analyse data. The description of data is one area of statistics: descriptive statistics. Data analysis involves the analysis of statistical data of one variable (univariate distribution). Data analysis also analyzes the relationship between two variables (bivariate distribution). Data analysis may involve the analysis of among more than two variables (multivariate distribution). The presentation of data can either be tabular and graphical. In this part, both the description and computation, illustration, using any easily available computer softwares are to be presented side by side.

WHAT DO ECONOMISTS DO

In a certain sense, economics is really what economists do. Two major tasks on which economists spend their energies are: First, observing facts and trends in data and, second, explaining the relationship(s) between them in a cause-effect fashion.

Observing Facts and Trends in Data

A fact is a verifiable observation or phenomenon. It is an observed fact that retail gasoline prices in California are higher than elsewhere in the U.S. The problem is then to explain why and how this is so. A fact(s) is usually presented in terms of variable(s). A variable is a quantity of something which varies and the researcher is interested in. There are two types of variables: discrete (or discontinuous) and continuous variables. When a variable(s) is continuous, the researcher can identify a trend in it. Examples of trend are a linear trend, parabolic or quadratic trend.

There are several publications containing fact and trends of economic variables on the overall U.S. economic conditions. The first is the Statistical Abstract of US and

Economic Report of the President, the annual publication by the Council of Economic Advisors. Two other well-known publications on more current economic situations are Economic Indicators, a monthly publication prepared by the Council of Economic Advisors and Economic Trends, another monthly publication by the Federal Reserve Bank of Cleveland. To obtain specific information on some specific aspect of the economy, say health care or energy, one needs to consult specialized publications.

Some examples of interesting facts and trends in the US economy during the last two decades are:

- Health care spending in the U.S. has increased faster than other spending categories than before.
- The Budget deficit and national debt have increased faster than before.
- Foreign exchange rates have fluctuated more widely than before.

When a fact or phenomenon is measured (quantified), organized, and presented in a desirable fashion, we call them data. Data provide factual information. They are usually regarded as reliable subject to the implicit understanding that there may be some sampling or measurement errors in the process of collection.

To learn how to work with macroeconomics data on the U.S. economy, consider the following set of major economics data on the U. S. economy for the period from 1960 through 1995. Home Work Assignment.

Spread sheet data and How to Use Excel for Data Analysis [Ed's work]

Measurement of variables in level, first difference, index numbers, percentage changes, per capital measures, converting nominal to real variables, fixed weight vs. chain weight GDP measure, adjustments to data (seasonal adjustments, various moving averages, normalized variables *etc.* Use of spread sheet (Excel) to manipulate data and graph.

When we examine facts and trend in the variable (s), we often present information in a table or in a graph. When we examine facts and trend, we need to be mindful of the distinction between correlation vs. causation between the two variables.

There are a number of important issues relating to measurement of variables, and these issues will be taken up later chapter on data and data analysis.

Explaining and Predicting the Relationship

Even though fact or phenomena provides useful factual knowledge, fact alone is not interesting.

Even though facts are interesting, they are not very useful by themselves.

As C.S. Peirce expressed so clearly, "a person can stare stupidly at phenomena; but in the absence of imagination they will not connect themselves together in any rational way"* More interesting and useful are raising analytical questions regarding facts. With the first fact on rapid increase in health-care spending, one can ask why and how health-care spending increased faster than spending in other sectors? Relating to this question, one can also ask why and how the price of health-care increased faster than other prices?

Observing another fact, say the large fluctuations in foreign exchange rates, one may ask whether or not, and why and how a change in the international monetary arrangement might affect the pattern of change in exchange rate. To answer this question, we need to have a measure for various exchange rates. They are bilateral versus multilateral, or the effective exchange rate, and the nominal versus real exchange rate.

Given the first fact, the problem is to explain why and how retail gasoline prices have increased during this particular period and to explain why and how retail gasoline prices in California are substantially higher than in the United States. Explaining the relationship between the share of health care expenditure out of GDP requires the use of the scientific method. Therefore, it is necessary for us to be familiar with concepts of research, such as hypothesis, theory, and model. We also have to learn operational terms of scientific method, such as variables, assumptions, parameters, and functional form.

SCIENTIFIC VS. NON-SCIENTIFIC METHODS

Having finished a brief discussion of the meaning of research and related terms, we are now ready to take up the discussion of the scientific method. Let us start by stating that there are essentially two methods of obtaining knowledge: scientific and non-scientific methods.

Non-Scientific Method

The second method of obtaining knowledge is the scientific method. This method of gaining knowledge is learning by reasoning. It is considered today to be the most reliable method of gaining knowledge. In contrast with the scientific method, the validity of knowledge obtained by scientific method can be subject to testing.

Let's explore the meaning of these three characteristics. First, scientific method is *objective*, not in the sense of being value free. But it is objective in the sense that the analysts are not biased or prejudiced or subject to personal whims. Second, scientific method is *logical* in the sense that science follows logical reasoning.

Logical reasoning is thinking in reasonable fashion. It is sufficient to point out two types of reasoning process, namely deductive and inductive logic. A full discussion on the logical process will be presented below. Third and finally, the search for truth in science is *systematic*. This means that researchers follow a systematic set of procedures through which knowledge is gained.

Deductive and Inductive Logic

Let us now turn our attention to the logical process in scientific method. First, note that there are two types of reasoning process, namely deductive and inductive logic.

Deductive logic is the process of reasoning from general conditions or premises using assumptions to specific conclusions. Economic theory rests largely on deductive logic. We establish a set of assumptions about conditions and behaviors to arrive at conclusions through logical process. [example]

Examples: Utility maximization in the consumer behaviour; profit maximization in the producer behaviour.

Inductive logic is reasoning from the specific outcomes to a generalized conclusion. This is usually done by observing many individual experiences and cases to formulate a general conclusion. The inductive logic of reasoning is followed in most empirical economic research.

Hypothesis, Theory, and Law

We *deduce* from facts and assumptions to a *hypothesis*. “A *hypothesis* is a tentative assertion of a relationship between factors or events that is subject to verification or rejection.” In short, a hypothesis is a testable proposition of the relationship between or among variables.

Hypothesis is a single statement that attempts to explain a single interesting or puzzling phenomenon. In other words, a hypothesis is a testable proposition on an interesting or puzzling phenomenon. It usually takes the form of an educated guess or conjecture. Usually the hypothesis is based on facts and assumptions.

Theory is a whole system of thought (or systematic explanation) that refers to many phenomena and whose parts are related to one another in deductive, logical form.

Law: A theory that has been subjected to extensive testing over time and across space, and that has won virtually universal acceptance, is called a *law*.

For instance, the law of supply and demand refers to the commonly observed phenomena that, in a free market, the forces of supply and demand generally push the price towards its equilibrium level, the price at which the quantity supplied and quantity demanded are equal. Another example is the law of diminishing marginal returns.

Language of Scientific Method

In the previous section, we first learned that the essential element of science lies in its method of analysis, and that the three main characteristics of the scientific method are that it is objective, logical, and systematic.

Then we learned about the process of scientific method beginning from, say, deductive logic to hypothesis, from hypothesis to theory, and from theory to law. Now we need to learn the operational aspects of the scientific method by focusing on terms and concepts of research. In particular, we need to learn basic concepts such as assumptions, variables, parameters, and functional forms. Finally, we need to learn something about *ceteris paribus*.

Assumption

In our study of gasoline price hike, we first assume that consumers are "rational" in that their decision-making in the purchase of gasoline is consistent with maximization of consumer satisfaction. Likewise, suppliers of gasoline (retail gasoline merchant) are also assumed to be rational in that their decision-making in pricing, inventory, *etc.* is consistent with profit maximization. We theorize that the retail gasoline price is jointly determined by the forces of demand and supply of gasoline.

On the demand side, the quantity of gasoline demanded depends inversely on the retail price of gasoline, positively consumer income, and a host of other factors. On the supply side, the quantity of gasoline supplied depends positively on the price of gasoline price, cost of production (including price of crude oil and refinery cost), and a host of other factors, including the gasoline tax and environmental cost.

Once we introduce the role of assumption in the research, it appropriate to further introduce related basic concepts of research: variables, functional relationship and parameters.

Variables in General

A variable is a quantity of something which varies and you are interested in. Price of gasoline is a variable to an economic analyst studying the recent gasoline price increase in 1996 but not to most motorists and not even to an economic analyst studying the relation between stock and bond prices.

The researcher chooses his or her variables. Choosing variables correctly is one of the first essential step of carrying out research. Therefore, choose variables with extreme care. To choose variable correctly, one has to know the two types of variables: the dependent variable and independent variable.

The Dependent Variable

The dependent variable is that quantity whose change the researcher wants to find out, explain, or predict. In the cause-effect relationship, the effect variable is the dependent variable.

In the study of the demand for gasoline, the quantity of gasoline demanded is the dependent variable, because the quantity of gasoline demanded changes in response to changes in gasoline price, consumer income, and other demand side factors. This researcher is ultimately interested in measuring the impacts of these changes on gasoline price. In some cases, however, the dependent variable is not quantity, but represents a qualitative choice.

The Independent Variable

The independent variable is a variable whose effect upon the dependent variable one is trying to understand, explain, and predict. In the cause-effect relationship, it is the

cause variable. Using the study of the demand for gasoline, the independent variables are the price of gasoline, consumer income, and other variables such as fuel efficiency and population characteristics.

The Parameter

The concept of parameter is a bit more tricky to define. A parameter is a quantity measuring the response of the dependent variable to change in the independent variable, and is usually assumed to remain constant during the period of study.

To illustrate, Figure shows the hypothetical relationship between the quantity of gasoline demanded and gasoline price over time. On the vertical axis, the quantity of gasoline demanded measured in gallons. On the horizontal axis, the price of gasoline is measured in U.S. dollars per gallon. The scatter plot of cross represents the combination of the quantity of gasoline demanded and corresponding gasoline price in each year over the period. The line is drawn through the scatter plot of the quantity of gasoline demanded and gasoline price as closely as possible. The line represents a best guess of the average relationship between the observed quantity of gasoline demanded and corresponding gasoline price during the period of study.

The Functional Form

The algebraic relationship $Q_g^d = a + bP_g$ is linear relationship between the two variables Q^d and P . It is linear both in the parameters and variables. Parameters like a^2 or $1/a$ are non-linear; variables like Q^2 and $1/Q$ are examples of non-linear.

When there is a relationship between two variables, we say that "y(dependent variable) is a function of x(independent variable)" in the mathematical language. By this we mean to say that the magnitude of y depends on the magnitude of x. But when the dependent variable y is determined by not only by x but also by other independent variables, we say that y is a function of x_1, x_2 , and x_3 . Or in short $y = f(x_1, x_2, x_3)$.

In the gasoline demand example, the quantity of gasoline demanded (Q_g^d) depends upon (or is a linear function of) price of gasoline (P_g) and consumer income (Y). Algebraically, we write: $Q_g^d = f(P_g, Y)$.

Linear functional form of this relationship can be written as $Q^d = a + bP_g + cY$. Since parameters a, b, and c enter the equation are linear, the relationship is linear. But the relationship may be other than linear functional forms. They include the log-linear model, semi-log models, and reciprocal models.

Most - but not all- economic relationship is the causal relationship between the dependent variable and one or more independent variable. The value of the dependent variable y that is observed for subject in the study depends on three elements: First, it depends on the level of the independent variable; second on the influence of the independent variable(s) on the dependent variable, and finally on the influences of all other variables that may act upon the value of the subject but you are not interested in

and thus excluded in the specification of the functional relationship. These excluded variables are assumed to remain the same during the period of study.

RELATIONSHIP, CORRELATION AND CAUSATION

When variables are related in certain fashion, there are two kinds of relationships. When one variable is related with another variable, we say that there exists correlation between the two variables. More precisely, *correlation* or more precisely *simple correlation* measures the degree of a linear association between the variables. When one measures correlation among more than two variables, one measures *partial correlation*. When analysts are interested in measuring the degree of correlation. If they move together in the same direction, there is a positive correlation; if they move in opposite directions, there is negative correlation.

The share of health care spending in GDP and the relative price of health care are positively correlated. More examples of correlation.

When there exists a cause and effect relationship between two or more variables, we say that causation runs from *cause* variable to *effect* variable. *Causation* then means changes in one variable brings about changes in another variable. The cause-effect relationship is often called the dependence relationship. Using the example of share of health-care spending out of GDP and relative price of health-care, health care share is the effect or dependent variable; and the relative price of health care is cause and independent variable. Does that mean that the health care share is always the dependent variable and the relative price of health care is always the independent variable? In other words, is the dependence or causality inherent in the variable themselves? Obviously, answer is no.

But then *how* does a researcher decide on which variable depends on what variable? It is deliberately chosen and imposed by researchers, usually from theory or existing studies on the subject. Importance of determining the causal-relationship in research: Art of specifying causal relationship? Experience and knowledge?

In the study of gasoline demand, the quantity of gasoline demanded is assumed to depend on the price of gasoline. There are two reasons for the direction of causality running from the gasoline price to the quantity of gasoline demanded, but not the other way around. When we study the individual demand for gasoline, the price of gasoline is determined by aggregate decisions of all consumers and suppliers. When there are many consumers in the market, a reasonable assumption, the role played by an individual consumer is small enough to be ignored in the determination of retail gasoline price. Analysts then are interested in studying how an individual consumer respond to changes in gasoline price.

When economists are interested in studying the feedback effect from aggregate quantity of gasoline demanded to gasoline price, they build simultaneous equation model where the forces of aggregate demand and supply are allowed to interact in determining the gasoline price. -two stage method of estimating gasoline price. Regression analysis

to study the dependence relationship. Most economists consider economics as a science because they follow the scientific method. Nobel prize in natural science (physics, chemistry, and medicine). Some economists argue that their discipline does not qualify as science.

Regardless of the merit or demerit of whether economics should be viewed as science or not, it is true that economic research uses the scientific method. It is good for us to learn the language and process of the scientific method.

GRAPHICAL ILLUSTRATION

Relationships among the key concepts of research, such as variables, parameters, and functional forms can be easily illustrated using algebraic relationship with graphical illustration.

In algebraic terms, the formula for a straight line is Q_g^d equals a plus bP_g . More concisely, $Q_g^d = a + bP_g$ where Q_g^d is the price of gasoline and P_g is the price of gasoline and where a and b are parameters, representing the intercept and slope of the algebraic relationship. The intercept term tells us what the demand for gasoline would be if the price of gasoline is zero. But this is a mechanical interpretation of the intercept, and it is reasonable to interpret the intercept as the net influence of other variables in the relationship, namely consumer income and other variables in the case of gasoline demand. The slope coefficient “ b ” tells us how much the quantity of gasoline demanded would change as consumer income change on the average.

Organizational Research and Management

FUNCTIONS OF MANAGEMENT

Management operates through various functions, often classified as planning, organizing, leading/motivating and controlling.

- *Planning*: Deciding what has to happen in the future (today, next week, next month, next year, over the next five years, *etc.*) and generating plans for action.
- *Organizing*: Making optimum use of the resources required to enable the successful carrying out of plans.
- *Leading/Motivating*: Exhibiting skills in these areas for getting others to play an effective part in achieving plans.
- *Controlling*: Monitoring-checking progress against plans, which may need modification based on feedback.

Management of a business concern is a highly complex phenomenon and it is not possible to consider all its facets at the same moment. In the process of managing, a manager performs a variety of functions to accomplish the pre-determined goals. The authorities and the scholars of management differ widely about managerial functions and their classification. A widely accepted classification was given quite early by Henri Fayol. He described them as follows:

- (i) Forecasting and planning,
- (ii) Organising,
- (iii) Commanding,
- (iv) Co-ordinating, and
- (v) Controlling.

Later, several authorities have been adding other functions to the list. Dale and Michelon have described them as planning, organising, staffing, directing, controlling, innovation, presentation, communication.

Greenwood suggests planning, organising, staffing, directing, controlling and decision-making. Massies states them as decision-making, planning, organising, staffing, controlling, and communicating. Alien describes them as planning, co-ordinating, motivating, communicating and organising. According to Terry, they are: planning, organising, staffing, actuating, and controlling. According to Koontz and O'Donnell, "the most useful method of classifying managerial functions is to group them around the activities of planning, organising, staffing, directing and controlling".

However, according to them, in practice it is not always possible "to slice all managerial activities neatly into these categories, since the functions tend to coalesce." Similar views have been expressed by C.S. George when he states that: "The management process, therefore, is not a series of separate functions (planning, organising, directing, controlling) which can be performed independently; it is a composite process made up of these individual ingredients. No one function can be performed without involving the others, and by employing his composite, managers create an environment conducive to the co-ordinated participation of team members."

Moreover, a distinction has to be made between management functions and the operating tasks or business functions it performs, *i.e.*, manufacturing, purchasing, selling, financing, accounting *etc.* Normally and theoretically there may be some sequential order in which these functions are performed. In other words, planning comes first and organising, staffing, directing, co-ordinating and controlling follow it. But each of these functions is inter-linked and interwoven in actual practice, and a manager need not follow this order. He solves the problems as and when they arise without giving much attention to this order. He can move easily from one 'function to another and devote attention to the most pressing problem before him. It has been rightly pointed out that "Managing is, after all, a systematic network and not a sequentially undertaken set of duties."

We must recognise that there is a Principle of Universality of Managerial Functions. Management at different levels—top, middle, or lower—perform the same set of functions. Similarly, these very functions have to be performed in every type of undertaking, business or non-business, small or large, trading or manufacturing, philanthropic or profit-making, partnership or joint stock company, university, hotel, club, military or Government departments. As managers, they all are engaged in getting things done through others and are performing the same set of functions. According to

Koontz and O'Donnell, this principle has many implications, but two of them are very important. *Firstly*, it means that “anything significant that is said about the functions of one manager applies to all managers”. *Secondly*, it implies that “managerial knowledge and experience are transferable from department to department and from enterprise to enterprise; merchandising executives may be shifted to manufacturing; or the military commander to peaceful pursuits; or foremen from flour milling to warehousing; and production managers to sales.”

Having discussed the nature and classification of various functions of management, let us now discuss each of these functions, namely, planning, organising, staffing, directing, coordinating and controlling, in detail. Planning is the most basic and important of all management functions and precedes all other functions or management activities. Every manager plans before he acts.

According to George R. Terry, it is “the selecting and relating of facts in the visualization and formation of proposed activities believed necessary to achieve desired results”. In the words of Koontz and O'Donnell, “Plans involve selecting enterprise objectives and departmental goals and determining ways of reaching them. Plans thus provide a rational approach to pre-selected objectives.” They further observe that “Planning is deciding in advance what to do, when to do it, and who is to do it. Planning bridges the gap from where we are to where we want to go. It makes it possible for things to occur which would not otherwise happen.”

Planning thus is intellectual or mental process to foresee the future events and decide about the future course of actions and the manner in which they have to be taken. It necessarily involves thinking about; different lines of actions, and would imply looking ahead and conceptualizing the future events and deciding in advance the various alternatives. Of course, the plan has to be linked with the objectives of the firm. Since there may be several possibilities, a series of plans may have to be developed. But all these plans need not be implemented. However, the management must be ready, in advance, to tackle all the future possibilities.

A plan may be formal or informal. It is formal when it is reduced to writing. It would be advantageous to prepare a formal plan specially when a large number of actions has to be taken in future. This is necessary to avoid oversight or inattention by the management. An informal plan is one which is not reduced to writing but is kept only in the mind of the manager. If the number of actions to be taken is less and the actions have to be taken in a short period informal planning may be usefully adopted. Plans may be short-term or long-term depending upon the period which is kept in view while formulating a plan. Similarly, there may be single-use or standing plans. A single-use plan is one which sets a course of action for a specific situation and is used up once the particular goal is reached, whereas the standing plan is one which is designed to be used over and over again.

Planning is to be done at all levels of organisation, but the lower-level management tends to derive its own plans from the plan of higher-level management. However, higher-level management spends more time and energy in planning than lower-level

management. Higher-level management keeps broader objective of the company as a whole and takes a long-run view whereas others keep in mind only immediate objective and sectional approach in planning.

THE MANAGEMENT OF RESEARCH AND DEVELOPMENT

Ideas concerning the management of research and development have undergone rapid development in the last few years and several new management methods have come into use in companies. For eighty years or more R&D activity expanded in medium and large companies, but it has perhaps now reached saturation level. Scientists and engineers in companies often pursued research of their own devising while having only slight awareness of the commercial requirements.

At the same time corporate managers tended to leave R&D to its own devices, both from a feeling that this was something they did not understand, and because R&D issues are never so urgent as the problems of production start-ups and market launches. This was often unwise because the products being launched determined the futures of the companies, yet were sometimes selected for development by R&D without due attention by senior management. Company spending on R&D varies from 0.1 percent of turnover to over twenty percent, with an average of perhaps four percent. Despite its importance R&D has received scant attention in management teaching and textbooks. R&D management almost never features in business school syllabuses or faculty lists, or in the indexes of standard management texts. It is however covered in really excellent handbooks of management.

NEW THINKING

Pressure for change has come from the deep worldwide recession of the early nineties causing companies to look closely at all classes of expenditure, and also from rather belated recognition of technology and innovation as competitive elements. It has been recognised that although competition does indeed take place on the classic grounds of efficiency, price, promotion and marketing, ownership of a technology can also give a profound advantage.

Mastery of the difficult art of making silicon chips restricts that business to a few, while legal ownership of drug formulations enables pharmaceutical companies to make monopoly profits. Both of these advantages originate in the R&D laboratories of the companies concerned. Innovation is of course broader than technology development by R&D, and can occur for example when business advantage is gained simply by buying-in new technology machinery from outside. Innovation is a concern of all the company, and the role of R&D in this is best defined simply as that part of innovation which happens to be done by the people in the R&D department.

Major contributions to new thinking about the management of R&D are the concepts of “third generation R&D” brought together by the consultancy company Arthur D Little, the Stage-Gate system of managing new product development, portfolio management theory of R&D projects and the methodology of technology foresight. These provide difficult but convincing tools for managing an activity that has caused much anguish in the past, and they enable constructive dialogue to take place between R&D and the rest of the company.

THE NATURE OF R&D

R&D work is intrinsically risky. Almost by definition R&D tries out new ideas to see how well they work, seeks information that is as yet unknown, and hopes for ideas as yet unthought of. This means that success cannot be commanded, and most R&D is managed in terms of probabilities.

The laboratory is gambler territory. It is useful to contrast R&D with design, where the assumption is that all the knowledge required exists, and if the design is competent the product will work.

Products that are designed tend to be of modest profitability because the knowledge required is likely to be fairly widely available. By moving into the unknown, R&D accepts risk in return for the chance of greater reward. Managers of R&D have to be able to live with a lot of failure.

MARKETING TECHNIQUES

Marketing is the process of performing market research, selling products and/or services to customers and promoting them via advertising to further enhance sales. It generates the strategy that underlies sales techniques, business communication, and business developments. It is an integrated process through which companies build strong customer relationships and create value for their customers and for themselves.

Marketing is used to identify the customer, to satisfy the customer, and to keep the customer. With the customer as the focus of its activities, it can be concluded that *marketing management* is one of the major components of business management. Marketing evolved to meet the stasis in developing new markets caused by mature markets and overcapacities in the last 2-3 centuries. The adoption of marketing strategies requires businesses to shift their focus from production to the perceived needs and wants of their customers as the means of staying profitable.

The term *marketing concept* holds that achieving organizational goals depends on knowing the needs and wants of target markets and delivering the desired satisfactions. It proposes that in order to satisfy its organizational objectives, an organization should anticipate the needs and wants of consumers and satisfy these more effectively than competitors.

BASIC METHODS OF MARKET RESEARCH

While there are many ways to perform market research, most businesses use one or more of five basic methods: surveys, focus groups, personal interviews, observation, and field trials. The type of data you need and how much money you're willing to spend will determine which techniques you choose for your business.

Surveys: With concise and straightforward questionnaires, you can analyze a sample group that represents your target market. The larger the sample, the more reliable your results will be.

- In-person surveys are one-on-one interviews typically conducted in high-traffic locations such as shopping malls. They allow you to present people with samples of products, packaging, or advertising and gather immediate feedback. In-person surveys can generate response rates of more than 90 percent, but they are costly.
With the time and labour involved, the tab for an in-person survey can run as high as \$100 per interview.
- Telephone surveys are less expensive than in-person surveys, but costlier than mail. However, due to consumer resistance to relentless telemarketing, convincing people to participate in phone surveys has grown increasingly difficult. Telephone surveys generally yield response rates of 50 to 60 percent.
- Mail surveys are a relatively inexpensive way to reach a broad audience. They're much cheaper than in-person and phone surveys, but they only generate response rates of 3 percent to 15 percent. Despite the low return, mail surveys remain a cost-effective choice for small businesses.
- Online surveys usually generate unpredictable response rates and unreliable data, because you have no control over the pool of respondents. But an online survey is a simple, inexpensive way to collect anecdotal evidence and gather customer opinions and preferences.

Focus groups: In focus groups, a moderator uses a scripted series of questions or topics to lead a discussion among a group of people. These sessions take place at neutral locations, usually at facilities with videotaping equipment and an observation room with one-way mirrors. A focus group usually lasts one to two hours, and it takes at least three groups to get balanced results.

Personal interviews: Like focus groups, personal interviews include unstructured, open-ended questions. They usually last for about an hour and are typically recorded. Focus groups and personal interviews provide more subjective data than surveys. The results are not statistically reliable, which means that they usually don't represent a large enough segment of the population. Nevertheless, focus groups and interviews yield valuable insights into customer attitudes and are excellent ways to uncover issues related to new products or service development.

Observation: Individual responses to surveys and focus groups are sometimes at odds with people's actual behaviour. When you observe consumers in action by videotaping them in stores, at work, or at home, you can observe how they buy or use a product. This gives you a more accurate picture of customers' usage habits and shopping patterns.

Field trials: Placing a new product in selected stores to test customer response under real-life selling conditions can help you make product modifications, adjust prices, or improve packaging. Small business owners should try to establish rapport with local store owners and Web sites that can help them test their products.

MANAGEMENT'S USE OF MARKETING RESEARCH

In spite of the rapid growth of marketing research, many companies still fail to use it sufficiently or correctly. Several factors can stand in the way of its greater utilization.

1. A narrow conception of marketing research. Many managers see marketing research as only a fact-finding operation. The marketing researcher is supposed to design a questionnaire, choose a sample conduct interviews, and report results, often without being given a careful definition of the problem or of the decision alternatives facing management. As a result, some fact finding fails to be useful. This reinforces management's idea of the limited usefulness of some marketing research.
2. Uneven calibre of marketing researchers. Some managers view marketing research as little better than a clerical activity and reward it as such. Poorly qualified marketing researchers are hired, and their weak training and deficient creativity lead to unimpressive results. The disappointing results reinforce management's prejudice against expecting too much from marketing research. Management continues to pay low salaries, perpetuating the basic difficulty.
3. Late and occasional erroneous findings by marketing research. Managers want quick results that are accurate and conclusive. But good marketing research takes time and money. If they can't perceive the difference between quality and shoddy research, managers become disappointed, and they lower their opinion of the value of marketing research. This is especially a problem in conducting marketing research in foreign countries.
4. Intellectual differences. Intellectual divergences between the mental styles of line managers and marketing researchers often get in the way of productive relationships. The marketing researcher's report may seem abstract, complicated, and tentative, while the line manager wants concreteness, simplicity, and certainty. Yet in the more progressive companies, marketing researchers are increasingly being included as members of the product management team, and their influence on marketing strategy is growing.

MATCHING QUALITATIVE RESEARCH TO THE RESEARCH OBJECTIVES

Qualitative research can be used as an *exploratory* tool. Where there is uncertainty about a subject, and the researcher does not know what detailed questions to ask, a few depth interviews or groups may be sufficient to provide an understanding and explanations which answer the problem. Imagine that you are a manager of a hospital, eager to improve the standard of patient care.

Your 'clients' include the young, the mentally ill, the old and the very poorly. What they think of your service is decisive in guiding how you develop in the future – but how do you find out what they think?

The very young, the very old and the mentally ill may be unable to tell you. And when do you interview them? If you do so while they are in the hospital, it could bias the result as their treatment may not yet be over and, in any case, they may find it hard to be critical during the period of medical care. It may be better to interview soon after they return home, even though this is the more expensive option.

THINK ABOUT

The next time you talk to someone in business, pay attention to their body language. What does it tell you? What other clues surrounding the person tell you about them such as the books on their shelves, the watch they are wearing, their clothes, and the way they organise themselves? How much more can you learn about this person and their attitudes from paying attention to non-verbal leakage?

Designing a quantitative study for checking on standards of patient care is likely to be complicated and costly. Instead of focusing on the patients themselves, an understanding of the issues could be obtained from people who can speak on their behalf. Group discussions with general practitioners in the hospital's territory could provide a distilled view of the strengths and weaknesses of the hospital and individual departments. Group discussions with GPs could be quickly convened and the results available within a couple of weeks.

A couple of focus groups is not a sizeable study but they could provide a signpost as to where the real problems lie and give focus to any subsequent quantitative research.

In the same vein, researchers may be faced with a marketing problem which needs investigation. Problem solving requires an understanding of the cause and effects and these could be flushed out by discussion in groups or depths. When used in this capacity, qualitative research is a *diagnostic* tool. Typical subjects for analysis could be a downturn in sales, a loss of market share or an increase in complaints. A major UK manufacturer of domestic heating boilers looked at the latest trade association figures and saw that it had suffered a loss of share. This had not been apparent from the company's sales figures which were holding up quite well in a market which was buoyant. Four group

discussions with plumbers who install boilers were sufficient to show that a major competitor was heavily discounting and using direct mail to promote its boilers – a method of promotion which was concealed from the competing suppliers.

Very often, qualitative research is used for *creative* inspiration and guidance. For example, if an advertising agency wants ideas for a new campaign it may get these from the interaction of thoughts arising in focus groups. The creative power of qualitative research need not be confined to teasing out ideas for advertising; it can be used to guide all types of innovative work such as new product development and branding.

A manufacturer of crisps was concerned that the health food trend would eventually affect his market and wanted to know in which way he could respond. Qualitative research was commissioned to find out if, with some modification, crisps could be positioned as a health food.

Qualitative research can also be used to *evaluate* ideas. Focus groups or depth interviews can be used to find out what people think of a different presentation of an advert or pack design. Of course, there will be no large numbers to substantiate the evaluation but the views of the small number of respondents may be sufficiently convincing.

A cable manufacturer wanted to steal a march in the competitive field of house wiring and designed a pack which looked very different to the conventional reels which electricians had used for years.

The pack dispensed the cable from the centre of the reel and so offered a number of advantages. However, electricians are very much driven by habit and group discussions showed if there would be any resistance to the new product and how to overcome it.

FRAMEWORK FOR CHOOSING A RESEARCH DESIGN

One reason why secondary research sources are often omitted is institutional. Most research is carried out by market research agencies which for various reasons find problems in offering desk research as a profit earning service. In this respect in-house research can have an edge over buying-in.

Once desk research is completed (and assuming it does not yield all the information required for the project) primary research or fieldwork can be planned to fill in the gaps.

THINK ABOUT

When did you last visit a commercial library? What do you think you could obtain from a library that you cannot obtain from the internet? Are you aware that some libraries will carry out short desk research assignments for you for a small fee? If you had £5,000 to spend, how much could you find out by desk research?

How much could you find out by primary research?

Decisions in primary research design.

When contemplating the research method for primary data, the following should be considered:

- Whether the nature of the information sought is primarily quantitative or qualitative.

Sampling issues including:

- The universe to be covered *e.g.*: all adults, housewives, buyers of specific products in consumer markets and comparable aspects of industrial markets.
- Sectors or sub-groupings of the whole universe which are of specific interest and need to be considered in sampling design.
- Sampling method and size (number of interviews).
- Data collection methods – *e.g.* face to face or phone interviews, postal survey, observation *etc.*
- How the resulting data will be analysed – particularly relevant for more sophisticated techniques such as conjoint analysis.

The basis of which of the above should be covered in the design, depends on the research objectives, the information coverage, and the accuracy sought. Or at least this should be so in theory. In practice other factors have to be taken into account and especially the resources available including the budget.

For example, the ideal method may be face to face interviews with a sample of 1,000 people in their home. However, but for practical reasons (speed and cost) street interviews with 200 may be used instead. There is, of course, some point beyond which compromises are a danger to the accuracy of the result. What budget should be made available for the research project?

This is not a question of arguing that the budget should be whatever is needed to meet the research objectives at the required accuracy level. It is more a question of what funds are available or can be afforded for the project relative to other calls on business expenditure. Furthermore, even if cash is freely available, there are other considerations and especially the amount at risk in the decision which the research is to guide. For example, a research budget of £30,000 may be well worth spending if it is to guide a decision that entails capital expenditure of £5 million. However, if the decision has much lower cost implications, the value of doing the research will be less and obviously there is no point to spending £30,000 on research to decide whether to invest in a project entailing only this level of expenditure. For this reason, many businesses cannot afford to test the effectiveness of their advertising campaigns because the cost of the research is more than the campaigns themselves.

Apart from a cash budget, the major resource required for a research project is suitably qualified people. An in-house researcher may be able to carry out desk research but would find it very difficult to carry out extensive fieldwork and for this would need the help of an agency.

A research plan needs a timetable. The deadline for the project may well be driven by external events such as a board meeting or input for a marketing plan.

The timeline available for the research often influences the research methods. Face to face interviews may be preferred but telephone interviews are quicker.

Certainly good research can be carried out within a short timetable but beyond some point, quality will be compromised if the timetable is squeezed too much.

THE CASE OF THE SOFTONE LIGHT BULBS

At this stage the reader may find it helpful to think about a case study that incorporated a number of different design issues that have been discussed.

Philips Lighting is an established player in the light bulb market where it faced a number of challenges:

- Consumers are not very interested in the product
- They are sold at low prices
- They have a strong seasonality of sales (winter months) and low growth
- There is considerable competition and margins are low
- Light bulbs have low brand awareness and there is a strong own brand presence.

A product development from the Philips research laboratories was a technique for electrostatic coating the inside of a lamp to produce a softer light.

The additions of pigments offered the opportunity to produce light with a hint of colour. The new product was given the brand name *Softone*. Philips decided to use this product to stimulate interest in a commodity market and to help build its position as an innovative lighting supplier. Market research played a crucial role at all stages in the life cycle of this product development.

The first market research gateway was to establish that the product had significant consumer appeal, especially at a premium price. If the results of the research proved negative, it would be back to the technical drawing board to improve the product features or the launch would be dropped.

As is often the case with new product research, there is considerable pressure for the research to show a green light as the new product will have gained a momentum with many people in the organization banking on its success. The market researchers had to use a method that would deliver a true understanding of the new lamp's appeal and that would stand up to the political attacks it could face if the results were less than positive.

The research method chosen to test the new product's appeal was a survey of 200 face to face interviews with a quota sample of lamp buyers in two locations several hundred miles apart (Dewsbury in Yorkshire and Southampton in Hampshire). Respondents were recruited in shopping malls and brought into "halls". A sample of 200 could be expected to yield an accuracy of + or – 7% (ie if 50% of the sample expected to buy the lamp, the true level among the whole population might be expected

to be between 43% and 57%). Restricting the sample to only two locations ruled out any true regional analysis but would at least indicate whether major geographical differences existed (it was assumed they did not).

Quotas were imposed to ensure that a mix of demographic groups representing the population as a whole was covered. In a hall test of this type and without quotas, there would be an over representation of lower social grades and older people.

The results from this preliminary research were positive and although not many people bought conventional coloured lamps, the interest in the pastel concept was high with 79% saying that they would be very or quite likely to buy the new product. A small price premium was acceptable to most people. It should be pointed out that it would be dangerous to extrapolate the high interest into a propensity to buy figure but it did suggest that there was a significant body of interest and it was worth going forward to the next stage.

Instead of carrying out the hall test, the research team could have opted for a more qualitative approach using focus groups. However, they believed that groups on their own would not have given them a feel for the size of the demand and some quantitative method would still have been needed. In the interests of budget and speed, the groups were skipped. Focus groups did play a part in the next phase of the research as the marketing team needed help with ideas for the pack design.

Philip's design agency developed various alternative pack concepts for evaluation and these were tested in four focus groups. Women aged 22 to 45 (the main group of lamp buyers) were recruited to the groups which were split between the North (two groups) and the South (two groups). It was also decided that the group respondents should be restricted to those with a strong interest in home decoration and colour coordination in their homes.

The focus groups confirmed the high level of interest in the new range of lamps and gave some interesting pointers for the pack designers. The brand *Softone* worked well on its own without the need to emphasise the words pastel or stronger colours. In fact, it was thought most appealing to communicate that the lamps offered just a hint of(colour) because subtlety in lighting was considered paramount. Philips backed the Softone launch with both television and press advertising.

This was going to incur a substantial sum and again, market research was called upon to help guide the decisions. Three alternative TV advertising styles were developed to the *animatics* stage (moving story boards) and the press adverts were developed in *storyboard* form (artist prepared visuals without the detail of the final artwork). Research had to establish the effectiveness of the alternative TV and press treatments in terms of impact, memorability and the interest in the Softone product.

It was thought that focus groups would not be a suitable method to test the advertising concepts as there could be contamination in respondents' views caused by the debate that would be sure to take place. It was considered better to carry out face to face depth interviews and for this the researchers used a *mini-hall test*.

Halls were taken in four locations and at each hall 40 people were shown the TV ads and 40 were shown the press ads. The interviews lasted half an hour and asked questions such as:

- What sort of product is being advertised? Is it a *new* product or just another coloured light bulb?
- What is the impact of the advert? Would it stop the reader/viewer and catch their attention?
- What is the message of the advert?
- Is the advert believable?
- Is the product for the respondents? Would they buy it?
- Where would it be used?

Having seen the adverts, the great majority of respondents were enthusiastic about Softone, they saw it as innovative and many were keen to have it in their homes. None of the adverts had a particularly good impact but there were some strong pointers as to how to make improvements.

The more direct approaches in the adverts were considered important for a novel product where guidance in home use was thought to be needed. One of the TV ads and one of the press ads was stronger than the others and these were the ones that were modified by the agency for final use. (In theory the revised ads should have been tested again but it was essential not to delay the campaign any further so no further testing took place).

The campaign was extremely important to Philips and it was considered important that its effectiveness should be measured in terms of brand awareness, whether the product had been bought (and intentions to repurchase) and the recall of the content of the ads which were used. In the choice of the tracking design, the results from each stage of the research had to be capable of yielding awareness levels and other measures that showed a true difference and not one that could be within the bounds of sampling error. A sample size of 200 was considered the minimum in each of the 12 TV regions which meant that a national sample of 2,400 lamp buyers would need to be interviewed in each wave of the test. Philips knew from previous research that amongst a random sample of the adult population, two thirds would have bought a lamp within the last six months and therefore be classed as “lamp buyers” for the purposes of the survey. This meant that a minimum sample of 3,500 adults would have to be approached to achieve the 2,400 interviews.

The study aimed to track awareness and behaviour over a period of three years and this required nine stages – each of them with the large sample of 2,400 interviews. This would have been a very costly exercise if nine ad hoc studies had been carried out. Omnibus surveys offered the solution. Omnibus surveys run regularly, and subscribers can buy space on the questionnaire on a pay as you go basis.

Using this tracking measure, Philips was able to see that the recall of its Philips/Softone brand doubled from 15% to over 30% in the three years of the campaign.

Sometimes market research is used in an ad hoc way to guide a business decision. In the Philips Softone case study we saw how research was used continuously over a number of years to ensure the successful goal of building a significant share in the lighting market with a new and innovative brand of lamp.

The case study also illustrates that there are no methods or research designs that have to be used to meet an objective. Research designs are often a compromise between the trade-off of the accuracy of the information required against the time and cost of carrying out the research. We saw that Philips research team used a mixture of research designs:

- Hall tests to establish that the lamp concept was attractive and would be successful.
- Focus groups to get ideas for the pack design.
- Depth interviews to test TV and press ads.
- Omnibus interviews to track brand recall.

Market research followed the Softone product through a number of stages of its life cycle. In the Softone case study, there was very little secondary research used because Philips already had a good knowledge of the lighting industry. For other companies there may be a considerable use made of secondary research in the early stages of a product's life cycle.

EVOLUTION OF CLASSICAL APPROACH TO MANAGEMENT

Traditional process of learning is either through observation and experiment. Nature or environment is considered uniform and when we observe certain phenomenon or events uniformly leading to the same result or results, we conclude a cause and effect relationship between the two. This is learning by observation or in other words by experience. Earlier thinkers on management followed this approach in developing theories of management.

Learning principally is through empirical process and through analysis of the data collected through observation. Draw the principles of management by looking at and analysing the jobs that all managers commonly do. This approach served as a starting point for pioneers on management science to verify the validity and improve the applicability of the principles and practices of management. Analysis of observed data is what constitute a case study. The observational method of case study helps arriving at logical conclusions about past experience and to test the same as standards for future events.

The German sociologists, Max Weber followed the classical approach and developed his theory of Bureaucracy, which portrays the structure and design of organisation characterised by a hierarchy of authority, formalised rules and regulations that serve to guide the coordinated functioning of an organization.

BASIC POSTULATES OF THE CLASSICAL APPROACH BY MAX WEBER

1. Management of an organization is considered as a chain of inter-related functions. The study of the scope and features of these functions, the sequence through which these are performed and their inter-relationship leads one to draw principles of management suitable for universal application
2. Learning principles of management is done through the past experiences of actual practicing managers
3. As business environment consists of uniform cycles exhibiting an underlying unity of realities, functions and principles of management derived through process of empirical reasoning are suitable for universal application
4. Emerging new managers through formal education and case study can develop skill and competency in management concepts and practices
5. The classical approach also recognised the importance of economic efficiency and formal organizational structure as guiding pillars of management effectiveness.
6. Business activity is based on economic benefit. Organizations should therefore control economic incentives

NEOCLASSICAL THEORY OF MANAGEMENT

There are 3 neoclassical theories:

Human Relations theory: Explains the modern advancement of Human Relations Management theory which takes into account human factors like the employer-employee relationship. Human relations theory is largely seen to have been born as a result of the Hawthorne experiments which Elton Mayo conducted at the Western Electrical Company.

The important strand in the development of modern management was the increase in attention to the human factors, which has become known as the 'human relations school of management.' The core aspect of Human Relations Theory is that, when workers were being observed and included in the research, they felt more important and valued by the company. As a result, their productivity levels went up significantly. This represented a significant departure from many of the classical theories, particularly Fordism, as it went against the notion that management needed to control workers, and remove their autonomy at every step. Instead, it showed that by engaging with workers and considering their requirements and needs, company's could benefit from increased productivity.

Behavioral theory: The behavioral management theory is often called the human relations movement because it addresses the human dimension of work. Behavioral theorists believed that a better understanding of human behaviour at work, such as motivation, conflict, expectations, and group dynamics, improved productivity.

The theorists who contributed to this school viewed employees as individuals, resources, and assets to be developed and worked with — not as machines, as in the past. Several individuals and experiments contributed to this theory.

Social systems theory.: Developed by Niklas Luhmann is an option for the theoretical foundation of Human Resource Management (HRM). After clarifying the advantages of using a grand (social) theory as the basic theoretical perspective, the roots of this social systems theory - the deterministic view of systems as machines, the open systems approach and non-linear systems theory - are addressed.

Based on the view of social systems as autopoietically closed systems, five major contributions to a theoretical foundation of HRM are identified: (1) the conceptualisation of organising and managing human resources as social processes, thus overcoming an individualistic angle; (2) the new importance of individuals as essential element in the system's environment; (3) the abstention from far reaching or highly unrealistic assumptions about the 'nature' of human beings; (4) the interaction between various levels and units of analysis built into the theory which is essential for comprehensive and in-depth analyses of HR phenomena and (5) the openness for additional theories for which social systems theory provides the overall framework. George Elton Mayo was in charge of certain experiments on human behaviour carried out at the Hawthorne Works of the Western Electric company in Chicago between 1924 and 1927. His research findings have contributed to organizational development in terms of human relations and motivation theory.

Elton Mayo's contributions came as part of the Hawthorne studies, a series of experiments that rigorously applied classical management theory only to reveal its shortcomings. The Hawthorne experiments consisted of two studies conducted at the Hawthorne Works of the Western Electric Company in Chicago from 1924 to 1932. The first study was conducted by a group of engineers seeking to determine the relationship of lighting levels to worker productivity. Surprisingly enough, they discovered that worker productivity increased as the lighting levels decreased — that is, until the employees were unable to see what they were doing, after which performance naturally declined. A few years later, a second group of experiments began. Harvard researchers Mayo and F. J. Roethlisberger supervised a group of five women in a bank wiring room.

They gave the women special privileges, such as the right to leave their workstations without permission, take rest periods, enjoy free lunches, and have variations in pay levels and workdays. This experiment also resulted in significantly increased rates of productivity.

In this case, Mayo and Roethlisberger concluded that the increase in productivity resulted from the supervisory arrangement rather than the changes in lighting or other associated worker benefits. Because the experimenters became the primary supervisors of the employees, the intense interest they displayed for the workers was the basis for the increased motivation and resulting productivity. Essentially, the experimenters became a part of the study and influenced its outcome. This is the origin of the term Hawthorne effect, which describes the special attention researchers give to a study's subjects and the impact that attention has on the study's findings.

RESEARCH POLICY AND RESEARCH MANAGEMENT

There has been a clear increase in the role of the private sector in agricultural R&D, both profit and non-profit. This is due to two forces. On the demand side, it has resulted from democratization and decentralization of governance, increasing organization of civil society, and structural adjustment policies which have reduced the role and the budgetary capacity of the public sector. This has led to the privatization of many agricultural services including variety development, seeds, animal health services, marketing boards, water delivery, and technology transfer. On the supply side, advances in biotechnology, stronger intellectual property protection, improved information systems that facilitate access to scientific and market information, and more open market policies have expanded the range of technologies whose benefits can be appropriated by the private sector. For technologies which remain public goods, the supply side has also frequently been privatized under funding by foundations or national and international grants. Transfer of services to the private sector has been unevenly successful and the distribution of benefits from the change has been unequally distributed across the farm population, with some sectors benefiting while others have been left unattended. The CG focus on research management consequently needs to be broadened to understand the whole spectrum of public, non-profit, and private actors involved. Steps have been taken in this direction, with ISNAR working with NGOs, and looking at the complementary roles of the private and public sector, particularly in Latin America.

The panel consequently recommends that the CG mandate on research policy and research management be broadened to capture not just the traditional public management dimensions of R&D, irrigation, and forestry but also the public-private interface, the co-production of public goods by public and private sectors, and the role and management problems of NGOs and other non-profit organizations of civil society.

POLICY AND MANAGEMENT DECISIONS

We observed that most CG policy research had addressed the question of policy impact and policy advice, while little had been done on policy making and policy-making advice. Yet, the full payoff from policy impact analysis requires that it be imbedded within policy making analysis. This is because the political feasibility of policy recommendations requires assessing policy making.

For instance, recommendations of decentralized governance for participatory rural development may be vacuous, as shown by the experience with community development in India, without an understanding of the decision-making process at the local level. The policy-making game is fundamental for both positive analysis and for policy-making advice. Equally, important is to assess the credibility of policy in the sense of time consistency. If policy making institutions are not credible, for instance a Central Bank

which is not autonomous from government and need to assume deficits created by soft budgets constraints, neither is monetary policy credible.

The need for political economy analysis also applies to management research. It is important in water management since much of the planned allocation of water is perverted by rent seeking. Patterns of rent seeking need to be understood to improve the efficiency of existing systems, to facilitate the transition to water markets, and to design effective water users' associations. It is also important in research management since rent seeking pressures distort the socially optimal or CG-optimal allocation of resources. In addition, not all resources that enter a research system contribute to research, with extensive deviations of funds in response to rent seeking forces in some countries.

This Panel is informed of previous recommendations made by internal and external reviewers as well as of past attempts to develop this important area of research and of modest ongoing initiatives. It recommends that every effort be made to continue to explore possibilities of making headways in research on the political economy of policy and management decisions.

RESEARCH ON INSTITUTIONS

An area of economic theory where there have been major advances in the last fifteen years is the "new" institutional economics. These theoretical advances have allowed to bring behaviour into traditional institutional economics and to thus understand the determinants of the choice of institutions for efficiency purposes.

This body of theory postulates that technological and institutional changes are brought about to reduce transactions and transformation costs (North and Wallis, 1994). In particular, market failures due to transactions costs or to problems of imperfect information induce the quest for institutional alternatives that allow to achieve the objectives sought (production, consumption, or exchanges) more efficiently than through markets. This theory has been highly successful in the field of agrarian institutions, for instance to explain the choice of land tenure contracts (sharecropping versus fixed rent), optimum farm size (family versus commercial farms), labour relations (permanent versus seasonal labour; time versus piece rate wages), vertical integration, the Grafting of informal financial institutions (group lending, savings and loan cooperatives), the organization of cooperative users associations (water users' associations, service cooperatives, social forestry schemes), *etc.* It has also been important in the field of the environment, for instance for the definition of rules for the management of common property resources and the definition of incentive contracts for the conservation of resources. Finally, it has also been highly successful in the field of industrial organization and management to answer such questions as why firms are organized the way they are, how to design incentive compatible regulatory agencies, and how to design incentive systems in firms and organizations.

In general, use of this approach has been seriously lagging in CG research. Notable exceptions include work on intra-household food distribution and inequality (*e.g.*,

Haddad and Kanbur, 1990), and work on mutual insurance schemes and consumption smoothing using the ICRISAT panel data (but largely done by non-CG scientists). We are aware of the fact that some of the research on institutional economics has been highly academic, with little of relevance for normative policy purposes and risks of engaging in marginal issues for the CGIAR. Yet, this is not the rule. Very profitable uses have been made of this approach, ranging from socioeconomic research on household behaviour under imperfect markets, to the Grafting of agrarian institutions in response to the vacuum left by state contraction, and to the design of research and management institutions for technology and water among others. In management research, institutional issues arise in the design of the optimum interface between private and public roles. For germplasm conservation, questions of incentive structures for in-situ conservation require the design of implementing and regulatory agencies.

This Panel recommends that all Centres make a serious efforts to get better acquainted with the potential that the “new” institutional economics offers for their socioeconomic, management, and policy research. It suggests that scientists concerned in different centres collaborate in systematically exploring areas where worthwhile advances could be made using this approach.

COMMON PROPERTY RESOURCES

The theme of common property resources appears in the research agenda of many Centres. Indeed, this is an important issue for many resources with which the System is dealing: open access and community crop and range lands, irrigation systems, community forestry, and open access fisheries. The topic has become very fashionable in universities, in part because it is quite interesting from a theoretical standpoint. However, there is little good empirical work. Like for village and household studies, the CG could make a distinct contribution by organizing and sustaining data collection on the impact of common property on performance. Elinor Ostrom’s work has attracted attention because she was able to collaborate with strong theoreticians while keeping the theoretical exercises focused on relevant questions. The CG could pursue this example, seeking collaborations with universities where there is good theory and helping make the link to empiricism and policy on the types of questions relevant for the CG.

The current large emphasis given to research on common property resources in the System seems fully warranted and this Panel suggests that this effort should be pursued, particularly in terms of solid empirical research. There is somewhat of a contradiction between the mandate given by the CG to its Centres, namely to give priority to strategic research, *i.e.*, to research which is producing international public goods, and policy research since policy is fundamentally an issue for which the field of application is the nation state, with all its specificities and idiosyncracies. Cross-country policy studies in which IFPRI engages are appropriate to isolate the common dimensions of policy to a number of countries and to recommend a set of generic policy adjustments. In general, IFPRI’s research has shied away from close intermingling with particular country

governments, except at a fairly abstract level. If generic research ends up extracting generic policy conclusions and making generic policy recommendations, much of the potential value of research for policy advice may be lost. This is because, the bulk of policy advice is highly country specific as well as specific to a particular contextual instance (conjuncture), for example in terms of the particular coalition of forces that prevails at the moment when policy advice is made. There are several solutions to this dilemma. One is to consider that IFPRI's role is more to demonstrate how to do policy research and to suggest an array of alternative policy options than to make country specific policy recommendations. If this is the case, IFPRI must be careful that the research teaching function is indeed fulfilled and that specific countries are able to extract the country level policy recommendations that derive from the analysis. Leaving this to the initiative of the reader of a generic study is too much to expect. The second solution is to engage in collaborative research, as IFPRI does, explicitly giving to country counterparts the responsibility of extracting country-specific policy conclusions. This requires more than simple co-participation of nationals to the research effort. It also means that IFPRI must assist in the national effort at extracting specific policy conclusions. This can be done in the context of preparing materials for national policy fora, as has indeed sometimes been done.

A related issue is that many of the policy studies recommend partial policy adjustments without resetting them in the country's global budgetary and political processes that establish the opportunity cost of these recommendations. For instance, most of the research on infrastructure simply recommends that more is better than less, but without establishing the opportunity cost of additional investment in infrastructure, and without asking from where funds would be coming. Again, correcting this would require developing a comprehensive view of the country's options and the processes whereby these options are selected and implemented. It would consequently require getting more immersed in the country's policy problems and the constraints under which policy options must be established.

The implication is that more comprehensive country-level policy analyses may be needed. This would significantly recast the definition of research priorities, for instance focusing on countries engaging in significant policy reforms, say Mexico or Vietnam, and proceeding with methodologies for comprehensive assessment of the set of policy options open to them in a particular time frame.

The Panel does not claim to have the answer to these difficult questions. It recommends that IFPRI and other Centres involved in policy and management research devote some thinking to the issue of generic versus country-specific policy studies before the next EPR and that this issue be addressed by the EPR itself.

MICRODATA

Many CG Centres and NARSs have conducted household-farm or farm surveys, for instance ICRISAT for household-farm and IRRI for farm surveys. These data sets vary

in quality and in comparability. Low-quality data are essentially useless. Sometimes, low quality is due to limited design and scope. For example, the T&V Extension Systems all collect Monitoring and Evaluation data. To this Panel's knowledge, none of these data bases has led to published work, even though millions of dollars have been spent in Africa. Sometimes, the low quality is due to poor management and control (you cannot manage these from Washington). We are not advocating that existing data be centralized. On the other hand, there clearly exists a need to better document existing data bases, both biological and socioeconomic, and to advertize opportunities for collaborative analysis, particularly for post-graduate students seeking data for their theses.

There is also a need for a more systematic process of regular surveying through the CG-NARSs system, including and especially through panel data. This is again an instance where the outposting of IFPRI senior staff for joint projects involving primary data would be mutually beneficial. There is now enough experience to proceed to a case format in panel data collection-with local variations and special studies. Agricultural research and extension systems have a stake in measuring farmers. So far, this has not been possible, but it should be possible. In general, economists are running out of data and theory has made advances which have not, for this reason, been followed by empirical validation.

POLICY AND MANAGEMENT RESEARCH IN THE SYSTEM

The overall importance of social science research in the CGIAR and the relative importance of its three components-socioeconomic, policy, and management research. These data are derived from answers to the questionnaire sent to Centres for this study. The overall allocation of CG scientists to the social sciences is 17.4% and the corresponding budget allocation is 14.8%. This indicates that the social sciences have assumed a major role in the System, raising the question, whether this share seems to be too high or not. The smaller share of budgets compared to scientists presumably reflects the fact that social scientists are cheaper because they do not require extensive laboratory facilities. Social scientists are not only engaged in research but also importantly in outreach, including institutional development, and in training activities.

In addition to IFPRI, IIMI, and ISNAR which are primarily social science oriented, some other Centres are heavily vested in the social sciences. This includes CIFOR (34.8% of its scientists), ICRAF (17.3%), ICARDA (16.4%), ILRI (14%), CIP (13.5%), ICLARM (12.5%), and CIAT (10.2%). The larger crop improvement and primarily biophysical centres such as CIMMYT (8.1%), IRRI (8.3%), and ICRISAT (8.9%) have lower shares of their scientific staff in the social sciences. In the "Other Centres", socioeconomic is vastly more important than policy and management research: overall the distribution of scientists is 8.3% to socioeconomic, 3.2% to policy, and 0.4% to

management. In the three social science centres, policy is the dominant activity due to IFPRI's weight (45.3% of all scientists), followed by management (9.8%), and socioeconomics (9.2%). These three social science centres account for 103 of the System's 937 scientists, or 12.4% of the total.

POLICY RESEARCH

The lead Centre for policy research is IFPRI with a budget of US\$9.3 million in core funds and US\$4.5 million in complementary funds, or a total budget of US\$13.8 million in 1994. At that date, IFPRI had 32 research fellows (including the DG, division directors, and a visiting research fellow) and a total of 30 social scientists. Research is organized in four divisions: environment and production technology, markets and structural studies, food consumption and nutrition, and trade and macroeconomics. Institution strengthening is managed by an outreach division. The MTP focuses on four areas of research: accelerated growth and transformation of agriculture, natural resource management policies, market economic reforms and trade policies, and household food security and nutrition. It is implementing this programme through 17 multicountry programmes that focus on strategic research issues with expected large payoffs. This research is wide ranging, covering subjects from macro problems of structural adjustment and its impact on agriculture to intra-household and nutritional effects of credit programmes and technological change policies. IFPRI is a highly visible institution, with the 20/20 Vision for Food, Agriculture, and the Environment programme as a case in point. It has generated an intense flow of publications and manages a large number of collaborative research programmes with foreign institutions and with other CG Centres. In general, the relevance and intensity of IFPRI's contributions have been highly praised in the economic profession and international community. Its scientific credibility is well established and it is widely known in LDC policy circles. According to medium-term plans and answers to the questionnaire, other Centres with involvement in policy research are as follows:

- CIAT has a small commitment to policy research, identifying only two areas of work: the dynamics of land use, in collaboration with IFPRI, and the policy determinants of sustainability.
- CIFOR has a significant commitment to policy research, with 44% of its social scientists devoted to this activity and 6.5% of the Centre's total research budget. However, it stresses the tendency of its scientists to work on socioeconomic and policy issues in an interrelated fashion, thus making separation between the two difficult. Themes addressed include the determinants of diffusion of technical improvements in forestry, the policy determinants of sustainability of forests resources, policy options to increase employment and income from forests, equity in the distribution of benefits from forestry, and extra-sectoral influences on tropical forests including the role of the policy-making process. Specific research projects include the

formulation and implementation of African forest policies, international comparisons of the intersectoral linkages with tropical forests, links between policies and livelihoods in the Amazon forest margins, and reforestation policies on degraded lands in Asia.

- CIMMYT does not have projects with a main focus on policy research, but much of its socioeconomic research has policy components. Examples of sub-projects with a policy content are analysis of the determinants of adoption of resource-conserving cropping practices in Mexico and Central America (with IFPRI and CIAT) and preparation of materials for workshops on policy reforms in Central America.
- CIP similarly does not consider that it has distinct policy, management, and socioeconomic projects, but that these activities are all integral parts of larger, umbrella-like multi-disciplinary projects, with differential weights given to the three social science components according to the nature of each project. Projects with an important policy component include the characterization of constraints and opportunities for potato and sweet potato production in the Andes, evaluation of the impact and sustainability of potato production technologies, determinants of the expanding utilization of potato in less developed countries (with IFPRI), and post-harvest management of Andean food commodities.
- ICARDA has a large share of its scientists devoted to the social sciences (16.4%) although only 2.7% are engaged in policy research. However, boundaries between socioeconomic, management, and policy research are again stated not to be clear. Projects with a significant policy component include analysis of land tenure and public management of common property resources including rangeland and groundwater resources, management of natural resources at the farm level with emphasis on the behavioural implications of distortive policy incentives, and constraints on adoption of improved practices, particularly for range management, including the role of inappropriate incentives that lead to misuse of resources. It collaborates with IFPRI in a regional programme for integrated crop/livestock production in WANA.
- ICLARM has significant socioeconomic research but very limited explicit policy activity. Research with policy content includes the development of systems for the co-management of fisheries, and integrated resource management in small-scale mixed-enterprise farming systems. It plans to collaborate with IFPRI on themes of common property resources and fisheries policy.
- ICRAF allocates 15% of its budget and 17.3% of its scientists to social science research. It has an explicit policy research programme where the main themes are the identification and analysis of policy constraints to adoption of agroforestry technologies and the development of alternative policies to facilitate adoption.

- ICRISAT strongly stresses the need for and existence of a continuum between socioeconomic and policy research. It has had a significant programme in the socioeconomic analysis of farming systems, household behaviour, and village economies, with important policy implications. It collaborates with IFPRI on resource management research.
- IITA has a small social science programme but 41% of the social science budget is allocated to policy research. Policy research is contained in projects addressing the determinants of adoption of improved practices and systems in moist savanna agroecosystems (in collaboration with ILRI), and opportunities for improving welfare while arresting resource degradation in humid forest zone development (in collaboration with ICRAF).
- ILRI (formerly ILCA) has been actively involved in policy research with 8 senior social scientists and 4 associate social scientists, and 15% of its budget allocated to socioeconomic (12%) and policy (3%) research. However, socioeconomic and policy research are again difficult to separate. ILRI's policy programme focuses on four research themes: market and consumption of livestock products, animal health and disease control policies, macroeconomic reforms and livestock sector development, and resource management policy. It has defined four areas of collaboration with IFPRI: production systems in Sub-Saharan Africa; macro-economic reforms, regional integration, and the livestock sector in West Africa; reforms for animal-health input markets; and determinants of competitiveness of and demand for domestic dairy products in Sub-Saharan Africa.
- While mainly devoted to management research, IIMI also engages in policy research with 15.4% of its social science research budget allocated to this activity. Projects with a dominant policy component include the strengthening of irrigation management in Egypt and improvement of the national water management programme in India.
- IPGRI has a small social science research budget (6%), but 35% of it is devoted to policy research. Themes with policy content include the analysis of policy issues relating to genetic resource management, national and international structures, and intellectual property rights issues relating to plant genetic resources and their conservation and access; and feasibility study on options for a multilateral system for plant genetic resources conservation and use.
- IRRI devotes 7.2% of its budget to social science research but only 1% to policy research. The latter includes long-term projections and policy implications of rice supply and demand in Asia (with IFPRI); and analysis of the roles of technology, price, and non-price factors in the development of rice economies of Asia.

- WARDA allocates 8% of its budget to social science research, half of it for policy analysis. Policy research includes the analysis of comparative advantages of rice production systems in West Africa and the analysis of pesticide use and policies in Côte d'Ivoire.

While principally dedicated to management research, ISNAR and IIMI are also contributing to policy research. A full characterization of policy research in the System would need to identify these contributions as well.

MANAGEMENT RESEARCH

Centres specialized in management research are ISNAR for research on research policy and on the management of research, and IIMI for research on water management.

- ISNAR tackles several management issues in its programmes. Some of these are not “management” in a narrow sense, such as research programme design, which is concerned with both macro-level priority setting that belongs to policy, and with institute level scientific programme management. Other aspects that have received considerable attention are monitoring and evaluation systems generally, as well as some specific thematic issues, such as organizing for natural resource management research. ISNAR’s programme on “Management of Organizations and Resources” gave considerable attention to the development and installation in several countries of a NARO management information system (INFORM). Other aspects of ISNAR’s research include the management of human (including gender), financial, physical, and information resources by NARSs.
- IIMI engages in a wide range of aspects of water management research that includes assessment and improvement of the performance of irrigated agriculture, sector-level management of irrigated agriculture, improvement of public irrigation organizations, local management of irrigation systems, and the sustainable management of water delivery and disposal. Cross-cutting themes include gender issues in the performance of irrigated agriculture, health and environmental problems related to irrigation, analysis of technological options, and development of a global data base on irrigated agriculture. An important feature of IIMI’s management work is that, following extensive privatizations and devolution of management to users, it has expanded its research beyond the public management of irrigation systems to the role of water users associations and the determinants of cooperative behaviour in the management of common-property water resources.
- Among other Centres, most declare not to have a specific management research component, even though management aspects may enter indirectly through socioeconomic and policy research. Centres with no explicit management research are CIAT, CIMMYT, ICLARM, ICRAF, ICRISAT, IITA, ILRI, and WARDA.

- CIFOR devotes a small fraction (1.3%) of its budget to management research. This research includes a survey on research manpower in the South African development community, and identification of the policy and management barriers to application of forestry research results.
- CIP has minimal management research that mainly enters through projects with socioeconomic or policy objectives. This is for instance the case with research on characterization of constraints and opportunities for potato production in the Andes.
- ICARDA is conducting institutional research on property rights and public/local institutions that regulate use of rangelands (in cooperation with IFPRI). It is seeking support to expand its research on the public management of underground water for irrigation.
- IPGRI does research on the design, organization, and management of plant-genetic resource activities in NARSs.
- IRRI conducts management research on the setting of priorities in rice research. This includes analysis of yield gaps, identification of research priorities and estimation of payoffs from research, and assessment of rice-research capacity in the Asian NARSs.

While principally engaged in policy research, IFPRI also conducts management research. This includes public management research which is an important component of public policy research. In addition, IFPRI has become increasingly involved in research on research management, water management, and forestry management, generally in collaboration with the corresponding specialized Centres, namely ISNAR, IIMI, and CIFOR.

SOCIOECONOMIC RESEARCH

This is the most important area of social science research in the Centres other than IFPRI (policy research) and ISNAR and IIMI (management research). The nature of this research has been analysed in another TAC special study and we only make general reference to this type of research principally because it is both essential for policy and management research and often hardly separable from these two. Even IFPRI, ISNAR, and IIMI do considerable socioeconomic research since analysis of the behaviour of consumers, producers, households, communities, associations, managers, rent seekers, and policy makers are fundamental cornerstones of policy and management research.

In principle, socioeconomic research conducted by Centres where biophysical research is dominant has as a purpose to guide research resource allocation towards priority areas, help in technology design, assess the determinants of adoption of new technologies, and determine the impact of diffusion of these technologies on welfare and the environment. For this type of research, the biophysical centres have a comparative advantage over IFPRI as they can most easily mobilize the necessary interdisciplinarity with biological scientists. This research has an important service function in giving

guidelines to researchers and the Centres. In fact, socioeconomic research covers a wide array of issues, showing that it has become an integral component of research programmes as research questions and the CO mandate have become increasingly more complex and holistic. No longer is socioeconomic research introduced as a prop to the definition and diffusion of technological change, but it has become an undissociable component of the search for solutions to complex interdisciplinary problems. This is, of course, the main reason why social science research has acquired growing importance in the CG's total research portfolio.

It is worth noticing that 17 out of 78 scientists devoted to socioeconomic research in the CGIAR, or 22%, are at ICRAF. This percentage raises to 25% when IFPRI, ISNAR, and IIMI are excluded. The Panel could not look into the reasons for this disproportionately high presence of socioeconomic research at this particular Center.

Some examples of the wide array of issues addressed by Centres' socioeconomic research is indicated by the following answers to the Centres' questionnaire:

- CIAT focuses mainly on the analysis of prototype production systems and participatory research methods.
- CIFOR conducts research on household strategies and community differentiation in the production of non-traditional forest products in Sri Lanka, household economics and forest use in Brazil, interactions between environment and population in Thailand, forests and food security, income generation and incentives for forest management, and the social dimensions of sustainability criteria.
- CIMMYT's socioeconomic research addresses questions of global maize research and the maize seed industries, the conservation of germplasm diversity, impact assessment of research in wheat and maize, priority setting and research resource allocation (with IFPRI), the determinants of adoption of long-term conservation investments, and the diagnosis of natural resource management issues.
- CIP's socioeconomic research is integrated in the same topics that are part of its policy research programme mentioned above.
- ICARDA's socioeconomic research focuses on the economic analysis of production systems and the ex ante and ex post analysis of new technology options. Projects address integrated crop/livestock production, water resource management, water harvesting, and farmers' participation in and use of local knowledge at breeding barley for specific adaptation.
- ICLARM conducts socioeconomic research on the dissemination and evaluation of genetically improved fish strains, the bioeconomic modelling of capture fisheries, and bioeconomic models of coastal aquaculture systems.
- ICRAF's socioeconomic research includes the development of typologies of land-use systems and farmers, the development of ecological-economic models which predict the adoption potential and sustainability of improved agroforestry

practices, and the evaluation of the actual ecological, social, cultural, and economic impacts of adoption of agroforestry practices.

- IITA's socioeconomic research includes improving postharvest systems, the diversification of humid forest production systems, improvement of yam-based systems, and the determinants of cassava productivity under various ecological conditions.
- ILRI's research includes a wide range of themes focusing on the economic and environmental impacts of livestock disease control, the economics of mixed crop-livestock systems, the socioeconomic analysis of livestock production and natural resource management, the economics of peri-urban dairy systems, and impact assessment and priority setting studies.
- IPGRI's research addresses the role of gender in plant genetic resource conservation and use, documentation of indigenous knowledge for farmers and scientists, and the effect of conservation and use of germplasm on socioeconomic development through agricultural development.
- IRRI's socioeconomic research focuses on the determinants of adoption of knowledge intensive rice technology, understanding the risks in rainfed rice farming, and the impact of rice technology on rural income distribution and poverty.
- WARDA's research agenda includes farmers' perceptions and adoption of new rice varieties, the economics of rice-based cropping systems, agricultural intensification and women, economic constraints to rice based rotations in the Sahel, differentials in economic efficiency for small and large rice farmers, and the rate of return to rice breeding research.

Ethical Decision Making in Research

Although codes, policies, and principals are very important and useful, like any set of rules, they do not cover every situation, they often conflict, and they require considerable interpretation. It is therefore important for researchers to learn how to interpret, assess, and apply various research rules and how to make decisions and to act in various situations. The vast majority of decisions involve the straightforward application of ethical rules. For example, consider the following case,

Case 1: The research protocol for a study of a drug on hypertension requires the administration of the drug at different doses to 50 laboratory mice, with chemical and behavioural tests to determine toxic effects. Tom has almost finished the experiment for Dr. Q. He has only 5 mice left to test. However, he really wants to finish his work in time to go to Florida on spring break with his friends, who are leaving tonight. He has injected the drug in all 50 mice but has not completed all of the tests. He therefore decides to extrapolate from the 45 completed results to produce the 5 additional results. Many different research ethics policies would hold that Tom has acted unethically by fabricating data. If this study were sponsored by a federal agency, such as the NIH, his actions would constitute a form of research misconduct, which the government defines as “fabrication, falsification, or plagiarism” (or FFP). Actions that nearly all researchers classify as unethical are viewed as misconduct. It is important to remember, however, that misconduct occurs only when researchers intend to deceive: honest errors related to sloppiness, poor record keeping, miscalculations, bias, self-deception, and even negligence do not constitute misconduct. Also, reasonable disagreements about research methods, procedures, and interpretations do not constitute research misconduct. Consider the following case:

Case 2: Dr. T has just discovered a mathematical error in a paper that has been accepted for publication in a journal. The error does not affect the overall results of his research, but it is potentially misleading. The journal has just gone to press, so it is too late to catch the error before it appears in print. In order to avoid embarrassment, Dr. T decides to ignore the error. Dr. T's error is not misconduct nor is his decision to take no action to correct the error. Most researchers, as well as many different policies and codes, including ECU's policies, would say that Dr. T should tell the journal about the error and consider publishing a correction or errata. Failing to publish a correction would be unethical because it would violate norms relating to honesty and objectivity in research.

There are many other activities that the government does not define as "misconduct" but which are still regarded by most researchers as unethical. These are called "other deviations" from acceptable research practices and include:

- Publishing the same paper in two different journals without telling the editors
- Submitting the same paper to different journals without telling the editors
- Not informing a collaborator of your intent to file a patent in order to make sure that you are the sole inventor
- Including a colleague as an author on a paper in return for a favour even though the colleague did not make a serious contribution to the paper
- Discussing with your colleagues confidential data from a paper that you are reviewing for a journal
- Trimming outliers from a data set without discussing your reasons in paper
- Using an inappropriate statistical technique in order to enhance the significance of your research
- Bypassing the peer review process and announcing your results through a press conference without giving peers adequate information to review your work
- Conducting a review of the literature that fails to acknowledge the contributions of other people in the field or relevant prior work
- Stretching the truth on a grant application in order to convince reviewers that your project will make a significant contribution to the field
- Stretching the truth on a job application or curriculum vita
- Giving the same research project to two graduate students in order to see who can do it the fastest
- Overworking, neglecting, or exploiting graduate or post-doctoral students
- Failing to keep good research records
- Failing to maintain research data for a reasonable period of time
- Making derogatory comments and personal attacks in your review of author's submission
- Promising a student a better grade for sexual favors
- Using a racist epithet in the laboratory

- Making significant deviations from the research protocol approved by your institution's Animal Care and Use Committee or Institutional Review Board for Human Subjects Research without telling the committee or the board
- Not reporting an adverse event in a human research experiment
- Wasting animals in research
- Exposing students and staff to biological risks in violation of your institution's biosafety rules
- Rejecting a manuscript for publication without even reading it
- Sabotaging someone's work
- Stealing supplies, books, or data
- Rigging an experiment so you know how it will turn out
- Making unauthorized copies of data, papers, or computer programmes
- Owning over \$10,000 in stock in a company that sponsors your research and not disclosing this financial interest
- Deliberately overestimating the clinical significance of a new drug in order to obtain economic benefits.

These actions would be regarded as unethical by most scientists and some might even be illegal. Most of these would also violate different professional ethics codes or institutional policies. However, they do not fall into the narrow category of actions that the government classifies as research misconduct. Indeed, there has been considerable debate about the definition of "research misconduct" and many researchers and policy makers are not satisfied with the government's narrow definition that focuses on FFP. However, given the huge list of potential offences that might fall into the category "other serious deviations," and the practical problems with defining and policing these other deviations, it is understandable why government officials have chosen to limit their focus.

Finally, situations frequently arise in research in which different people disagree about the proper course of action and there is no broad consensus about what should be done. In these situations, there may be good arguments on both sides of the issue and different ethical principles may conflict. These situations create difficult decisions for research known as ethical dilemmas. Consider the following case:

Case 3: Dr. Wexford is the principal investigator of a large, epidemiological study on the health of 5,000 agricultural workers. She has an impressive dataset that includes information on demographics, environmental exposures, diet, genetics, and various disease outcomes such as cancer, Parkinson's disease (PD), and ALS. She has just published a paper on the relationship between pesticide exposure and PD in a prestigious journal.

She is planning to publish many other papers from her dataset. She receives a request from another research team that wants access to her complete dataset. They are interested in examining the relationship between pesticide exposures and skin cancer. Dr. Wexford was planning to conduct a study on this topic.

Dr. Wexford faces a difficult choice. On the one hand, the ethical norm of openness obliges her to share data with the other research team. Her funding agency may also have rules that obligate her to share data. On the other hand, if she shares data with the other team, they may publish results that she was planning to publish, thus depriving her (and her team) of recognition and priority. It seems that there are good arguments on both sides of this issue and Dr. Wexford needs to take some time to think about what she should do. One possible option is to share data, provided that the investigators sign a data use agreement. The agreement could define allowable uses of the data, publication plans, authorship, *etc.*

The following are some steps that researchers, such as Dr. Wexford, can take to deal with ethical dilemmas in research:

What is the problem or issue? It is always important to get a clear statement of the problem. In this case, the issue is whether to share information with the other research team. What is the relevant information? Many bad decisions are made as a result of poor information. To know what to do, Dr. Wexford needs to have more information concerning such matters as university or funding agency policies that may apply to this situation, the team's intellectual property interests, the possibility of negotiating some kind of agreement with the other team, whether the other team also has some information it is willing to share, *etc.* Will the public/science be better served by the additional research?

What are the different options? People may fail to see different options due to a limited imagination, bias, ignorance, or fear. In this case, there may be another choice besides 'share' or 'don't share,' such as 'negotiate an agreement.'

How do ethical codes or policies as well as legal rules apply to these different options?

The university or funding agency may have policies on data management that apply to this case. Broader ethical rules, such as openness and respect for credit and intellectual property, may also apply to this case. Laws relating to intellectual property may be relevant.

Are there any people who can offer ethical advice? It may be useful to seek advice from a colleague, a senior researcher, your department chair, or anyone else you can trust (?). In the case, Dr. Wexford might want to talk to her supervisor and research team before making a decision.

After considering these questions, a person facing an ethical dilemma may decide to ask more questions, gather more information, explore different options, or consider other ethical rules. However, at some point he or she will have to make a decision and then take action. Ideally, a person who makes a decision in an ethical dilemma should be able to justify his or her decision to himself or herself, as well as colleagues, administrators, and other people who might be affected by the decision. He or she should be able to articulate reasons for his or her conduct and should consider the following questions in order to explain how he or she arrived at his or her decision:

- Which choice could stand up to further publicity and scrutiny?
- Which choice could you not live with?

- Think of the wisest person you know. What would he or she do in this situation?
- Which choice would be the most just, fair, or responsible?
- Which choice will probably have the best overall consequences?

After considering all of these questions, one still might find it difficult to decide what to do. If this is the case, then it may be appropriate to consider other ways of making the decision, such as going with one's gut feeling, seeking guidance through prayer or meditation, or even flipping a coin. Endorsing these methods in this context need not imply that ethical decisions are irrational or that these other methods should be used only as a last resort. The main point is that human reasoning plays a pivotal role in ethical decision-making but there are limits to its ability to solve all ethical dilemmas in a finite amount of time.

PROMOTING ETHICAL CONDUCT IN SCIENCE

Many of you may be wondering why you are required to have training in research ethics. You may believe that you are highly ethical and know the difference between right and wrong. You would never fabricate or falsify data or plagiarize. Indeed, you also may believe that most of your colleagues are highly ethical and that there is no ethics problem in research.

If you feel this way, relax. No one is accusing you of acting unethically. Indeed, the best evidence we have shows that misconduct is a very rare occurrence in research, although there is considerable variation among various estimates. The rate of misconduct has been estimated to be as low as 0.01% of researchers per year (based on confirmed cases of misconduct in federally funded research) to as high as 1% of researchers per year (based on self-reports of misconduct on anonymous surveys).

Clearly, it would be useful to have more data on this topic, but so far there is no evidence that science has become ethically corrupt. However, even if misconduct is rare, it can have a tremendous impact on research. Consider an analogy with crime: it does not take many murders or rapes in a town to erode the community's sense of trust and increase the community's fear and paranoia. The same is true with the most serious crimes in science, *i.e.* fabrication, falsification, and plagiarism. However, most of the crimes committed in science probably are not tantamount to murder or rape, but ethically significant misdeeds that are classified by the government as 'deviations.' Moreover, there are many situations in research that pose genuine ethical dilemmas.

Will training and education in research ethics help reduce the rate of misconduct in science? It is too early to tell. The answer to this question depends, in part, on how one understands the causes of misconduct. There are two main theories about why researchers commit misconduct. According to the "bad apple" theory, most scientists are highly ethical. Only researchers who are morally corrupt, economically desperate, or psychologically disturbed commit misconduct. Moreover, only a fool would commit misconduct because science's peer review system and self-correcting mechanisms will

eventually catch those who try to cheat the system. In any case, a course in research ethics will have little impact on “bad apples,” one might argue. According to the “stressful” or “imperfect” environment theory, misconduct occurs because various institutional pressures, incentives, and constraints encourage people to commit misconduct, such as pressures to publish or obtain grants or contracts, career ambitions, the pursuit of profit or fame, poor supervision of students and trainees, and poor oversight of researchers. Moreover, defenders of the stressful environment theory point out that science’s peer review system is far from perfect and that it is relatively easy to cheat the system.

Erroneous or fraudulent research often enters the public record without being detected for years. To the extent that research environment is an important factor in misconduct, a course in research ethics is likely to help people get a better understanding of these stresses, sensitize people to ethical concerns, and improve ethical judgement and decision making.

Misconduct probably results from environmental and individual causes, *i.e.* when people who are morally weak, ignorant, or insensitive are placed in stressful or imperfect environments. In any case, a course in research ethics is useful in helping to prevent deviations from norms even if it does not prevent misconduct. Many of the deviations that occur in research may occur because researchers simply do not know or have never thought seriously about some of the ethical norms of research. For example, some unethical authorship practices probably reflect years of tradition in the research community that have not been questioned seriously until recently. If the director of a lab is named as an author on every paper that comes from his lab, even if he does not make a significant contribution, what could be wrong with that? That’s just the way it’s done, one might argue. If a drug company uses ghostwriters to write papers “authored” by its physician-employees, what’s wrong about this practice? Ghost writers help write all sorts of books these days, so what’s wrong with using ghostwriters in research?

Another example where there may be some ignorance or mistaken traditions is conflicts of interest in research. A researcher may think that a “normal” or “traditional” financial relationship, such as accepting stock or a consulting fee from a drug company that sponsors her research, raises no serious ethical issues. Or perhaps a university administrator sees no ethical problem in taking a large gift with strings attached from a pharmaceutical company. Maybe a physician thinks that it is perfectly appropriate to receive a \$300 finder’s fee for referring patients into a clinical trial.

If “deviations” from ethical conduct occur in research as a result of ignorance or a failure to reflect critically on problematic traditions, then a course in research ethics may help reduce the rate of serious deviations by improving the researcher’s understanding of ethics and by sensitizing him or her to the issues. Finally, training in research ethics should be able to help researchers grapple with ethical dilemmas by introducing researchers to important concepts, tools, principles, and methods that can be useful in resolving these dilemmas. In fact, the issues have become so important that the NIH and NSF have mandated training in research ethics for graduate students.

COMPLEXITY THEORY AND SOCIAL RESEARCH

The implications of this reformation have been identified by the Gulbenkian Commission on Restructuring the Social Sciences (1996). Not only are the boundaries which constitute disciplines in the social sciences rendered irrelevant by this perspective, but it offers a challenge to the demarcation of the social from other sciences. This does not amount to prioritising any domain of knowledge. What we are dealing with here is not a new envy of physics founded on reductionism — quite the contrary. There are three points which need to be made before turning to the implications of complexity theory for the form and content of the social research programme.

The first is that we now have available an account of dynamics which centres on non-linear changes in the properties of systems as a whole rather than the linear trajectories of the elements which are located within those systems. As Prigogine and Stengers (1984) put it, we should replace mechanics with thermodynamics as our central analogy. The second, which follows from the first, is that systems are inherently evolutionary and that changes over time are not reversible — systems are essentially historical. This means that such systems have emergent properties and provides us with a way of conceptualising time which permits a proper understanding of the relationship between the temporal and the social. Finally, we have available a developed, sociologically located discussion of the meta-theoretical character of complexity theory which identifies it as a scientific ontology complementary to the philosophical ontology of critical realism developed by Bhaskar.

For social research the most important implication of this is that the long running problem of relating the quantitative and qualitative programmes no longer matters, apart from to those who deny any substantive reality to anything other than social actions and therefore regard all measurement as inherently reifying. The rest of us are now able to see the quantitative as itself inherently qualitative, given that the nonlinear and emergent character of social and other significant systems means that we can never establish general non-contextual laws and that the quantitative account itself is simply, but very usefully, a way of describing local contexts and transformation of systems as a whole.

This way of thinking corresponds very closely to the idea of the transformation of modes of production. The idea that systems are nonlinear means exactly that changes happen in them which are discontinuous and represent transformations of kind. Local quantitative descriptions of such bifurcation points are the measured history of such qualitative transformations of state.

Social scientists have responded to this new way of thinking in a variety of ways. One has been explicit imitation of the approaches of physical scientists. Such approaches involve the search for chaotic order in dynamic data sets measured across very large numbers of time points. This imitation of the method of physics is described in a set of chapters in Kiel and Elliott dealing with what they refer to as ‘mathematically rigorous

approaches to the statistical analysis of chaotic dynamics in social science data' (1996:9). Approaches including the use of spectral analysis, Lyapunov exponents (Brown 1996) and 'near neighbour techniques' are all intended to enable the identification of chaotic determinism in what appear to be purely random/stochastic systems. Spectral Analysis techniques involve the examination of the pattern of oscillations in a time series in relation to the frequency of those oscillations. They originated in the analysis of oscilloscope readings. The Lyapunov exponent is used to measure the extent to which small changes in initial conditions generate large scale divergence over time. Near neighbour techniques use the idea of attractors. If there are chaotic attractors, the next state of a system may be predicted by examining the history of its near neighbours because they are all drawn towards the same attractor.

These approaches are founded in the US centred tradition of the search for chaos rather than in the European interest in the emergence of order from chaos. Porush spells out the implications of the difference:

Although Prigogine's term "dissipative structures" hasn't sold as well (as chaos) it is in most senses much more accurate. First, it focuses on the dynamic system which undergoes the sudden transformation from apparently chaotic to apparently ordered (original emphasis) on the other side of the bifurcation point. Second, it implies the structure in Prigogine's mathematical model which specifies when such orderliness is not only possible, but likely to arise.

In other words there is not much point in searching for nonlinear equations which will serve to describe the deterministically chaotic form of a chaotic dynamic system and thereby provide a means of predicting the behaviour of that system. Moreover, chaos identification methods require the availability of very large runs of time specified measurements of a kind which are generally not available for social time series. However, people are doing this kind of thing and the cited references explain how.

A MUCH MORE USEFUL APPROACH

Another part of the chaos/complexity programme is simulation. The availability of computers permits both 'experimental mathematics' and the development, through simulations, of very elegant graphical representations of the behaviour of uninterpreted deterministic chaotic expressions.

These are simulations in which parameters are specified so as to describe the initial conditions of real systems, or at least of meaningful analogues of real systems. Given the general (but not absolute) impossibility of solving nonlinear equations which describe the evolutionary behaviour of dissipative systems, it has proved necessary to turn to what Reed and Harvey call 'iconological modelling' in which 'the gaze is more important than deductive logic in grasping the evolution of a chaotic structure.' (1996:310, emphasis in the original). There are relevant discussions of simulation in Gilbert and Doran (eds. 1994) and Gilbert and Conte (eds. 1995) and a discussion of Modelling Complex Phenomena in Lam and Naroditsky (eds. 1992).

A particular advantage of simulation noted by Gilbert and Conte (1995:5) is that it allows for the investigation of 'sociality', that is, of whole system properties corresponding exactly to Durkheim's conception of social facts, and allows for examination of their emergence and transformation over time. This is an important corrective to the overwhelming emphasis on the individual case in quantitative social science, where the individualistic focus involves an implicit and erroneous assumption that it is possible to derive the macro properties of social systems from the aggregation of the properties of the individuals constituting them.

Another way of doing social research in a complexity informed way is to think somewhat differently about the procedures and methods of both quantitative and qualitative social investigation as we are already doing them. The quantitative case is more straightforward. We have available a number of what the Economic and Social Research Council calls 'large and complex data sets', for example the products of annual UK General Household Surveys.

The 'complexity' in these datasets derives from their hierarchical character. At its simplest this means that they contain data about individuals within households. The data also constitute a time series and so what we have in effect is a series of snapshots of the nature of UK society describing individuals and households. With a panel study such as the British Household Panel Study we find that during successive passes the focus remains on the same individuals and households over time allowing for an examination of exact dynamic change.

We also have descriptions of higher system levels. From standard time series as published in Social Trends and Regional Trends we have accounts of the history of national and regional systems within which individuals are located within households. We can see how people's lives change within a changing social world.

A simple method of looking for system changes is the construction of time ordered classifications of the entities at any level within a hierarchical data set, such data sets being conceptualised as operationalizations of the nested systems described by Reed and Harvey (1996). The advantage of this is that the forms of social structures can be examined in relation to parameters which are characteristic of the system as a whole. Of particular interest is the relationship between the actual level of inequality in a social system, which, however operationalized, can only be a whole system property, and the forms of real socio-spatial division within that system. The relative neglect of numerical taxonomy techniques in analyses of social structure in general and of class forms in particular is puzzling. Time sequential classifications are a useful exploratory device. This can be done rather easily for successive time ordered data sets such as the General Household Survey cohorts using readily available cluster analysis procedures *e.g.* quick cluster in SPSS. Bryne (1997) has an example of such an approach using the invaluable Cleveland Social Survey which described household characteristics while locating households in neighbourhood zones within a single locality, and did so over an eighteen year period.

Several authors have proposed the use of logistic regression and loglinear procedures in analysing time orderable data sets. This approach has attractions because it represents a way of handling statistical interaction. Interaction is essentially equivalent to that demarcating characteristic of nonlinear systems, the breakdown of the property of superposition. It has been unfortunate that the emergent complexity generated by interaction has so often been reduced to terms in linear equations. A way of thinking informed by complexity leads us to treat procedures which enable the identification of significant interaction as qualitative exploratory devices rather than as ways of establishing law like models. In practice this is what social scientists really do (discussion of the use of G^2 as a criterion for model selection). Instead of generating models which seek to describe the determinants of outcomes, and writing interactions as terms in those models, we can see interactions as signs of the presence of complexity in general. This kind of approach is likely to be particularly fruitful with panel surveys in which interactions with time are included, these terms being interpreted as indicating nonlinear changes.

Assertions about the possibilities of qualitative analyses are even more speculative. However, given that procedures for computer based analyses of qualitative data are inherently founded around relational data bases, it does seem that in qualitative data sets which are in some way time ordered, we might use the perspective of emergent order with some profit. Certainly as Hayles (1990, 1991) has demonstrated there is considerable power in literary analyses based on these perspectives. An interesting example is provided by the work of Callaghan (in progress) who has interviewed a sample of young adults in Sunderland, both when they were 18 and when they were 23. The interviews can be analysed using Nud^oist, for example. The time ordering allows for the consideration of the existence of different types of trajectory over the transition to adult life. Such an approach allows for reflexive return to informants to assess their views about qualitative descriptions of processes of change.

The absolutely essential element in the approaches being suggested here is that the data are time ordered. We have to be able to examine processes of becoming. It is important to engage in *Analysing Social and Political Change* (Dale and Davies 1994), but from a perspective which prioritises whole system emergent properties. Otherwise we are reduced with the authors in that collection to linear models buttressing the time honoured individualistic fallacy of quantitative social science.

Structured Analysis and Design Methodology

The structured analysis and design methodology classification, also referred to as the data flow modeling methodology, emerged in the mid 1970s to complement structured programming, and today is one of the dominant methodologies being utilized for business-oriented information systems development, such as is done in management information systems (MIS) departments within businesses and governmental agencies.

In this methodology, the real world is described by the flow of data through an information system and the transformations of the data into information as the data move from input to storage to output. A structured analysis and design methodology, as its name implies, adds dimensions that allow it to be rigorous (formalized), repeatable, and measurable. In addition, a new dimension was added starting in the early 1980s—it can be enhanced by automated support called computer aided software engineering (CASE). The CASE acronym has evolved more recently into newer terms such as integrated/software development environment (IDE or SDE). Terms such as these continue to evolve in the hopes of maintaining relevance (and marketing hype) so that in a few years there may be other terms used to refer to this classification of automated software development tool suite. In the first edition of this text I referred to this type of software tool suite exclusively as CASE. In this edition I will mostly refer to this type of software tool suite as IDE or SDE (although a few carry-over CASE terms may still show up).

The predominant focus of the structured analysis and design methodology is to do the analysis and design utilizing a functional perspective (approach). In other words, the systems analyst uses a “*What does the system have to do?*” problem solving approach. With the introduction of CASE/IDE tools around 1984, computerized software support has emerged to provide assistance for drawing the methodology notations, validating and verifying the methodology models that the notations represent, and allowing management to monitor the progress of a project via computerized project management support.

For reference purposes, names such as Constantine, DeMarco, Gane, Hatley, Myers, Orr, Paige Jones, Palmer, Sarson, Stevens, Ward, Warnier (warn-yea), Yourdon, and others have made significant and documented contributions to the structured analysis and design methodology classification.

Tools such as data flow diagrams (DFDs), structure charts, Warnier Orr diagrams, Petri nets, and data dictionaries are often used in conjunction with the structured analysis and design methodology. Remember, conceptually there are only two dominant generic structured analysis and design methodologies (Yourdon, and Gane and Sarson), but businesses have in most cases created their own hybrid versions of the generic ones. A sample DFD that is the dominant model used in structured analysis and design.

THREE CLASSIC CHALLENGES IN INFORMATION SYSTEMS DEVELOPMENT

Three of the most significant challenges that have continuously been associated with the pervading analysis and design methodologies appear to be resolved or at least significantly reduced with an object oriented analysis and design notation and methodology. The first of the three challenges is that most other methodologies construct separate models for the functional and information views of the proposed information system Challenge “A”. In addition, a third component—behaviour or user-interaction—is becoming more important in today’s information systems and needs to be modeled as well. The popularity of interactive, graphical user interfaces and object oriented operating systems, such as Windows, Macintosh and others has shifted programming from a deterministic approach to an event driven approach. This introduces the need for a third view of the system—a behavioral view, and this view is rapidly becoming quite important.

The function and data multiple models have value in that each depicts some portion of the same information system. The difficulty lies in the fact that neither systems analysts nor IDE software tools have been able to completely validate and verify the consistency and accuracy of the integration of the two models simply because they each represent a different perspective of the information system. In object oriented systems analysis and design several integrated models are used to represent all three

views—function, information, and behaviour. This is a very significant move forward because more rigorous validation and verification of the integrated set of models is now possible.

The second challenge that has continuously plagued information systems development is the proverbial transition from analysis to design. Once again, in most of the older methodologies one or more analysis models are created and then a completely new model or set of models is drawn during design that communicates more specifically to the programmers for the impending programming effort. Analysis models have historically been a communication tool between the systems analyst and the user. The design models have historically been a communication tool between the systems analyst and the programmer. With the object oriented analysis and design notation and methodology there is no transition to design. What happens instead is a progressive expansion of the analysis models as they progress through design to include additional refinement details that address the programming task that is approaching. So systems analysts, users, and programmers deal with one model of the information system from analysis all the way through design, coding, testing, and implementation. The resulting information system can be validated back to the original analysis model because it is still the most valid model of the system even after the system is implemented.

The third information systems development challenge. Software developers have historically had two common mindsets: 1) I write better code than any other programmer, and 2) When I need to make a programming change I go directly to the source code and figure out where to make the change, regardless of documentation availability. The first of these two mindsets is probably a bit distorted and fosters a “start from scratch” software development philosophy rather than one that thinks, “Let’s try to reuse someone else’s code as much as possible before writing my own code.” The second mindset simply can consume large amounts of development time as one looks through hundreds if not thousands of lines of source code to make a change. Often, a programmer in the midst of making a software change decides to rewrite portions of the code because it is sub-standard (called “spaghetti code” in some software development circles).

The notion of reuse is becoming more and more “fashionable” (and economically [time and money] necessary). In addition, modification of models is proving to be less time-consuming than modifying source code assuming an IDE software development tool is being used to maintain the models and automatically generate source code for the software developers.

Object technology enabled via an object-oriented methodology coupled with object-oriented models of the proposed information system significantly assists in overcoming these three historical information systems development challenges. The object-oriented approach to developing information systems combines a robust set of inter-related object-oriented models that are more favorably integrated with each other starting with the static and dynamic models created during the early activities of the development lifecycle and culminating with the deployment models that could be created near the end of the

lifecycle. Object-oriented approach proponents argue that there is no transition from analysis to design or need to redraw analysis models to reflect physical design models as in the older methodologies. In my opinion, for now, the object-oriented approach is the best the software development industry has to offer until something better comes along—and something better eventually will come along.

CLASSIFICATION THEORY

Object oriented methodologies and programming are based on classification theory, a fundamental concept most of us learned prior to or during our first few years of elementary school.

The “common methods of organization” characteristic discussed earlier in this chapter is really about Classification Theory. Classification theory simply states that people constantly and consistently use three methods of organization to help clarify their thinking about something. The three methods are:

1. The differentiation of experience into particular objects and their characteristics—objects and their characteristics. An example is being able to distinguish between a tree and its size or spatial relationship to other things in our world, such as a rock, flower, bush, bicycle, and so on.
2. The distinction between whole objects and their component parts—wholes and parts. For example, a tree is made up of leaves, branches, and roots, while a bicycle is made up of wheels, rims, spokes, pedals, seat, and handlebars.
3. The formation of and the distinction between different classes of objects—classes or groups of objects. Examples include grouping trees (maple, elm, palm, eucalyptus) into a tree class; grouping buildings (homes, apartments, offices, churches, schools) into a building class.

People’s common understanding and use of classification theory is one strong reason why an object oriented perspective is a viable approach to analyzing, designing, and building information systems. People are already familiar with it. Learning something totally new is not required.

THE UNIFIED MODELING LANGUAGE (UML) AND AN OBJECT ORIENTED METHODOLOGY

Since their beginning in the late 1980s, several methodologists, researchers, practitioners and authors have put forth specific object oriented notations and methodologies. In one-way or another many of these people contributed and collaborated to help shape the UML into the worldwide standard that it is today. Formally adopted in November 1997, the UML includes a set of nine models and notations for each of the models along with a metamodel—that is, a model of the constructs in UML. The

complete set of UML models and notations is extensive; therefore, for the purpose of this text, a subset of the UML models and notation will be presented, discussed and used.

It is one thing for a group of professionals to create a worldwide uniform set of object-oriented models and notations (the UML). Many other disciplines, such as industrial architecture and electrical engineering, have had standard model and notation sets for many, many years. It is an entirely different—and far more challenging—thing to arrive at a worldwide object-oriented methodology standard. Remember our earlier discussion about generic methodologies—traditional, structured analysis and design, information modeling and object-oriented—and that most organizations tailor these generic ones to create customized versions? Although we in the software development industry overwhelmingly want to create standardized models utilizing standardized notations, we want to create and manipulate these notations and models in different ways—hence, different methodologies.

For the purpose of this text a simple object-oriented methodology will be presented, discussed and used. This simple methodology is based on aspects of a proposed standardized “unified process” and other methodologies. This book is not intended to be the definitive work on the UML nor the methodology to apply it. Interested readers are directed to the Booch, Jacobson, and Rumbaugh references at the end of the chapter for more definitive works. Part of the rationale for using a simple methodology in this book is that more than likely any software development organization you (go to) work for will train you in its own unique object-oriented methodology.

The simplified object-oriented methodology is presented here before giving an overview of the UML. Remember, the methodology is the way in which we will create and manipulate (use) the UML’s models and notations. The methodology consists of seven (7) activities as follows:

1. Identify the purpose of the information system.
2. Identify the primary actors and features of the information system.
3. Identify Use Cases and create a Use Case Diagram for the features.
4. Identify Objects and their Classes and create a Class Diagram.
5. Create Interaction/Scenario Diagrams.
6. Create detail logic for Operations.
7. Repeat activities 1-6 as necessary refining and fine-tuning the various diagrams.

The methodology is listed sequentially above, primarily to illustrate the different activities that are performed during systems analysis. What are not shown in this sequential list are the notions of parallelism, substitution, and omission. Parallelism gives the project manager license to perform activities in parallel when a systems development situation permits. Substitution gives the project manager license to substitute activities out of the standard order presented in the above list when a systems development situation permits. Likewise, omission gives the project manager license to omit one or more detail activities shown in the list when the situation permits.

A simple explanation of each of the methodology activities is presented here since each one is elaborated on in the following chapters.

Activity 1: Identify the purpose of the information system. This activity needs to be done prior to doing any object modeling of the problem domain. In fact, this activity should be done before any type of modeling is done. The user and the systems analyst need to identify, understand, and document the system's intended purpose. For example, a university course registration system purpose could be stated as, "The information system will allow authorized students to add, change, and delete their own course seat reservation." Sometimes, the purpose of the information system can also seem like a feature. When this seems to be the case, the system's purpose is usually broader in scope than any single feature. This activity combined with activity 2 below is analogous to identifying the information system's goals and objectives.

Activity 2: Identify the primary actors and features of the information system. Working and interacting with the user(s), the systems analyst should identify all of the significant (primary) actors and features associated with this new or changing information system. Often, this activity translates into identifying a list of just a few features (for example, less than a dozen) in a small-scale information system or a small-scale one that is undergoing a few changes, and dozens of features for a medium to large-scale information system or a medium to large-scale one that is undergoing significant revision. This activity combined with activity 1 above is analogous to identifying the information system's goals and objectives. A couple of features for a university course registration information system would be "register for course" and "drop a course." Actors represent roles that people or other information systems play with respect to this information system. A couple of actors for a university course registration information system would be "student", "faculty", and "registrar".

Features most often answer the following question, "What can actors do with the information system?" The emphasis is on functionality—what the system will do—rather than technology that addresses how the system will do it.

Activity 3: Identify Use Cases and create a Use Case Diagram for the features. Use cases parallel features—one for each feature. The individual use cases are grouped together to create a Use Case Diagram. This is the beginning point for using the UML and an object-oriented methodology. As mentioned before, activities 1 and 2 above should be done regardless of modeling, notation or methodology. A simple Use Case Diagram showing four use cases that could be part of a university course registration information system.

Activity 4: Identify Objects and their Classes and create a Class Diagram. Objects represent real-world persons, places, things or concepts (*e.g.*, an airline flight from Los Angeles to New York). A Class represents a group of objects that have similar characteristics. For example, there are about 30,000 student "objects" attending our university this semester; we could group them all together and call them the Student class. A Class Diagram shows the collection of all classes in the information system. The end of this chapter illustrates a class diagram for a Video Store information system.

Activity 5: Create Interaction/Scenario Diagrams. Each Use Case in a Use Case diagram is expanded to reveal the details of what it really takes to accomplish “register for a course”, “manage the curriculum”, *etc.* (for example).

This elaboration is called a scenario and it represents a “start-to-finish” series of actions (steps) necessary to accomplish the use case. One of two types of Interaction Diagrams is used for this elaboration—a Sequence Diagram or a Collaboration Diagram. Each of these diagrams is “dynamic” in nature in the sense that if you follow the diagram you will be able to envision “what it takes” to perform the feature that it is describing—registering for a course, for example. A portion of a Sequence Diagram for the “registering for a course” use case.

Activity 6: Create detail logic for Operations. Operations are the transformations and actions that an object or class is responsible for. Operations usually contain the business rules, policies, procedures, logic, and algorithms that are necessary for an information system to be useful. Operations are usually identified as part of identifying objects and classes (Activity 4) or as part of creating the interaction diagrams (Activity 5) or a combination of both. The names above the arrows on the sequence diagram. Each of these names represents operations—the detail actions that the class directly above the arrow’s “tip” is responsible for taking care of. Most of these operations need detail logic within them in order to do what their name implies they do, and this logic is entered during activity 6.

As mentioned earlier, each of the above activities will be discussed and illustrated in much more detail in the following chapters as we learn how to create information system models using the UML. The final section of this chapter presents an overview of a UML Class Diagram for a Video Store Information System.

A UML CLASS DIAGRAM FOR A VIDEO STORE INFORMATION SYSTEM

If you were asked to draw an office building or a home, could you do it? Probably so! Why could you draw an office building or a home? You could do so for at least two reasons: (1) You have a mental picture of what an office building or home looks like (even though your mental picture may be quite different from someone else’s mental picture of an office building or home), and (2) you already know the basic notation needed to draw an office building or home, such as lines, circles, rectangles, triangles, squares, windows, doors, patios, porches, roofs, and so on.

Now if you were asked to draw a picture of a payroll information system or even a university course registration information system using the Unified Modeling Language (UML), could you do it? Probably not, unless you (1) have some understanding of how a payroll information system or a university course registration system functions, and (2) know what the UML looks like—its notation and models—and have some minimal understanding of how to use the UML to draw the models of either of those information

systems. What's really required here? At least two things come to mind: (1) familiarity and understanding with a problem domain (*e.g.*, office building, home, payroll system, university course registration system), and (2) familiarity and understanding of specific notation and models. Familiarity with and understanding of a problem domain means that you have knowledge of a specific problem domain ranging from minimal knowledge all the way to being a subject matter expert (SME)—one who knows a great deal about a problem domain. Notation is a set of symbols used to communicate or represent something. An alphabet is a universal example of a notation. Familiarity with a notation implies that you know which notation symbols are available to use, and understanding of a notation implies that you know the “packaging” for integrating and combining the notation symbols into models. As you know, the real value of an alphabet is being able to combine the alphabetical symbols into meaningful words, sentences, paragraphs, *etc.* Likewise, being able to package the UML notations into the various UML models (diagrams) is its real value.

What follows now is a discussion of a simplistic, partially completed UML Class Diagram that represents the user requirements for a Video Store information system. The intent is to show you in advance what an object-oriented UML class diagram for a problem domain looks like so you can see “the big picture” before diving into all the details of the various UML models. A colleague once made the following comment about textbooks.

He said, “Textbooks are very good at explaining all of the ingredients of a tossed salad (lettuce, tomatoes, onions, carrots, celery, and so on), but few textbooks actually show you a completed tossed salad.” The intent of this section is to show you the completed “tossed salad” allowing for some license with respect to the object oriented UML class diagram model really being complete. As we walk through the following explanation you may find it helpful to draw upon your subject matter expertise as a customer of your favorite video rental store (assuming you have one).

A video sale/rental store information system is used for presenting the object-oriented user requirements UML class diagram model example. The example is incomplete from two perspectives. First, the details of the video store information system problem domain are omitted for simplification purposes. Second, the UML class diagram model that is presented is only showing the basics of the UML class diagram notation. As you might expect, the difficulty of presenting an example at this point in the book is that you need to focus your attention on the object oriented model's UML notation rather than on the details of the specific video store problem domain being depicted by the model. Hopefully, you have purchased or rented a video from a video store and, by doing so, have at least minimal knowledge of the video store problem domain. Some of you may work at a video store or have done so in the past, making you somewhat of an SME. If you fit into this classification, realize that the model presented here probably does not exactly represent the video store information system you have personal expertise with. “What makes systems analysis and design such a difficult activity” and the opening comments in this chapter that suggest that there are potentially many different solutions for every problem domain.

Since this is a model of a fabricated automated information system, please recognize that there could be many other variations to the way this model is being presented. This video store model (with much more detail) was created by a large team of three-dozen or so “aspiring systems analysts” (undergraduate students) working in the classroom for about two months during a semester. The instructor was serving dual roles acting as the senior systems analyst and simultaneously as the video store owner/manager (user). These requirements may or may not be the same as the ones you would come up with working with a different user in a similar problem domain. Remember, the intent here is not to make you subject matter experts for the video store problem domain, but to expose you to a partially completed UML class diagram model of user requirements so that you will know what one looks like “early in the game”.

The more commonly used UML class diagram notation symbols that are used in the Video Store class, 3.11a, 3.11b and 3.11c. The notation is simplistic in that there are only a few symbols, yet it is sophisticated enough to depict large and complex information systems, ones exhibiting functions, data, and behaviour. The symbols in the notation are the Class, Generalization Class relationship, Object association, Object Aggregation association, and Object Composition association.

To assist with the discussion of the Video Store’s UML class diagram and its notation. The UML class diagram model for the video store’s information system for purchasing, selling, and renting inventory items, such as video movies and games, VCRs, and concession items such as popcorn, sodas and candy. In addition, the information system keeps track of these same inventory items when they need to be ordered so that the store will have enough of them to sell or rent to its customers. The video store’s user requirements. The intent here is to show you the resulting UML class diagram model based on the user requirements that we assume have already been identified. You may recall that a user requirements model of a video store’s mission, goals, objectives, and tactics.

Simply stated, classes represent the things that an information system needs to know about within a specific problem domain. Classes can be any of people, places, things, or concepts, such as employee, vendor, store location, rental transaction, video, VCR, and so on. Some classes will never contain objects—we refer to these as abstract classes—while others will contain objects. Abstract classes are used to communicate Generalization (Specialization) relationships among classes. All other classes in the class diagram should contain objects although that is not a rule, just a good guideline. Objects are the actual instances of people, places, things, and concepts and are always associated with a class—in other words, objects can stand alone in a class diagram; they must belong to a class. For example, individual video objects associated with the Video class would be each specific movie video that is available for either rent or sale.

The Inventory, SaleItem, RentalItem, and Transaction are abstract classes because they represent two layers of Generalization class relationships. None of them have (or will ever have) object instances. Video, Game, Concession item, VCR, Employee,

Storelocation, Sales transaction, Rental transaction, Vendor, Member, Purchase order, SaleRentalLineItem, and POLineItem are all examples of classes that can have objects and will more than likely have one or more object instances. You might be thinking, “How do you determine what is a class with objects and what is an abstract class?” The determination of classes versus abstract classes often surfaces during user requirements discussions with the users or later on as the systems analyst refines the class diagram. Classes and abstract classes are treated similarly in most situations, since abstract classes are simply classes that are and always will be void of objects. The way to distinguish an abstract class from a class with objects on a UML class diagram is to look for the Generalization notation symbol. The class touching the arrowhead is abstract—but that is not a rule either, just a good guideline. Flexibility in this regard is important.

Classes have attributes or characteristics that describe them in more detail. For example, the Member class has attributes member Number, first Name, last Name, telephone, address, city, *etc.*

Classes have operations that the class is responsible for accomplishing. The Member class has checkOut Video, checkIn Video, and buyItem as its operations.

The next four UML class diagram notation symbols are the Generalization class relationship, Object association, Object Aggregation association, and Object Composition association. All four of these notation symbols are used to associate classes with other classes in the UML class diagram. All classes have responsibilities. One of their responsibilities answers the question, “Whom do I know?” The Generalization class relationship and the three object associations represent this responsibility in the UML class diagram. Each of these four relationship types is briefly discussed next.

The Generalization class relationship notation symbol is used to associate one generalization class with one or more specialization classes in a hierarchical parent child pattern type of relationship. For example, a SaleItem class, the generalization portion of the parent child pattern, may have several specialization classes, such as Video, Game and Concession Item. The generalization “parent child” pattern. Often specializations can be thought of as being a “type” or “kind” of the generalization. For example, a game is a type of sale item as well as a type of rental item. The Generalization relationship is a class relationship—as its name implies—and not a relationship between the objects of each of the classes involved in the relationship. The next three notations are used to relate objects to other objects within classes.

The Object association UML notation symbol is used to relate two class symbols with each other, however in reality the relationship is really between objects in each of the classes—hence the name Object association. The Object association relationship is used when two unrelated classes in the real-world need to be related in order for an information system to perform its work. Often this type of relationship is considered to be “weak” in the sense that the only reason the two classes are related is for purposes of this information system. For example, both SalesTransaction and RentalTransaction classes are connected to the Member class explicitly because members of the video

store perform these two types of transactions. Member object—people like you and me—are very different from SalesTransaction and RentalTransaction objects. Constraints are indicated on all three types of Object associations. For example, referring to the SalesTransaction to Member Object association, a specific Member may have zero or more (represented by “*” or “0..*”) SalesTransactions and a specific SalesTransaction is related to only one specific Member (represented by “1”). Some members never buy anything in a video store; they only rent and these constraints allow for this condition.

The Aggregation Object association (the clear/white diamond symbol) and the Composition Object association (the black diamond symbol) UML notation symbols are used to associate two class symbols in a “Whole Part” pattern type of relationship. The difference between these two associations and the “whole” object in the association is touching the diamond symbol while the “part” object in the association is touching the end of the line.

For now, a generic example of such an association is an Automobile class and an AutomobilePart class. The Automobile class would be the “whole”, and it would have several AutomobilePart “part” type classes such as Door, Window, Wheel, Seat, Gauge, and so on associated with it.

The PurchaseOrder class represents a “whole” portion of the whole part pattern and the POLineItem class represents the “part” portion of the pattern. Also the SaleRentalLineItem class is the “part” portion of both the SalesTransaction and the RentalTransaction classes. The constraint notations associated with these symbols.

Basically, this notation represents constraints placed on the object connection relationship. For example, the “1..*” notation next to the diamond symbol. Aggregation Object association notation is to indicate how many “whole” objects are associated with one “part” object. The “0..*” notation at the bottom of the same symbol is to indicate how many “part” objects are associated with one “whole” object. The Vendor class has a relationship with the PurchaseOrder class. Vendor represents businesses or individuals that the video store purchases inventory from. PurchaseOrder represents a business transaction and identifies inventory items that are purchased from a vendor. The Vendor to PurchaseOrder relationship constraints indicate that a specific Vendor may have zero or more (represented by the “*”) PurchaseOrders and a specific PurchaseOrder is related to only one specific Vendor (represented by the “1”).

UML class diagram omitted information about attributes and operations simply because I wanted to get the entire class. The attributes and operations. Additional diagram model details become important and necessary at some subjective point during systems analysis. The UML class diagram model has two additional levels of detail: (1) attributes, and (2) operations. Attributes and operations help communicate more about what data (attributes) are associated with the information system, and what functions (operations) are performed by the information system. In mind that these three figures are the exact same model, just with additional details. Due to the extent of the details, the model was divided into three sections in order to fit on the pages of this book. Using a computer

based IDE tool to draw this UML class diagram, you would have it all connected even though you might only see sections of the diagram at any moment in time on the computer's monitor (screen).

Attributes are characteristics that describe a class in more detail. Sometimes the name of a class, such as *StoreLocation* is not sufficient to communicate its intended purpose in the system. Looking at this class's attributes might help communicate its intended purpose. For example, *StoreLocation*'s attributes are *storeNumber*, *storeAddress*, *storeCity*, *storeState*, *storeZipcode*, and *storePhone*. The data values associated with these attributes indicate that the system needs to keep track of the video store's identification information for various reasons. One reason could be to print it on certain reports. Another reason could be the need to consolidate two or more video store's data together in the information system and still be able to electronically separate it into the data belonging to each store.

Are the attributes all of the attributes that this information system will have? Probably not! As the systems analyst progresses through analysis and into design, additional attributes may surface and become important to keep track of and would be added to the information system via a change to the applicable UML models with the user's approval.

Operations, as described earlier, represent the transformations, actions or functions that a particular class of objects must accomplish in order to help fulfill the overall mission of the information system. An operation is associated with one and only one class, the one in the real world most closely associated with the function to be performed. That is why there are more operations in the *Member* class than any of the other classes. Keep in mind that a member initiates much of what happens in a video store, and the "motto" of a class (object) is "I do it myself".

Are these all of the operations that this information system will have? Probably not! As the systems analyst progresses through analysis and into design, additional operations may surface and become important and required for the information system to accomplish its mission. These newly discovered operations would be added to the information system via a change to the appropriate UML models, similar to the way attributes are added, with the user's approval.

The several classes have no attributes and operations shown in them. For example, *Video*, *Game*, *ConcessionItem*, and *VCR* have no attributes and no operations listed in their respective class symbols. This is okay since each of these classes is the Specialization part of a Generalization class relationship. In a Generalization class relationship, all specialization classes inherit all the attributes and all the operations from the generalization.

Therefore, *ConcessionItem* has the same attributes and operations as *Inventory* and *SaleItem* combined. *VCR* has the same attributes and operations as *Inventory* and *RentalItem* combined. Finally, *Video* and *Game* have the same exact attributes and operations as *Inventory*, *SaleItem*, and *RentalItem* combined.

This concludes the review of the video store UML class diagram model. Remember, are object-oriented and represent the very important foundational and static structure (architecture) of the proposed video store's information system. There are still several other UML models to create that will collectively represent the "blueprints" for the video store's information system.

COMPARATIVE ANALYSIS OF INSTITUTIONS

The three cases presented here will be studied through comparative analysis. Comparison is considered to be 'so central to good analysis that the scientific method is unavoidably comparative'. Yet in the field of post-Sovietology, relatively few researchers have focused on the *comparative* analysis of *formal state institutions*. On comparing legislatures for example, Joel Ostrow remarks that 'most works in the field of comparative legislative studies either develop descriptive *typologies* of legislatures based on their characteristics and functions or elaborate *single case* studies. Little effort has been made to develop and apply criteria for explaining legislative behaviour or why some perform better than others.' (Ostrow 2000: 20). He mainly puts the blame with comparativists who, according to him, have not sufficiently embraced institutionalism as an approach. I endorse his claim that 'a comparative institutionalist approach offers much potential for the broader subfield of comparative politics. It embraces a bottom-up approach to theory-building that (...) is a refreshing departure from attempts at grand-theory and all of the accompanying limitations demonstrated by past and recent 'modernisation' to 'transitions to democracy' approaches' (Ostrow 2000: 21). In this way, the comparative analysis of institutional change in the Duma, MFA and Presidential Administration of the three selected countries (Russia, Ukraine and Belarus) can turn post-Sovietology into a more varied field of research (met meer relief; gevarieerder; genuanceerder).

When choosing the comparative method, distinctive methodological issues that arise in the systematic analysis of a small number of cases (small N: Russia, Ukraine, Belarus) can be reason for some concern, especially when studying large-scale political phenomena, as is the case here (institutional change in the Post Soviet area). One risks to end up in a situation where you have too many variables that are inherent to the study of these macropolitical phenomena, and with a limited number of cases to study these variables. This unbalanced situation can considerably (and unnecessarily) complicate the research.

The small N problem has generated a lot of discussion on how to approach the comparative study of a limited amount of cases most efficiently. Since its publication in 1971, Arend Lijpharts article '*Comparative Politics and Comparative Method*', has served as one of the strongest answers to this particular problem, by placing the comparative method in-between the case-study method on the one hand and the experimental and statistical method on the other, thereby attributing an intermediate function to the comparative method that allows it to assemble the best of these approaches

into the comparative approach. It offers a stronger basis for evaluating hypotheses than do case studies, and on the other hand the data requirements of comparative analysis are not so demanding as in experimental or statistical research (Collier 1991: 9).

Assembling the best of both worlds under the comparative denominator demonstrates the value of the method but does not immediately solve the small N problem. The well-known answer to this is *triangulation*; obtaining data from a wide range of different sources, using a variety of methods, investigators or theories; its main purposes being confirmation of certain hypotheses and achieving greater completeness in the study. This study endorses Denzin's multiple triangulation approach: MAYBE HERE SOME WORDS OF EXPLANATION? SEE DENZIN both theoretical and methodological triangulation will be applied; theoretical triangulation by approaching the research with diverse perspectives and hypotheses in mind (cfr. supra), and methodological triangulation by using two or more methods (e.g. survey and discourse analysis, cfr. infra) to measure same phenomenon will be applied.

Clifford Geertz' (1973, qtd in Collier 1991: 14) notion of '*thick description*' to unravel the underlying meaning of political phenomena rooted in particular contexts is especially relevant for this study. The phenomena of institutional isomorphism, and more importantly, the variations in isomorphism between the three selected countries are rooted in historical processes, like decisions taken during processes of institutional formation that still determine policy choices. A limited amount of cases is therefore important to explain the divergence in isomorphist phenomena in an exhaustive and thorough manner; too many cases would not allow us to inquire deep into the reasons for this divergence or variations..

A Comparative Analytical 'Research Cycle'

The 'great intellectual success of the school of comparative historical analysis' has also played an important role in legitimising a focus on a small-N (Collier 1991: 15). The studies of Paul Pierson, Theda Skocpol and Margaret Somers demonstrate that the comparative method is compatible with a path dependent approach.

Paul Pierson sees the small-N being gradually resolved by '... using the growing theoretical understanding of path dependent processes to generate more observable implications, for instance by focusing on intermediate stages in the processes (Pierson 2000a: 265, 2000b, Pierson a Skocpol 2002). He adds to this remark that '.... comparative politics has always been a field that emphasises critical junctures' – defining events that can encompass major institutional change.

Theda Skocpol and Margaret Somers (1980) on their part pay special attention to linking comparative politics to the path dependence approach. They identify three distinct logics of comparative history that form a *research cycle*: 1) parallel demonstration of theory, 2) contrast of contexts, 3) macro-causal analysis.

The *parallel demonstration of theory* approach compares case histories in order to prove that a certain theory or an explicitly delineated hypothesis can repeatedly

demonstrate its fruitfulness when applied to a series of ‘relevant historical trajectories’ (Skocpol and Somers 1980: 176). One looks for similarities among the cases in terms of common applicability of the theoretical arguments. It is characteristic for parallel demonstration of theory to develop theoretical models and hypotheses before going into the cases. One could say that Parallel Comparativists aim to ‘demonstrate that a theory similarly holds good from case to case: for them differences among cases are primarily contextual particularities against which to highlight the generality of the processes with which their theories are basically concerned’ (Skocpol and Somers 1980: 178).

The *contrast of contexts* approach reasons the other way round. It highlights the unique features of each case and aims to demonstrate how the particularities of each case affect social processes. This approach does not depart from any overarching theory, but allows themes to emerge and be subject to generalisation during the study of the cases. Special attention is paid to the particular historical context of the cases. The contrast of contexts confirm the path dependent notion that specific empires or countries each have their own developments that determine further outcomes and can be the cause of differences between several seemingly similar cases. Reinhard Bendix remarks about this approach:

‘By means of comparative analysis I want to preserve a sense of historical particularism as far as I can, while still comparing different countries. Rather than aim at broader generalisations and lose that sense, I ask the same or at least similar questions of divergent materials and so leave room for divergent answers. I want to make more transparent the divergence among structures of authority and among the ways in which societies have responded to the challenges implicit in the civilisational accomplishments of other countries.’ (Bendix 1976: 247, in Skocpol and Somers 1980: 180)

In similar fashion, the contrast of contexts approach helps to explain the divergence in the foreign policy choices of the countries under study by pointing out certain historical particularities and by stressing the importance of the knowledge of contexts. Yet unlike this study, contrast oriented comparativists do not aim to find new explanatory generalisations through comparative analysis.

Macro-causal analysis (also called multivariate hypothesis-testing) is the third approach suggested by Skocpol and Somers and unlike the contrast oriented comparativists, macro-analysts do aspire to ‘test the validity of existing theories and to develop new causal generalisations to replace invalidated ones’ (Skocpol and Somers 1980: 182) through comparative analysis. The main purpose of macro-causal analysis corresponds with the methodological aim of the second stage of this research: looking at sources of variation to make causal inferences about macro-level structures and processes. The comparisons of macro-analysts also appear to be more focused than their contrast oriented counterparts/colleagues, who look at whole contexts or histories of countries in their entirety; macro-analysts tend to ‘move back and forth between alternative explanatory hypotheses and comparisons of *relevant aspects* (...) of two or

more cases (Skocpol and Somers 1980: 182). This more focused and in-depth approach is in line with our specific comparative analysis and might encompass an even higher explanatory power than the contrast-oriented approach.

Although the parallel demonstration of theory, the contrast of contexts and macro-causal analysis are three distinct approaches, they are not mutually exclusive. Different academic works have sometimes combined (especially in pairs) these logics. What is more, together, these three approaches constitute a complementary system. They internalise both path dependent and comparative approaches by offering a model for comparing different historical contexts.

The approaches in triangular fashion, with the major logics of Somers' and Skocpol's comparative approach situated at the extremities of the triangle, connected by dimensions shared by these major logics. Dimension (a) encompasses the Parallel and Contrast oriented types of comparative historical analysis. Both approaches endorse the notion that 'prior general ideas – theories in the case of the Parallel approach; themes or questions in the case of the Contrast-oriented approach – are brought to bear on each individual case in the overall comparative-historical investigation'. Dimension (b) indicates a shared objective of Macro analysis and Parallel Comparativists in developing explanations – be it 'causal generalisations, or deductively elaborated theories'. One does not find this aspect present in the Contrast-oriented approach. The third dimension (c) of the triangle points out commonalities between the Contrast-oriented and Macro-analytic logics.

They share the feature that 'direct comparisons between or among historical cases are intrinsic to their respective manner of argumentation', whether it concerns highlighting the unique case contexts (Contrast-oriented) or making causal inferences (Macro-analytic). Since it applies theory to cases one by one, the Parallel approach lacks this common characteristic (Skocpol and Somers 1980: 87-189).

Following the earlier remark that different academic works have sometimes combined these logics, especially in pairs, this study will accordingly reflect features of the three distinct logics. Most notably the parallel comparison and macro-causal analysis logics will come to the fore. The two approaches overlap in their shared goal to develop explanations (causal inferences or theories), which is evidently one of the main aims of the study. Parallel comparison of history applies here given the three countries evident historical parallels, when Ukraine and Belarus were part of the Russian Empire, when all three countries were part of the Soviet Union, *etc.* More importantly, it also is relevant in the first stage of the research in which institutional definition and isomorphic tendencies are compared in all three countries parallelly using the same organisational characteristics. In the second stage of the study, the macro-analytic approach comes in: causal inferences will be made about processes of institutional change in the three selected countries and will then be compared, which eventually will result in the refining of isomorphist theory. Yet these approaches are not the only ones that relate to the topic of study [although they are the most relevant]. When looking into the causal mechanisms

of isomorphism and sources of divergence of the foreign policy choices in the second stage, the contrast of contexts approach surfaces and might prove to be an adequate tool for comparison.

The macro-analytic approach – which underscores the underlying idea of the research in its entirety – can take place through the *method of agreement*, in which different cases that have the phenomenon to be explained in common also have the hypothesised causal factors in common, even though other factors may vary between these cases.

This method corresponds with the well-known Most Similar Systems Design (MSSD) as put forward by Przeworski and Teune (1970), who describe it as: ‘... based on a belief that a number of theoretically significant differences will be found among similar systems and these differences can be used in explanation. The alternative design [MDSD], which seeks maximal heterogeneity in the sample of systems, is based on a belief in intersystemic differentiation, the [cases] will differ with regard to only a limited number of variables or relationships’ (Przeworski and Teune 1970: 39)

In MSSD, the comparison between relatively similar countries allows to neutralise certain differences in order to permit a better analysis of other differences. Not surprisingly, most scholars involved in regional studies take on this approach. The same goes for this research. The countries of the Slavic Core of the CIS have many similarities, not only on governmental or organisational level; as mentioned earlier, Ukraine, Russia and Belarus also hold numerous historical and cultural parallels. The comparable background of these countries allows us to focus on one specific aspect, *i.e.* institutional change resulting in disparate foreign policy orientations.

The small N problem cannot only be solved by Lijphart’s proposal of triangulation but also through looking for innovations in the comparative method over the last two decades. By endorsing Somers’ and Skocpol’s concept of a ‘research cycle’ – with special attention for the parallel demonstration of theory and macro-causal analysis comparisons, I intend to establish a profound comparative framework for the analysis of institutional change in the three selected countries. Even though Skocpol and Somers primarily see their research cycle applied in comparative history, the research cycle holds for comparative politics in general. It can therefore be employed in this study as well. Comparative analysis is rarely unequivocal, nor is it one-sided. On the contrary, comparative analysis is more likely multi-faceted and one risks to get confounded by the different aspects of analysis. In order not to become entangled in the numerous angles of the comparison but to on the contrary make a comparison that is structured and clear, I opt for the comparative analytical research cycle, which encompasses different comparative approaches that adequately cover the different aspects of this study. By connecting these three major logics of comparative analysis and forming pairs that represent different dimensions of the comparison, the diverse angles of the intended comparison in this research can be studied in a systematic manner. Skocpol and Somers’ comparative analytical research cycle approach thus/dan ook epitomises the underlying notion of this comparative study.

Institutional Change and Foreign Policy

After dwelling upon the specifics of comparative analysis, I will now concentrate in the following sections on *what* will be compared. The *topic* of comparison is institutional change as a consequence of the three countries' foreign policy orientations. This comparison will be made on the level of formal institutions (ministry of foreign affairs, parliament, presidential administration).

In some respects, this comparison is a peculiar one. First of all, institutional change has so far not often been analysed systematically, let alone in a comparative manner. When analysed, one mostly focuses on institutional change in one institution or one country, not in different institutions in several countries. However, the proposed model of isomorphism that will be further elaborated and applied in this research specifically permits a *systematic* comparison across countries and across institutions. A second and more important specificity of this comparison is the fact that studying the interaction between foreign policy and institutional change brings together two disciplines; political science and international relations (IR), or in other words, the study of a country's domestic policy and the study of its international relations. Studying the common ground of these two disciplines is perceived as an interesting challenge. Although few researchers combine studying the role of domestic *institutions* in foreign policy making, this perspective is not altogether new or under-researched. When taking this research perspective however, it is of paramount importance to leave the traditional sharp distinction between domestic and foreign policy analysis behind. Clarke and Smith make some interesting and pertinent observations about the link between domestic and foreign policy:

'Many writers on policy analysis have been greatly absorbed by the problems of understanding and analysing complexity, and there are many good studies. Curiously, most writers do not consider the international arena in this respect since they tend to make the implicit assumption that this arena is somehow qualitatively different. But this is mistaken.

The international arena in reality is a mixture of national and international elements, and is becoming more so as interdependence becomes more prevalent. It is in this international arena that all the problems of complexity are seen at their starkest; where some agencies will be foreign and operate under their own rules and norms; where implementation coalitions are absolutely crucial to the process (...)' (Clarke and Smith 1992: 179-180).

In straddling the boundary between two academic disciplines (IR and Political Science) one can however encounter some pitfalls. Each of these disciplines 'has its own corpus of assumptions, concepts and modes of analysis'. So in walking this fine line on the boundary of two disciplines, it is nonetheless important to choose a clear perspective from which to study the foreign policy, in order to avoid ending up with conflicting assumptions or modes of analysis of both disciplines united in one study. Brian White distinguishes between the two perspectives in the following manner:

'If the analyst views foreign policy behaviour from the perspective of IR, he or she will be predisposed to see elements of the international environment as the major determinant of foreign policy. A Political Science perspective, on the other hand, predisposes the analyst to highlight domestic determinants like governmental politics, pressure group activity and public opinion. The tendency is for analysis from this perspective to provide a more detailed, perhaps 'messier' picture of FP formulation. This is another dimension of the so-called 'level of analysis' problem that was touched on earlier' (White 1992: 7-8).

Although it should be stressed that both disciplines of political science and IR are included in this study by looking at the interaction between (domestic) institutional change and foreign policy, the main focus is on the analysis of the domestic political institutions of Russia, Ukraine and Belarus. This Political Science perspective implies the assumptions and modes of analysis of this discipline and aims at a more detailed analysis of the complex interdependencies between the institutions involved in foreign policy making. Still, the IR perspective of the international environment as a major determinant of foreign policy is definitely acknowledged in the model of isomorphism elaborate.

All these remarks and clarifications reveal the need for a balanced and finely tuned definition of foreign policy, which can reflect notions of both the IR and the political science disciplines touched upon in this study.

This implies no strict definition that focuses on territorial boundaries and separates foreign and domestic policy along geographic lines, but a more inclusive definition that describes foreign policy as 'the area of politics which bridges the all-important boundary between the nation state and its international environment', and where the distinction lies in the fact that foreign policy, like domestic policy, is formulated within the state, but, unlike domestic policy, is directed and must be implemented in the environment external to that state' (White 1992: 5, Wallace 1971:7).

The Organisational Process Model

In order to provide an adequate context for comparing the interaction between institutional change and foreign policy in Russia, Ukraine and Belarus, this study endorses the approach of Graham Allison, one of the most prominent scholars on foreign policy. He developed an innovative perspective on foreign policy that does not primarily focus on decision makers, but perceives *the institutions* and *bureaucracies* behind these decision makers as playing a crucial role in foreign policy, thereby treading the common ground between IR and political science: 'Studies of organisations have had little influence upon the existing literature of international affairs. Few specialists in international politics have studied organisation theory. It is only recently that organisation theorists have come to study *organisations as decision makers*' (Allison 1971:69).

Allison developed two models which could form an alternative to the traditional approach of foreign policy analysis that focuses on the actor-decision maker. Allison

sees the action that governments undertake on the level of foreign policy as *organisational* output rather than as the result of individual decision making. According to Allison, organisations as decision makers prevail over individual actors' decisions, which are also important but not predominant.

Opposing the behaviouralist view of foreign policy being determined by actors as rational decision makers (which he brands the Classical Rational Actor Model), Graham Allison proposed two new models in his 1971 book: the Organisational Process Model and the Bureaucratic Politics Model. Both models – often mentioned under the common denominator of the 'bureaucratic politics account' (as opposed to the rational actor model) - offered a way of explaining 'FP decision-making that is distinctly different from the 'rational-actor' account of the process. (...) Whereas the vast bulk of writing on FP behaviour has looked for explanations at the level of key-decision-makers who are assumed to behave rationally, the bureaucratic [and organisational] politics account look at the role of bureaucrats and the impact of organisational behaviour on the making and the implementation of decisions' (Smith 1992: 109).

Governments consist of different organisations (institutions) 'among which primary responsibility for particular tasks is divided' in order to respond to a wide spectrum of policy issues. Each organisation attends to a special set of issues and acts to resolve possible problems and to realise policy objectives. The organisational perspective 'builds on the premises of incrementalism by 'standard operating procedures' [SOPs, that are only] occasionally waived or breached. One of the main premises of this perspective is that although foreign policy actions are 'not necessarily determined without analysis or reason', it is particularly stressed that 'the analysis may not be a unified or overall one consistent with more formal assumptions.

Few important issues fall exclusively within the domain of a single organisation, especially on the level of foreign policy. The 'multiple facets of foreign affairs requires that problems be cut up and parcelled out to various organisations' – like the MFA, parliament and Presidential Administration included in this study. Organisations may well act perfectly logically and clearly within their own standard operating procedures and responsibilities, but there is no guarantee that their outputs will be consistent or compatible within the foreign policy process as a whole' (Allison 1971:67, 80; Clarke 1992: 51).

Graham Allison goes as far to state that 'a government consists of a conglomerate of semi-feudal, loosely allied organisations, each with a substantial life of its own' (Allison 1971: 67). The specificity of each organisation that contributes to a greater or lesser extent to the development of foreign policy priorities and directives might therefore imply some obstacles along the road to a coherent foreign policy position. One of the strong points of the organisational process model is its particular mindfulness for and sensitivity towards the specific internal aspects of these loosely allied domestic organisations that determine foreign policy. By not only paying attention to the formal standard operating procedures on the highest level, but also by being attentive to the

complex interplay between these institutions on the domestic level, this approach may help to uncover the complex machineries of the state and its different institutions that result in foreign policy actions.

The bureaucratic politics model goes one step further by focusing on lower-level institutions. According to this third model, Allison perceives foreign policy formation as a process of 'bargaining among regularised circuits among players positioned hierarchically within the government'. By stressing the role of individuals in these lower-level institutions as players in foreign policy making, Allison situates this model between Model I (Rational Actor Model) and Model II (Organisational Process Model). Unlike Model II, that focuses on organisations and hence perceives foreign policy decisions as organisational outputs, the bureaucratic politics model focuses on individual players in lower-level institutions that determine foreign policy through bargaining (games). But unlike Model I, the bureaucratic politics model 'sees no unitary actor but rather many actors as players – players who focus not on a single strategic issue but on many diverse intra-national problems as well; players who act in terms of no consistent set of strategic objectives but rather according to various conceptions of national, organisational and personal goals; players who make government decisions not by a single, rationale choice but by the pulling and hauling that is politics' (Allison 1971: 144).

The making of foreign policy is therefore not always merely a case of bargaining, but can at times be perceived as a struggle to overcome incompatibilities between different institutions or administrations. Clarke remarks felicitously here that 'officials are not unaware of the significance of their actions and, seeing national interests through the eyes of their role in their departments, will do all they can to have that view prevail over others. A foreign policy decision, therefore, may be the result not of any single view of the national interest, it may not have much to do with the intrinsic merits of the problem – even in times of crisis – but may merely represent the *only acceptable compromise* between competing bureaucratic interests (Clarke 1992: 51).

In sum, the bureaucratic politics model distinguishes itself by two traits: foreign policy making shares many characteristics with the formation of domestic policy. Therefore one should focus on the actors in the lower level institutions as well. Secondly, 'foreign policy is conceived (...) not as governmental choice or as organisational output, but as the outcome or 'resultant' of various bargaining games [or compromises] among the key players within the government' (White 1992: 16).

This innovative approach that perceives the making of foreign policy as not the exclusive work of high-level decision makers, but by proposing two models includes the organisational processes or pictures foreign policy formation as the bargaining between individuals in lower – level institutions or bureaucracies, had a significant impact and commanded a great deal of support. The models were applied and elaborated in studies aiming to explain specific workings of foreign policy decision-making. However, as most innovative concepts, these models were also prone to extensive

criticism. Interestingly, while bureaucratic politics has tended to dwarf the looser concept of organisational politics, it is simultaneously this third model that has received most criticism (Clarke 1992: 51).

One of the critiques Allison received is that his work is not altogether original and partly derives from the work of Kremlinologists (Horelick, Johnson and Steinbruner (1973), in Smith 1992: 113). This critique implied that Allison based his theoretical assumptions for the case study of American foreign policy during the Cuban Missile Crisis partly on studies of *Soviet* foreign policy. In the context of our research that intends to focus on post-Soviet foreign policy, this criticism is hardly relevant, but even rather convenient. Another persistent criticism claims that Allison's models cannot be applied to countries other than the United States. In his book on the Cuban Missile crisis, Allison did however attempt to explain not only US, but also USSR foreign policy by using his new models. Although Allison clearly intended his models to be applied to other countries, some claim that they cannot for example be applied to developing countries because 'they do not have enough stability of structure or form in their organisational routines or bargaining processes' (Migdal in Smith 1992: 120). Others however did apply these models to developing countries, without any major obstacles and 'finding them very useful'. A more stern criticism appeared to be that Allison's models do not apply to the Soviet Union Karen Dawisha claims that bureaucracy in the Soviet Union is different and one of a kind because of the ubiquitous influence of the Communist Party and its ideology (Dawisha in Smith 1992: 121), which makes studying the USSR's foreign policy on the level of bureaucracy a unique case that does not correspond with the bureaucratic politics model, where individuals of different bureaucracies bargaining with bureaucracies of other institutions about foreign policy formation.

Although Dawisha has a point here, this critique as well has become irrelevant because the Soviet system has in the mean time ceased to exist and the political situation in the former Soviet Union has changed profoundly. In the current institutional landscape of Russia, Ukraine and Belarus, there is definitely more room for bargaining and compromising in foreign policy making. Moreover, other authors are not so radical to write off Allison's models and retorted Dawisha's critique by stating that the models do in fact apply to the Soviet Union (enumerate authors). They even deem these models useful tools for the analysis of Soviet foreign policy, provided that they are modified and tailored to the specific (Soviet) situation. Applying bureaucratic politics and organisational process and bureaucratic politics models might even 'substantially illuminate certain neglected aspects of the Soviet decision-making process' (Valenta and Eidlin in Smith 1992: 122).

Therefore, unlike Dawisha, we can on the contrary claim that the fact that Allison's developed these models with not only the analysis of US, but also USSR foreign policy on his mind, can only be an advantage for this study (Valenta and Eidlin DATE). Their flexibility, applicability and innovative, institution-centred approach to foreign policy

analysis make these models particularly well-fit tools for analysing the topic and region of study – the Slavic core of the CIS – instead of the many models that are tailored to western institutions and derived from western-centred paradigms.

The bureaucratic politics approach has over the years received more criticism than the organisational process model, which ‘better stood the test of time and academic scrutiny. Partly in response to these criticisms, the two models were *amalgamated* and reformulated (...) in later years’, by both Allison as other scholars like Jerel Rosati who elaborated on his models (Smith 1992: 110, Rosati 1981). In my approach I endorse the remark by Michael Clarke that ‘bureaucratic politics is probably best regarded as a particular *variant*, applicable in precise circumstances, to the more generally ubiquitous perspective of organisational politics’ (Clarke 1992: 51-52). The organisational process model is more than fitting to study processes of isomorphism in Russia, Ukraine and Belarus.

Its perspective generally corresponds with the basic characteristics of isomorphism, and can therefore serve as a comprehensive context enabling us to carry out a comparative study of foreign policy on the level of *institutions*. With such (methodological) perspective of foreign policy analysis that is distinctly tailored to the specificities of the research topic (most importantly, studying foreign policy on the level of institutions instead of focusing on a few decision makers), this approach is bound to give us very useful and innovative insights about the interaction between foreign policy formation in the institutions of Russia, Ukraine and Belarus and isomorphic institutional change.

A Comparison of State Institutions

As stated before, this research focuses on assessing institutional change in the political systems of the Slavic Core of the CIS. The political system is perceived here as a set of related objects that are connected by many components, be it power, authority, or historical and cultural links (Mahler 1995: 12). This research focuses on a selection of executive and legislative institutions. As I mentioned earlier, there are several reasons for selecting *state institutions*: the study of formal institutions lends itself more to generalisations; also, political institutions are considered to be more enduring than developing informal institutions. Moreover, the study of formal structures allows observation and measurement more readily than do other approaches (Mahler 1995: 20). After explaining the choice of countries and the choice of institutions to be analysed, a last part of this chapter will touch upon a more functional aspect of the research, namely methodology and data collection.

Choice of Countries

The rationale behind the selection of Russia, Belarus and Ukraine is threefold. First of all, these three countries constitute a particular region that is *crucial in the*

contemporary European geopolitical landscape. Of the CIS region, these three countries matter the most for the enlarged European Union. Ukraine and Belarus matter because since 2004, they share a common border with the European Union and have to co-operate as new neighbours on a.o. border issues. Russia matters because of its dominant influence in the post-Soviet area, as well as for security and energy considerations. Although their proximity to the organisational field of the European Union determines the three countries' foreign policy, their membership of the maybe less developed but therefore not less significant organisation of the CIS should not be ignored. Although not amounting to the organisational level of the European Union, this much younger Eastern European construction that hopes to mirror the EU structures in the future does function as an organisational field (Malfliet 2002, Rontoyanni 2000, Stefes 2003). The foreign policy of the three countries is therefore also determined by this organisational field.

Secondly, the fact that these three countries constitute the *Slavic core of the CIS* connotes correlation and analogy. Historical, political, and cultural similarities between the three countries are manifold, and we will elaborate on these analogies. The affinities between Russia, Ukraine and Belarus allow us moreover to apply MSSD and single out the differences to be studied from these predominantly similar backgrounds

This leads us to the third and most important rationale for choosing these three states: the differences in their foreign policy orientation. When taking a first look at the Slavic Core, one does not immediately encounter any clear indicators that can explain discrepancies in foreign policies. However, close scrutiny of the particular context in which the institutional design and formation took place, as well as puncture points that triggered institutional change during the transition period, uncover a different picture that is of interest to this study.

INFORMATION MODELING METHODOLOGY

The information modeling methodology classification also referred to as the data modeling methodology or information engineering methodology, emerged in the early 1980s as businesses began to embrace database management systems. Today it too is a popular methodology used to assist in the development of business-oriented information systems.

An information modeling methodology subscribes to the notion that good engineering is simple engineering. It approaches the development of information systems predominantly from an information perspective rather than a functional perspective, as does the structured analysis and design methodology.

The real world is described by its data and the data's attributes and their relationships to each other. As with the structured analysis and design methodology, information modeling adds dimensions that allow it, too, to be rigorous (formalized), repeatable, measurable and automated. The predominant focus of this methodology is to do the analysis and design from an information perspective. In other words, the systems analyst

uses a “*What information does the system have to be able to provide the user?*” problem solving approach. Today there are only a handful of purchasable IDE technology products that fully support an information modeling methodology.

For reference purposes, names such as Peter Chen, James Martin, and Ian Palmer among others have made significant and documented contributions to the information modeling methodology classification. Tools and techniques such as an entity relationship diagram business area analysis (BAA), and information models are often used in conjunction with the information modeling methodology.

Object Oriented Methodology

The object oriented systems analysis and design methodology classification emerged in the mid to late 1980s as businesses began to seriously consider object oriented-programming languages for developing and implementing systems. Even though Simula, circa 1966, is credited as being the first object oriented language, popular object oriented (or object-enabled/enhanced) languages such as Smalltalk, C++, Objective C, and Eiffel came into their own in the 1980s and Java was introduced by Sun Microsystems in mid-1995. A few other object-enhanced products made their way to the marketplace in the 1990s – perhaps most notably Visual Basic, Delphi and PowerBuilder. All of these object oriented languages approach programming from a significantly different paradigm than previous programming languages. Rather than follow the structured, deterministic, and sequential programming paradigm associated with languages such as COBOL, Fortran, C, Basic, PL/I, Ada, Algol and others, these languages follow the approach pioneered by Simula based on objects, attributes, responsibilities, and messages.

Often history repeats itself. Such is the case with object oriented analysis and design. Just as structured analysis and design emerged as a complement to structured programming in the 1970s, object oriented analysis and design emerged as a complement to object oriented programming in the 1980s. The problem solving strategy inherent in object oriented programming is to approach the problem from an object (*e.g.*, person, place, thing or concept) perspective rather than the functional perspective of traditional and structured methodologies or the information perspective of an information engineering methodology. Because of this, object oriented programming began to capture mind- and market-share as a dominant programming paradigm mostly in computer science, software engineering and scientific programming environments while business-oriented programming remained somewhat skeptical of its value to them. With the increased emphasis on graphical user interfaces (GUI) and software that runs on distributed and heterogeneous, client server (multi-tier) computer hardware platforms across the Internet even business-oriented programming is shifting to object-orientation.

Since object oriented programming’s problem solving strategy is unique, the need for an object oriented systems analysis and design approach leading up to the programming task makes sense. The older, more mature methodologies have limitations in representing a model of the information system that can readily be implemented

using an object oriented programming language primarily due to their focus on functions or data, not objects. This is not a statement that the older methodologies are inferior. Rather, they are just different and were well suited for the job they were intended for.

Object oriented analysis and design has good and bad news associated with it. The bad news is that there is yet another new methodology in the marketplace. The good news is that much can be borrowed from the other pervading methodologies and applied to the object oriented methodology, which some believe makes it more evolutionary than revolutionary. In other words, experienced systems analysts need not throw out all their knowledge and experience acquired using the other methodologies, which would tend to demoralize the veterans in the field. Rather than the object oriented analysis and design methodology being revolutionary, it is evolutionary in the “borrowing from what we already know” sense. The methodology notations and subsequent models it represents are new, yet similar to other notations. The most difficult aspect of learning an object oriented methodology and notation for experienced systems analysts is the transition from a functional or information (data) centered problem solving strategy to that of an object problem solving strategy.

Moving from a “function think” or “data think” problem solving strategy to an “object think” strategy is not an easy one. For example, just think of the difficulty you would have retraining yourself to type using a Dvorak keyboard layout, which is significantly different from the standard Qwerty keyboard layout that we are all familiar with!

Study, practice, experience, and time will all be important for experienced systems analysts to make a successful transition in their thinking about the problem domain and then determining requirements and documenting them using an object oriented notation and methodology.

The use of the object oriented methodology (or any other for that matter) for the novice systems analyst or trainee should be less difficult due to the limited experience he or she has with other methodologies. In other words, novices have less experience and habits to retrain. Someone who does not know how to type using a keyboard has little or no keyboarding memory and habits to retrain and could likely learn the Dvorak keyboard with little difficulty.

For reference purposes, names such as Booch, Coad, Cox, Firesmith, Henderson Sellers, Jacobson, Mellor, Meyer, Odell, Rumbaugh, Shlaer, Wirfs Brock, and Yourdon among others have made significant and documented contributions to the object oriented methodology classification. Tools and the techniques that support them such as dynamic and static object oriented models, sequence, collaboration and state diagrams, and use cases are the basis for object-oriented modeling. In November 1997 the Unified Modeling Language (UML) became a universally recognized standard of the Object Management Group (OMG). The UML is a set of models and notations that give software developers a standard way to visualize, specify, construct, document and communicate the artifacts of an information system. The UML will be used later in this book for object-oriented modeling. As we move through time the UML continues to evolve and be refined.

FOUNDATIONAL CHARACTERISTICS OF AN OBJECT ORIENTED METHODOLOGY

The systems analysis and design is a complex process. Accommodating this complex process requires systems analysts to exploit commonly understood characteristics or principles for managing complexity. As cited earlier, today's information systems are complex and involve tens of thousands, hundreds of thousands, or millions of lines of programming code. For example, some knowledgeable software developers have estimated that Microsoft's popular Windows Operating System has more than 35 million lines of programming source code—Wow! That's a lot! In fact the systems being developed today are far more sophisticated than those of just five years ago. A personal computer word processor was "feature poor" back then compared to the ones currently available for purchase. Shrink wrap software has moved into the competitive arena of hundreds of thousands or even millions of units in commercial sales. As a result, each new version or release of the software must exceed the features and benefits of the older version. Not only that, new releases often must exceed the capability of their strongest software competitor's currently selling software version.

Some common characteristics for managing complexity during systems analysis and design are available for systems analysts to use to one degree or another regardless of the SDLC used to develop the information system. At least eight characteristics or principles for managing complexity are also considered foundational and are generally accepted characteristics of object oriented analysis, design, and programming. Each of these common characteristics is presented here for your consideration. You may already be familiar with many of them because they often have a much broader applicability to your own personal life, not just to software development. They are:

1. Common methods of organization
2. Abstraction
3. Encapsulation or information hiding
4. Inheritance
5. Polymorphism
6. Message communication
7. Associations
8. Reuse.

Not all object oriented books discuss all eight of these characteristics. However, because I believe they are important, I have included them here. These eight concepts are not necessarily new to object oriented systems development as other older methodologies incorporate one or more of them in some way. However, together all eight form the fundamental building blocks for object oriented technologies. Each is discussed next.

Common methods of organization assist with organizing an information systems model as well as the software that is ultimately written. The common methods of organization that are applicable here are:

- a. Objects and attributes or characteristics. For example, you could be the object and your name, address, height, weight, colour of eyes, date of birth, and so on are some of your attributes or characteristics.
- b. Wholes and parts. For example, your television set is the “whole” and the individual parts that make up the television set are the “parts.” A personal computer system—the whole—usually consists of a box with all kinds of hardware in it, a printer, a monitor, a keyboard, and a mouse—the parts.
- c. Groups and members. Conceptually this is similar to wholes and parts, but its examples are often different and do not seem to fit neatly into a “wholes and parts” classification. For example, a computer club would be a group and the individual members of the computer club would be the members. Your Systems Analysis and Design course would be the group and the students in the course would be the members.

People in general are quite familiar and comfortable with the above common methods of organization; therefore they will be able to related to our Systems Analysis and Design models that take advantage of these concepts.

Abstraction is the principle of ignoring those aspects of a problem domain that are not relevant to the current purpose in order to concentrate more fully on those that are. Abstraction is a concept that you use every day. It has to do with the amount of detail you care to get involved in. In systems analysis and design, this is called levels of abstraction. Let’s say that your friend decides to bake a cake. He offers you the choices of both helping him bake the cake and then helping him eat it or just helping him eat it. If you choose the first option, you are demonstrating a detailed level of abstraction because you have chosen to get involved with the details of making a cake. If you choose the second option, you are demonstrating a generalized level of abstraction because you chose not to deal with the details of baking the cake, only eating it. Telling your father that you love him is abstract, or generalized, while telling him one or more of the reasons why you love him is specific and thus a more detailed level of abstraction.

In systems analysis and design, abstraction is used to identify essential information system requirements while simultaneously postponing or eliminating non-essential aspects. An abstraction intentionally ignores some qualities, attributes, or functions of an information system in order to focus attention on others. An abstraction is also a summary, covering the highlights and leaving out the details. It also omits the pieces of the system that are not necessary for understanding the system at a given level of detail. Maps typically represent a good example of abstraction. You can get maps of the United States, individual states, counties, cities, zip codes, and so on. Each map cited here in succession is less abstract and hence more detailed than the prior one. In addition, a city map intentionally leaves off the details of the surrounding county, state, and country that it is a part of in order to allow you to focus on the details of the city. Another example of abstraction would be to view an aerial picture of your entire university campus. Such a picture would not have detailed information regarding a specific office

on a specific floor within a specific building on the campus. Abstraction has been a common characteristic in all of the prevailing information systems analysis and design methodologies. Therefore, experienced systems analysts can continue to utilize this system building characteristic as they transition to an object oriented methodology.

Encapsulation or information hiding is the notion that a software component (module, subroutine, operation, method, and so on) should isolate or hide a single design decision. This notion is based on the work of David Parnas dating back to the early 1970s. An example of programming encapsulation may illustrate this notion best.

The creation and subsequent use of a single software component to look up student account numbers in a student database isolates this lookup feature from the rest of the software that makes up the information system. Each time the information system needs to look up a student account number, this routine is used, since the logic to look up student account numbers has been encapsulated within this one routine. If a new lookup routine is created, tested, and used each time there is a new requirement to look up a student account in the database, two potentially negative things happen. First, prior work—software construction and testing as a minimum—is replicated. Second, additional maintenance activity will be required for the information system should it ever be necessary to alter the student lookup routine(s).

In systems analysis and design, systems analysts decompose the problem domain into small, encapsulated units. These analysis and design decisions eventually become software modules. Encapsulation helps to localize their volatility when changes and maintenance are required after they have become software modules.

A further elaboration of encapsulation or information hiding is any mechanism that allows certain portions of the information system to be “hidden.” This can be useful in at least two different situations:

- a. When it is important that people only use or have access to a certain subset of the complete information system. For example, while an information system is being developed, development team members may be assigned to work on a certain part of the system and not allowed access to other parts of the system being worked on by other team members. Once the information system is operational, certain users may only be allowed to use a few of the features of the information system while other users may have access to many more features.
- b. To purposely prevent other components of the information system from being aware of or unable to take advantage of certain other components in the system. This situation addresses another aspect of encapsulation—assignment of responsibilities. Just as in real life you have certain responsibilities, a specific component of an information system has its responsibility, say dispensing cash in an ATM, while other components of the same system have their responsibilities, which exclude the dispensing of cash since that function is done by another component.

Another way to think about encapsulation is from a programming perspective, which has long since advocated the creation and use of small blocks of function specific code (e.g., procedure, subroutine, paragraph, and so on) that can be assembled into a working computer programme in a modular fashion, much like the plug compatible stereo components in homes today. Each of these blocks has encapsulated within it one or a few design functions.

For example, each of the following functions could be represented by a small block of computer programming code and then bundled together to perform a higher level function, such as registering for courses for this coming school year — “get student identification (Id) number,” “validate student Id number,” “check student Id number for outstanding fines,” “get course Id to add to student’s schedule,” “determine if a seat is available for this course Id,” “determine student Id number’s eligibility to take this course Id,” and so on.

Finally, encapsulation has been a common characteristic in all of the prevailing information systems methodologies. Therefore, experienced systems analysts can continue to utilize this system building characteristic as they transition to an object oriented methodology.

There is one subtle difference. With the older, more mature methodologies, encapsulation was usually limited to encapsulation of functions separately from encapsulation of data. With the object oriented methodology, encapsulation incorporates both functions and data together into objects. This notion will be discussed in more detail later in the book.

Inheritance is a mechanism for expressing similarity. Just as you inherited certain physical features from your mother and certain ones from your father, so also can information system components inherit certain things from related components. A hierarchical or parent child relationship exists between a group called Person and groups of Faculty, Administrator, and Student. We purposely are using the singular for each of these words in keeping with the notation that is used throughout the book, but keep in mind that Administrator means one or more administrators, as does student and faculty.

A sibling relationship exists between Faculty, Administrator, and Student. There are certain characteristics that can be associated with Person and thus inherited by all the children in the hierarchy—Faculty, Administrator, and Student—such as name, address, phone number, and date of birth to name a few.

Other characteristics may be unique to a single sibling. For example, office hours may only be applicable to Faculty, job title may only be applicable to Administrator, and grade point average may only be applicable to Student.

These characteristics would be associated with the specific child node in the hierarchy. Similarly, there are certain things that a Person can do that would be common to Faculty, Administrator, and Student, such as eating at the cafeteria and walking on campus. Likewise, there are certain things that are unique to each of the children nodes. For example, a Faculty person may assign grades and add students to the class, whereas an

Administrator person may manage people and key in administrative data to the computer, while a Student person could register for courses and take an exam. These actions would be associated with the specific child node that each belongs to.

As is usually the case in real life, inheritance in systems development is a one way street—down the hierarchy—from the “parent” node to each of the “children” (siblings) nodes. However, observe three arrows—each pointing upwards towards the “parent.” The explanation for this is that each of Faculty, Administrator and Student “extends from” or “is an extension of” or “is a specialization of” or “inherits from” the Person—hence the upward-pointing arrows.

Polymorphism in the general sense is the ability to take on different forms. For example H₂O can take on three forms—water, steam, or ice. In a sense, you could say that you are polymorphic with respect to your behaviour when approaching a traffic signal in your car. You respond differently depending on the status of the signal light. In software, for example, the *print* operation associated with textual characters and numbers can print textual characters and numbers “because it knows how to do this”. The print operation associated with graphic symbols and images can print graphic symbols and images “because it knows how to do this”. Each of the *print* operations just described will print something on a printer in the abstract sense; however, at the detail level each of these *print* operations does its job a little differently because what is being printed is of a different type—textual characters and numbers, graphic symbols, and images. Because of the differences at the detail level, the print operation is considered polymorphic. Polymorphism in computer programming isn’t new, but its use during object oriented analysis and design by systems analysts probably is.

Message communication is the way objects in an object oriented methodology communicate with each other. It is similar to a subroutine call with parameters or a paragraph or procedure call in some other conventional programming languages such as Fortran, COBOL, C, or Pascal. This concept is widely understood by practicing systems analysts so it, too, can be directly transferred into an object oriented methodology by them.

Associations are useful to assist with relating things in an information system to each other. Things that happen at the same time could be associated as well as things that happen under similar conditions. For example, VISA, MasterCard, Discover, and other credit card companies send out periodic statements showing us our balance owing. These companies often take this opportunity to insert a number of advertisements along with the statement, since the postage is often the same with or without the advertisements.

Reuse. Everywhere you look in your world you see reuse in one form or another. The same power cord can often be used for your computer, your monitor, or your printer. Apartment windows often come in standard sizes, and dozens of a certain size window are used in a large apartment complex. The can of orange juice you bought last time probably looks just like the one you bought today. How many shirts or blouses do you have that reuse the same type of button? How often do you reuse a textbook that was

used by some other student before you? Have you ever reused a videocassette to record over some other event that you recorded at a previous time?

Reuse is a much talked about concept in systems analysis and design, but one which is still struggling to make a significant contribution to the development process due primarily to two factors. The first factor is the retraining of the mind-set of the analysts and programmers who are accustomed to creating their own systems models and code. This translates to the “not invented here” mentality held by some systems analysts and programmers. In other words, they want to create and debug their own “whatever” rather than use someone else’s “whatever” that does the same thing (and is already debugged I might add!). Management can address and overcome this problem given proper amounts of time, retraining, and other motivators for the systems analysts and programmers who are resistant to reuse. The second factor relates to the administrative factors that must be established for setting up a library of reusable models and code. This is no easy task for a variety of reasons including strategy to accomplish, personal, organizational, cultural, and legal aspects. As used in our everyday world, reuse can take one of three forms: (1) *sharing*, (2) *copying or cloning*, or (3) *adjusting*. Each is discussed here.

An example of sharing is when you use a textbook and then sell it back to the bookstore, which in turn resells it to another student. Or everyone living in or visiting your apartment or home shares (reuses) the telephone you have. In software development, a subroutine, module, paragraph, procedure, service, or file definition are all potentially reusable. A small procedure could be created that only looks up student identification numbers in a student database and then responds with valid or invalid. Such a procedure could be used over and over again, intact, within an information system.

An example of copying or cloning is what is done every day in the manufacturing industry. An automobile model is prototyped and then moved into production to produce thousands of that model. In software development, we could have a paragraph of a COBOL programme that we copy and put into another COBOL programme, giving two paragraphs that are identical.

An example of adjusting is a variation on the copying or cloning. Using the automobile example, as a certain model moves down the manufacturing assembly line, standard features for this automobile are assembled (copying) as well as different combinations of options are assembled such as air conditioner, CD player, spoiler, tinted glass, and so on (adjusting). Granted that these options are standard reusable components just assembled in different combinations in the particular automobile. However, the final product, the automobile, is a variation on the standard make and model. A variation of the adjusting type of reuse is the situation in which the standard item is actually changed and then used in some other application. For example, a computer modem cable can have two of its wires physically swapped within the cable housing, and then this cable becomes a different kind of modem cable called a null modem cable—an adjustment to the standard cable.

An example of adjusting in systems analysis and design is when a programmer locates a subroutine, module, or paragraph, and so on from a library of reusable components. The subroutine becomes the programmer's beginning point for creating a new subroutine, somewhat similar to the original library version. Then the programmer begins making adjustments to the subroutine, such as removing some of the code in the subroutine, changing other parts of the code, and adding new code to this new subroutine to meet a specific purpose. The end result is a new subroutine. The eight characteristics described above play pivotal parts in the object-oriented systems analysis and design methodology. This is indeed good news because most people—software developers and users of computers—are already familiar with most of them.

Marketing Research Process

There are basically four steps in the marketing research process that franchisors must follow:

- Define the problem and research objectives.
- Develop a research plan for collecting information.
- Implement the research plan by collecting and analyzing the data.
- Interpret and report the findings.

When defining the problem and research objectives, it is important that the franchisors doing the research work closely with the decision makers to determine what research is needed and how it will be used.

If this step is skipped, the results generally will not address the major problem. In a franchise setting, the research people must work closely with those people in the franchise who handle recruitment of potential franchisees. They must also work with the franchisees themselves to understand the needs of the market.

Marketing research is designed to reduce uncertainty and provide knowledge for the franchisor. Thus, marketing research will help determine what is going right and what went wrong. Research can help the franchisor understand consumers' buying motivations and determine the strengths and weaknesses of the products or services.

When developing a research plan, there are three objectives of market research to remember:

- *Exploratory:* To gather primary information that will better define problems and suggest improvements.

- *Descriptive*: To describe the market potential for a product or the demographics and attitudes of consumers who buy the products.
- *Causal*: To understand the cause and effect relationships in market practices.

The three major functional roles in marketing research are descriptive, diagnostic, and predictive.

The descriptive role focuses on gathering and presenting statements of fact. For example, what has been the sales trend for the franchise outlet during the last five years? What are the consumers' attitudes towards the product or service? The second role of research goes beyond description and is diagnostic.

A researcher attempts to explain the data. What was the impact on sales when we introduced the new product? Have sales increased as a result of the new product line introduced? The final function of research would be predictive. Will this individual operate a franchise in a manner consistent with franchise philosophy? Can we predict how a new advertising campaign will affect sales? If we replace the manager, will we improve on employee morale and performance?

The third stage of a marketing research project is implementing the research plan by collecting and analyzing the data. The franchisor must determine the best instrument to use to collect data for the project. Interviews, focus groups, and surveys can all be used to collect data, but the costs, ease, and depth of information vary from one method to another. For example, when conducting research about a franchisee's goals, more in-depth information will probably be required than can be provided by a simple questionnaire.

The fourth step in marketing research is interpreting and reporting the findings where conclusions are drawn to the franchisors. The purpose of this research is not to overwhelm management but provide information which will be useful for making decisions.

MARKETING RESEARCH CHARACTERISTICS

First, marketing *research is systematic*. Thus systematic planning is required at all the stages of the marketing research process. The procedures followed at each stage are methodologically sound, well documented, and, as much as possible, planned in advance. Marketing research uses the scientific method in that data are collected and analysed to test prior notions or hypotheses. Marketing research is *objective*. It attempts to provide accurate information that reflects a true state of affairs. It should be conducted impartially. While research is always influenced by the researcher's research philosophy, it should be free from the personal or political biases of the researcher or the management. Research which is motivated by personal or political gain involves a breach of professional standards. Such research is deliberately biased so as to result in predetermined findings. The motto of every researcher should be, "Find it and tell it like it is." The objective nature of marketing research underscores the importance of ethical considerations.

MARKET RESEARCH

One of the most typical functions of a marketing research programme is to understand the habits and attitudes of customers. This information, when provided to franchisees by the franchisor, can build stronger channel relationships. The customer profile is often conducted through exit interviews as well as a general survey.

The exit interview generally is conducted as customers leave a store. At this time, their attitudes about the product or service are measured. In addition, perception indicators can indicate how individuals felt about the quality of the product or service being sold compared to competitors' offerings. Also, demographic information will provide valuable information about the customer segment currently visiting the franchisee's outlet. With the customer profile for each outlet, franchisors may be able to tailor their franchise programme to better meet the unique needs of each franchisee, hence reducing conflict within the relationship.

Additionally, marketing research may be conducted with focus groups which will allow for greater insights into the end consumer. Focus groups may indicate the strengths or weaknesses of a particular company, product, or service. The focus groups may easily be divided up into heavy users, moderate users, light users, or non-users of a particular franchise outlet. Marketing research can play an integral part in proper decision making in a franchising organization. More and more franchisors are recognizing the benefits of implementing a marketing research programme.

Along with the commonly recognized benefits of marketing research, franchisors should also recognize that marketing research can be an effective tool in building strong relationships with its franchisees.

First, marketing research can help franchisors select franchisees whose goals are consistent with system-wide growth. Second, franchisors can conduct customer research and share the findings with franchisees.

Marketing research provides information which franchisors can use to properly determine the profile for both the successful franchisee and the target customer. When both the franchisee and customer profiles have been developed, the franchisor will be better able to select franchisees who can manage the franchising outlet and serve the customer. Marketing research becomes one tool to reducing unnecessary conflict within franchise channels.

The increasing efforts by marketers to target diverse groups of consumers call for a closer examination of the ethical implications of market segmentation and differentiated marketing. Previous research suggests that marketers and consumers often differ in their perceptions of marketing ethics. Based on contingency theory, this research proposes an integrated framework—which includes the nature of the product, consumer characteristics, and market selection—to analyse the ethical complexities of the marketing exchange. Interactions among these factors lead to various contingencies with different ethical implications for marketing managers and public policy makers. Marketers should assess consumer interests and the ethics of marketing programmes

before their implementation. In the last several decades, targeting distinctive consumer segments with differentiated marketing has been a popular strategy among many marketers. The distinctive nature of various consumer groups such as children, the elderly, women, and ethnic minorities has made them attractive market segments. However, market segmentation and targeted marketing have, from time to time, met with tremendous difficulties. Targeting potentially harmful products at vulnerable and disadvantaged consumers such as children, the elderly, and inner-city residents has received negative publicity and been subjected to damaging litigation. The increasing willingness of some large corporations to exploit vulnerable consumers indicates unfair treatment of these consumers and a lack of justice in the marketplace.

On the other hand, cases of discrimination and redlining still occur in certain product and market areas, such as the discrimination directed at minority consumers by insurance and mortgage companies. Disenchanted consumers are increasingly rallying consumer interest groups to put pressure on firms that they consider to be predatory or discriminatory in their marketing practices. Companies such as R. J. Reynolds, Prudential Insurance, and Texaco Oil Company have suffered from tarnished images, consumer boycotts, and court penalties of hundreds of millions of dollars.

Although the modern marketing concept emphasizes its mission to satisfy consumer needs and wants, that promise, in reality, is sometimes lost or misplaced, resulting in outcomes that are not in the best interests of either the customers or society. A lack of understanding of the ethical issues associated with market segmentation and selection has contributed to these problems, which have tremendous social and economic costs. While targeting harmful products at vulnerable consumers has received harsh criticism, restricting the marketing of certain products and labeling some consumers as vulnerable are considered equally troublesome, suggesting that the ethical implications of marketing practices are complicated.

Although researchers have examined the ethics of market segmentation and selection, effort is lacking in synthesizing various issues to provide a holistic understanding of the ethical implications of the marketing exchange. Based on contingency theory, we integrate previous research and propose a three-dimensional framework—which includes the nature of the product, consumer characteristics, and market selection—to analyse the ethics of market segmentation and related marketing strategies. Interactions among these variables lead to various scenarios with different ethical implications for marketing management and public policy making.

Particular attention is given to areas in which marketers and consumers differ in their perceptions of ethical propriety with respect to the nature of the product and market selection. To avoid ethical conflicts in marketing, we suggest that companies consider consumers' interest and assess the ethical implications of their marketing plans before implementation. In the marketing field, market segmentation and the differentiation of the marketing mix have become so prevalent that they are almost synonymous with competitive strategies.

While targeted marketing has largely been beneficial to both consumers and marketers, cases of consumer discontent from time to time raise questions about the ethical implications of market segmentation and targeted marketing, most notably the targeting of harmful products at vulnerable consumers, such as targeting alcohol and cigarettes at inner-city consumers and churning insurance policies to the elderly. Meanwhile, the opposite of targeting—the exclusion of certain consumers from a company's offerings—is just as controversial.

These problems are not unique to marketers of consumer goods. Business service providers and public and nonprofit organizations are not immune to such challenges. Organizations, including educational institutions and government agencies, that cater to certain segments of society have also become the targets of public scrutiny and face the prominent possibility of consumer discontent and legal actions.

While consumers and marketers continue to struggle with these complicated issues and seek viable solutions, researchers have studied the ethics of market segmentation and market selection. Recent studies have focused on the ethical implications of targeted marketing of harmful products at vulnerable consumers.

TARGETED MARKETING

Targeted marketing, as a popular marketing strategy, refers to the concentrated marketing of a product to a segment of consumers due to the attractiveness of the group, in terms of such factors as its size and growth rate. Theoretically speaking, there is nothing inherently wrong with targeted marketing. When marketers promote a product beneficial to a group of consumers, targeted marketing is largely ethical and welcome by consumers.

For instance, some restaurants and hotels have special promotion programmes aimed at children, and indirectly at their parents, with inexpensive toys and special vacation packages. In recent years, however, many cases of targeting potentially harmful products at vulnerable consumers have raised ethical concerns in terms of justice and fairness, such as targeting sweepstakes at the elderly and handguns at women. Even indirect and subtle targeting of potentially harmful products at vulnerable consumers has received criticism, such as targeting children with R-rated movies and using animal characters to promote cigarettes and alcohol.

PRODUCT HARM

Many products in the marketplace, such as wholesome food and medicine, are beneficial or at least beneficial to the target group. However, products that are intended to be beneficial have from time to time been found harmful to some consumers, resulting in "market failures".

For instance, unless a pharmaceutical company tests a new product on everyone (instead of limited clinical trials), it is difficult to determine whether the product will

cause any harm. The key issue here is the marketer's knowledge of the product's potential harm, even to a small group of people. If a company discovers that a small percentage of people may suffer negative reactions from its products and it immediately takes corrective measures, the company may be deemed as acting ethically, believing that it should take responsibility for the incident as part of its business liability.

If the company had prior knowledge of the side effect and it took no remedial measures upon such discovery, it would be considered as intentionally harming the consumers.

Having prior knowledge of the harmfulness of a product and withholding that information have been used as evidence against companies, such as in the Dow Corning case of silicone breast implants and the legal battle against tobacco companies.

However, the evaluation of a product's harmfulness is not always clear-cut. Even the marketing of seemingly beneficial and harmless products can sometimes take an unexpected turn, because such products can be harmful to consumers who have known characteristics. Some products, even though not inherently harmful, can be potentially harmful to consumers due to abuse or misuse.

For instance, targeting alcoholic beverages at poor inner-city consumers is particularly problematic, as this segment already suffers from a greater number of alcohol-related health and social problems than the general population. Thus, ethical evaluations of many products depend on their interaction with consumer characteristics and marketing practices.

VULNERABLE CONSUMERS

Recently, consumer vulnerability has drawn much attention in studies of marketing ethics. In numerous legal cases, the court system in the U.S. has defined vulnerable consumers as a group of people who, due to various idiosyncrasies, are sensitive and susceptible to the potential negative effects associated with using a particular product. For instance, a small group of people, due to their body biology, may suffer side effects from using certain medications. Moreover, children do not have the same level of knowledge, experience, or maturity as adults to process commercial information. Many elderly people, due to their physical and/or mental conditions, also face challenges as consumers. In addition, some consumers may be prone to addiction or compulsion while others may be disadvantaged due to their social and economic conditions.

Such examples of targeting vulnerable consumers include the targeting of high interest loans of credit cards to consumers with poor credit histories and less financial sophistication. Thus, product harmfulness is heightened for consumers at a risk or disadvantage, and poses ethical concerns.

DISCRIMINATION

While much recent research has examined the ethics of targeting harmful products to vulnerable consumers, some researchers continue to investigate the opposite case:

the exclusion of certain consumers in marketing. Discrimination in marketing refers to the practice of denying access to products to a group of consumers due to their racial or ethnic background, age, gender, or other characteristics. Discrimination may also have to do with discrepancies in product quality and variety, and the terms of exchange such as pricing and payment method. Since the 1960s, many studies have investigated pricing discrimination against the poor and minority consumers. Despite the tremendous progress in the last few decades, discrimination and redlining have persisted in areas such as housing, mortgage lending, and financial services. Here, the important gauge points are the equivalence or proportionality in marketing intensity and the disparate treatment of consumers.

Existing studies have made significant progress in conceptualizing the ethics of market segmentation and differentiated marketing. In several important respects, their findings have elevated our understanding of the ethical implications of the marketing exchange. First, these studies emphasize that the concern with the ethical consequences of marketing practices in the United States is well grounded in the fundamental beliefs in social and moral equity, fairness, and justice for all people.

Second, researchers have emphasized the ethical risks inherent in market segmentation and differentiated marketing. Dunfee, Smith and Ross suggest that due to the boundary-expanding nature of marketing, ethical concerns may arise in certain situations when marketers cross such boundaries. For instance, marketing products to youth that are intended for mature consumers, such as alcoholic beverages, leads to different ethical evaluations.

According to Star, the social discontent and ethical concerns associated with marketing stem from the functional limitations of the marketing concept. Contemporary marketing practices often rely on market segmentation, which divides consumers into submarkets based on characteristics such as age, gender, race, and nationality. To provide unique values to consumers and to avoid head-on competition, each market segment may require a unique marketing mix. Thus, segmentation and subsequent marketing practices often imply differential albeit unequal treatment of various consumer groups, and raise ethical concerns.

Third, extant studies have found that ethical concerns or dilemmas often arise when a marketer's perception is in conflict with that of consumers or the public. In fact, the customer interface represents the area of marketing in which conflicts in the perception of the ethical propriety of marketing practices frequently occur. Thus, marketers must study the perception of the public when formulating marketing strategies in order to avoid potential misunderstandings.

NEED FOR INTEGRATION

The review of literature reveals several deficiencies in the existing research. First, previous studies have focused on one or another issue of the marketing exchange and lack the integration of disparate perspectives. While much recent research has

concentrated on targeting harmful products at vulnerable consumers, the practice of excluding some consumer groups is equally controversial. However, discussion of these two topics has largely been separate in the existing literature.

Theoretically speaking, these two issues are closely related to each other because discrimination and redlining have as much to do with market segmentation and selection as targeted marketing, but in opposite directions. Moreover, insufficient effort has been expended to examine the many positive marketing activities that target disadvantaged groups, such as charity, corporate sponsorship, and public service programmes.

Although the existing literature condemns unethical marketing practices, it has not offered consistent guidelines to determine exactly when the marketing of certain products to some consumers becomes ethically problematic. There are also questions as to whether some consumer characteristics are the causes of vulnerability or simply its correlates. For instance, when controlling for education and income, the effect of vulnerability among ethnic groups may be less pronounced.

Thus, several authors have cautioned against the use of race- or ethnicity-based vulnerability. Government intervention to restrict marketing may have some merit, and at the same time finds strong opponents among those who support the free market and free speech. Often, consumer advocates and industry spokespeople find themselves in deadlocks in the heated debate about the ethics of market segmentation and selection.

Recent studies have revealed complex interactions among products, consumer characteristics, and marketing practices in ethical evaluations of marketers, and have recommended that marketers should tread carefully when venturing into certain market segments.

Despite these research efforts and the many theories of marketing ethics that have emerged, a framework for comprehensive analysis of the ethical implications of market selection and related marketing strategies remains elusive. Noting the manifold issues and perspectives, several researchers stress the need for a normative theory of ethics about market selection and marketing strategies. A more integrated approach to analyzing the ethical implications of the marketing exchange can help chart a coherent discourse on these critical issues and a better understanding of marketing ethics for managers and public policy makers.

A CONTINGENCY APPROACH

Since the early 1980s, businesses and researchers have devoted much attention to the ethical implications of business practices, which has signaled the arrival of the “ethics era” in which consumer sovereignty is given more emphasis than marketers’ interests, and there has been a decline of the caveat emptor assumption. A creative variety of frameworks and theories has been developed within the marketing ethics literature, such as the general theory of marketing ethics, moral decision-making theory, and social contracts theory. These theories typically rely on one or more of the classical theories, such as Kantian ethics and perspectives of rights, duties, and justice.

Existing theories of marketing ethics have developed various “tests” to examine the ethicality of marketing practices. Some of these ethical tenets are related to product harm, consumer characteristics, and market selection. For instance, Laczniaik and Murphy’s framework of ethical reasoning elaborates several such tests. The motive test asks whether the intent of the contemplated action is harmful. The consequence test establishes whether any major damage to people or organizations will result from the contemplated action.

The justice test asks whether the proposed action leaves another person or group less well off, and whether this person or group is already a member of a relatively underprivileged class. Although these tests have pragmatic appeals, there may be conflicting responses to their questions due to multiple stakeholders in the marketing exchange.

Despite the frequent ethical problems that involve consumers, few theories have considered the role of consumer perceptions in marketing ethics. The exceptions include the consumer sovereignty test and distributive justice. The consumer sovereignty test (comprising three dimensions) requires marketers to examine whether consumers are vulnerable or disadvantaged, perhaps due to age, education, or income.

Marketers need to establish whether a target market has the capability to understand the benefits and risks associated with a product, has sufficient information to judge whether their expectations for purchases will be fulfilled, and has the choice of going elsewhere. The theory of distributive justice suggests that the unequal and differential treatment of consumers, when it leads to detrimental effects on their well-being, particularly of those who are disadvantaged, may result in a perceived lack of fairness and justice. However, most normative theories of marketing ethics that have emerged in the last two decades have focused on either defining the fundamental principles of ethics to develop guidelines, or on the moral development and ethical reasoning of managers.

Although this approach undoubtedly has merit in assisting marketers to reflect on the ethics of their decisions, it is less satisfactory in providing definitive ethical evaluations of specific practices. Meanwhile, others have proposed universal codes of marketing ethics, such as those of the American Marketing Association, that deal with various ethical issues in different marketing areas such as advertising, sales management, and marketing research. However, they do not specifically address the complex interplay among consumer characteristics, product types, and marketing strategies.

CONTINGENCY THEORY

Dunfee, Smith, and Ross contend that the pluralistic approaches in current marketing ethics research reflect the inadequacy of general normative theories of marketing ethics to handle the rich and complex context of the marketing function. Thompson argues that making decisions on the basis of the context-independent general principles while disregarding the contextual details can result in socially irresponsible, and even detrimental, consequences.

Although most marketing ethics theories, particularly the classic utilitarian (teleological) theories of ethics, consider the interests of various parties and complex situations, they have been criticized as too abstract and general to provide adequate guidance for managers. Due to the great number of ethical issues inherent in customer relationships and marketing functions, one set of universal rules or principles of ethics may be too simplistic to apply to all marketing situations.

To provide clear guidelines for ethical decision-making in marketing, several researchers have emphasized the role of various cultural, organizational and environmental factors in ethical evaluations of marketing activities, including the contextualist approach and the social contracts theory. The contextualist approach stresses that the sources of ethical concerns for marketers come from the interests of different stakeholders and affect the evaluation of the ethical justifiability of a particular marketing activity.

According to the social contracts theory, ethical concerns arise when one community's legitimate norms (*e.g.*, those of marketers) conflict with those of another community (*e.g.*, consumers or the larger society). Therefore, a theory that considers different stakeholder interests and distinguishes various marketing scenarios and their ethical evaluations would be the most fruitful.

One of the theoretical approaches well suited to analyzing complex interactions among various dimensions of a phenomenon is contingency theory. Rooted in organizational behaviour research by Herbert Simon, contingency theory deals with the effect of interactions among various organizational and environmental factors on performance outcomes. According to Zeithaml, Varadarajan, and Zeithaml, contingency theory emphasizes the importance of situational influences on the management of organizations and questions the existence of a single best way to manage.

Contingency theory contributes to practical management by encouraging managers to:

- Identify important contingency variables that distinguish between contexts,
- Group similar contexts based on the variables, and
- Determine the most effective solution for each group.

The contingency approach is particularly helpful in analyzing the ethics of marketing activities. Based on contingency theory, we reject the possibility that a single set of criteria can determine marketing ethics in all circumstances.

Instead, we examine the interactions between the contextual variables and emphasize the analysis of the ethical implications of specific scenarios. As the most frequent ethical conflict that faces marketing managers involves attempting to balance corporate interests against those of consumers, a useful framework should first examine how a marketer's perception of marketing ethics interacts with that of the consumer, who is as an integral and indispensable party to and a key stakeholder of the marketing exchange process. Marketing is often considered as an exchange between marketers and consumers that aims to satisfy consumer needs and maximize the return on investment for shareholders.

Thus, there is an inevitable and omnipresent tension between marketers' interests and those of consumers. This conflict forms the basis for different positions on the ethics continuum of marketing practices.

However, placing consumers' interests against those of marketers on the ethics continuum may be too simplistic, because it may imply that marketing is a zero-sum game and reject the possibility of a win-win outcome. Both marketers and consumers may form their perceptions of the ethics of specific marketing scenarios according to ethical principles such as rights, justice, fairness, and equity. In many cases, marketers and consumers agree on the ethical evaluations of certain marketing scenarios and raise no ethical concerns. For instance, both groups agree in principle that providing beneficial products to consumers is ethical as well as desirable and responsible.

However, existing research has documented significant gaps between marketers and consumers in their ethical philosophies as well as ethical evaluations of specific marketing practices. From time to time, a marketer's perception may conflict with that of the public, which results in ethical concerns and, perhaps, disapproving behaviour by the public such as protests and boycotts.

For instance, consumers may believe that reduced pricing for medications such as AZT for the critically ill is ethical and laudable, but manufacturers may consider it unacceptable and unethical based on their obligations to shareholders. In other cases, while consumers perceive targeting harmful products at vulnerable consumers as unethical and exploitative, some marketers consider any restriction of marketing as an infringement of their legal right to free speech under the First Amendment. Thus, consumers' ethical evaluations of a particular marketing situation may be negative, while the marketer's perceptions are based on different moral priorities.

Given such complexities, this research integrates previous studies, applies contingency theory to analyse the ethical implications of the marketing exchange, and explains how ethical concerns arise in various marketing situations based on the agreement of lack thereof in the perceptions of ethics between marketers and consumers. Several recent studies have used two-dimensional frameworks and focused on the targeting of harmful products at vulnerable consumers. In our three-dimensional model, we include three concepts as the contextual variables, *i.e.*, the nature of the product, consumer characteristics, and market selection. Furthermore, we examine three categories in each variable. In the following sections, we define each of the three contingency variables and discuss how interactions among these dimensions distinguish between various contexts, and how ethical conflicts or dilemmas may arise under certain circumstances.

THE STRUCTURE OF THE MARKET RESEARCH INDUSTRY

Market research is bought by companies to help market the goods and services they produce and by government organisations to assist policy making.

THINK ABOUT

What decisions are made in your organization that could benefit from market research? What are the drivers that result in market research being used in your organization, and what are the barriers that stop it being used?

Research by UK companies and organisations is just over £1 billion per annum. There is a significant expenditure on research with business to business respondents. Remember many consumer companies such as Philips Lighting, or Shell, have significant turnover with businesses and carry out a good deal of business to business research. The percentage of companies that commission market research and have no consumer face is actually quite small and around only 10% of the total pie. The value of ad hoc and continuous market research bought-in from specialist market research suppliers and excludes research carried out by non-specialists such as management consultancies. Nor does it include the value of work carried out “in house” by people who do their own collection of data. Spending on market research by pharmaceutical companies has doubled in the last five years and now leads all other sectors.

The demand for market research services has grown rapidly through the late years of the twentieth century though there are signs that some sectors of this youthful industry are now maturing. As well as the variation in demand for market research by sector, there are enormous differences in the spending on market research between user organisations and in the way they organise the market research function internally. At the top, in terms of expenditure, are organisations which each spend several millions every year on market research. These include some of the very largest commercial companies and the Government (albeit spread through many departments and agencies).

Research buyers of this scale, often have central market research departments which act as service providers to the line management throughout their organisations. These departments not only act as professional buyers of market research but also carry out the analysis, interpretation and communication of the data within their organisations. There is a trade association representing the interests of these larger buyers of market research – the *Association Of Users Of Research Agencies (AURA)*.

If a company is not large enough to justify its own market research specialist, the buying and control of research may be carried out by business managers who need the information. Apart from market research bought by individual organisations to meet their own needs, there is a significant expenditure by industry groups collectively. Trade associations may commission ad-hoc or continuous research and in some areas there are special joint research programmes to meet industry wide information needs – media research is the most notable example of this with much audience and readership data obtained in this way.

As well as users and suppliers of market research, there are some types of organisations which fall somewhere in between. The most significant of this group are advertising agencies who commission research on behalf of their own clients and may build-in such as advertising testing and evaluation into the planning of major campaigns.

The largest advertising agencies often have their own research departments staffed by professionals and involved in the development of sophisticated techniques for media and advertising linked research (Admap is a monthly journal specialising in this area).

THE MARKET RESEARCH SOCIETY CODE OF PRACTICE

The *Market Research Society (MRS)* in the UK is the largest body of market research professionals in the world. The MRS has established a “code of practice” that covers the ethical aspects of market research, responsibilities to fellow members of the bodies setting the codes, clients, survey respondents and the public at large. The MRS code is taken seriously by all professional researchers and even if they are not personally members of the Society, they are likely to subscribe to the principles embodied in this code. A copy of the code is obtainable from the MRS and research users should be familiar with its main provisions since it affects and to some extent restricts the user/supplier relationship. Some important aspects of the Code in this respect include the following:

- Information can only be collected from respondents by fair means. Respondents must be honestly and comprehensively told that the information is for research purposes and that their participation is entirely voluntary. They are asked to give their consent. The only exception to this is observation, including mystery shopping, where the observed behaviour is public (*e.g.* shoppers looking in a window).
- Information given by respondents is confidential and may not be passed, in an identifiable form, to anyone outside the agency carrying out the research. This includes the client. Confidentiality even extends to the identity of respondents.

This requirement, therefore, largely excludes using formal market research (as understood by the MRS) to build up personal details of potential customers and producing lists or databases that can be used in subsequent marketing. The requirement for confidentiality can, however, be relaxed with the freely given and express permission of the respondent at the time of the interview and there are some differences in interpretation for business respondents.

- Equally if the research sponsor – the client – asks for confidentiality and it is promised, it must be respected.
- Furthermore, the results of research carried out for specific clients is confidential to that client and may not be disclosed to others or used to the benefit of other clients.
- The Code also sets standards for reporting the results of research including that any results must be supported by adequate data. Research agencies cannot,

for example, be asked to lend their name to promotion claims (..the results of market research shows that our brand was rated consistently better...) which are untrue or not backed by research data.

- Agencies are required to safeguard all data to meet the requirements for confidentiality and ensure that records are kept for a reasonable time to allow queries arising from the research to be answered.

TRENDS IN MARKETING RESEARCH

Marketing research, as a specialized field of activity, is relatively new. As recently as 1920, firms actively engaged in specific marketing research activities, carried on in a department created for such a purpose, were rare indeed. The Curtis Publishing Company and the General Motors Corporation are outstanding early examples. Even by 1940 it was still the rather exceptional company that recognized marketing research as a specific function. Early marketing research activity tended to be narrowly restricted, in line with common business attitudes about the then prevailing concept of the job of marketing—namely, to dispose of the output of field and factory through selling effort. Hence, pioneering marketing research activity, as carried on within business firms, tended to emphasize the collection of information about markets and about the selling function of marketing.

Since about 1950, the number of companies with specialized marketing research departments has greatly increased. It is now common to find persons with the title Director of Marketing Research among medium sized and large manufacturing companies, public utilities, publishing companies, advertising agencies, wholesaling corporations, retailing chains, financial institutions, transportation companies, and many other kinds of enterprises.

Not only has the number of marketing research practitioners increased greatly, but also the scope of their interests and activities has enlarged and become diversified. Whereas early marketing research tended to be confined to market characteristics and to analyses of selling effort, it is now commonplace for it to deal with buying policies, merchandise handling, location of facilities, transportation costs and methods, inventory levels and controls, order processing methods, risk management, credit and financial management, product development and testing, and general marketing administration.

The widening and enlarging scope of marketing research activities reflects an increasing acceptance of a broad and customer-oriented conception of marketing, compared with earlier and rather provincial attitudes that tended to equate marketing with selling effort. As the conception of marketing by business leaders broadens to coincide with that held by academicians, the scope of marketing research in business will correspondingly expand.

The nature of contemporary marketing research activity has also been influenced considerably by various trends relating to approaches, techniques, methods, and sources. First, there has been growing awareness that every business firm generates a great deal

of internal information that can be utilized in solving marketing problems. Second, external sources of published and readily available marketing information have multiplied manifold and have been characterized by substantial improvement in quality of data. It is no small task for a research practitioner merely to keep informed about such information. Third, many of the crude survey methods that were a common attribute of early marketing research have been supplanted by more advanced techniques borrowed from other areas in the social sciences. A fourth trend has been the greater use of mathematical models that involve sophisticated techniques of quantitative analysis and which require modern electronic data processing equipment to handle vast quantities of numerical information that could not practically be processed in any other way.

SOCIAL AND PUBLIC ASPECTS

While marketing research is commonly and properly regarded as one of the most significant marketing functions within progressive business firms, certain of its broader implications and applications must not be overlooked. First, as will be conspicuously evident in the following major section, various governmental agencies are actively engaged in the marketing research function, as providers of a vast stream of highly valued marketing information which could never be collected and processed by any individual enterprise.

Second, marketing research often provides the factual basis for certain types of public policy. Illustrative is information collected for the purpose of anticipating effects of proposed antitrust or trade regulation legislation.

Other examples include data gathered by public agencies in regard to credit controls, price controls in periods of emergency, agricultural production control or price support programmes, public aids to business recovery in periods of recession, and so on. Such activities are not confined to the federal government but are also common at the state and local levels, especially in connection with programmes to stimulate industrial development, modify taxation, regulate trade, or forecast revenues which may be dependent upon future market conditions. Third, activities of a marketing research nature are often carried on in connection with solving operating problems of public agencies. One example is extensive research conducted within the armed forces on the problem of logistics of supply, which is similar in many respects to the management of physical distribution functions in industry. Another example would be demand analysis for purposes of deciding upon the desirability of proposed price changes in certain state-owned liquor stores.

SOURCES OF MARKETING INFORMATION

There is an all-too-prevalent idea that marketing research is concerned primarily with the collection of information through the conduct of surveys of ultimate consumers or business establishments. While numerous specialized marketing research

organizations engage principally in work of this kind, by far the greater part of the quantitative data and qualitative information used in the research activities of marketing establishments is available from internal, governmental, and private information sources. It is believed that more widespread application of research procedures is not so much affected by the nonexistence of relevant information as it is by the fact that businessmen and marketing students generally are not well informed regarding the sources from which such information may be obtained.

INTERNAL INFORMATION

The types of marketing information generated within the normal course of business in the typical enterprise are myriad. For the most part, however, they may be grouped into three categories, as follows:

- Data about customer characteristics or behaviour,
- Information about the distribution or allocation of marketing effort, and
- Facts which aid in evaluating marketing performance.

The kind of information available about customer characteristics or behaviour naturally varies from one type of business to another. In department store retailing, detailed information about the socioeconomic attributes of credit customers is on file in credit applications. Such information can be used to construct frequency distributions of customers according to age level, income, occupation, location of residence, or many other factors.

Through analyses of charge account purchases, much can be learned about the extent to which customers with certain socioeconomic characteristics patronize different departments, respond to special sales, use different kinds of store services, and so on. Wholesale establishments, on the basis of information in accounting records, can readily determine the number and sales volume significance of customers according to location, size, class of trade designation, and other pertinent characteristics.

Opportunities to learn more about the way in which a firm distributes or allocates its marketing effort are also manifold. In a manufacturing or wholesaling firm with an outside selling organization, data from salesmen's call report sheets may be tabulated periodically to ascertain how many sales calls are made upon various classes or types of customers. Through systematic approaches to cost allocation, a firm can determine how much of its total economic resources are used for various categories of its business, such as merchandise lines, salesmen's territories, and classes of customers.

In order to evaluate performance, internal sources of marketing information are often supplemented with data available from external sources. For example, a manufacturer of carpeting can compare his own sales of each type of carpeting with comparable industry statistics for total production or sales. In this manner it may be possible to ascertain specific segments of the business in which the firm is gaining or losing out, in terms of relative share of market position, and to pinpoint specific trouble spots that call for corrective action or improved marketing approaches.

A manufacturer of electrical appliances may find that his sales, by broad geographic divisions of the country, should coincide approximately with the per cent of the total number of wired homes in each such region. Comparison of actual sales with those that would be expected on the basis of such an hypothesis may be indicative of particular areas where performance is weak or unsatisfactory. Internal information is also used for evaluative purposes when standards are established. For example, marketing research activity may reveal the number of calls per day that salesmen operating under given conditions should normally make. Such a ratio, along with others that indicate standard or ideal levels of performance in different types of selling activities or responsibilities, aids in evaluating performance from a number of viewpoints and in diagnosing difficult or troublesome situations in such a manner that the nature of remedial measures is rather clearly indicated.

The foregoing examples are merely a token illustration of the wide variety of types of marketing information which is or can be available in the business enterprise. Often such information is adequate to solve a number of kinds of marketing problems. In most instances, however, sole reliance upon internal information is unduly introspective and reveals little about opportunities which were or might be missed or about significant industry or environmental trends that may be developing. Thus, all business enterprises, at least to some extent, rely also upon various kinds of information available from external sources.

GOVERNMENT SOURCES

In some way or another, government agencies are involved in the performance of all of the marketing functions. This involvement ordinarily is of a regulatory or facilitating nature, as indicated by references to various types of legislation in connection with the other marketing functions. Marketing information, however, presents a rather unusual situation in the sense that the government assumes a much more active role in the performance of the function.

This participation is based upon the premise that the structure and functioning of free markets are dependent upon millions of decisions made by countless individuals in various kinds of marketing agencies. All such decisions are based upon some kind of information, which may be highly accurate or, in some cases, quite unreliable. Obviously, the decisions can be no sounder than the information upon which they are based, and the functioning of the whole economy can be no more efficient than the aggregate of the individual judgments which enter into it.

Thus, in the interest of marketing efficiency and in behalf of the general welfare, the government engages in various kinds of programmes to provide on a widely disseminated basis adequate information relating to marketing and other business activity. The amount and variety of such information are so great and significant that complete books have been devoted to the subject. Thus, the present discussion is to be regarded as merely illustrative.

Census Information

Through periodic censuses conducted by the federal government, detailed information is collected by a complete enumeration of some of the most important sectors of the national economy. Census data are provided for a variety of reasons and are widely used for a large number of business, economic, political, and other social and individual purposes. From the standpoint of marketing interest, they supply most of our basic knowledge about the institutional structure for marketing and the composition of national and local area markets. The Census of Agriculture, scheduled to be taken every five years, provides an inventory of the nation's resources for agricultural production. In it are to be found complete data pertaining to the manner in which agricultural resources are used and the kinds and amounts of production on our some 6 million farms. Data are compiled and presented in such a way that many kinds of information are available for counties and often for minor civil divisions within counties. Such statistics are of great importance not only in connection with agricultural marketing, but in delineating various kinds of markets for industrial goods which are purchased by farmers.

Most of our knowledge of the marketing structure has come from the various Censuses of Business which have been taken since 1929. Liberal use has been made of statistics from these Censuses throughout this volume. Coverage is provided for wholesale, retail, and service trades. Among the kinds of available information are data on the following major subjects: number of establishments, sales volume, merchandise inventories, credit sales, payroll, operations of multiunit organizations, legal form of organization, sales volume classifications, tabulations by various forms or methods of operation, and operating expenses for wholesale trade. A considerable quantity of data for individual lines of trade is presented for cities, standard metropolitan areas, counties, and states. Such information is invaluable in determining trends by lines of business, by method of operation of marketing institutions, and by regions or local areas.

The Census of Manufactures has been taken periodically since 1810, usually at five-year intervals. It provides for a complete enumeration of the activities of all manufacturing establishments. Information is available, partly on a national and partly on a local area basis, relating to value of shipments, value added by manufacture, input of labour, consumption of fuel and important materials, value of inventories, expenditures for plant and equipment, and for thousands of individual commodities, the value of manufacturing production. Such statistics provide benchmark data indicative of changes in the volume of manufacturing and demand for individual commodities, changes in the number of establishments engaged in particular types of manufacturing, and regional or local area shifts in industrial location. This census is the best available source of information for determining the magnitude and location of the markets for many kinds of goods purchased by industry.

Censuses of Housing and Population are taken decennially and the information thus secured is of great marketing value. National and local area statistics from these Censuses are used to determine the size and character of markets in terms of number of residents

or consumer units (households). Valuable information on growth trends is available by comparisons with data from former censuses which, in the case of some subjects, date back to 1790. For many marketing purposes, it is extremely important to know much about “kinds of people” as well as their number. Data are available, even for small local areas, for such population characteristics as sex, age, colour or race, nativity, urban-rural residence, marital status, educational level, occupational and industrial classification of those in the labour force, family income, family status, and general characteristics of housing.

Current Marketing Facts

Between the intervals of the various kinds of censuses, current information is needed. Various government bureaus and administrative agencies devote much effort to monthly, quarterly, or annual compilations of statistics, collected on the basis of sample studies and covering generally the same areas as reported in the censuses.

The U.S. Bureau of the Census issues a large number of short publications each year in a series entitled “*Current Population Reports*” that includes, among others, the following: *Population Characteristics*, which covers items such as marital status, household characteristics, and population mobility; *Population Estimates*, which provide data on the size of the population nationally and by states, current estimates by age, sex, or colour, and population projections for future years; and *Consumer Income*, which reveals the current income of persons and families.

An *Annual Survey of Manufactures* and monthly or quarterly *Facts for Industry Releases* are published by the Bureau of the Census to provide businessmen with information on short-term developments. For many specific industries, data are presented on manufacturing production, shipments, consumption of materials, unfilled orders, stocks, and utilization of machinery and equipment. Such information guides marketing executives in determining policies that should be followed in regard to current inventories, or in deciding what kinds of markets merit special sales attention.

The *Survey of Current Business*, which is published monthly by the Office of Business Economics of the U.S. Department of Commerce, is one of the most highly valued sources of current marketing information. It usually contains about 40 pages of statistical data, and several special issues each year are more comprehensive in their coverage, as are the biennial supplement on “*Business Statistics*,” the regular “*Annual Review Number*”, and the “*National Income Number*”. Among the more significant facts regularly reported are industrial production, national and consumer income, personal consumption expenditures, price indexes, sales and inventories in wholesale and retail trade by kinds of business, advertising activities, transportation and communications activity, and production by industrial classifications.

The *Federal Reserve Bulletin*, published monthly by the Board of Governors of the Federal Reserve System, contains detailed statistical information on department store sales and stocks by regions of the country and by departmental breakdowns. Among

other current facts of marketing significance are data on consumer credit extensions and indebtedness, business finance, foreign trade, and prices. Regular articles in this publication deal with summaries of business conditions and provide interpretations of current conditions that are available in judging probable future developments.

The *Monthly Labour Review* and other publications of the U.S. Bureau of Labour Statistics are of interest to marketing, chiefly because of the data they contain on wholesale prices, retail prices, cost-of-living index, housing construction, nonresidential construction, and changes in the labour force.

Much of the information collected by the aforementioned and other government agencies is published annually in summary form in the *Statistical Abstract of the United States*, which is valuable not only for the facts published therein, but also for the source notes indicating the agencies which collect other kinds of relevant marketing information.

Certain state bureaus are a source of information to businessmen. The highway departments of the different states, for example, furnish information on proposed new construction, the kind of materials which will be used, dates on which contracts will be let, *etc.* Municipal offices, including those of the buildings inspectors, have records which are of interest. Contractors are interested in knowing what building permits have been issued and to whom, for such information guides them in their sales activities. Manufacturers of certain kinds of household and building equipment are also interested in building permits. Records of transfers of deeds may be very important, as are other court records, including wills filed for probate, marriage licenses, records of attachments, dealer licenses issued, automobile licenses, and similar items.

Special Information Services for Agricultural Commodities

Government activity in collecting and disseminating marketing information has been of unusual significance in the case of agricultural commodities. Data pertaining to the current condition of more than 150 principal crops and livestock products are furnished regularly by the Crop Reporting Service of the Agricultural Estimates Division of the U.S. Department of Agriculture. Crop reports include estimates of the acreages that farmers intend to plant, acres planted for harvest, monthly estimates of probable production based on crop conditions and probable yields per acre, and production actually harvested.

In the case of livestock and poultry production, estimates are made of the number of animals on farms and the intentions of farmers with respect to subsequent seasons. Such information is collected through the cooperation of 41 state governmental agencies or agricultural colleges. Each year, on the average, about three-quarters of a million farmers and other persons make reports which enter into the compilation of the published data.

Information relating to current supply and price conditions in approximately 100 important wholesale market centers is provided by the Market News Service of the U.S. Department of Agriculture. Daily information about market conditions for staple

agricultural commodities, fresh fruits and vegetables, meat and livestock is collected by reporters in each market. Local market offices all over the United States are connected by about 11,000 miles of teletype circuits so that each local market office is in contact with all others several times daily. Through a consolidation of reports, five large area offices prepare and release information to the press and radio three times daily.

Data from such reports are regularly featured as news items on broadcasts from about 1,500 radio and television stations and are published on the market pages of some 1,000 newspapers. Through such facilities, traders in any one market know about price and supply conditions in that market, in other markets, and at important shipping points, as well as the volume of unloadings or diversions in transit at each principal terminal market.

In addition to the foregoing types of information releases, the U.S. Department of Agriculture regularly publishes a number of so-called "situation reports" which provide a rather complete picture of the current and prospective supply, demand, and price situation for the agricultural economy as a whole and for important individual commodities.

The character of these publications is suggested by the following titles: *The Demand and Price Situation*, *The Agricultural Situation*, *The National Food Situation*, *The Farm Cost Situation*, *The Farm Income Situation*, and *The Marketing and Transportation Situation*.

It is sometimes erroneously assumed that marketing information disseminated through agencies of the U.S. Department of Agriculture is of interest principally to farmers. Actually, such information is used no more extensively by the farmer, and in some cases less extensively, than by speculators on the major commodity exchanges, assemblers or local market buying middlemen, primary handlers or processors such as cotton gins or tobacco warehouses, manufacturing companies such as bakeries or tobacco manufacturing companies, wholesale distributors, large retailing organizations, and agencies that provide ultimate consumers with buying information.

Special Government Studies of Interest to Marketing

In the foregoing paragraphs, attention has been directed to information that is *regularly* compiled and disseminated by government agencies. There is, however, a wide range of special studies which are made at times and upon occasions that the public interest seems to justify.

So voluminous is the marketing information made available by the government and other sources that the U.S. Department of Commerce publishes a monthly magazine, *Marketing Information Guide*, which usually contains more than 100 annotated references to selected current marketing studies, and in this manner brings them to the attention of interested parties who subscribe to the service.

Also, in order to bring marketing information to widespread attention, bibliographies are published and periodically revised by the U.S. Small Business Administration.

Illustrative of the types of topics for which comprehensive and annotated bibliographies are available are the following: *Wholesaling, Retailing, Shopping Centers, Marketing Research Procedures, and Distribution Cost Analysis*.

MARKETING INFORMATION THROUGH COOPERATION

Much marketing information reaches business establishments as a result of cooperative activity through which the members of some group share it. Two significant variations include trade associations and cooperative marketing organizations.

Trade Associations

Trade associations representing retailers, wholesalers, and manufacturers in the same line of business have long played a very important role in gathering and publishing information which is of value to their members.

Through regular and special publications, reports upon various kinds of research activities are made available to members and, in some cases, to the general public.

Surveys of typical operating ratios, studies of business techniques and procedures, the dissemination of current news of interest to the trade, and educational activities are among the numerous functions of the more important associations.

Cooperative Marketing Organizations

One of the important services of cooperative marketing organizations is that of collecting and disseminating marketing information to promote orderly marketing for the benefit of the membership.

An example is provided by Florida Citrus Mutual, an organization with a membership of about 7,000 citrus fruit growers. Growers are divided into some 40 council groups. Officers are invited to headquarters in advance of each marketing season.

At this time, officials present all available information relating to supply and demand conditions which will have a bearing upon the average price for the growers' fruit during the season. Such information is provided for each type of utilization, including fruit for fresh shipment, canned juice, and frozen concentrate. Throughout the season, a special weekly publication is sent to all members, advising them of all up-to-date information deemed to be of significance in the marketing of fruit available for disposition.

Private Sources

Public and cooperative sources of marketing information are supplemented by a wide variety of private sources. Some of the more important of these are indicated in the following paragraphs.

Trade and Professional Magazines

One of the most significant sources consists of trade or professional magazines or business papers. Most people have seen a few such publications as a matter of ordinary experience, but very few persons have any real knowledge of the vast number that is published regularly. In one source, over 2,500 different trade or professional magazines are indexed and described. Some of these are of rather general interest and are read by executives and operating personnel in a variety of positions in all kinds of industries. Examples include *The Journal of Marketing*, *Business Week*, *Distribution Age*, *Fortune*, and the *Harvard Business Review*.

Certain magazines contain information of special significance to those charged with the responsibility of a particular function, which is the case with the periodicals *Credit and Financial Management*, *Sales Management*, and *Purchasing*. Many of them are primarily of interest to those in some particular line of trade or type of industry as suggested by the titles *Frozen Food Age*, *Men's Wear*, *Steel*, *Oil and Gas Journal*, *Hardware Age*, or *Building Supply News*. Some are of national significance, while others are of mere local or sectional interest as illustrated by the titles *Kansas City Grocer*, *New England Electrical News*, or *Michigan Drug Journal*. Trade magazines are of interest for several reasons.

First, they contain discussions of research activities and expert or editorial appraisals of current marketing developments. Second, many provide continuing feature columns or sections which contain reports of trade statistics, legislative developments or other matters. Third, the trade advertising that they contain is informative in itself. Finally, many of the major publications have special annual issues, of which the following are of unusual significance:

Sales Management, "Survey of Buying Power Issue", published annually on May 10. Contains current estimates of population, disposable family income, retail sales in total and by five store groups for cities over 10,000 population, counties, metropolitan areas, and states; data on industrial marketing potentials and wholesale trade for states and leading counties.

Printers' Ink, "Advertisers' Guide to Marketing", published annually in October. Provides information summarized from marketing studies, advertising expenditures by major companies and by class of media, operations of advertising agencies, trends in various types of sales promotion as point of-purchase material, premiums, films, packaging; contains summaries of legal or regulatory developments in the field of marketing.

Industrial Marketing, "Market Data and Directory Number", published annually in June. For each of some 90 industrial markets (e.g., apparel manufacturing, automotive, aviation, baking, etc.) provides information on current trends, what each industry buys, additional sources of information about the industry, current statistics on sales, production, number of plants, geographical distribution of the industry, trade publications and other magazines in each industrial classification. *Advertising Age*, "Annual Market

Data Section”. Contains an annotated description of many items of marketing information or statistics available from media, associations, and other groups. Compilation is classified under the following headings: national markets, farm markets, regional and local markets, distribution markets, industrial markets, professional markets, Canadian markets, and international markets.

Advertising Media

Many national magazines, some city newspapers, and radio and television stations publish research studies containing much valuable marketing analysis material. Such studies often contain quantitative information about the size of the market served by the media, qualitative characteristics of the consumer units in the market, their buying power, and the kind of appeals that can be used to reach them. Special studies are often made to determine buying habits, preferences, or attitudes of consumers. Illustrative are the “Consumer Analysis” studies, published annually by the *Milwaukee Journal* and a number of other newspapers. Information is tabulated on brand preferences of consumers for hundreds of advertised products, the kind of stores in which the consumers purchase each type of product, and the extent of distribution for each product in stores in the community. Some newspapers also publish lists of local retail stores which are arranged in route order, so that they can be used by salesmen in making personal calls. Some national magazines publish their circulation and other market figures by local areas in a manner that is helpful to manufacturers in establishing sales potentials, in delineating salesmen’s territories, or in allocating local advertising expenditures.

University Bureaus of Business Research

Bureaus of business research have been established in some 45 universities for the purpose of obtaining and publishing information of value to the business community. Among the better known are those of Harvard, Ohio State, Michigan, Minnesota, Illinois, Indiana, Washington, and Texas. The bureau at Harvard is well known for its studies on the operation of certain types of wholesale and retail institutions, especially its annual reports on the operating results of department stores and limited price variety stores. The bureau at Ohio State has published a number of volumes dealing with various kinds of management problems in several wholesale and retail lines of business. Most of the bureaus have some regular publication which deals with business conditions in general and with sales trends and marketing problems, either nationally or in the state or region principally served by the university.

Marketing Research Companies

Increasing attention to the importance of the marketing research function has led to the development of a large number of institutions that specialize in this activity and do work for regular or occasional clients, usually on a fee basis. Many of these organizations

are primarily engaged in conducting surveys and polls of consumer or trade opinion or attitudes. Some of them have a number of branch offices and employ a substantial staff equipped to handle large-scale interviewing surveys in all parts of the country. Certain companies provide a continuing kind of information service available on a subscription basis. Among the kinds of marketing information which can be purchased in this manner are reports of sales by brands of products as ascertained by audits in retail stores; readership of newspaper advertisements in metropolitan papers; and extent of the audience reached by broadcast advertising media. Some agencies are highly specialized in terms of the service rendered, confining their activities to problems of product package design, to market testing, or to the testing of personnel for placement in marketing or selling positions. Other research organizations offer a very wide range of services and have expert marketing consultants capable of setting up a plan for doing research on any kind of a marketing problem and provide the machinery for doing the research work and reporting on it to their clients.

Other Private Sources

A variety of business institutions whose primary business is something other than the supplying of marketing information are nevertheless actively engaged in the performance of this function. As has been explained in earlier parts of this text, marketing information is one of the important services rendered for customers by brokers, selling agents, wholesalers, commodity exchanges, advertising agencies, resident buying offices, and mercantile credit organizations. Other kinds of companies that often make studies pertaining to the market areas they serve are banks, public utilities, railroads, and large motor freight carriers.

MODELS IN MARKETING RESEARCH

Conjoint analysis is a statistical technique used in market research to determine how people value different features that make up an individual product or service. The objective of conjoint analysis is to determine what combination of a limited number of attributes is most influential on respondent choice or decision making. A controlled set of potential products or services is shown to respondents and by analysing how they make preferences between these products, the implicit valuation of the individual elements making up the product or service can be determined. These implicit valuations (utilities or part-worths) can be used to create market models that estimate market share, revenue and even profitability of new designs.

Conjoint originated in mathematical psychology and was developed by marketing professor Paul Green at the University of Pennsylvania and Data Chan. Other prominent conjoint analysis pioneers include professor V. “Seenu” Srinivasan of Stanford University who developed a linear programming (LINMAP) procedure for rank ordered data as well as a self-explicated approach, Richard Johnson (founder of Sawtooth Software)

who developed the Adaptive Conjoint Analysis technique in the 1980s and Jordan Louviere (Ph.D., University of Iowa) who invented and developed Choice-based approaches to conjoint analysis and related techniques such as MaxDiff.

Today it is used in many of the social sciences and applied sciences including marketing, product management, and operations research. It is used frequently in testing customer acceptance of new product designs, in assessing the appeal of advertisements and in service design. It has been used in product positioning, but there are some who raise problems with this application of conjoint analysis.

Conjoint analysis techniques may also be referred to as multiattribute compositional modelling, discrete choice modelling, or stated preference research, and is part of a broader set of trade-off analysis tools used for systematic analysis of decisions. These tools include Brand-Price Trade-Off, Simalto, and mathematical approaches such as evolutionary algorithms or Rule Developing Experimentation.

MARKETING RESEARCH MIX

The term Marketing research mix (or the “Mr. Mix”) was created in 2004 and published in 2007. It was designed as a framework to assist researchers to design or evaluate marketing research studies. The name was deliberately chosen to be similar to the Marketing Mix-it also has four Ps. Unlike the marketing mix these elements are sequential and they match the main phases that need to be followed. These four Ps are: Purpose; Population; Procedure and Publication.

Purpose

The purpose of the research is the reason why it is being done. The word “purpose” is useful because it has a wide coverage. It can be specifically defined or it can be loosely explained. The wide term also embraces studies to gather marketing intelligence, where the manager’s role is to scan the environment for useful data, and there may be no specific objective.

Many marketers avoid the term hypothesis entirely and tend to use the words “research objective” or “aim”. The word hypothesis (plural, hypotheses) is quite different from the terms mentioned above. It is an essential starting point for quantitative researchers, but takes a lesser role for qualitative researchers. The term is deeply rooted in the history of scientific thought. In statistics we expend much time and energy to generate hypotheses, to test hypotheses, and to reject them. Some people argue that we should only test one hypothesis; others say we should test several. In hypothesis testing we create a statement, which may be true or false, this statement is a “proposition”-we propose that something may be the case. If it is right then we accept it. If it is not right, if it is “wrong” then we reject it.

The first step is to formulate the null-hypothesis, abbreviated to H_0 . This is usually intended to be rejected. Another carefully constructed hypothesis is the alternative

hypothesis or the H1; this is actually called the ‘research hypothesis’. After these have been articulated the researcher can design a research programme to test the hypotheses. When the results are received, they are examined against the prediction of the null hypothesis. The basic idea is to use this possible explanation and then look for data to support the explanation (or not). It is best to spend as much time as possible on the hypothesis: it is the research question, and it determines how the study is carried out. It determines the design because it defines the problem.

The subject of hypothesis testing has been debated heavily for many years and there are suggestions that misuse of null hypothesis significance testing is widespread and damaging (Finch et al. 2001), at least in psychological research. It is relevant to cite an article on hypothesis testing in marketing research by Lawrence (1982) who tells us: “Practical survey researchers (realise) that, in many cases, no adequate theory exists for setting up hypotheses in advance”. The article continues: “Drawing one-off hypotheses out of the air offers no solution to the problem. Researchers will be guided by their own ideas, experiences, hunches”.

Population

When considering any market sector we need to ask “Who is involved in this marketplace?” Who are the players? Who should be the centre of the investigation and where are those subjects? This area considers the target audience, customer or player; the users or non-users. Who will become the respondent or informant? Should we contact all players or just some of them? Should we carry out a census or a sample: should respondents be selected by probability or non-probability methods?

An important concept for primary research is sampling. We choose to interview or observe people who we think will give us the information that will solve our problems. So in choosing our research method, we need to consider whom we select and how we select. This applies to qualitative research, with only a few people, and quantitative research with many people. Much emphasis in marketing research is on the end user, but “experts” can bridge the gap between primary and secondary data. An expert may be someone who has been in the business for many years. This part of the Mr. Mix involves identifying relevant sampling frames.

Procedure

When considering the procedure the key question is “How should the study be conducted?” Will it be qualitative or quantitative? This area also covers the question of timing: when will the fieldwork take place?

The best research starts by looking at secondary data, this information already exists. The two basic sources: internal (within an organisation) and external, published by someone outside have become easier to access in recent years. Information Technology, with Intranets and the Internet, has improved our ability to find such data. If secondary

data doesn't solve the problem then original data (primary data) is sought. It is useful to think of different primary methods in these terms: we can ask people what they are doing; we can watch them or detect what they have done by counting or we can manipulate some variables to discover the effect. This creates three categories: questioning; observation and experimentation.

Primary data collection techniques can be subdivided into: interviewer-administered or respondent-administered; direct or indirect; personal or impersonal. Processing data, analysis and interpretation are essential parts of the procedure. Detailed examination of the appropriate tools used in the data collection needs to take place.

Publication

Under the heading of "Publication" the key questions are: Who is the audience for the results? What should be communicated? When and how should they be communicated? Research is of no use if findings are kept within the research team; similarly commercially sensitive information will have no competitive advantage if placed in the public domain. Choices need to be made on how publication takes place. Will a written report be created? Will tabulations be provided? Will a personal presentation take place? Who should be allowed sight of the results?

There are many different readers of research reports and these audiences all have very different expectations. Reporting must be personalised, writing and presentation style must be customised and adapted to the user. At one extreme there is the general public.

There are many reasons why research is reported to the "mass consumer". It may be a government report that has been commissioned to be in the public interest: concerning health, welfare, transport and so on. It may be a consumer report: consumer watchdog reports are of great interest to the man on the street, so we find the Which? Magazine and similar bodies have enabled the layman appreciate survey findings. Editors of periodicals regularly commission research for editorial reasons, so the results may become part of an article for mass consumption. The research agency may report directly to the public on web pages, by e-mail or by post, this is because it is now common to offer a short summary report to a respondent as a gesture of goodwill, an incentive, a thank you for co-operating in the research. Research findings may appear as part of a promotional campaign, appealing to the consumer's need to know that indeed this is a best seller ("nine out of ten cat-owners prefer..."). Then there are smaller audiences such as managers who are anxious to receive a report in order to make instant decisions. Additionally there are managers who will benefit from the information much later when the report is consulted as secondary data in the future.

MYSTERY SHOPPING

Mystery shopping or a mystery consumer is a tool used by Mystery Shopping Providers and market research companies to measure quality of retail service or gather specific

information about products and services. Mystery Shopping should be performed by a person, so called Mystery Shopper, who is unknown to the establishment she/he is evaluating.

Mystery shoppers posing as normal customers perform specific tasks—such as purchasing a product, asking questions, registering complaints or behaving in a certain way—and then provide detailed reports or feedback about their experiences. Mystery shopping was standard practice by the early 1940s as a way to measure employee integrity.

Tools used for mystery shopping assessments range from simple questionnaires to complete audio and video recordings. Mystery shopping can be used in any industry, with the most common venues being retail stores, hotels, movie theatres, restaurants, fast food chains, banks, gas stations, car dealerships, apartments, health clubs and health care facilities. In the UK mystery shopping is increasingly used to provide feedback on customer services provided by local authorities, and other non-profit organizations such as housing associations and churches.

Methodology

When a client company hires a company providing mystery shopping services, a survey model will be drawn up and agreed to which defines what information and improvement factors the client company wishes to measure. These are then drawn up into survey instruments and assignments that are allocated to shoppers registered with the mystery shopping company. The details and information points shoppers take note of typically include:

- Number of employees in the store on entering
- How long it takes before the mystery shopper is greeted
- The name of the employees
- Whether or not the greeting is friendly, ideally according to objective measures
- The questions asked by the shopper to find a suitable product
- The types of products shown
- The sales arguments used by the employee
- Whether or how the employee attempted to close the sale
- Whether the employee suggested any add-on sales
- Whether the employee invited the shopper to come back to the store
- Cleanliness of store and store associates
- Speed of service
- Compliance with company standards relating to service, store appearance, and grooming/presentation.

Shoppers are often given instructions or procedures to make the transaction atypical to make the test of the knowledge and service skills of the employees more stringent or specific to a particular service issue (known as scenarios).

For instance, mystery shoppers at a restaurant may pretend they are lactose-intolerant, or a clothing store mystery shopper could inquire about gift-wrapping services.

Not all mystery shopping scenarios include a purchase. While gathering information, shoppers usually blend in to the store being evaluated as regular shoppers.

They may sometimes be required to take photographs or measurements, return purchases, or count the number of products, seats, people during the visit. A timer or a stopwatch may be required. In some states in the USA, mystery shoppers must also be licensed as private investigators in order to perform some of the tasks. After the visit the shopper submits the data collected to the mystery shopping company, which reviews and analyses the information, completing quantitative or qualitative statistical [analysis] reports on the data for the client company. This enables measurement against the previously defined criteria.

Statistics

The mystery shopping industry had an estimated value of nearly \$600 million in the United States in 2004, according to a 2005 report commissioned by the Mystery Shopping Providers Association (MSPA). Companies that participated in the report experienced an average growth of 11.1 percent from 2003 to 2004, compared to an average growth of 12.2 percent. The report estimates more than 8.1 million mystery shops were conducted in 2004. The report represents the first industry association attempt to quantify the size of the mystery shopping industry. Similar surveys are available for European regions where mystery shopping is becoming more embedded into company procedures. As a measure of its importance, customer/patient satisfaction is being incorporated more frequently into executive pay. A study by a U.S. firm found more than 55% of hospital chief executive officers surveyed in 2005 had “some compensation at risk,” based on patient satisfaction, up from only 8% to 20% a dozen years ago.”

CBC Television's news magazine programme Marketplace ran a segment on this topic during a January 2001 episode.

Ethics

The Trade Organization for Mystery Shopping Providers, MSPA has defined a Code of Professional Standards and Ethics Agreement for Mystery Shopping Providers and for Mystery Shoppers. MSPA has also defined Standards for Mystery Shopping. The Standard is available in a full version and an Abstract version. The Abstract is available in 32 languages. Other organizations that have defined standards for Mystery Shopping are; ESOMAR, MRS and MRA.

USA

In June 2008 the American Medical Association's Council on Ethical and Judicial Affairs released a recommendation on the use of “secret shopper patients”. The

Recommendation: “Physicians have an ethical responsibility to engage in activities that contribute to continual improvements in patient care. One method for promoting such quality improvement is through the use of secret shopper ‘patients’ who have been appropriately trained to provide feedback about physician performance in the clinical setting.” The most widely used set of professional guidelines and ethics standards for the Market Research industry is ISO 20252 ratified in 2006.

Scams

There is a fraudulent confidence trick (a form of advance fee fraud) perpetrated on people in several countries who wish to be mystery shoppers. A person is sent a money order, often from Western Union, or check for a larger sum than a mystery purchase they are required to make, with a request to deposit it into their bank account, use a portion for a mystery purchase and their fee, and wire the remainder through a wire transfer company such as Western Union or MoneyGram; the money is to be wired immediately as response time is being evaluated. The check is fraudulent, and is returned unpaid by the victim’s bank, after the money has been wired. One scam involved fraudulent websites using a misspelled URL to advertise online and in newspapers under a legitimate company’s name. Valid mystery shopping companies do not normally send their clients a check prior to work being completed, and their advertisements usually include a contact person and phone number. Some fraudulent checks can be identified by a financial professional. On February 3, 2009 The Internet Crime Complaint Center issued a warning on this scam. A legitimate company that occasionally sends prepayment for large transactions says “We do occasionally fund upfront for very large spend purchases but we use cheques or direct bank transfers which should mean you can see when they are cleared and so can be sure you really do have the money.”

CLASSIFICATION OF MARKETING RESEARCH

Organizations engage in marketing research for two reasons: (1) to identify and (2) solve marketing problems. This distinction serves as a basis for classifying marketing research into problem identification research and problem solving research.

Problem identification research is undertaken to help identify problems which are, perhaps, not apparent on the surface and yet exist or are likely to company image, market characteristics, sales analysis, short-range forecasting, long range forecasting, and business trends research. Research of this type provides information about the marketing environment and helps diagnose a problem. For example, The findings of problem solving research are used in making decisions which will solve specific marketing problems.

The Stanford Research Institute, on the other hand, conducts an annual survey of consumers that is used to classify persons into homogeneous groups for segmentation purposes. The National Purchase Diary panel (NPD) maintains the largest diary panel in the United States.

Standardized services are research studies conducted for different client firms but in a standard way. For example, procedures for measuring advertising effectiveness have been standardized so that the results can be compared across studies and evaluative norms can be established. The Starch Readership Survey is the most widely used service for evaluating print advertisements; another well-known service is the Gallup and Robinson Magazine Impact Studies. These services are also sold on a syndicated basis.

- Customized services offer a wide variety of marketing research services customized to suit a client's specific needs. Each marketing research project is treated uniquely.
- Limited-service suppliers specialize in one or a few phases of the marketing research project. Services offered by such suppliers are classified as field services, coding and data entry, data analysis, analytical services, and branded products. Field services collect data through mail, personal, or telephone interviewing, and firms that specialize in interviewing are called field service organizations. These organizations may range from small proprietary organizations which operate locally to large multinational organizations with WATS line interviewing facilities. Some organizations maintain extensive interviewing facilities across the country for interviewing shoppers in malls.
- Coding and data entry services include editing completed questionnaires, developing a coding scheme, and transcribing the data on to diskettes or magnetic tapes for input into the computer. NRC Data Systems provides such services.
- Analytical services include designing and pretesting questionnaires, determining the best means of collecting data, designing sampling plans, and other aspects of the research design. Some complex marketing research projects require knowledge of sophisticated procedures, including specialized experimental designs, and analytical techniques such as conjoint analysis and multidimensional scaling. This kind of expertise can be obtained from firms and consultants specializing in analytical services.
- Data analysis services are offered by firms, also known as tab houses, that specialize in computer analysis of quantitative data such as those obtained in large surveys. Initially most data analysis firms supplied only tabulations (frequency counts) and cross tabulations (frequency counts that describe two or more variables simultaneously). With the proliferation of software, many firms now have the capability to analyze their own data, but, data analysis firms are still in demand.
- Branded marketing research products and services are specialized data collection and analysis procedures developed to address specific types of marketing research problems. These procedures are patented, given brand names, and marketed like any other branded product.

THE FRANCHISE DILEMMA

The franchise dilemma is that franchisors have two distinct and different target markets, including:

1. The prospective franchisee
2. The end consumer.

Unlike most major retail or service outlets, the franchisor does not sell directly to an end consumer but through a franchising organization. The franchises are generally owned and operated by franchisees who actually run, manage, or supervise the outlet stores. It is, therefore, important for the franchisor to first develop a success profile for prospective franchisees.

Using Research to Create a Franchisee Profile

The successful franchisee profile is best developed by devising a preference research model. Such an approach involves research using both personal interviews and a general survey instrument. The personal interview is generally completed with existing successful franchisees. The purpose of the interview is to measure franchisees' attitudes towards their managing experience, as well as general demographic information. The franchisees selected for the study are generally among the top quartile or top third in sales or other success criteria as determined by the franchisors.

In addition to the personal interview, a general survey is made of all franchisees concerning what they believe would be a success profile for franchisees. This general survey is often developed by first using focus groups. The focus group generally includes 6-10 people gathered together for a few hours with a trained interviewer to talk about franchisees, managers, and successful management techniques.

These individuals may be the franchisor's franchisees, but they may also include other managers and owners of different types of businesses, as long as they have some understanding of management and ownership responsibilities. The trained interviewer generally begins with broad questions before moving to specific issues. The interviewer encourages open discussion, hoping that the interactions between the participants bring out actual experiences and recommendations for successful franchisees. These discussions are generally taped for future playback and study.

Once a focus group has determined critical factors that lead to successful franchisees, a survey of franchisees can begin. The purpose of the survey is to obtain primary data concerning attitudes towards the franchisor and the franchise system, and goals of the franchisee, as well as demographics. It is with this information that the franchisor can begin to understand where problems with relationships with their franchisees are occurring.

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